

$$(\nabla \times \mathbf{V}) \cdot \mathbf{n} = -\frac{d\Gamma}{dS}$$

Figure 2.39 Relation between vorticity and circulation.

irrotational everywhere within the contour of integration (i.e., if $\nabla \times \mathbf{V} = 0$ over any surface bounded by C), *then* $\Gamma = 0$. A related result is obtained by letting the curve C shrink to an infinitesimal size, and denoting the circulation around this infinitesimally small curve by $d\Gamma$. Then, in the limit as C becomes infinitesimally small, Equation (2.137) yields

$$d\Gamma = -(\nabla \times \mathbf{V}) \cdot \mathbf{dS} = -(\nabla \times \mathbf{V}) \cdot \mathbf{n}\, dS$$

or

$$\boxed{(\nabla \times \mathbf{V}) \cdot \mathbf{n} = -\frac{d\Gamma}{dS}} \tag{2.138}$$

where dS is the infinitesimal area enclosed by the infinitesimal curve C. Referring to Figure 2.39, Equation (2.138) states that at a point P in a flow, the component of vorticity normal to dS is equal to the negative of the "circulation per unit area," where the circulation is taken around the boundary of dS.

EXAMPLE 2.8

For the velocity field given in Example 2.4, calculate the circulation around a circular path of radius 5 m. Assume that u and v given in Example 2.4 are in units of meters per second.

■ Solution

Since we are dealing with a circular path, it is easier to work this problem in polar coordinates, where $x = r\cos\theta$, $y = r\sin\theta$, $x^2 + y^2 = r^2$, $V_r = u\cos\theta + v\sin\theta$, and $V_\theta = -u\sin\theta + v\cos\theta$. Therefore,

$$u = \frac{y}{x^2 + y^2} = \frac{r\sin\theta}{r^2} = \frac{\sin\theta}{r}$$

$$v = -\frac{x}{x^2 + y^2} = -\frac{r\cos\theta}{r^2} = -\frac{\cos\theta}{r}$$

$$V_r = \frac{\sin\theta}{r}\cos\theta + \left(-\frac{\cos\theta}{r}\right)\sin\theta = 0$$

$$V_\theta = -\frac{\sin\theta}{r}\sin\theta + \left(-\frac{\cos\theta}{r}\right)\cos\theta = -\frac{1}{r}$$

$$\mathbf{V} \cdot \mathbf{ds} = (V_r \mathbf{e}_r + V_\theta \mathbf{e}_\theta) \cdot (dr\, \mathbf{e}_r + r\, d\theta\, \mathbf{e}_\theta)$$

$$= V_r\, dr + r V_\theta\, d\theta = 0 + r \left(-\frac{1}{r} \right) d\theta = -d\theta$$

Hence,
$$\Gamma = -\oint_C \mathbf{V} \cdot \mathbf{ds} = -\int_0^{2\pi} -d\theta = \boxed{2\pi \text{ m}^2/\text{s}}$$

Note that we never used the 5-m diameter of the circular path; in this case, the value of Γ is independent of the diameter of the path.

2.14 STREAM FUNCTION

In this section, we consider two-dimensional steady flow. Recall from Section 2.11 that the differential equation for a streamline in such a flow is given by Equation (2.118), repeated below

$$\frac{dy}{dx} = \frac{v}{u} \tag{2.118}$$

If u and v are known functions of x and y, then Equation (2.118) can be integrated to yield the algebraic equation for a streamline:

$$f(x, y) = c \tag{2.139}$$

where c is an arbitrary constant of integration, with different values for different streamlines. In Equation (2.139), denote the function of x and y by the symbol $\bar{\psi}$. Hence, Equation (2.139) is written as

$$\bar{\psi}(x, y) = c \tag{2.140}$$

The function $\bar{\psi}(x, y)$ is called the *stream function*. From Equation (2.140) we see that the equation for a streamline is given by *setting the stream function equal to a constant* (i.e., c_1, c_2, c_3, etc). Two different streamlines are illustrated in Figure 2.40; streamlines ab and cd are given by $\bar{\psi} = c_1$ and $\bar{\psi} = c_2$, respectively.

There is a certain arbitrariness in Equations (2.139) and (2.140) via the arbitrary constant of integration c. Let us define the stream function more precisely in order to reduce this arbitrariness. Referring to Figure 2.40, let us define the

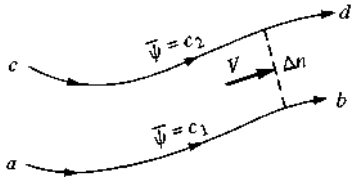


Figure 2.40 Different streamlines are given by different values of the stream function.

numerical value of $\bar{\psi}$ such that the *difference* $\Delta\bar{\psi}$ between $\bar{\psi} = c_2$ for streamline cd and $\bar{\psi} = c_1$ for streamline ab is equal to the *mass* flow between the two streamlines. Since Figure 2.40 is a two-dimensional flow, the mass flow between two streamlines is defined *per unit depth perpendicular to the page*. That is, in Figure 2.40 we are considering the mass flow inside a streamtube bounded by streamlines ab and cd, with a rectangular cross-sectional area equal to Δn times a unit depth perpendicular to the page. Here, Δn is the normal distance between ab and cd, as shown in Figure 2.40. Hence, mass flow between streamlines ab and cd per unit depth perpendicular to the page is

$$\Delta\bar{\psi} = c_2 - c_1 \tag{2.141}$$

The above definition does not completely remove the arbitrariness of the constant of integration in Equations (2.139) and (2.140), but it does make things a bit more precise. For example, consider a given two-dimensional flow field. Choose one streamline of the flow, and give it an arbitrary value of the stream function, say, $\bar{\psi} = c_1$. Then, the value of the stream function for any other streamline in the flow, say, $\bar{\psi} = c_2$, is fixed by the definition given in Equation (2.141). Which streamline you choose to designate as $\bar{\psi} = c_1$ and what numerical value you give c_1 usually depend on the geometry of the given flow field, as we see in Chapter 3.

The equivalence between $\bar{\psi} =$ constant designating a streamline, and $\Delta\bar{\psi}$ equaling mass flow (per unit depth) between streamlines, is natural. For a steady flow, the mass flow inside a given streamtube is constant along the tube; the mass flow across any cross section of the tube is the same. Since by definition $\Delta\bar{\psi}$ is equal to this mass flow, then $\Delta\bar{\psi}$ itself is constant for a given streamtube. In Figure 2.40, if $\bar{\psi}_1 = c_1$ designates the streamline on the bottom of the streamtube, then $\bar{\psi}_2 = c_2 = c_1 + \Delta\bar{\psi}$ is also constant along the top of the streamtube. Since by definition of a streamtube (see Section 2.11) the upper boundary of the streamtube is a streamline itself, then $\psi_2 = c_2 =$ constant must designate this streamline.

We have yet to develop the most important property of the stream function, namely, derivatives of $\bar{\psi}$ yield the flow-field velocities. To obtain this relationship, consider again the streamlines ab and cd in Figure 2.40. Assume that these streamlines are close together (i.e., assume Δn is small), such that the flow velocity V is a constant value across Δn. The mass flow through the streamtube per unit depth perpendicular to the page is

$$\Delta\bar{\psi} \equiv \rho V\, \Delta n(1)$$

or

$$\frac{\Delta\bar{\psi}}{\Delta n} = \rho V \tag{2.142}$$

Consider the limit of Equation (2.142) as $\Delta n \to 0$:

$$\rho V = \lim_{\Delta n \to 0} \frac{\Delta\bar{\psi}}{\Delta n} \equiv \frac{\partial\bar{\psi}}{\partial n} \tag{2.143}$$

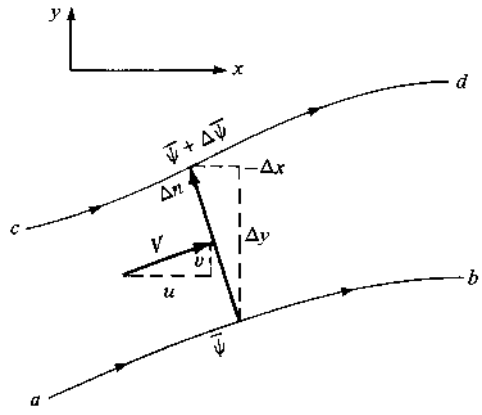


Figure 2.41 Mass flow through Δn is the sum of the mass flows through Δy and $-\Delta x$.

Equation (2.143) states that if we know $\bar{\psi}$, then we can obtain the product (ρV) by differentiating $\bar{\psi}$ in the direction *normal* to V. To obtain a practical form of Equation (2.143) for cartesian coordinates, consider Figure 2.41. Notice that the directed normal distance Δn is equivalent first to moving upward in the y direction by the amount Δy and then to the left in the negative x direction by the amount $-\Delta x$. Due to conservation of mass, the mass flow through Δn (per unit depth) is equal to the sum of the mass flows through Δy and $-\Delta x$ (per unit depth):

$$\text{Mass flow} = \Delta\bar{\psi} = \rho V\, \Delta n = \rho u\, \Delta y + \rho v(-\Delta x) \tag{2.144}$$

Letting cd approach ab, Equation (2.144) becomes in the limit

$$d\bar{\psi} = \rho u\, dy - \rho v\, dx \tag{2.145}$$

However, since $\bar{\psi} = \bar{\psi}(x, y)$, the chain rule of calculus states

$$d\bar{\psi} = \frac{\partial\bar{\psi}}{\partial x}dx + \frac{\partial\bar{\psi}}{\partial y}dy \tag{2.146}$$

Comparing Equations (2.145) and (2.146), we have

$$\boxed{\rho u = \frac{\partial\bar{\psi}}{\partial y}} \tag{2.147a}$$

$$\boxed{\rho v = -\frac{\partial\bar{\psi}}{\partial x}} \tag{2.147b}$$

Equations (2.147a and b) are important. If $\bar{\psi}(x, y)$ is known for a given flow field, then at any point in the flow the products ρu and ρv can be obtained by differentiating $\bar{\psi}$ in the directions normal to u and v, respectively.

If Figure 2.41 were to be redrawn in terms of polar coordinates, then a similar derivation yields

$$\boxed{\rho V_r = \frac{1}{r}\frac{\partial \bar{\psi}}{\partial \theta}} \tag{2.148a}$$

$$\boxed{\rho V_\theta = -\frac{\partial \bar{\psi}}{\partial r}} \tag{2.148b}$$

Such a derivation is left as a homework problem.

Note that the dimensions of $\bar{\psi}$ are equal to mass flow per unit depth perpendicular to the page. That is, in SI units, $\bar{\psi}$ is in terms of kilograms per second per meter perpendicular to the page, or simply kg/(s · m).

The stream function $\bar{\psi}$ defined above applies to both compressible and incompressible flow. Now consider the case of incompressible flow only, where $\rho =$ constant. Equation (2.143) can be written as

$$V = \frac{\partial(\bar{\psi}/\rho)}{\partial n} \tag{2.149}$$

We define a new stream function, for incompressible flow only, as $\psi \equiv \bar{\psi}/\rho$. Then Equation (2.149) becomes

$$V = \frac{\partial \psi}{\partial n}$$

and Equations (2.147) and (2.148) become

$$\boxed{u = \frac{\partial \psi}{\partial y}} \tag{2.150a}$$

$$\boxed{v = -\frac{\partial \psi}{\partial x}} \tag{2.150b}$$

and

$$\boxed{V_r = \frac{1}{r}\frac{\partial \psi}{\partial \theta}} \tag{2.151a}$$

$$\boxed{V_\theta = -\frac{\partial \psi}{\partial r}} \tag{2.151b}$$

The incompressible stream function ψ has characteristics analogous to its more general compressible counterpart $\bar{\psi}$. For example, since $\bar{\psi}(x, y) = c$ is the equation of a streamline, and since ρ is a constant for incompressible flow, then $\psi(x, y) \equiv \bar{\psi}/\rho =$ constant is also the equation for a streamline (for incompressible flow only). In addition, since $\Delta\bar{\psi}$ is mass flow between two streamlines (per unit depth perpendicular to the page), and since ρ is mass per unit volume, then physically $\Delta\psi = \Delta\bar{\psi}/\rho$ represents the *volume flow* (per unit depth) between two streamlines. In SI units, $\Delta\psi$ is expressed as cubic meters per second per meter perpendicular to the page, or simply m^2/s.

In summary, the concept of the stream function is a powerful tool in aerodynamics, for two primary reasons. Assuming that $\bar{\psi}(x, y)$ [or $\psi(x, y)$] is known through the two-dimensional flow field, then:

1. $\bar{\psi} = \text{constant}$ (or $\psi = \text{constant}$) gives the equation of a streamline.
2. The flow velocity can be obtained by differentiating $\bar{\psi}$ (or ψ), as given by Equations (2.147) and (2.148) for compressible flow and Equations (2.150) and (2.151) for incompressible flow. We have not yet discussed how $\bar{\psi}(x, y)$ [or $\psi(x, y)$] can be obtained in the first place; we are assuming that it is known. The actual determination of the stream function for various problems is discussed in Chapter 3.

2.15 VELOCITY POTENTIAL

Recall from Section 2.12 that an irrotational flow is defined as a flow where the vorticity is zero at every point. From Equation (2.129), for an irrotational flow,

$$\boldsymbol{\xi} = \nabla \times \mathbf{V} = 0 \tag{2.152}$$

Consider the following vector identity: if ϕ is a scalar function, then

$$\nabla \times (\nabla \phi) = 0 \tag{2.153}$$

that is, the curl of the gradient of a scalar function is identically zero. Comparing Equations (2.152) and (2.153), we see that

$$\boxed{\mathbf{V} = \nabla \phi} \tag{2.154}$$

Equation (2.154) states that for an *irrotational* flow, there exists a scalar function ϕ such that the velocity is given by the gradient of ϕ. We denote ϕ as the *velocity potential*. ϕ is a function of the spatial coordinates; that is, $\phi = \phi(x, y, z)$, or $\phi = \phi(r, \theta, z)$, or $\phi = \phi(r, \theta, \Phi)$. From the definition of the gradient in cartesian coordinates given by Equation (2.16), we have, from Equation (2.154),

$$u\mathbf{i} + v\mathbf{j} + w\mathbf{k} = \frac{\partial \phi}{\partial x}\mathbf{i} + \frac{\partial \phi}{\partial y}\mathbf{j} + \frac{\partial \phi}{\partial z}\mathbf{k} \tag{2.155}$$

The coefficients of like unit vectors must be the same on both sides of Equation (2.155). Thus, in cartesian coordinates,

$$\boxed{u = \frac{\partial \phi}{\partial x} \quad v = \frac{\partial \phi}{\partial y} \quad w = \frac{\partial \phi}{\partial z}} \tag{2.156}$$

In a similar fashion, from the definition of the gradient in cylindrical and spherical coordinates given by Equations (2.17) and (2.18), we have, in cylindrical coordinates,

$$V_r = \frac{\partial \phi}{\partial r} \quad V_\theta = \frac{1}{r}\frac{\partial \phi}{\partial \theta} \quad V_z = \frac{\partial \phi}{\partial z} \tag{2.157}$$

and in spherical coordinates,

$$V_r = \frac{\partial \phi}{\partial r} \quad V_\theta = \frac{1}{r}\frac{\partial \phi}{\partial \theta} \quad V_\Phi = \frac{1}{r \sin\theta}\frac{\partial \phi}{\partial \Phi} \tag{2.158}$$

The velocity potential is analogous to the stream function in the sense that derivatives of ϕ yield the flow-field velocities. However, there are distinct differences between ϕ and $\bar{\psi}$ (or ψ):

1. The flow-field velocities are obtained by differentiating ϕ in the same direction as the velocities [see Equations (2.156) to (2.158)], whereas $\bar{\psi}$ (or ψ) is differentiated normal to the velocity direction [see Equations (2.147) and (2.148), or Equations (2.150) and (2.151)].
2. The velocity potential is defined for irrotational flow only. In contrast, the stream function can be used in either rotational or irrotational flows.
3. The velocity potential applies to three-dimensional flows, whereas the stream function is defined for two-dimensional flows only.[2]

When a flow field is irrotational, hence allowing a velocity potential to be defined, there is a tremendous simplification. Instead of dealing with the velocity components (say, u, v, and w) as unknowns, hence requiring three equations for these three unknowns, we can deal with the velocity potential as one unknown, therefore requiring the solution of only one equation for the flow field. Once ϕ is known for a given problem, the velocities are obtained directly from Equations (2.156) to (2.158). This is why, in theoretical aerodynamics, we make a distinction between irrotational and rotational flows and why the analysis of irrotational flows is simpler than that of rotational flows.

Because irrotational flows can be described by the velocity potential ϕ, such flows are called *potential flows*.

In this section, we have not yet discussed how ϕ can be obtained in the first place; we are assuming that it is known. The actual determination of ϕ for various problems is discussed in Chapters 3, 6, 11, and 12.

EXAMPLE 2.9

Calculate the velocity potential for the flow field over a wavy wall given in Example 2.1.

■ Solution

From Example 2.1, the equations for u and v are given by

$$u = V_\infty \left[1 + \frac{h}{\beta}\frac{2\pi}{\ell}\left(\cos\frac{2\pi x}{\ell}\right) e^{-2\pi\beta y/\ell} \right] \tag{2.35}$$

[2] $\bar{\psi}$ (or ψ) can be defined for axisymmetric flows, such as the flow over a cone at zero degrees angle of attack. However, for such flows, only two spatial coordinates are needed to describe the flow field (see Chapter 6).

and

$$v = -V_\infty h \frac{2\pi}{\ell} \left(\sin \frac{2\pi x}{\ell} \right) e^{-2\pi\beta y/\ell} \tag{2.36}$$

From Equation (2.156),

$$u = \frac{\partial \phi}{\partial x} \quad \text{and} \quad v = \frac{\partial \phi}{\partial y}$$

To find an expression for ϕ, we first integrate u with respect to x and then integrate v with respect to y, as follows

$$\phi = \int \frac{\partial \phi}{\partial x} dx = \int u \, dx \tag{E2.13}$$

Substituting Equation (2.35) into (E2.13), we have

$$\phi = \int \left[V_\infty + \frac{V_\infty h}{\beta} \frac{2\pi}{\ell} \left(\cos \frac{2\pi x}{\ell} \right) e^{-2\pi\beta y/\ell} \right] dx$$

$$\phi = V_\infty x + \frac{V_\infty h}{\beta} \frac{2\pi}{\ell} \left(\sin \frac{2\pi x}{\ell} \right) \left(\frac{2\pi}{\ell} \right)^{-1} e^{-2\pi\beta y/\ell}$$

$$\phi = V_\infty x + \frac{V_\infty h}{\beta} \left(\sin \frac{2\pi x}{\ell} \right) e^{-2\pi\beta y/\ell} + f(y) \tag{E2.14}$$

The function of $f(y)$ is added to Equation (E2.14) in the spirit of a "constant of integration" since Equation (E2.13) is an integral just with respect to x. Indeed, since $u = \frac{\partial \phi}{\partial x}$, when Equation (E2.14) is differentiated with respect to x, the derivative of $f(y)$ with respect to x is zero, and we recover Equation (2.35) for u. Also, we have

$$\phi = \int \frac{\partial \phi}{\partial y} dy = \int v dy \tag{E2.15}$$

Substituting Equation (2.36) into (E2.15), we have

$$\phi = \int \left[-V_\infty h \frac{2\pi}{\ell} \left(\sin \frac{2\pi x}{\ell} \right) e^{-2\pi\beta y/\ell} \right] dy$$

$$\phi = -V_\infty h \frac{2\pi}{\ell} \left(\sin \frac{2\pi x}{\ell} \right) e^{-2\pi\beta y/\ell} \left(-\frac{\ell}{\partial \pi \beta} \right)$$

$$\phi = V_\infty \frac{h}{\beta} \left(\sin \frac{2\pi x}{\ell} \right) e^{-2\pi\beta y/\ell} + g(x) \tag{E2.16}$$

where $g(x)$ in Equation (E2.16) is again a "constant of integration" since Equation (E2.15) is an integral just with respect to y. When Equation (E2.16) is differentiated with respect to y, the derivative of $g(x)$ with respect to y is zero, and we recover Equation (2.36) for v.

Compare Equations (E2.14) and (E2.16). These are two equations for the same velocity potential. Hence, in Equation (E2.14), $f(y) = 0$ and in Equation (E2.16), $g(x) = V_\infty x$. Thus, the velocity potential for the flow over the wavy wall in Example 2.1 is

$$\phi = V_\infty x + \frac{V_\infty h}{\beta} \left(\sin \frac{2\pi x}{\ell} \right) e^{-2\pi\beta y/\ell} \tag{E2.17}$$

Recall that a velocity potential exists for any irrotational flow. In Example 2.7 we proved that the flow field over the wavy wall shown in Figure 2.17 as specified in Example 2.1 is irrotational. Hence a velocity potential exists for that flow. Equation (E2.17) is the velocity potential.

2.16 RELATIONSHIP BETWEEN THE STREAM FUNCTION AND VELOCITY POTENTIAL

In Section 2.15, we demonstrated that for an irrotational flow $\mathbf{V} = \nabla\phi$. At this stage, take a moment and review some of the nomenclature introduced in Section 2.2.5 for the gradient of a scalar field. We see that a line of constant ϕ is an isoline of ϕ; since ϕ is the velocity potential, we give this isoline a specific name, *equipotential line*. In addition, a line drawn in space such that $\nabla\phi$ is tangent at every point is defined as a gradient line; however, since $\nabla\phi = \mathbf{V}$, this gradient line is a *streamline*. In turn, from Section 2.14, a streamline is a line of constant $\bar{\psi}$ (for a two-dimensional flow). Because gradient lines and isolines are perpendicular (see Section 2.2.5, Gradient of a Scalar Field), then equipotential lines (ϕ = constant) and streamlines ($\bar{\psi}$ = constant) are mutually perpendicular.

To illustrate this result more clearly, consider a two-dimensional, irrotational, incompressible flow in cartesian coordinates. For a streamline, $\psi(x, y) =$ constant. Hence, the differential of ψ along the streamline is zero; that is,

$$d\psi = \frac{\partial\psi}{\partial x}dx + \frac{\partial\psi}{\partial y}dy = 0 \tag{2.159}$$

From Equation (2.150*a* and *b*), Equation (2.159) can be written as

$$d\psi = -v\,dx + u\,dy = 0 \tag{2.160}$$

Solve Equation (2.160) for dy/dx, which is the slope of the ψ = constant line, that is, the slope of the streamline:

$$\left(\frac{dy}{dx}\right)_{\psi=\text{const}} = \frac{v}{u} \tag{2.161}$$

Similarly, for an equipotential line, $\phi(x, y) =$ constant. Along this line,

$$d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy = 0 \tag{2.162}$$

From Equation (2.156), Equation (2.162) can be written as

$$d\phi = u\,dx + v\,dy = 0 \tag{2.163}$$

Solving Equation (2.163) for dy/dx, which is the slope of the ϕ = constant line (i.e., the slope of the equipotential line), we obtain

$$\left(\frac{dy}{dx}\right)_{\phi=\text{const}} = -\frac{u}{v} \tag{2.164}$$

Combining Equations (2.161) and (2.164), we have

$$\left(\frac{dy}{dx}\right)_{\psi=\text{const}} = -\frac{1}{(dy/dx)_{\phi=\text{const}}} \tag{2.165}$$

Equation (2.165) shows that the slope of a ψ = constant line is the negative reciprocal of the slope of a ϕ = constant line (i.e., streamlines and equipotential lines are mutually perpendicular).

2.17 HOW DO WE SOLVE THE EQUATIONS?

This chapter is full of mathematical equations—equations that represent the basic physical fundamentals that dictate the characteristics of aerodynamic flow fields. For the most part, the equations are either in partial differential form or integral form. These equations are powerful and by themselves represent a sophisticated intellectual construct of our understanding of the fundamentals of a fluid flow. However, the equations by themselves are not very practical. They must be *solved* in order to obtain the actual flow fields over specific body shapes with specific flow conditions. For example, if we are interested in calculating the flow field around a Boeing 777 jet transport flying at a velocity of 800 ft/s at an altitude of 30,000 ft, we have to obtain a *solution* of the governing equations for this case—a solution that will give us the results for the dependent flow-field variables p, ρ, $\mathbf{V}$, etc., as a function of the independent variables of spatial location and time. Then we have to squeeze this solution for extra practical information, such as lift, drag, and moments exerted on the vehicle. How do we do this? The purpose of the present section is to discuss two philosophical answers to this question. As for practical solutions to specific problems of interest, there are literally hundreds of different answers to this question, many of which make up the content of the rest of this book. However, all these solutions fall under one or the other of the two philosophical approaches described next.

2.17.1 Theoretical (Analytical) Solutions

Students learning any field of physical science or engineering at the beginning are usually introduced to nice, neat analytical solutions to physical problems that are simplified to the extent that such solutions are possible. For example, when Newton's second law is applied to the motion of a simple, frictionless pendulum, students in elementary physics classes are shown a closed-form analytical solution for the time period of the pendulum's oscillation, namely,

$$T = 2\pi\sqrt{\ell/g}$$

where T is the period, ℓ is the length of the pendulum, and g is the acceleration of gravity. However, a vital assumption behind this equation is that of *small amplitude* oscillations. Similarly, in studying the motion of a freely falling body in a gravitational field, the distance y through which the body falls in time t after

release is given by

$$y = \tfrac{1}{2} g t^2$$

However, this result neglects any aerodynamic drag on the body as it falls through the air. The above examples are given because they are familiar results from elementary physics. They are examples of theoretical, closed-form solutions—straightforward algebraic relations.

The governing equations of aerodynamics, such as the continuity, momentum, and energy equations derived in Sections 2.4, 2.5, and 2.7, respectively, are highly nonlinear, partial differential, or integral equations; to date, no general analytical solution to these equations has been obtained. In lieu of this, two different philosophies have been followed in obtaining useful solutions to these equations. One of these is the theoretical, or analytical, approach, wherein the physical nature of certain aerodynamic applications allows the governing equations to be simplified to a sufficient extent that analytical solutions of the simplified equations can be obtained. One such example is the analysis of the flow across a normal shock wave, as discussed in Chapter 8. This flow is one-dimensional, that is, the changes in flow properties across the shock take place only in the flow direction. For this case, the y and z derivatives in the governing continuity, momentum, and energy equations from Sections 2.4, 2.5, and 2.7 are identically zero, and the resulting one-dimensional equations, which are still *exact* for the one-dimensional case being considered, lend themselves to a direct analytical solution, which is indeed an exact solution for the shock wave properties. Another example is the compressible flow over an airfoil considered in Chapters 11 and 12. If the airfoil is thin and at a small angle of attack, and if the freestream Mach number is not near one (not transonic) nor above five (not hypersonic), then many of the terms in the governing equations are small compared to others and can be neglected. The resulting simplified equations are linear and can be solved analytically. This is an example of an *approximate* solution, where certain simplifying assumptions have been made in order to obtain a solution.

The history of the development of aerodynamic theory is in this category—the simplification of the full governing equations apropos a given application so that analytical solutions can be obtained. Of course this philosophy works for only a limited number of aerodynamic problems. However, classical aerodynamic theory is built on this approach and, as such, is discussed at some length in this book. You can expect to see a lot of closed-form analytical solutions in the subsequent chapters, along with detailed discussions of their limitations due to the approximations necessary to obtain such solutions. In the modern world of aerodynamics, such classical analytical solutions have three advantages:

1. The act of developing these solutions puts you in intimate contact with all the physics involved in the problem.
2. The results, usually in closed-form, give you direct information on what are the important variables, and how the answers vary with increases or decreases in these variables. For example, in Chapter 11 we will obtain a

simple equation for the compressibility effects on lift coefficient for an airfoil in high-speed subsonic flow. The equation, Equation (11.52), tells us that the high-speed effect on lift coefficient is governed by just M_∞ alone, and that as M_∞ increases, then the lift coefficient increases. Moreover, the equation tells us *in what way* the lift coefficient increases, namely, inversely with $(1 - M_\infty^2)^{1/2}$. This is powerful information, albeit approximate.

3. Finally, the results in closed-form provide simple tools for rapid calculations, making possible the proverbial "back of the envelope calculations" so important in the preliminary design process and in other practical applications.

2.17.2 Numerical Solutions—Computational Fluid Dynamics (CFD)

The other general approach to the solution of the governing equations is numerical. The advent of the modern high-speed digital computer in the last third of the twentieth century has revolutionized the solution of aerodynamic problems and has given rise to a whole new discipline—computational fluid dynamics. Recall that the basic governing equations of continuity, momentum, and energy derived in this chapter are either in integral or partial differential form. In Anderson, *Computational Fluid Dynamics: The Basics with Applications,* McGraw-Hill, 1995, computational fluid dynamics is defined as "the art of replacing the integrals or the partial derivatives (as the case may be) in these equations with discretized algebraic forms, which in turn are solved to obtain *numbers* for the flow field values at discrete points in time and/or space." The end product of computational fluid dynamics (frequently identified by the acronym CFD) is indeed a collection of numbers, in contrast to a closed-form analytical solution. However, in the long run, the objective of most engineering analyses, closed form or otherwise, is a quantitative description of the problem (i.e., *numbers*).

The beauty of CFD is that it can deal with the full non-linear equations of continuity, momentum, and energy, in principle, without resorting to any geometrical or physical approximations. Because of this, many complex aerodynamic flow fields have been solved by means of CFD which had never been solved before. An example of this is shown in Figure 2.42. Here we see the unsteady, viscous, turbulent, compressible, separated flow field over an airfoil at high angle of attack (14° in the case shown), as obtained from Reference 56. The freestream Mach number is 0.5, and the Reynolds number based on the airfoil chord length (distance from the front to the back edges) is 300,000. An instantaneous streamline pattern that exists at a certain instant in time is shown, reflecting the complex nature of the separated, recirculating flow above the airfoil. This flow is obtained by means of a CFD solution of the two-dimensional, unsteady continuity, momentum, and energy equations, including the full effects of viscosity and thermal conduction, as developed in Sections 2.4, 2.5, and 2.7, *without* any further geometrical or physical simplifications. The equations with all the viscosity and thermal conduction terms explicitly shown are developed in Chapter 15; in this form, they are frequently labeled as the *Navier-Stokes equations*. There is no analytical solution

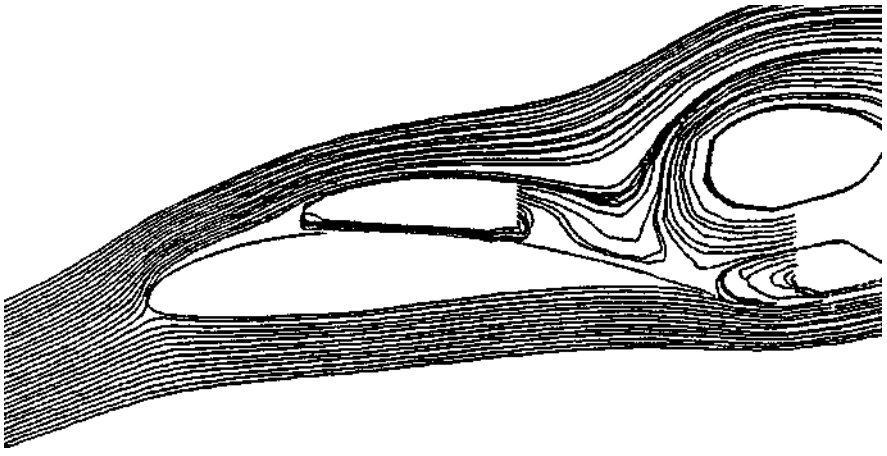


Figure 2.42 Calculated streamline pattern for separated flow over an airfoil. Re = 300,000, $M_\infty = 0.5$, angle of attack = 14°.

for the flow shown in Figure 2.42; the solution can only be obtained by means of CFD.

Let us explore the basic philosophy behind CFD. Again, keep in mind that CFD solutions are completely *numerical solutions,* and a high-speed digital computer must be used to carry them out. In a CFD solution, the flow field is divided into a number of *discrete* points. Coordinate lines through these points generate a *grid,* and the discrete points are called *grid points*. The grid used to solve the flow field shown in Figure 2.42 is given in Figure 2.43; here, the grid is wrapped around the airfoil, which is seen as the small white speck in the center-left of the figure, and the grid extends a very large distance out from the airfoil. This large extension of the grid into the main stream of the flow is necessary for a subsonic flow, because disturbances in a subsonic flow physically feed out large distances away from the body. We will learn why in subsequent chapters. The black region near the airfoil is simply the computer graphics way of showing a very large number of closely spaced grid points near the airfoil, for better definition of the viscous flow near the airfoil. The flow field properties, such as p, ρ, u, v, etc., are calculated just at the discrete grid points, and nowhere else, by means of the numerical solution of the governing flow equations. This is an inherent property that distinguishes CFD solutions from closed-form analytical solutions. Analytical solutions, by their very nature, yield closed-form equations that describe the flow as a function of continuous time and/or space. So we can pick any one of the infinite number of points located in this continuous space, feed the coordinates into the closed-form equations, and obtain the flow variables at that point. Not so with CFD, where the flow-field variables are calculated only at discrete grid points. For a CFD solution, the partial derivatives or the integrals, as the case may be, in the governing flow equations are *discretized,* using the flow-field variables at grid points only.

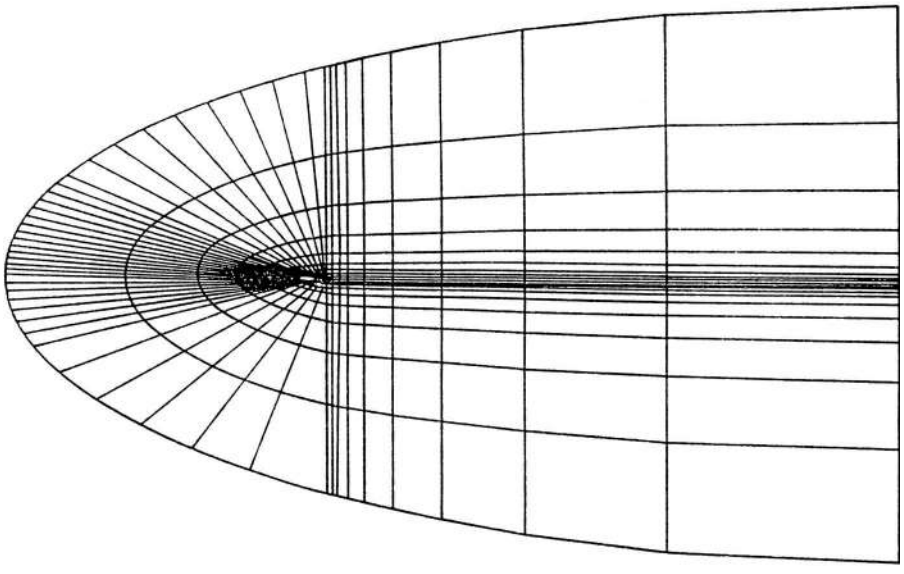


Figure 2.43 The grid used for the numerical solution of the flow over the airfoil in Figure 2.40. The airfoil is the small speck in the middle-left of the figure.

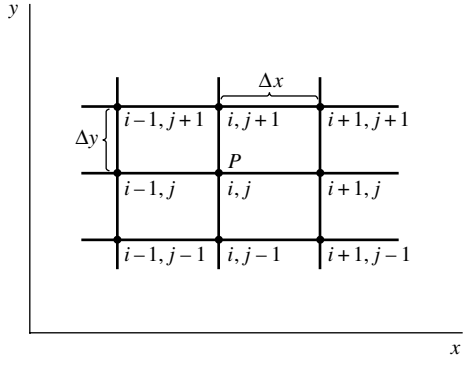


Figure 2.44 An array of grid points in a uniform, rectangular grid.

How is this discretization carried out? There are many answers to this equation. We will look at just a few examples, to convey the ideas.

Let us consider a partial derivative, such as $\partial u/\partial x$. How do we discretize this partial derivative? First, we choose a uniform rectangular array of grid points as shown in Figure 2.44. The points are identified by the index i in the x direction, and the index j in the y direction. Point P in Figure 2.44 is identified as point (i, j). The value of the variable u at point j is denoted by $u_{i,j}$. The value of u at the point immediately to the right of P is denoted by $u_{i+1,j}$ and that immediately

to the left of P by $u_{i-1,j}$. The values of u at the points immediately above and below point P are denoted by $u_{i,j+1}$ and $u_{i,j-1}$, respectively. And so forth. The grid points in the x direction are separated by the increment Δx, and the y direction by the increment Δy. The increments Δx and Δy must be uniform between the grid points, but Δx can be a different value than Δy. To obtain a discretized expression for $\partial u/\partial x$ evaluated at point P, we first write a Taylor series expansion for $u_{i+1,j}$ expanded about point P as:

$$u_{i+1,j} = u_{i,j} + \left(\frac{\partial u}{\partial x}\right)_{i,j} \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_{i,j} \frac{(\Delta x)^2}{2} + \left(\frac{\partial^3 u}{\partial x^3}\right)_{i,j} \frac{(\Delta x)^3}{6} + \left(\frac{\partial^4 u}{\partial x^4}\right)_{i,j} \frac{(\Delta x)^4}{24} + \cdots \tag{2.166}$$

Solving Equation (2.166) $(\partial u/\partial x)_{i,j}$, we have

$$\left(\frac{\partial u}{\partial x}\right)_{i,j} = \underbrace{\frac{u_{i+1,j} - u_{i,j}}{\Delta x}}_{\text{Forward difference}} - \underbrace{\left(\frac{\partial^2 u}{\partial x^2}\right)_{i,j} \frac{\Delta x}{2} - \left(\frac{\partial^3 u}{\partial x^3}\right)_{i,j} \frac{(\Delta x)^2}{6} + \cdots}_{\text{Truncation error}} \tag{2.167}$$

Equation (2.167) is still a mathematically exact relationship. However, if we choose to represent $(\partial u/\partial x)_{i,j}$ just by the algebraic term on the right-hand side, namely,

$$\left(\frac{\partial u}{\partial x}\right)_{i,j} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} \qquad \textit{Forward difference} \tag{2.168}$$

then Equation (2.168) represents an *approximation* for the partial derivative, where the error introduced by this approximation is the truncation error, identified in Equation (2.167). Nevertheless, Equation (2.168) gives us an algebraic expression for the partial derivative; the partial derivative has been *discretized* because it is formed by the values $u_{i+1,j}$ and $u_{i,j}$ at *discrete* grid points. The algebraic difference quotient in Equation (2.168) is called a *forward difference,* because it uses information ahead of point (i, j), namely $u_{i+1,j}$. Also, the forward difference given by Equation (2.168) has *first-order accuracy* because the leading term of the truncation error in Equation (2.167) has Δx to the first power.

Equation (2.168) is not the only discretized form for $(\partial u/\partial x)_{i,j}$. For example, let us write a Taylors series expansion for $u_{i-1,j}$ expanded about point P as

$$u_{i-1,j} = u_{i,j} + \left(\frac{\partial u}{\partial x}\right)_{i,j} (-\Delta x) + \left(\frac{\partial^2 u}{\partial x^2}\right)_{i,j} \frac{(-\Delta x)^2}{2} + \left(\frac{\partial^3 u}{\partial x^3}\right)_{i,j} \frac{(-\Delta x)^3}{6} + \cdots \tag{2.169}$$

Solving Equation (2.169) for $(\partial u/\partial x)_{i,j}$, we have

$$\left(\frac{\partial u}{\partial x}\right)_{i,j} = \underbrace{\frac{u_{i,j} - u_{i-1,j}}{\Delta x}}_{\text{Rearward difference}} + \underbrace{\left(\frac{\partial^2 u}{\partial x^2}\right)_{i,j} \frac{\Delta x}{2} - \left(\frac{\partial^3 u}{\partial x^3}\right)_{i,j} \frac{(\Delta x)^2}{6} + \cdots}_{\text{Truncation error}} \tag{2.170}$$

Hence, we can represent the partial derivative by the rearward difference shown in Equation (2.170), namely,

$$\left(\frac{\partial u}{\partial x}\right)_{i,j} = \frac{u_{i,j} - u_{i-1,j}}{\Delta x} \qquad \textit{Rearward difference} \tag{2.171}$$

Equation (2.171) is an approximation for the partial derivative, where the error is given by the truncation error labeled in Equation (2.170). The rearward difference given by Equation (2.171) has first-order accuracy because the leading term in the truncation error in Equation (2.170) has Δx to the first power. The forward and rearward differences given by Equations (2.168) and (2.171), respectively, are equally valid representations of $(\partial u/\partial x)_{i,j}$, each with first-order accuracy.

In most CFD solutions, first-order accuracy is not good enough; we need a discretization of $(\partial u/\partial x)_{i,j}$ that has at least second-order accuracy. This can be obtained by subtracting Equation (2.169) from Equation (2.166), yielding

$$u_{i+1,j} - u_{i-1,j} = 2\left(\frac{\partial u}{\partial x}\right)_{i,j} \Delta x + \left(\frac{\partial^3 u}{\partial x^3}\right)_{i,j} \frac{(\Delta x)^3}{3} + \cdots \tag{2.172}$$

Solving Equation (2.172) for $(\partial u/\partial x)_{i,j}$, we have

$$\left(\frac{\partial u}{\partial x}\right)_{i,j} = \underbrace{\frac{u_{i+1,j} - u_{i-1,j}}{2\Delta x}}_{\text{Central difference}} - \underbrace{\left(\frac{\partial^3 u}{\partial x^3}\right)_{i,j} \frac{(\Delta x)^2}{3} + \cdots}_{\text{Truncation error}} \tag{2.173}$$

Hence, we can represent the partial derivative by the *central difference* shown in Equation (2.173), namely

$$\left(\frac{\partial u}{\partial x}\right)_{i,j} = \frac{u_{i+1,j} - u_{i-1,j}}{2\Delta x} \qquad \textit{Central difference} \tag{2.174}$$

Examining Equation (2.173), we see that the central difference expression given in Equation (2.174) has *second-order accuracy,* because the leading term in the truncation error in Equation (2.173) has $(\Delta x)^2$. For most CFD solutions, second-order accuracy is sufficient.

So this is how it works—this is how the partial derivatives that appear in the governing flow equations can be discretized. There are many other possible discretized forms for the derivatives; the forward, rearward, and central differences obtained above are just a few. Note that Taylor series have been used to obtain these discrete forms. Such Taylor series expressions are the basic foundation of *finite-difference* solutions in CFD. In contrast, if the integral form of

the governing flow equations are used, such as Equations (2.48) and (2.64), the individual integrals terms can be discretized, leading again to algebraic equations that are the basic foundation of *finite-volume* solutions in CFD.

EXAMPLE 2.10

Consider a one-dimensional, unsteady flow, where the flow-field variables such as ρ, u, etc. are functions of distance x and time t. Consider the grid shown in Figure 2.45, where grid points arrayed in the x direction are denoted by the index i. Two rows of grid points are shown, one at time t and the other at the later time $t + \Delta t$. In particular, we are interested in calculating the unknown density at grid point i at time $t + \Delta t$, denoted by $\rho_i^{t+\Delta t}$. Set up the calculation of this unknown density.

■ Solution

Note in Figure 2.45 that the dashed loop (called the computational module) contain the grid points $i - 1$, i, and $i + 1$ at time t, where the flow field is known, and the grid point i at time $t + \Delta t$, where the flow field is unknown. From the continuity equation, Equation (2.52), repeated below

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0 \tag{2.52}$$

written for unsteady, one-dimensional flow, we have

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} = 0 \tag{2.175}$$

Rearranging Equation (2.175),

$$\frac{\partial \rho}{\partial t} = -\frac{\partial (\rho u)}{\partial x}$$

or

$$\frac{\partial \rho}{\partial t} = -\rho \frac{\partial u}{\partial x} - u \frac{\partial \rho}{\partial x} \tag{2.176}$$

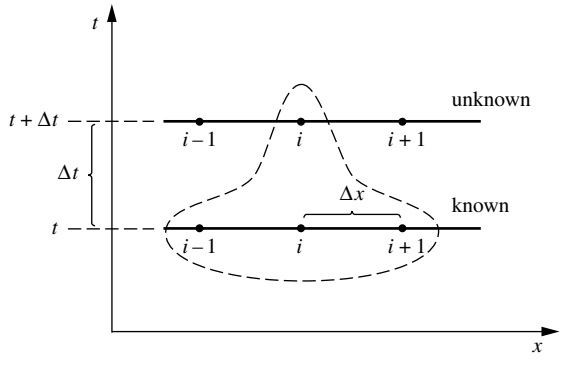


Figure 2.45 Computational module. The calculation of unknown properties at point i at time $(t + \Delta t)$ from known properties at points $i - 1, i, i + 1$ at time t.

In Equation (2.176), replace $\partial\rho/\partial t$ with a forward difference in time, and $\partial u/\partial x$ and $\partial\rho/\partial x$ with central differences in space, centered around grid point i

$$\frac{\rho_i^{t+\Delta t} - \rho_i^t}{\Delta t} = -\rho_i^t \left(\frac{u_{i+1}^t - u_{i-1}^t}{2\Delta x} \right) - u_i^t \left(\frac{\rho_{i+1}^t - \rho_{i-1}^t}{2\Delta x} \right) \tag{2.177}$$

Equation (2.177) is called a *difference equation;* it is an approximate representation of the original partial differential equation, Equation (2.176), where the error in the approximation is given by the sum of the truncation errors associated with each of the finite differences used to obtain Equation (2.177). Solving Equation (2.177) for $\rho_i^{t+\Delta t}$

$$\rho_i^{t+\Delta t} = \rho_i^t - \frac{\Delta t}{2\Delta x}\left(\rho_i^t u_{i+1}^t - \rho_i^t u_{i-1}^t + u_i^t \rho_{i+1}^t - u_i^t \rho_{i-1}^t\right) \tag{2.178}$$

In Equation (2.178), all quantities on the right-hand side are known values at time t. Hence, Equation (2.178) allows the direct calculation of the unknown value, $\rho_i^{t+\Delta t}$, at time $t + \Delta t$.

This example is a simple illustration of how a CFD solution to a given flow can be set up, in this case for an unsteady, one-dimensional flow. Note that the unknown velocity and internal energy at grid point i at time $t + \Delta t$ can be calculated in the same manner, writing the appropriate difference equation representations for the x component of the momentum equation, Equation (2.113a), and the energy equation, Equation (2.114).

The above example looks very straightforward, and indeed it is. It is given here only as an illustration of what is meant by a CFD technique. However, do not be misled. Computational fluid dynamics is a sophisticated and complex discipline. For example, we have said nothing here about the accuracy of the final solutions, whether or not a certain computational technique will be stable (some attempts at obtaining numerical solutions will go unstable—blow up—during the course of the calculations), and how much computer time a given technique will require to obtain the flow-field solution. Also, in our discussion we have given examples of some relatively simple grids. The generation of an appropriate grid for a given flow problem is frequently a challenge, and grid generation has emerged as a subdiscipline in its own right within CFD. For these reasons, CFD is usually taught only in graduate-level courses in most universities. However, in an effort to introduce some of the basic ideas of CFD at the undergraduate level, I have written a book, Reference 7, intended to present the subject at the most elementary level. Reference 7 is intended to be read *before* students go on to the more advanced books on CFD written at the graduate level. In the present book, we will, from time to time, discuss some applications of CFD as part of the overall fundamentals of aerodynamics. However, this book is not about CFD; Reference 7 is.

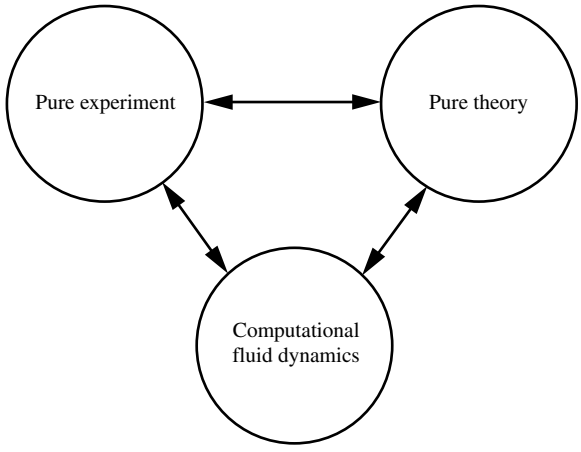


Figure 2.46 The three equal partners of modern aerodynamics.

2.17.3 The Bigger Picture

The evolution of our intellectual understanding of aerodynamics is over 2500 years old, going all the way back to ancient Greek science. The aerodynamics you are studying in this book is the product of this evolution. (See Reference 62 for an in-depth study of the history of aerodynamics.) Relevant to our current discussion is the development of the experimental tradition in fluid dynamics, which took place in the middle of the seventeenth century, principally in France, and the introduction of rational analysis in mechanics pioneered by Isaac Newton towards the end of the same century. Since that time, up until the middle of the twentieth century, the study and practice of fluid dynamics, including aerodynamics, has dealt with pure experiment on one hand and pure theory on the other. If you were learning aerodynamics as recently as, say 1960, you would have been operating in the "two-approach world" of theory and experiment. However, computational fluid dynamics has revolutionized the way we study and practice aerodynamics today. As sketched in Figure 2.46, CFD is today an *equal* partner with pure theory and pure experiment in the analysis and solution of aerodynamic problems. This is no flash in the pan—CFD will continue to play this role indefinitely, for as long as our advanced human civilization exists. Also, the double arrows in Figure 2.46 imply that today each of the equal partners constantly interact with each other—they do not stand alone, but rather help each other to continue to resolve and better understand the "big picture" of aerodynamics.

2.18 SUMMARY

Return to the road map for this chapter, as given in Figure 2.3. We have now covered both the left and right branches of this map and are ready to launch into the solution of practical aerodynamic problems in subsequent chapters. Look at

each block in Figure 2.3; let your mind flash over the important equations and concepts represented by each block. If the flashes are dim, return to the appropriate sections of this chapter and review the material until you feel comfortable with these aerodynamic tools.

For your convenience, the most important results are summarized below:

Basic Flow Equations

Continuity equation

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \, d\mathcal{V} + \oiint_{S} \rho \mathbf{V} \cdot \mathbf{dS} = 0 \tag{2.48}$$

or

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0 \tag{2.52}$$

or

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{V} = 0 \tag{2.108}$$

Momentum equation

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \mathbf{V} \, d\mathcal{V} + \oiint_{S} (\rho \mathbf{V} \cdot \mathbf{dS})\mathbf{V} = -\oiint_{S} p \, \mathbf{dS} + \iiint_{\mathcal{V}} \rho \mathbf{f} \, d\mathcal{V} + \mathbf{F}_{\text{viscous}} \tag{2.64}$$

or

$$\frac{\partial (\rho u)}{\partial t} + \nabla \cdot (\rho u \mathbf{V}) = -\frac{\partial p}{\partial x} + \rho f_x + (\mathcal{F}_x)_{\text{viscous}} \tag{2.70a}$$

$$\frac{\partial (\rho v)}{\partial t} + \nabla \cdot (\rho v \mathbf{V}) = -\frac{\partial p}{\partial y} + \rho f_y + (\mathcal{F}_y)_{\text{viscous}} \tag{2.70b}$$

$$\frac{\partial (\rho w)}{\partial t} + \nabla \cdot (\rho w \mathbf{V}) = -\frac{\partial p}{\partial z} + \rho f_z + (\mathcal{F}_z)_{\text{viscous}} \tag{2.70c}$$

or

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \rho f_x + (\mathcal{F}_x)_{\text{viscous}} \tag{2.113a}$$

$$\rho \frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \rho f_y + (\mathcal{F}_y)_{\text{viscous}} \tag{2.113b}$$

$$\rho \frac{Dw}{Dt} = -\frac{\partial p}{\partial z} + \rho f_z + (\mathcal{F}_z)_{\text{viscous}} \tag{2.113c}$$

(*continued*)

Energy equation

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho\left(e+\frac{V^2}{2}\right)d\mathcal{V} + \iint_{S} \rho\left(e+\frac{V^2}{2}\right)\mathbf{V}\cdot\mathbf{dS} \tag{2.95}$$
$$= \iiint_{\mathcal{V}} \dot{q}\rho\, d\mathcal{V} + \dot{Q}_{\text{viscous}} - \iint_{S} p\mathbf{V}\cdot\mathbf{dS} + \iiint_{\mathcal{V}} \rho(\mathbf{f}\cdot\mathbf{V})\, d\mathcal{V} + \dot{W}_{\text{viscous}}$$

or

$$\frac{\partial}{\partial t}\left[\rho\left(e+\frac{V^2}{2}\right)\right] + \nabla\cdot\left[\rho\left(e+\frac{V^2}{2}\right)\mathbf{V}\right] = \rho\dot{q} - \nabla\cdot(p\mathbf{V}) + \rho(\mathbf{f}\cdot\mathbf{V}) + \dot{Q}'_{\text{viscous}} + \dot{W}'_{\text{viscous}} \tag{2.96}$$

or

$$\rho\frac{D(e+V^2/2)}{Dt} = \rho\dot{q} - \nabla\cdot(p\mathbf{V}) + \rho(\mathbf{f}\cdot\mathbf{V}) + \dot{Q}'_{\text{viscous}} + \dot{W}'_{\text{viscous}} \tag{2.114}$$

Substantial derivative

$$\frac{D}{Dt} \equiv \underbrace{\frac{\partial}{\partial t}}_{\substack{\text{local}\\ \text{derivative}}} + \underbrace{(\mathbf{V}\cdot\nabla)}_{\substack{\text{convective}\\ \text{derivative}}} \tag{2.109}$$

A streamline is a curve whose tangent at any point is in the direction of the velocity vector at that point. The equation of a streamline is given by

$$\mathbf{ds}\times\mathbf{V} = 0 \tag{2.115}$$

or, in cartesian coordinates,

$$w\,dy - v\,dz = 0 \tag{2.117a}$$
$$u\,dz - w\,dx = 0 \tag{2.117b}$$
$$v\,dx - u\,dy = 0 \tag{2.117c}$$

The vorticity $\boldsymbol{\xi}$ at any given point is equal to twice the angular velocity of a fluid element $\boldsymbol{\omega}$, and both are related to the velocity field by

$$\boldsymbol{\xi} = 2\boldsymbol{\omega} = \nabla\times\mathbf{V} \tag{2.129}$$

When $\nabla\times\mathbf{V} \neq 0$, the flow is rotational. When $\nabla\times\mathbf{V} = 0$, the flow is irrotational.

Circulation Γ is related to lift and is defined as

$$\Gamma \equiv -\oint_C \mathbf{V} \cdot \mathbf{ds} \tag{2.136}$$

Circulation is also related to vorticity via

$$\Gamma \equiv -\oint_C \mathbf{V} \cdot \mathbf{ds} = -\iint_S (\nabla \times \mathbf{V}) \cdot \mathbf{dS} \tag{2.137}$$

or

$$(\nabla \times \mathbf{V}) \cdot \mathbf{n} = -\frac{d\Gamma}{dS} \tag{2.138}$$

The stream function $\bar{\psi}$ is defined such that $\bar{\psi}(x, y) =$ constant is the equation of a streamline, and the difference in the stream function between two streamlines $\Delta\bar{\psi}$ is equal to the mass flow between the streamlines. As a consequence of this definition, in cartesian coordinates,

$$\rho u = \frac{\partial \bar{\psi}}{\partial y} \tag{2.147a}$$

$$\rho v = -\frac{\partial \bar{\psi}}{\partial x} \tag{2.147b}$$

and in cylindrical coordinates,

$$\rho V_r = \frac{1}{r}\frac{\partial \bar{\psi}}{\partial \theta} \tag{2.148a}$$

$$\rho V_\theta = -\frac{\partial \bar{\psi}}{\partial r} \tag{2.148b}$$

For incompressible flow, $\psi \equiv \bar{\psi}/\rho$ is defined such that $\psi(x, y) =$ constant denotes a streamline and $\Delta\psi$ between two streamlines is equal to the volume flow between these streamlines. As a consequence of this definition, in cartesian coordinates.

$$u = \frac{\partial \psi}{\partial y} \tag{2.150a}$$

$$v = -\frac{\partial \psi}{\partial x} \tag{2.150b}$$

and in cylindrical coordinates

$$V_r = \frac{1}{r}\frac{\partial \psi}{\partial \theta} \tag{2.151a}$$

$$V_\theta = -\frac{\partial \psi}{\partial r} \tag{2.151b}$$

The stream function is valid for both rotational and irrotational flows, but it is restricted to two-dimensional flows only.

The velocity potential ϕ is defined for irrotational flows only, such that

$$\mathbf{V} = \nabla\phi \tag{2.154}$$

In cartesian coordinates

$$u = \frac{\partial\phi}{\partial x} \quad v = \frac{\partial\phi}{\partial y} \quad w = \frac{\partial\phi}{\partial z} \tag{2.156}$$

In cylindrical coordinates,

$$V_r = \frac{\partial\phi}{\partial r} \quad V_\theta = \frac{1}{r}\frac{\partial\phi}{\partial\theta} \quad V_z = \frac{\partial\phi}{\partial z} \tag{2.157}$$

In spherical coordinates,

$$V_r = \frac{\partial\phi}{\partial r} \quad V_\theta = \frac{1}{r}\frac{\partial\phi}{\partial\theta} \quad V_\Phi = \frac{1}{r\sin\theta}\frac{\partial\phi}{\partial\Phi} \tag{2.158}$$

An irrotational flow is called a potential flow.

A line of constant ϕ is an equipotential line. Equipotential lines are perpendicular to streamlines (for two-dimensional irrotational flows).

2.19 PROBLEMS

2.1 Consider a body of arbitrary shape. If the pressure distribution over the surface of the body is constant, prove that the resultant pressure force on the body is zero. [Recall that this fact was used in Equation (2.77).]

2.2 Consider an airfoil in a wind tunnel (i.e., a wing that spans the entire test section). Prove that the lift per unit span can be obtained from the pressure distributions on the top and bottom walls of the wind tunnel (i.e., from the pressure distributions on the walls above and below the airfoil).

2.3 Consider a velocity field where the x and y components of velocity are given by $u = cx/(x^2 + y^2)$ and $v = cy/(x^2 + y^2)$ where c is a constant. Obtain the equations of the streamlines.

2.4 Consider a velocity field where the x and y components of velocity are given by $u = cy/(x^2 + y^2)$ and $v = -cx/(x^2 + y^2)$, where c is a constant. Obtain the equations of the streamlines.

2.5 Consider a velocity field where the radial and tangential components of velocity are $V_r = 0$ and $V_\theta = cr$, respectively, where c is a constant. Obtain the equations of the streamlines.

2.6 Consider a velocity field where the x and y components of velocity are given by $u = cx$ and $v = -cy$, where c is a constant. Obtain the equations of the streamlines.

2.7 The velocity field given in Problem 2.3 is called *source flow,* which will be discussed in Chapter 3. For source flow, calculate:

a. The time rate of change of the volume of a fluid element per unit volume.

b. The vorticity.

Hint: It is simpler to convert the velocity components to polar coordinates and deal with a polar coordinate system.

2.8 The velocity field given in Problem 2.4 is called *vortex flow,* which will be discussed in Chapter 3. For vortex flow, calculate:

a. The time rate of change of the volume of a fluid element per unit volume.

b. The vorticity.

Hint: Again, for convenience use polar coordinates.

2.9 Is the flow field given in Problem 2.5 irrotational? Prove your answer.

2.10 Consider a flow field in polar coordinates, where the stream function is given as $\psi = \psi(r, \theta)$. Starting with the concept of mass flow between two streamlines, derive Equations (2.148*a* and *b*).

2.11 Assuming the velocity field given in Problem 2.6 pertains to an incompressible flow, calculate the stream function and velocity potential. Using your results, show that lines of constant ϕ are perpendicular to lines of constant ψ.

2.12 Consider a length of pipe bent into a U-shape. The inside diameter of the pipe is 0.5 m. Air enters one leg of the pipe at a mean velocity of 100 m/s and exits the other leg at the same magnitude of velocity, but moving in the opposite direction. The pressure of the flow at the inlet and exit is the ambient pressure of the surroundings. Calculate the magnitude and direction of the force exerted on the pipe by the airflow. The air density is 1.23 kg/m^3.

2.13 Consider the subsonic compressible flow over the wavy wall treated in Example 2.1. Derive the equation for the velocity potential for this flow as a function of x and y.

2.14 In Example 2.1, the statement is made that the streamline an infinite distance above the wall is straight. Prove this statement.

PART 2

Inviscid, Incompressible Flow

In Part 2, we deal with the flow of a fluid that has constant density—incompressible flow. This applies to the flow of liquids, such as water flow, and to low-speed flow of gases. The material covered here is applicable to low-speed flight through the atmosphere—flight at a Mach number of about 0.3 or less. ■

CHAPTER 3

Fundamentals of Inviscid, Incompressible Flow

Theoretical fluid dynamics, being a difficult subject, is for convenience, commonly divided into two branches, one treating of frictionless or perfect fluids, the other treating of viscous or imperfect fluids. The frictionless fluid has no existence in nature, but is hypothesized by mathematicians in order to facilitate the investigation of important laws and principles that may be approximately true of viscous or natural fluids.

Albert F. Zahm, 1912
(Professor of aeronautics, and developer of the first aeronautical laboratory in a U.S. university, The Catholic University of America)

PREVIEW BOX

Here we go again—fundamentals, and yet more fundamentals. However, in this chapter we focus on the fundamentals of a specific class of flow—inviscid, incompressible flow. Actually, such flow is a myth on two accounts. First, in real life there is always friction to some greater or lesser degree, so in nature there is, strictly speaking, no inviscid flow. Second, every flow is compressible to some greater or lesser degree, so in nature there is, strictly speaking, no incompressible flow. In engineering, however, if we were always "strictly speaking" about everything, we would not be able to analyze anything. As you progress in your studies, you will find that virtually every engineering analysis involves approximations about the physics involved.

In regard to the material in this chapter, there are a whole host of aerodynamic applications that are *so close* to being inviscid and incompressible that we readily make that assumption and obtain amazingly accurate results. Here you will find out how low-speed wind tunnels work. You will discover how to measure the velocity of a low-speed flow using a basic

instrument called a *Pitot tube*. Pitot tubes are installed on virtually all airplanes to measure their flight velocity—something as important and basic as having the speedometer in your automobile. You will discover why a baseball curves when the pitcher puts spin on it, and why a golf ball will sometimes hook or slice when a golfer hits it.

And there is a lot more. In the category of fundamentals, you will be introduced to Bernoulli's equation, far and away the most famous equation in fluid dynamics; it relates velocity and pressure from one point to another in an inviscid, incompressible flow. You will learn how to calculate the flow over a circular body shape, and how to calculate and plot the precise streamline shapes for this flow—something that is kind of fun to do, and that has importance further down the line in our study of aerodynamics.

So this chapter is a mixture of fundamentals and applications. Add these fundamentals to your growing inventory of basic concepts, and enjoy the applications.

3.1 INTRODUCTION AND ROAD MAP

The world of practical aviation was born on December 17, 1903, when, at 10:35 A.M., and in the face of cold, stiff, dangerous winds, Orville Wright piloted the Wright Flyer on its historic 12-s, 120-ft first flight. Figure 3.1 shows a photograph of the Wright Flyer at the instant of lift-off, with Wilbur Wright running along the right side of the machine, supporting the wing tip so that it will not drag the sand. This photograph is the most important picture in aviation history; the event it depicts launched the profession of aeronautical engineering into the mainstream of the twentieth century.[1]

The flight velocity of the Wright Flyer was about 30 mi/h. Over the ensuing decades, the flight velocities of airplanes steadily increased. By means of more powerful engines and attention to drag reduction, the flight velocities of airplanes rose to approximately 300 mi/h just prior to World War II. Figure 3.2 shows a typical fighter airplane of the immediate pre-World War II era. From an aerodynamic point of view, at air velocities between 0 and 300 mi/h the air density remains essentially constant, varying by only a few percent. Hence, the aerodynamics of the family of airplanes spanning the period between the two photographs shown in Figures 3.1 and 3.2 could be described by *incompressible flow*. As a result, a huge bulk of experimental and theoretical aerodynamic results was acquired over the 40-year period beginning with the Wright Flyer—results that applied to incompressible flow. Today, we are still very interested in incompressible aerodynamics because most modern general aviation aircraft still fly at speeds below 300 mi/h; a typical light general aviation airplane is shown in Figure 3.3. In addition to low-speed aeronautical applications, the principles of incompressible flow apply to the flow of fluids, for example, water flow through pipes, the motion of submarines and ships through the ocean, the design of wind turbines (the modern term for windmills), and many other important applications.

[1] See Reference 2 for historical details leading to the first flight by the Wright brothers.

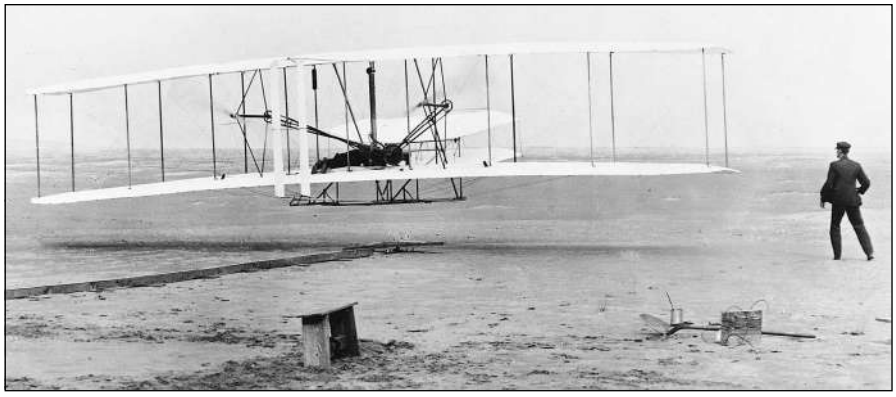


Figure 3.1 Historic photograph of the first successful heavier-than-air powered manned flight, achieved by the Wright brothers on December 17, 1903.



Figure 3.2 The Seversky P-35. *(Courtesy of the U.S. Air Force.)*

For all the above reasons, the study of incompressible flow is as relevant today as it was at the time of the Wright brothers. Therefore, Chapters 3 to 6 deal exclusively with incompressible flow. Moreover, for the most part, we ignore any effects of friction, thermal conduction, or diffusion; that is, we deal with *inviscid* incompressible flow in these chapters.[2] Looking at our spectrum of aerodynamic

[2] An inviscid, incompressible fluid is sometimes called an *ideal fluid,* or *perfect fluid.* This terminology will not be used here because of the confusion it sometimes causes with "ideal gases" or "perfect gases" from thermodynamics. This author prefers to use the more precise descriptor "inviscid, incompressible flow," rather than ideal fluid or perfect fluid.



Figure 3.3 Cessna Model 425 Conquest © Digital Vision/Getty RF

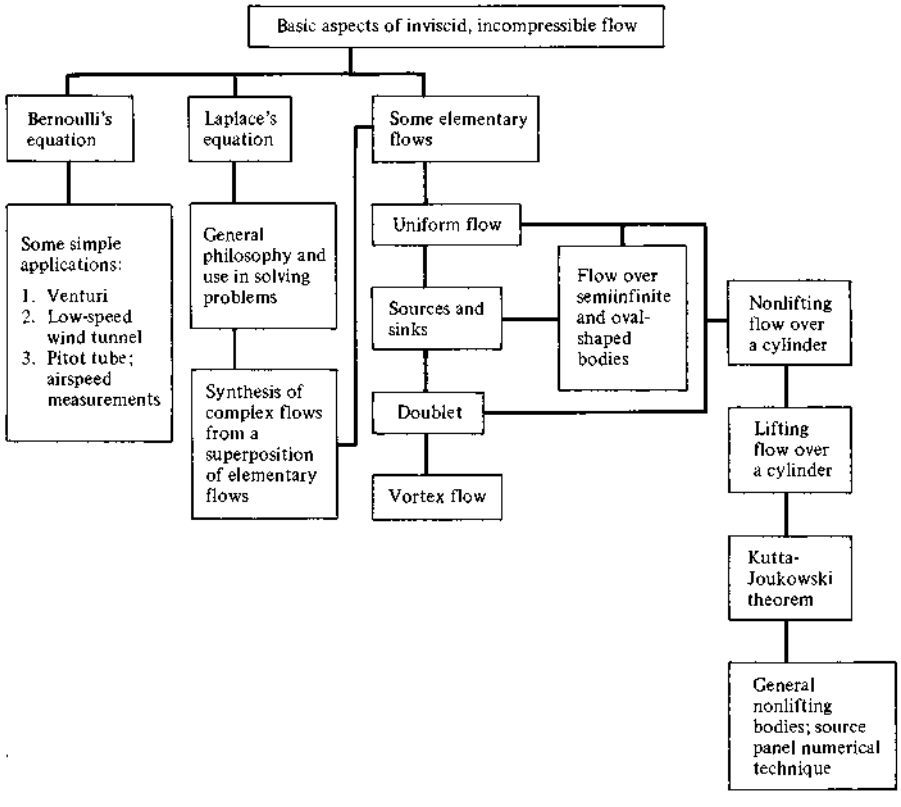


Figure 3.4 Road map for Chapter 3.

flows as shown in Figure 1.44, the material contained in Chapters 3 to 6 falls within the combined blocks D and E.

The purpose of this chapter is to establish some fundamental relations applicable to inviscid, incompressible flows and to discuss some simple but important flow fields and applications. The material in this chapter is then used as a launching pad for the airfoil theory of Chapter 4 and the finite wing theory of Chapter 5.

A road map for this chapter is given in Figure 3.4. There are three main avenues: (1) a development of Bernoulli's equation, with some straightforward applications; (2) a discussion of Laplace's equation, which is the governing equation for inviscid, incompressible, irrotational flow; and (3) the presentation of some elementary flow patterns, how they can be superimposed to synthesize both the nonlifting and lifting flow over a circular cylinder, and how they form the basis of a general numerical technique, called the *panel technique,* for the solution of flows over bodies of general shape. As you progress through this chapter, occasionally refer to this road map so that you can maintain your orientation and see how the various sections are related.

3.2 BERNOULLI'S EQUATION

As will be portrayed in Section 3.19, the early part of the eighteenth century saw the flowering of theoretical fluid dynamics, paced by the work of Johann and Daniel Bernoulli and, in particular, by Leonhard Euler. It was at this time that the relation between pressure and velocity in an inviscid, incompressible flow was first understood. The resulting equation is

$$p + \tfrac{1}{2}\rho V^2 = \text{ const}$$

This equation is called *Bernoulli's equation,* although it was first presented in the above form by Euler (see Section 3.19). Bernoulli's equation is probably the most famous equation in fluid dynamics, and the purpose of this section is to derive it from the general equations discussed in Chapter 2.

Consider the x component of the momentum equation given by Equation (2.113a). For an inviscid flow with no body forces, this equation becomes

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x}$$

or

$$\rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} + \rho w \frac{\partial u}{\partial z} = -\frac{\partial p}{\partial x} \tag{3.1}$$

For steady flow, $\partial u/\partial t = 0$. Equation (3.1) is then written as

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x} \tag{3.2}$$

Multiply Equation (3.2) by dx:

$$u\frac{\partial u}{\partial x}dx + v\frac{\partial u}{\partial y}dx + w\frac{\partial u}{\partial z}dx = -\frac{1}{\rho}\frac{\partial p}{\partial x}dx \tag{3.3}$$

Consider the flow along a streamline in three-dimensional space. The equation of a streamline is given by Equations (2.117*a* to *c*). In particular, substituting

$$u\,dz - w\,dx = 0 \tag{2.117b}$$

and

$$v\,dx - u\,dy = 0 \tag{2.117c}$$

into Equation (3.3), we have

$$u\frac{\partial u}{\partial x}dx + u\frac{\partial u}{\partial y}dy + u\frac{\partial u}{\partial z}dz = -\frac{1}{\rho}\frac{\partial p}{\partial x}dx \tag{3.4}$$

or

$$u\left(\frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz\right) = -\frac{1}{\rho}\frac{\partial p}{\partial x}dx \tag{3.5}$$

Recall from calculus that given a function $u = u(x, y, z)$, the differential of u is

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz$$

This is exactly the term in parentheses in Equation (3.5). Hence, Equation (3.5) is written as

$$u\,du = -\frac{1}{\rho}\frac{\partial p}{\partial x}dx$$

or

$$\frac{1}{2}d(u^2) = -\frac{1}{\rho}\frac{\partial p}{\partial x}dx \tag{3.6}$$

In a similar fashion, starting from the y component of the momentum equation given by Equation (2.113*b*), specializing to an inviscid, steady flow, and applying the result to flow along a streamline, Equations (2.117*a* and *c*), we have

$$\frac{1}{2}d(v^2) = -\frac{1}{\rho}\frac{\partial p}{\partial y}dy \tag{3.7}$$

Similarly, from the z component of the momentum equation, Equation (2.113*c*), we obtain

$$\frac{1}{2}d(w^2) = -\frac{1}{\rho}\frac{\partial p}{\partial z}dz \tag{3.8}$$

Adding Equations (3.6) through (3.8) yields

$$\frac{1}{2}d(u^2 + v^2 + w^2) = -\frac{1}{\rho}\left(\frac{\partial p}{\partial x}dx + \frac{\partial p}{\partial y}dy + \frac{\partial p}{\partial z}dz\right) \tag{3.9}$$

However,

$$u^2 + v^2 + w^2 = V^2 \tag{3.10}$$

and

$$\frac{\partial p}{\partial x}dx + \frac{\partial p}{\partial y}dy + \frac{\partial p}{\partial z}dz = dp \tag{3.11}$$

Substituting Equations (3.10) and (3.11) into (3.9), we have

$$\frac{1}{2}d(V^2) = -\frac{dp}{\rho}$$

or

$$\boxed{dp = -\rho V\, dV} \tag{3.12}$$

Equation (3.12) is called *Euler's equation*. It applies to an inviscid flow with no body forces, and it relates the change in velocity along a streamline dV to the change in pressure dp along the same streamline.

Equation (3.12) takes on a very special and important form for incompressible flow. In such a case, $\rho =$ constant, and Equation (3.12) can be easily integrated between any two points 1 and 2 along a streamline. From Equation (3.12), with $\rho =$ constant, we have

$$\int_{p_1}^{p_2} dp = -\rho \int_{V_1}^{V_2} V\, dV$$

or

$$p_2 - p_1 = -\rho\left(\frac{V_2^2}{2} - \frac{V_1^2}{2}\right)$$

or

$$\boxed{p_1 + \tfrac{1}{2}\rho V_1^2 = p_2 + \tfrac{1}{2}\rho V_2^2} \tag{3.13}$$

Equation (3.13) is *Bernoulli's equation,* which relates p_1 and V_1 at point 1 on a streamline to p_2 and V_2 at another point 2 on the same streamline. Equation (3.13) can also be written as

$$\boxed{p + \tfrac{1}{2}\rho V^2 = \text{const} \qquad \text{along a streamline}} \tag{3.14}$$

In the derivation of Equations (3.13) and (3.14), no stipulation has been made as to whether the flow is rotational or irrotational—these equations hold along a streamline in either case. For a general, rotational flow, the value of the constant in Equation (3.14) will change from one streamline to the next. However, if the flow is irrotational, then Bernoulli's equation holds between *any* two points in the flow, not necessarily just on the same streamline. For an irrotational flow, the constant in Equation (3.14) is the same for all streamlines, and

$$\boxed{p + \tfrac{1}{2}\rho V^2 = \text{const} \qquad \text{throughout the flow}} \tag{3.15}$$

The proof of this statement is given as Problem 3.1.

The physical significance of Bernoulli's equation is obvious from Equations (3.13) to (3.15); namely, *when the velocity increases, the pressure decreases, and when the velocity decreases, the pressure increases.*

Note that Bernoulli's equation was derived from the momentum equation; hence, it is a statement of Newton's second law for an inviscid, incompressible flow with no body forces. However, note that the dimensions of Equations (3.13) to (3.15) are energy per unit volume ($\frac{1}{2}\rho V^2$ is the kinetic energy per unit volume). Hence, Bernoulli's equation is also a relation for mechanical energy in an incompressible flow; it states that the work done on a fluid by pressure forces is equal to the change in kinetic energy of the flow. Indeed, Bernoulli's equation can be derived from the general energy equation, such as Equation (2.114). This derivation is left to the reader. The fact that Bernoulli's equation can be interpreted as either Newton's second law or an energy equation simply illustrates that the energy equation is redundant for the analysis of inviscid, incompressible flow. For such flows, the continuity and momentum equations suffice. (You may wish to review the opening comments of Section 2.7 on this same subject.)

The strategy for solving most problems in inviscid, incompressible flow is as follows:

1. Obtain the velocity field from the governing equations. These equations, appropriate for an inviscid, incompressible flow, are discussed in detail in Sections 3.6 and 3.7.
2. Once the velocity field is known, obtain the corresponding pressure field from Bernoulli's equation.

However, before treating the general approach to the solution of such flows (Section 3.7), several applications of the continuity equation and Bernoulli's equation are made to flows in ducts (Section 3.3) and to the measurement of airspeed using a Pitot tube (Section 3.4).

EXAMPLE 3.1

Consider an airfoil in a flow at standard sea level conditions with a freestream velocity of 50 m/s. At a given point on the airfoil, the pressure is 0.9×10^5 N/m^2. Calculate the velocity at this point.

■ Solution

At standard sea level conditions, $\rho_\infty = 1.23$ kg/m^3 and $p_\infty = 1.01 \times 10^5$ N/m^2. Hence,

$$p_\infty + \tfrac{1}{2}\rho V_\infty^2 = p + \tfrac{1}{2}\rho V^2$$

$$V = \sqrt{\frac{2(p_\infty - p)}{\rho} + V_\infty^2} = \sqrt{\frac{2(1.01 - 0.9) \times 10^5}{1.23} + (50)^2}$$

$$\boxed{V = 142.8 \text{ m/s}}$$

EXAMPLE 3.2

Consider the inviscid, incompressible flow of air along a streamline. The air density along the streamline is 0.002377 slug/ft^3, which is standard atmospheric density at sea level. At point 1 on the streamline, the pressure and velocity are 2116 lb/ft^2 and 10 ft/s, respectively. Further downstream, at point 2 on the streamline, the velocity is 190 ft/s. Calculate the pressure at point 2. What can you say about the relative change in pressure from point 1 to point 2 compared to the corresponding change in velocity?

■ Solution

From Equation (3.13),

$$p_1 + \tfrac{1}{2}\rho V_1^2 = p_2 + \tfrac{1}{2}\rho V_2^2$$

Hence,

$$\begin{aligned} p_2 &= p_1 + \tfrac{1}{2}\rho\left(V_1^2 - V_2^2\right) \\ p_2 &= 2116 + \tfrac{1}{2}(0.002377)[(10)^2 - (190)^2] \\ &= 2116 + \tfrac{1}{2}(0.002377)(100 - 36100) \\ &= 2116 - 42.8 = \boxed{2073.2\ \text{lb/ft}^2} \end{aligned}$$

Note: As the flow velocity increases from 10 ft/s to 190 ft/s along the streamline, the pressure decreases from 2116 lb/ft^2 to 2073.1 lb/ft^2. This is a factor of 19 increase in velocity, for only a factor of 42.8/2116, or 0.02 decrease in pressure. Stated another way, only a 2 percent decrease in the pressure creates a 1900 percent increase in the flow velocity. This is an example of a general characteristic of low-speed flows—only a small change in pressure is required to obtain a large change in velocity. You can sense this in the weather around you. Only a small barometric change from one location to another can create a strong wind.

3.3 INCOMPRESSIBLE FLOW IN A DUCT: THE VENTURI AND LOW-SPEED WIND TUNNEL

Consider the flow through a duct, such as that sketched in Figure 3.5. In general, the duct will be a three-dimensional shape, such as a tube with elliptical or rectangular cross sections which vary in area from one location to another. The flow through such a duct is three-dimensional and, strictly speaking, should be analyzed by means of the full three-dimensional conservation equations derived in Chapter 2. However, in many applications, the variation of area $A = A(x)$ is moderate, and for such cases it is reasonable to *assume* that the flow-field properties are uniform across any cross section, and hence vary only in the x direction. In Figure 3.5, uniform flow is sketched at station 1, and another but different uniform flow is shown at station 2. Such flow, where the area changes as a function of x and all

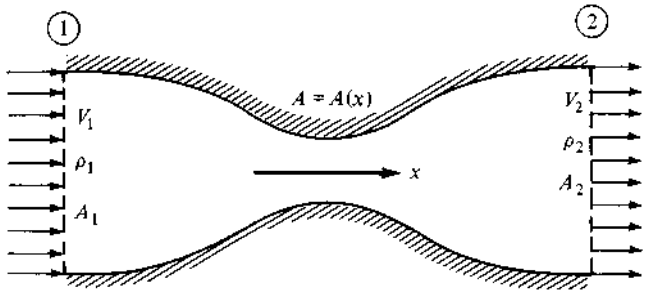


Figure 3.5 Quasi-one-dimensional flow in a duct.

the flow-field variables are assumed to be functions of x only, that is, $A = A(x)$, $V = V(x)$, $p = p(x)$, etc., is called *quasi-one-dimensional flow.* Although such flow is only an approximation of the truly three-dimensional flow in ducts, the results are sufficiently accurate for many aerodynamic applications. Such quasi-one-dimensional flow calculations are frequently used in engineering. They are the subject of this section.

Consider the integral form of the continuity equation written below:

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \, d\mathcal{V} + \oiint_{S} \rho \mathbf{V} \cdot \mathbf{dS} = 0 \tag{2.48}$$

For steady flow, this becomes

$$\oiint_{S} p\mathbf{V} \cdot \mathbf{dS} = 0 \tag{3.16}$$

Apply Equation (3.16) to the duct shown in Figure 3.5, where the control volume is bounded by A_1 on the left, A_2 on the right, and the upper and lower walls of the duct. Hence, Equation (3.16) is

$$\iint_{A_1} \rho \mathbf{V} \cdot \mathbf{dS} + \iint_{A_2} \rho \mathbf{V} \cdot \mathbf{dS} + \iint_{\text{wall}} \rho \mathbf{V} \cdot \mathbf{dS} = 0 \tag{3.17}$$

Along the walls, the flow velocity is tangent to the wall. Since by definition **dS** is perpendicular to the wall, then along the wall, $\mathbf{V} \cdot \mathbf{dS} = 0$, and the integral over the wall surface is zero; that is, in Equation (3.17),

$$\iint_{\text{wall}} \rho \mathbf{V} \cdot \mathbf{dS} = 0 \tag{3.18}$$

At station 1, the flow is uniform across A_1. Noting that **dS** and **V** are in opposite directions at station 1 (**dS** always points *out* of the control volume by definition),

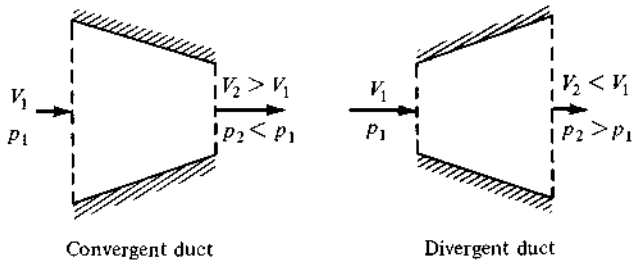


Figure 3.6 Incompressible flow in a duct.

we have in Equation (3.17)

$$\iint_{A_1} p\mathbf{V} \cdot \mathbf{dS} = -\rho_1 A_1 V_1 \tag{3.19}$$

At station 2, the flow is uniform across A_2, and since **dS** and **V** are in the same direction, we have, in Equation (3.17),

$$\iint_{A_2} \rho\mathbf{V} \cdot \mathbf{dS} = \rho_2 A_2 V_2 \tag{3.20}$$

Substituting Equations (3.18) to (3.20) into (3.17), we obtain

$$-\rho_1 A_1 V_1 + \rho_2 A_2 V_2 + 0 = 0$$

or

$$\boxed{\rho_1 A_1 V_1 = \rho_2 A_2 V_2} \tag{3.21}$$

Equation (3.21) is the quasi-one-dimensional continuity equation; it applies to both compressible and incompressible flow.[3] In physical terms, it states that the mass flow through the duct is constant (i.e., what goes in must come out). Compare Equation (3.21) with Equation (2.43) for mass flow.

Consider *incompressible* flow only, where ρ = constant. In Equation (3.21), $\rho_1 = \rho_2$, and we have

$$\boxed{A_1 V_1 = A_2 V_2} \tag{3.22}$$

Equation (3.22) is the quasi-one-dimensional continuity equation for incompressible flow. In physical terms, it states that the volume flow (cubic feet per second or cubic meters per second) through the duct is constant. From Equation (3.22), we see that if the area decreases along the flow (convergent duct), the velocity increases; conversely, if the area increases (divergent duct), the velocity decreases. These variations are shown in Figure 3.6; they are fundamental consequences of

[3] For a simpler, more rudimentary derivation of Equation (3.21), see chapter 4 of Reference 2. In the present discussion, we have established a more rigorous derivation of Equation (3.21), consistent with the general integral form of the continuity equation.

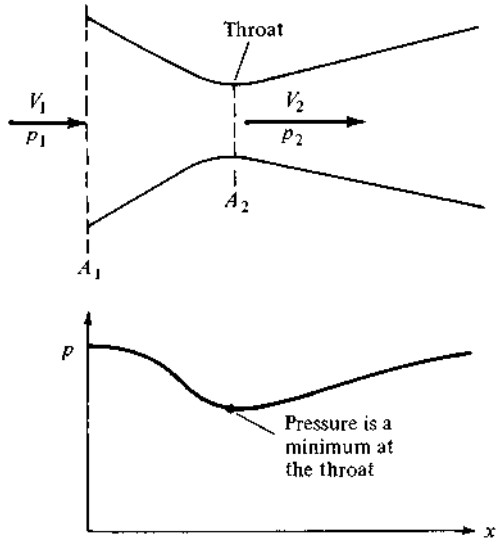


Figure 3.7 Flow through a venturi.

the incompressible continuity equation, and you should fully understand them. Moreover, from Bernoulli's equation, Equation (3.15), we see that when the velocity increases in a convergent duct, the pressure decreases; conversely, when the velocity decreases in a divergent duct, the pressure increases. These pressure variations are also shown in Figure 3.6.

Consider the incompressible flow through a convergent-divergent duct, shown in Figure 3.7. The flow enters the duct with velocity V_1 and pressure p_1. The velocity increases in the convergent portion of the duct, reaching a maximum value V_2 at the minimum area of the duct. This minimum area is called the *throat.* Also, in the convergent section, the pressure decreases, as sketched in Figure 3.7. At the throat, the pressure reaches a minimum value p_2. In the divergent section downstream of the throat, the velocity decreases and the pressure increases. The duct shown in Figure 3.7 is called a *venturi;* it is a device that finds many applications in engineering, and its use dates back more than a century. Its primary characteristic is that the pressure p_2 is lower at the throat than the ambient pressure p_1 outside the venturi. This pressure difference $p_1 - p_2$ is used to advantage in several applications. For example, in the carburetor of an automobile engine, there is a venturi through which the incoming air is mixed with fuel. The fuel line opens into the venturi at the throat. Because p_2 is less than the surrounding ambient pressure p_1, the pressure difference $p_1 - p_2$ helps to force the fuel into the airstream and mix it with the air downstream of the throat.

In an application closer to aerodynamics, a venturi can be used to measure airspeeds. Consider a venturi with a given inlet-to-throat area ratio A_1/A_2, as shown in Figure 3.7. Assume that the venturi is inserted into an airstream that

has an unknown velocity V_1. We wish to use the venturi to measure this velocity. With regard to the venturi itself, the most direct quantity that can be measured is the pressure difference $p_1 - p_2$. This can be accomplished by placing a small hole (a pressure tap) in the wall of the venturi at both the inlet and the throat and connecting the pressure leads (tubes) from these holes across a differential pressure gage, or to both sides of a U-tube manometer (see Section 1.9). In such a fashion, the pressure difference $p_1 - p_2$ can be obtained directly. This measured pressure difference can be related to the unknown velocity V_1 as follows. From Bernoulli's equation, Equation (3.13), we have

$$V_1^2 = \frac{2}{\rho}(p_2 - p_1) + V_2^2 \tag{3.23}$$

From the continuity equations, Equation (3.22), we have

$$V_2 = \frac{A_1}{A_2} V_1 \tag{3.24}$$

Substituting Equation (3.24) into (3.23), we obtain

$$V_1^2 = \frac{2}{\rho}(p_2 - p_1) + \left(\frac{A_1}{A_2}\right)^2 V_1^2 \tag{3.25}$$

Solving Equation (3.25) for V_1, we obtain

$$\boxed{V_1 = \sqrt{\frac{2(p_1 - p_2)}{\rho[(A_1/A_2)^2 - 1]}}} \tag{3.26}$$

Equation (3.26) is the desired result; it gives the inlet air velocity V_1 in terms of the measured pressure difference $p_1 - p_2$ and the known density ρ and area ratio A_1/A_2. In this fashion, a venturi can be used to measure airspeeds. Indeed, historically the first practical airspeed indicator on an airplane was a venturi used by the French Captain A. Eteve in January 1911, more than 7 years after the Wright brothers' first powered flight. Today, the most common airspeed-measuring instrument is the Pitot tube (to be discussed in Section 3.4); however, the venturi is still found on some general aviation airplanes, including home-built and simple experimental aircraft.

Another application of incompressible flow in a duct is the low-speed wind tunnel. The desire to build ground-based experimental facilities designed to produce flows of air in the laboratory which simulate actual flight in the atmosphere dates back to 1871, when Francis Wenham in England built and used the first wind tunnel in history.[4] From that date to the mid-1930s, almost all wind tunnels were designed to produce airflows with velocities from 0 to 250 mi/h. Such low-speed

[4] For a discussion on the history of wind tunnels, see chapter 4 of Reference 2.

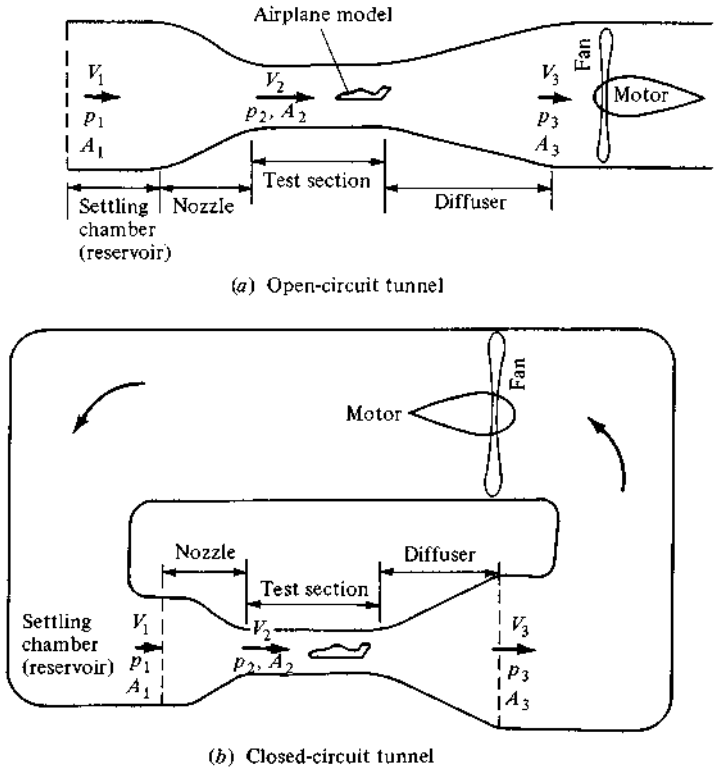


Figure 3.8 (*a*) Open-circuit tunnel. (*b*) Closed-circuit tunnel.

wind tunnels are still much in use today, along with a complement of transonic, supersonic, and hypersonic tunnels. The principles developed in this section allow us to examine the basic aspects of low-speed wind tunnels, as follows.

In essence, a low-speed wind tunnel is a large venturi where the airflow is driven by a fan connected to some type of motor drive. The wind-tunnel fan blades are similar to airplane propellers and are designed to draw the airflow through the tunnel circuit. The wind tunnel may be open circuit, where the air is drawn in the front directly from the atmosphere and exhausted out the back, again directly to the atmosphere, as shown in Figure 3.8*a*; or the wind tunnel may be closed circuit, where the air from the exhaust is returned directly to the front of the tunnel via a closed duct forming a loop, as shown in Figure 3.8*b*. In either case, the airflow with pressure p_1 enters the nozzle at a low velocity V_1, where the area is A_1. The nozzle converges to a smaller area A_2 at the test section, where the velocity has increased to V_2 and the pressure has decreased to p_2. After flowing over an aerodynamic model (which may be a model of a complete airplane or part of an airplane such as a wing, tail, engine, or nacelle), the air passes into a diverging duct called a *diffuser,* where the area increases to A_3, the velocity decreases to

V_3, and the pressure increases to p_3. From the continuity equation (3.22), the test-section air velocity is

$$V_2 = \frac{A_1}{A_2} V_1 \tag{3.27}$$

In turn, the velocity at the exit of the diffuser is

$$V_3 = \frac{A_2}{A_3} V_2 \tag{3.28}$$

The pressure at various locations in the wind tunnel is related to the velocity by Bernoulli's equation:

$$p_1 + \tfrac{1}{2}\rho V_1^2 = p_2 + \tfrac{1}{2}\rho V_2^2 = p_3 + \tfrac{1}{2}\rho V_3^3 \tag{3.29}$$

The basic factor that controls the air velocity in the test section of a given low-speed wind tunnel is the pressure difference $p_1 - p_2$. To see this more clearly, rewrite Equation (3.29) as

$$V_2^2 = \frac{2}{\rho}(p_1 - p_2) + V_1^2 \tag{3.30}$$

From Equation (3.27), $V_1 = (A_2/A_1)V_2$. Substituting into the right-hand side of Equation (3.30), we have

$$V_2^2 = \frac{2}{\rho}(p_1 - p_2) + \left(\frac{A_2}{A_1}\right)^2 V_2^2 \tag{3.31}$$

Solving Equation (3.31) for V_2, we obtain

$$\boxed{V_2 = \sqrt{\frac{2(p_1 - p_2)}{\rho[1 - (A_2/A_1)^2]}}} \tag{3.32}$$

The area ratio A_2/A_1 is a fixed quantity for a wind tunnel of given design. Moreover, the density is a known constant for incompressible flow. Therefore, Equation (3.32) demonstrates conclusively that the test-section velocity V_2 is governed by the pressure difference $p_1 - p_2$. The fan driving the wind-tunnel flow creates this pressure difference by doing work on the air. When the wind-tunnel operator turns the "control knob" of the wind tunnel and adjusts the power to the fan, he or she is essentially adjusting the pressure difference $p_1 - p_2$ and, in turn, adjusting the velocity via Equation (3.32).

In low-speed wind tunnels, a method of measuring the pressure difference $p_1 - p_2$, hence of measuring V_2 via Equation (3.32), is by means of a manometer as discussed in Section 1.9. In Equation (1.56), the density is the density of the liquid in the manometer (*not* the density of the air in the tunnel). The product of density and the acceleration of gravity g in Equation (1.56) is the weight per unit volume of the manometer fluid. Denote this weight per unit volume by w.

Referring to Equation (1.56), if the side of the manometer associated with p_a is connected to a pressure tap in the settling chamber of the wind tunnel, where the pressure is p_1, and if the other side of the manometer (associated with p_b) is connected to a pressure tap in the test section, where the pressure is p_2, then, from Equation (1.56),

$$p_1 - p_2 = w\Delta h$$

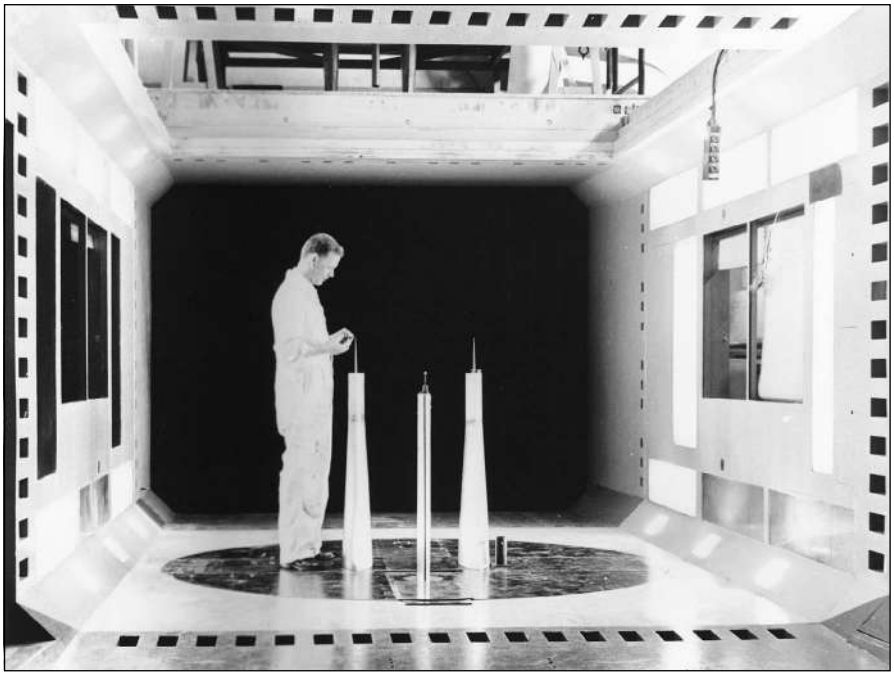
where Δh is the difference in heights of the liquid between the two sides of the manometer. In turn, Equation (3.32) can be expressed as

$$V_2 = \sqrt{\frac{2w\Delta h}{\rho[1 - (A_2/A_1)^2]}}$$

The use of manometers is a straightforward mechanical means to measure pressures. They are time-honored devices that date back to their invention by the Italian mathematician Evangelista Torricelli in 1643. The French engineer Gustave Eiffel used manometers to measure pressures on the surface of wing models mounted in his wind tunnel in Paris in 1909, initiating the use of manometers in wind tunnel work throughout much of the twentieth century. Today, in most wind tunnels manometers have been replaced by an array of electronic pressure-measuring instruments. We discuss manometers here because they are part of the tradition of aerodynamics, and they are a good example of an application in fluid statics.

In many low-speed wind tunnels, the test section is vented to the surrounding atmosphere by means of slots in the wall; in others, the test section is not a duct at all, but rather, an open area between the nozzle exit and the diffuser inlet. In both cases, the pressure in the surrounding atmosphere is impressed on the test-section flow; hence, $p_2 = 1$ atm. (In subsonic flow, a jet that is dumped freely into the surrounding air takes on the same pressure as the surroundings; in contrast, a supersonic free jet may have completely different pressures than the surrounding atmosphere, as we see in Chapter 10.)

Keep in mind that the basic equations used in this section have certain limitations—we are assuming a quasi-one-dimensional inviscid flow. Such equations can sometimes lead to misleading results when the neglected phenomena are in reality important. For example, if $A_3 = A_1$ (inlet area of the tunnel is equal to the exit area), then Equations (3.27) and (3.28) yield $V_3 = V_1$. In turn, from Equation (3.29), $p_3 = p_1$; that is, there is no pressure difference across the entire tunnel circuit. If this were true, the tunnel would run without the application of any power—we would have a perpetual motion machine. In reality, there are losses in the airflow due to friction at the tunnel walls and drag on the aerodynamic model in the test section. Bernoulli's equation, Equation (3.29), does not take such losses into account. (Review the derivation of Bernoulli's equation in Section 3.2; note that viscous effects are neglected.) Thus, in an actual wind tunnel, there is a pressure loss due to viscous and drag effects, and $p_3 < p_1$. The



(*a*)

Figure 3.9 (*a*) Test section of a large subsonic wind tunnel; the Glenn L. Martin Wind Tunnel at the University of Maryland. *(Courtesy of Dr. Jewel Barlow, University of Maryland.)*

function of the wind-tunnel motor and fan is to add power to the airflow in order to increase the pressure of the flow coming out of the diffuser so that it can be exhausted into the atmosphere (Figure 3.8*a*) or returned to the inlet of the nozzle at the higher pressure p_1 (Figure 3.8*b*). Photographs of a typical subsonic wind tunnel are shown in Figure 3.9*a* and *b*.

EXAMPLE 3.3

Consider a venturi with a throat-to-inlet area ratio of 0.8 mounted in a flow at standard sea level conditions. If the pressure difference between the inlet and the throat is 7 lb/ft^2, calculate the velocity of the flow at the inlet.

■ Solution

At standard sea level conditions, $\rho = 0.002377$ slug/ft^3. Hence,

$$V_1 = \sqrt{\frac{2(p_1 - p_2)}{\rho[(A_1/A_2)^2 - 1]}} = \sqrt{\frac{2(7)}{(0.002377)[(\frac{1}{0.8})^2 - 1]}} = \boxed{102.3 \text{ ft/s}}$$



(*b*)

Figure 3.9 (*continued*) (*b*) The power fan drive section of the Glenn L. Martin Wind Tunnel. *(Courtesy of Dr. Jewel Barlow, University of Maryland.)*

EXAMPLE 3.4

Consider a low-speed subsonic wind tunnel with a 12/1 contraction ratio for the nozzle. If the flow in the test section is at standard sea level conditions with a velocity of 50 m/s, calculate the height difference in a U-tube mercury manometer with one side connected to the nozzle inlet and the other to the test section.

■ Solution

At standard sea level, $\rho = 1.23$ kg/m^3. From Equation (3.32),

$$p_1 - p_2 = \frac{1}{2}\rho V_2^2 \left[1 - \left(\frac{A_2}{A_1}\right)^2\right] = \frac{1}{2}(50)^2(1.23)\left[1 - \left(\frac{1}{12}\right)^2\right] = 1527 \text{ N/m}^2$$

However, $p_1 - p_2 = w\Delta h$. The density of liquid mercury is 1.36×10^4 kg/m^3. Hence,

$$w = (1.36 \times 10^4 \text{ kg/m}^3)(9.8 \text{ m/s}^2) = 1.33 \times 10^5 \text{ N/m}^2$$

$$\Delta h = \frac{p_1 - p_2}{w} = \frac{1527}{1.33 \times 10^5} = \boxed{0.01148 \text{ m}}$$

EXAMPLE 3.5

Consider a model of an airplane mounted in a subsonic wind tunnel, such as shown in Figure 3.10. The wind-tunnel nozzle has a 12-to-1 contraction ratio. The maximum lift coefficient of the airplane model is 1.3. The wing planform area of the model is 6 ft^2. The lift is measured with a mechanical balance that is rated at a maximum force of 1000 lb; that is, if the lift of the airplane model exceeds 1000 lb, the balance will be damaged. During a given test of this airplane model, the plan is to rotate the model through its whole range of angle of attack, including up to that for maximum C_L. Calculate the maximum pressure difference allowable between the wind-tunnel settling chamber and the test section, assuming standard sea level density in the test section (i.e., $\rho_\infty = 0.002377$ slug/ft^3).

■ Solution

Maximum lift occurs when the model is at its maximum lift coefficient. Since the maximum allowable lift force is 1000 lb, the freestream velocity at which this occurs is obtained from

$$L_{\max} = \tfrac{1}{2}\rho_\infty V_\infty^2 S C_{L,\max}$$

or

$$V_\infty = \sqrt{\frac{2L_{\max}}{\rho_\infty S C_{L,\max}}} = \sqrt{\frac{(2)(1000)}{(0.002377)(6)(1.3)}} = 328.4 \text{ ft/s}$$

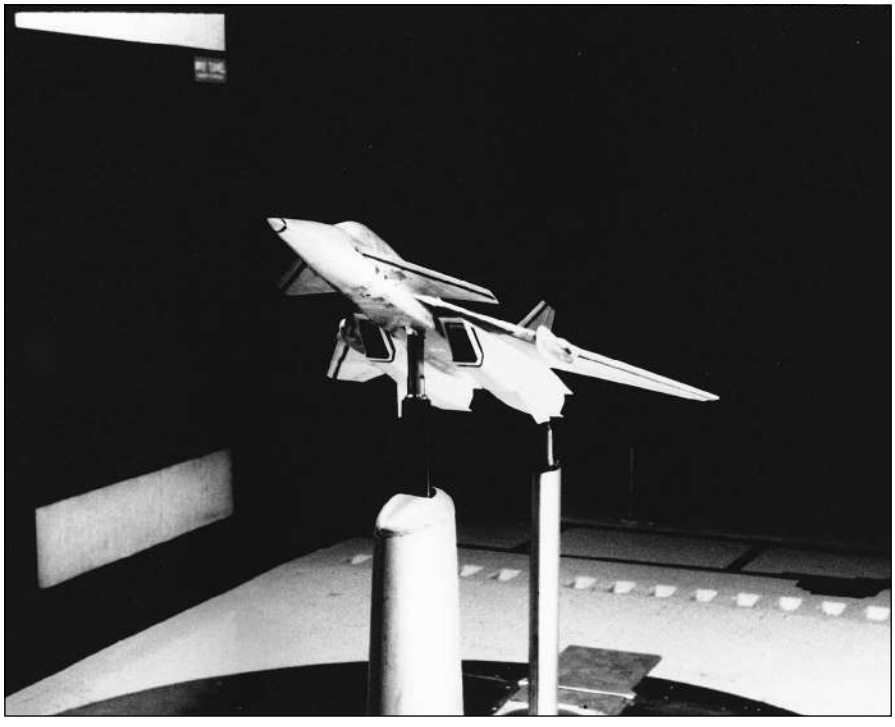


Figure 3.10 Typical model installation in the test section of a large wind tunnel. The Glenn L. Martin Wind Tunnel at the University of Maryland.

From Equation (3.32),

$$p_1 - p_2 = \frac{1}{2}\rho_\infty V_\infty^2 \left[1 - \left(\frac{A_2}{A_1}\right)^2\right]$$

$$= \frac{1}{2}(0.002377)(328.4)^2 \left[1 - \left(\frac{1}{12}\right)^2\right] = \boxed{127.3 \text{ lb/ft}^2}$$

EXAMPLE 3.6

a. The flow velocity in the test section of a low-speed subsonic wind tunnel is 100 mph. The test section is vented to the atmosphere, where atmospheric pressure is 1.01×10^5 N/m^2. The air density in the flow is the standard sea-level value of 1.23 kg/m^3. The contraction ratio of the nozzle is 10-to-1. Calculate the reservoir pressure in atmospheres.

b. By how much must the reservoir pressure be increased to achieve 200 mph in the test section of this wind tunnel? Comment on the magnitude of this increase in pressure relative to the increase in test-section velocity.

■ Solution

a. Miles per hour is not a consistent unit for velocity. To convert to m/s, we note that 1 mi = 1609 m, and 1 h = 3600 s. Hence.

$$1\frac{\text{mi}}{\text{h}} = \left(1\frac{\text{mi}}{\text{h}}\right)\left(\frac{1609\,\text{m}}{1\,\text{mi}}\right)\left(\frac{1\,\text{h}}{3600\,\text{s}}\right) = 0.447\,\text{m/s}$$

(*Note:* This is the author's iron-clad method for carrying out conversion from one set of units to another. Take the original mile/hour, and multiply it by an equivalent "unity." Since 1609 m is the same distance as 1 mi, then the ratio (1609 m/1 mi) is essentially "unity," and since 1 h is the same time as 3600 s, then the ratio (1 h/3600 s) is essentially "unity." Multiplying (mi/h) by these two equivalent unity ratios, the miles cancel, and the hours cancel, and we are left with the proper number of meters per second.) Therefore,

$$V_2 = 100\,\text{mph} = (100)(0.447) = 44.7\,\text{m/s}.$$

From Equation (3.31),

$$p_1 - p_2 = \frac{\rho}{2}V_2^2\left[1 - \left(\frac{A_2}{A_1}\right)^2\right]$$

$$p_1 - p_2 = \frac{1.23}{2}(44.7)^2\left[1 - \left(\frac{1}{10}\right)^2\right] = 0.01217 \times 10^5\,\text{N/m}^2$$

Thus,

$$p_1 = p_2 + 0.01217 \times 10^5 = 1.01 \times 10^5 + 0.01217 \times 10^5$$

$$p_1 = \boxed{1.022 \times 10^5\ \text{N/m}^2}$$

In atm, $p_1 = 1.022 \times 10^5/1.01 \times 10^5 = \boxed{1.01\ \text{atm}}$

b. $V_2 = 200\,\text{mph} = (200)(0.447) = 89.4\text{m/s}.$

$$p_1 - p_2 = \frac{\rho}{2}V_2^2\left[1 - \left(\frac{A_2}{A_1}\right)^2\right] = \frac{1.23}{2}(89.4)^2(0.99) = 0.0487 \times 10^5\,\text{N/m}^2$$

$$p_1 = 1.01 \times 10^5 + 0.0487 \times 10^5 = \boxed{1.059 \times 10^5\ \text{N/m}^2}$$

In atm,

$$p_1 = 1.059 \times 10^5/1.01 \times 10^5 = \boxed{1.048\ \text{atm}}$$

Comparing this result with part (a) above, we observe that to achieve a doubling of the test-section flow velocity from 100 mph to 200 mph, the reservoir pressure needed to be increased by only 0.038 atm (i.e., by 3.8 percent). This reinforces the general trend noted in Example 3.2, namely, that in a low-speed flow, a small pressure change results in a large velocity change.

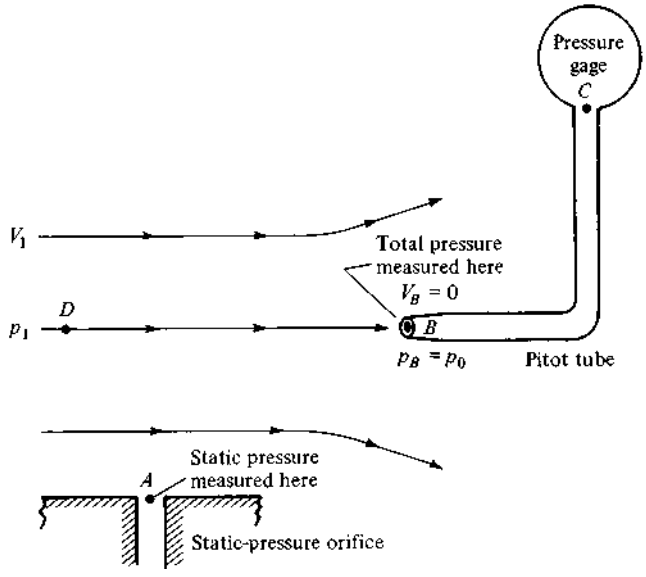


Figure 3.11 Pitot tube and a static pressure orifice.

3.4 PITOT TUBE: MEASUREMENT OF AIRSPEED

In 1732, the Frenchman Henri Pitot was busy trying to measure the flow velocity of the Seine River in Paris. One of the instruments he used was his own invention—a strange-looking tube bent into an L shape, as shown in Figure 3.11. Pitot oriented one of the open ends of the tube so that it faced directly into the flow. In turn, he used the pressure inside this tube to measure the water flow velocity. This was the first time in history that a proper measurement of fluid velocity was made, and Pitot's invention has carried through to the present day as the *Pitot tube*—one of the most common and frequently used instruments in any modern aerodynamic laboratory. Moreover, a Pitot tube is the most common device for measuring flight velocities of airplanes. The purpose of this section is to describe the basic principle of the Pitot tube.[5]

Consider a flow with pressure p_1 moving with velocity V_1, as sketched at the left of Figure 3.11. Let us consider the significance of the pressure p_1 more closely. In Section 1.4, the pressure is associated with the time rate of change of momentum of the gas molecules impacting on or crossing a surface; that is, pressure is clearly related to the motion of the molecules. This motion is very random, with molecules moving in all directions with various velocities. Now imagine that you hop on a fluid element of the flow and ride with it at the velocity V_1. The gas molecules, because of their random motion, will still bump

[5] See chapter 4 of Reference 2 for a detailed discussion of the history of the Pitot tube, how Pitot used it to overturn a basic theory in civil engineering, how it created some controversy in engineering, and how it finally found application in aeronautics.

into you, and you will feel the pressure p_1 of the gas. We now give this pressure a specific name: the *static* pressure. Static pressure is a measure of the purely random motion of molecules in the gas; it is the pressure you feel when you ride along with the gas at the local flow velocity. All pressures used in this book so far have been static pressures; the pressure p appearing in all our previous equations has been the static pressure. In engineering, whenever a reference is made to "pressure" without further qualification, that pressure is always interpreted as the *static* pressure. Furthermore, consider a boundary of the flow, such as a wall, where a small hole is drilled perpendicular to the surface. The plane of the hole is parallel to the flow, as shown at point A in Figure 3.11. Because the flow moves over the opening, the pressure felt at point A is due only to the random motion of the molecules; that is, at point A, the static pressure is measured. Such a small hole in the surface is called a *static pressure orifice,* or a *static pressure tap*.

In contrast, consider that a Pitot tube is now inserted into the flow, with an open end facing directly into the flow. That is, the plane of the opening of the tube is perpendicular to the flow, as shown at point B in Figure 3.11. The other end of the Pitot tube is connected to a pressure gage, such as point C in Figure 3.11 (i.e., the Pitot tube is closed at point C). For the first few milliseconds after the Pitot tube is inserted into the flow, the gas will rush into the open end and will fill the tube. However, the tube is closed at point C; there is no place for the gas to go, and hence after a brief period of adjustment, the gas inside the tube will stagnate; that is, the gas velocity inside the tube will go to zero. Indeed, the gas will eventually pile up and stagnate *everywhere* inside the tube, including at the open mouth at point B. As a result, the streamline of the flow that impinges directly at the open face of the tube (streamline DB in Figure 3.11) sees this face as an obstruction to the flow. The fluid elements along streamline DB slow down as they get closer to the Pitot tube and go to zero velocity right at point B. Any point in a flow where $V = 0$ is called a *stagnation point* of the flow; hence, point B at the open face of the Pitot tube is a stagnation point, where $V_B = 0$. In turn, from Bernoulli's equation we know the pressure increases as the velocity decreases. Hence, $p_B > p_1$. The pressure at a stagnation point is called the *stagnation* pressure, or *total* pressure, denoted by p_0. Hence, at point B, $p_B = p_0$.

From the above discussion, we see that two types of pressure can be defined for a given flow: static pressure, which is the pressure you feel by moving with the flow at its local velocity V_1, and total pressure, which is the pressure that the flow achieves when the velocity is reduced to zero. In aerodynamics, the distinction between total and static pressure is important; we have discussed this distinction at some length, and you should make yourself comfortable with the above paragraphs before proceeding further. (Further elaboration on the meaning and significance of total and static pressure will be made in Chapter 7.)

How is the Pitot tube used to measure flow velocity? To answer this question, first note that the total pressure p_0 exerted by the flow at the tube inlet (point B) is impressed throughout the tube (there is no flow inside the tube; hence, the pressure everywhere inside the tube is p_0). Therefore, the pressure gage at point C reads p_0. This measurement, in conjunction with a measurement of the static pressure p_1 at point A, yields the difference between total and static pressure, $p_0 - p_1$,

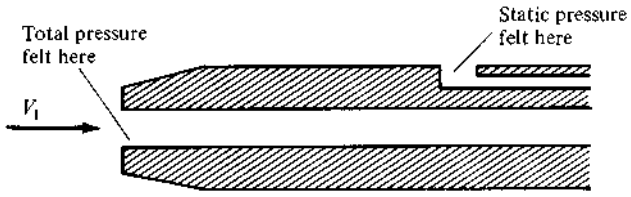


Figure 3.12 Pitot-static probe.

and it is this pressure *difference* that allows the calculation of V_1 via Bernoulli's equation. In particular, apply Bernoulli's equation between point A, where the pressure and velocity are p_1 and V_1, respectively, and point B, where the pressure and velocity are p_0 and $V = 0$, respectively:

$$p_A + \tfrac{1}{2}\rho V_A^2 = p_B + \tfrac{1}{2}\rho V_B^2$$

or

$$p_1 + \tfrac{1}{2}\rho V_1^2 = p_0 + 0 \tag{3.33}$$

Solving Equation (3.33) for V_1, we have

$$\boxed{V_1 = \sqrt{\frac{2(p_0 - p_1)}{\rho}}} \tag{3.34}$$

Equation (3.34) allows the calculation of velocity simply from the measured difference between total and static pressure. The total pressure p_0 is obtained from the Pitot tube, and the static pressure p_1 is obtained from a suitably placed static pressure tap.

It is possible to combine the measurement of both total and static pressure in one instrument, a *Pitot-static probe,* as sketched in Figure 3.12. A Pitot-static probe measures p_0 at the nose of the probe and p_1 at a suitably placed static pressure tap on the probe surface downstream of the nose.

In Equation (3.33), the term $\frac{1}{2}\rho V_1^2$ is called the *dynamic pressure* and is denoted by the symbol q_1. The grouping $\frac{1}{2}\rho V^2$ is called the dynamic pressure by definition and is used in all flows, incompressible to hypersonic:

$$q \equiv \tfrac{1}{2}\rho V^2$$

However, for incompressible flow, the dynamic pressure has special meaning; it is precisely the difference between total and static pressure. Repeating Equation (3.33), we obtain

$$\underset{\text{static pressure}}{p_1} + \underset{\text{dynamic pressure}}{\tfrac{1}{2}\rho V_1^2} = \underset{\text{total pressure}}{p_0}$$

or

$$p_1 + q_1 = p_0$$

or

$$\boxed{q_1 = p_0 - p_1} \tag{3.35}$$

It is important to keep in mind that Equation (3.35) comes from Bernoulli's equation, and thus holds for *incompressible flow only*. For compressible flow, where Bernoulli's equation is not valid, the pressure difference $p_0 - p_1$ is *not* equal to q_1. Moreover, Equation (3.34) is valid for incompressible flow only. The velocities of compressible flows, both subsonic and supersonic, can be measured by means of a Pitot tube, but the equations are different from Equation (3.34). (Velocity measurements in subsonic and supersonic compressible flows are discussed in Chapter 8.)

At this stage, it is important to repeat that Bernoulli's equation holds for incompressible flow only, and therefore any result derived from Bernoulli's equation also holds for incompressible flow only, such as Equations (3.26), (3.32), and (3.34). Experience has shown that some students when first introduced to aerodynamics seem to adopt Bernoulli's equation as the gospel and tend to use it for all applications, including many cases where it is not valid. Hopefully, the repetitive warnings given above will squelch such tendencies.

EXAMPLE 3.7

An airplane is flying at standard sea level. The measurement obtained from a Pitot tube mounted on the wing tip reads 2190 lb/ft^2. What is the velocity of the airplane?

■ Solution

Standard sea level pressure is 2116 lb/ft^2. From Equation (3.34), we have

$$V_1 = \sqrt{\frac{2(p_0 - p_1)}{\rho}} = \sqrt{\frac{2(2190 - 2116)}{0.002377}} = \boxed{250 \text{ ft/s}}$$

EXAMPLE 3.8

In the wind-tunnel flow described in Example 3.5, a small Pitot tube is mounted in the flow just upstream of the model. Calculate the pressure measured by the Pitot tube for the same flow conditions as in Example 3.5.

■ Solution

From Equation (3.35),

$$\begin{aligned} p_0 &= p_\infty + q_\infty = p_\infty + \tfrac{1}{2}\rho_\infty V_\infty^2 \\ &= 2116 + \tfrac{1}{2}(0.002377)(328.4)^2 \\ &= 2116 + 128.2 = \boxed{2244 \text{ lb/ft}^2} \end{aligned}$$

Note in this example that the dynamic pressure is $\frac{1}{2}\rho_\infty V_\infty^2 = 128.2$ lb/ft^2. This is less than 1 percent larger than the pressure difference $(p_1 - p_2)$, calculated in Example 3.5, that is required to produce the test-section velocity in the wind tunnel. Why is $(p_1 - p_2)$ so close to the test-section dynamic pressure? *Answer:* Because the velocity in the settling chamber V_1 is so small that p_1 is close to the total pressure of the flow. Indeed, from Equation (3.22),

$$V_1 = \frac{A_2}{A_1} V_2 = \left(\frac{1}{12}\right)(328.4) = 27.3 \text{ ft/s}$$

Compared to the test-section velocity of 328.4 ft/s, V_1 is seen to be small. In regions of a flow where the velocity is finite but small, the local static pressure is close to the total pressure. (Indeed, in the limiting case of a fluid with zero velocity, the local static pressure is the same as the total pressure; here, the concepts of static pressure and total pressure are redundant. For example, consider the air in the room around you. Assuming the air is motionless, and assuming standard sea level conditions, the pressure is 2116 lb/ft^2, namely, 1 atm. Is this pressure a static pressure or a total pressure? *Answer:* It is *both*. By the definition of total pressure given in the present section, when the local flow velocity is itself zero, then the local static pressure and the local total pressure are exactly the same.)

EXAMPLE 3.9

Consider the P-35 shown in Figure 3.2 cruising at a standard altitude of 4 km. The pressure sensed by the Pitot tube on its right wing (as seen in Figure 3.2) is 6.7×10^4 N/m^2. At what velocity is the P-35 flying?

■ Solution

From Appendix D, at a standard altitude of 4 km, the freestream static pressure and density are 6.166×10^4 N/m^2 and 0.81935 kg/m^3, respectively. The Pitot tube measures the total pressure of 6.7×10^4 N/m^2. From Equation (3.34),

$$V_1 = \sqrt{\frac{2(p_0 - p_1)}{\rho}} = \sqrt{\frac{2(6.7 - 6.166) \times 10^4}{0.81935}} = \boxed{114.2 \text{ m/s}}$$

Note: From the conversion factor between miles per hour and m/s obtained in Example 3.6, we have

$$V_1 = \frac{114.2}{0.447} = 255 \text{ mph}$$

EXAMPLE 3.10

The P-35 in Example 3.9 experiences a certain dynamic pressure at its cruising speed of 114.2 m/s at an altitude of 4 km. Now assume the P-35 is flying at sea level. At what velocity must it fly at sea level to experience the same dynamic pressure?

■ Solution

At $V_1 = 114.2$ m/s and at a standard altitude of 4 km, where $\rho = 0.81935$ kg/m^3,

$$q_1 = \tfrac{1}{2}\rho V_1^2 = \tfrac{1}{2}(0.81935)(114.2)^2 = 5.343 \times 10^3 \text{N/m}^2$$

For the airplane to experience the same dynamic pressure at sea level where $\rho = 1.23$ kg/m^3, its new velocity, V_e, must satisfy

$$q_1 = \tfrac{1}{2}\rho V_e^2$$

$$5.343 \times 10^3 = \tfrac{1}{2}(1.23) V_e^2$$

or,

$$V_e = \sqrt{\frac{2(5.343 \times 10^3)}{1.23}} = \boxed{93.2 \text{ m/s}}$$

In Example 3.10 the velocity of 93.2 m/s is the *equivalent airspeed*, V_e, of the airplane flying at an altitude of 4 km at a true airspeed of 114.2 m/s. The general definition of equivalent airspeed is as follows. Consider an airplane flying at some true airspeed at some altitude. Its *equivalent airspeed* at this condition is defined as the velocity at which it would have to fly at standard sea level to *experience the same dynamic pressure*. In Example 3.10, we have the P-35 flying at an altitude of 4 km at a true airspeed of 114.2 m/s, and simultaneously at an equivalent airspeed of 93.2 m/s.

DESIGN BOX

The configuration of the Pitot-static probe shown in Figure 3.12 is a schematic only. The design of an actual Pitot-static probe is an example of careful engineering, intended to provide as accurate an instrument as possible. Let us examine some of the overall features of Pitot-static probe design.

Above all, the probe should be a long, streamlined shape such that the surface pressure over a substantial portion of the probe is essentially equal to the freestream static pressure. Such a shape is given in Figure 3.13*a*. The head of the probe, the nose at which the total pressure is measured, is usually a smooth hemispherical shape in order to encourage smooth, streamlined flow downstream of the nose. The diameter of the tube is denoted by d. A number of static pressure taps are arrayed radially around the

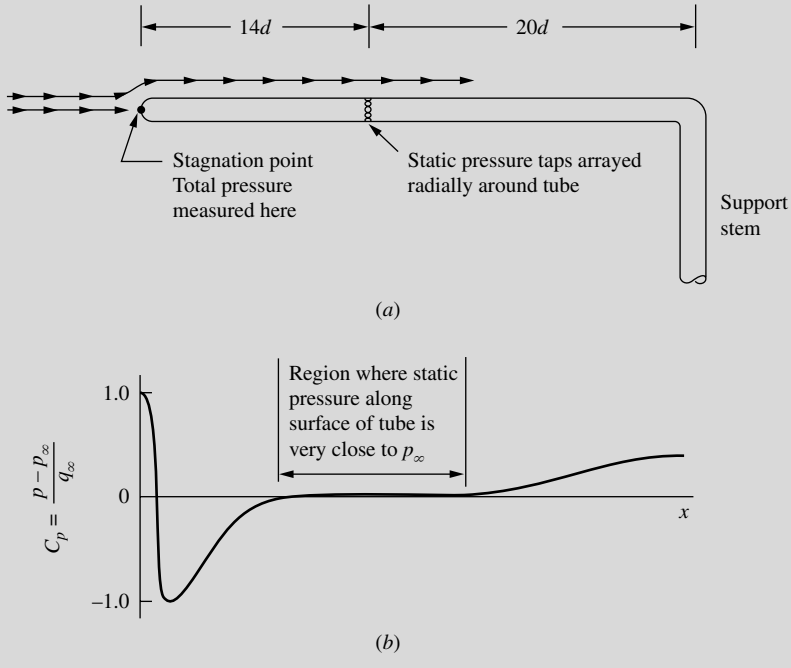


Figure 3.13 (*a*) Pitot-static tube. (*b*) Schematic of the pressure distribution along the outer surface of the tube.

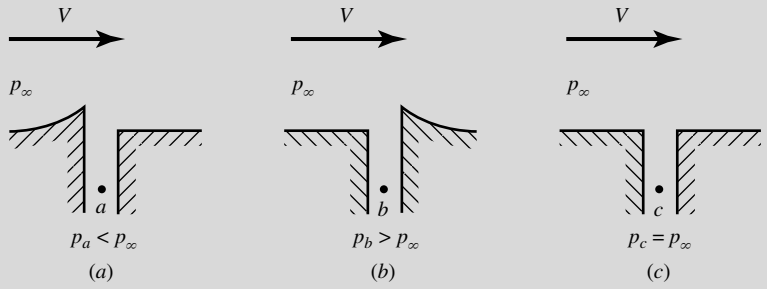


Figure 3.14 Schematic of static pressure taps. (*a*) and (*b*) Poor design. (*c*) Proper design.

circumference of the tube at a station that should be from 8*d* to 16*d* downstream of the nose, and at least 16*d* ahead of the downstream support stem. The reason for this is shown in Figure 3.13*b*, which gives the axial distribution of the pressure coefficient along the surface of the tube. From the definition of pressure coefficient given in Section 1.5, and from Bernoulli's equation in the form of Equation (3.35), the pressure coefficient at a stagnation point for incompressible flow is given by

$$C_p = \frac{p - p_\infty}{q_\infty} = \frac{p_0 - p_\infty}{q_\infty} = \frac{q_\infty}{q_\infty} = 1.0$$

Hence, in Figure 3.13*b* the C_p distribution starts out at the value of 1.0 at the nose, and rapidly drops as the flow expands around the nose. The pressure decreases below p_∞, yielding a minimum value of $C_p \approx -1.25$ just downstream of the nose. Further downstream the pressure tries to recover and approaches a value nearly equal to p_∞ at some distance (typically about 8*d*) from the nose. There follows a region where the static pressure along the surface of the tube is very close to p_∞, illustrated by the region where $C_p = 0$ in Figure 3.13*b*. This is the region where the static pressure taps should be located, because the surface pressure measured at these taps will be essentially equal to the freestream static pressure p_∞. Further downstream, as the flow approaches the support stem, the pressure starts to increase above p_∞. This starts at a distance of about 16*d* ahead of the support stem. In Figure 3.13*a*, the static pressure taps are shown at a station 14*d* downstream of the nose and 20*d* ahead of the support stem.

The design of the static pressure taps themselves is critical. The surface around the taps should be smooth to insure that the pressure sensed inside the tap is indeed the surface pressure along the tube. Examples of poor design as well as the proper design of the pressure taps are shown in Figure 3.14. In Figure 3.14*a*, the surface has a burr on the upstream side; the local flow will expand around this burr, causing the pressure sensed at point *a* inside the tap to be less than p_∞. In 3.14*b*, the surface has a burr on the downstream side; the local flow will be slowed in this region, causing the pressure sensed at point *b* inside the tap to be greater than p_∞. The correct design is shown in Figure 3.14*c*; here, the opening of the tap is exactly flush with the surface, allowing the pressure sensed at point *c* inside the tap to be equal to p_∞.

When a Pitot-static tube is used to measure the speed of an airplane, it should be located on the airplane in a position where it is essentially immersed in the freestream flow, away from any major influence of the local flow field around the airplane itself. An example of this can be seen in Figure 3.2, where a Pitot-static probe can be seen mounted near the right wing tip of the P-35, extending into the freestream ahead of the wing. A similar wing-mounted probe is shown in the planview (top view) of the North American F-86 in Figure 3.15.

Today, many modern airplanes have a Pitot tube mounted at some location on the fuselage, and the measurement of p_∞ is obtained independently from a properly placed static pressure tap somewhere else on the fuselage. Figure 3.16 illustrates a fuselage-mounted Pitot tube in the nose region of the Boeing

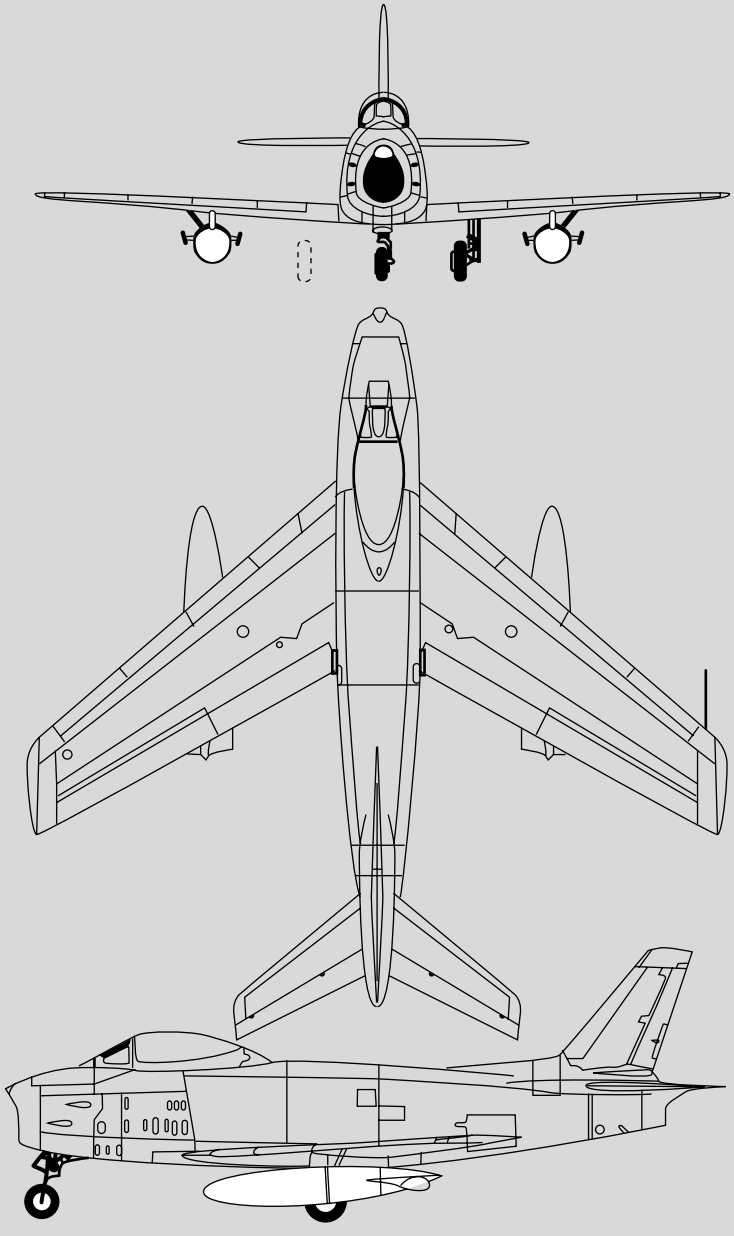


Figure 3.15 Three views of the North American F-86H. Note the wing-mounted Pitot-static tube.

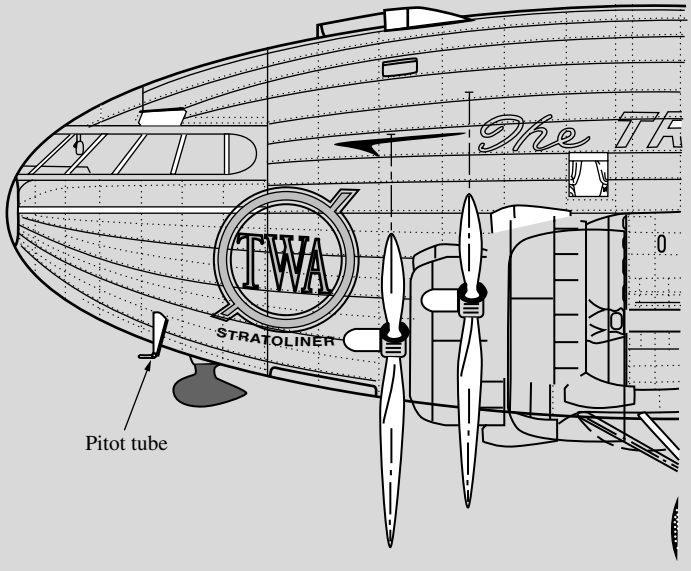


Figure 3.16 Nose-mounted Pitot tube on the Boeing Stratoliner. (Stratoliner detail courtesy of Paul Matt, Alan and Drina Abel, and Aviation Heritage, Inc., with permission.)

Stratoliner, a 1940s vintage airliner. When only a Pitot measurement is required, the probe can be much shorter than a Pitot-static tube, as can be seen in Figure 3.16. In this type of arrangement, the location of the static pressure tap on the surface of the fuselage is critical; it must be located in a region where the surface pressure on the fuselage is equal to p_∞. We have a pretty good idea where to locate the static pressure taps on a Pitot-static tube, as shown in Figure 3.13*a*. But the proper location on the fuselage of a given airplane must be found experimentally, and it is different for different airplanes. However, the basic idea is illustrated in Figure 3.17, which shows the measured pressure coefficient distribution over a streamlined body at zero angle of attack. There are two axial stations where $C_p = 0$ (i.e., where the surface pressure on the body equals p_∞). If this body were an airplane fuselage, the static pressure tap should be placed at one of these two locations. In practice, the forward location, near the nose, is usually chosen.

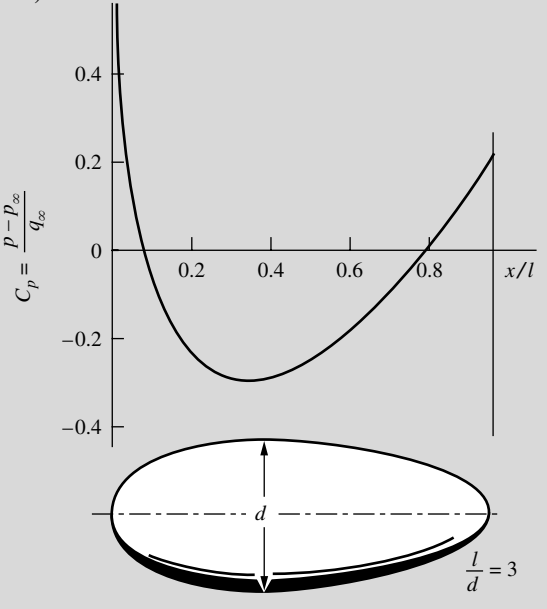


Figure 3.17 Experimentally measured pressure coefficient distribution over a streamlined body with a fineness ratio (length-to-diameter ratio) of 3. Zero angle of attack. Low-speed flow.

Finally, we must be aware that none of these instruments, no matter where they are located, are perfectly accurate. In particular, misalignment of the

probe with respect to the freestream direction causes an error which must be assessed for each particular case. Fortunately, the measurement of the total pressure by means of a Pitot tube is relatively insensitive to misalignment. Pitot tubes with hemispherical noses, such as shown in Figure 3.13*a*, are insensitive to the mean flow direction up to a few degrees. Pitot tubes with flat faces, such as illustrated in Figure 3.12, are least sensitive. For these tubes, the total pressure measurement varies only 1 percent for misalignment as large as 20°. For more details on this matter, see Reference 65.

3.5 PRESSURE COEFFICIENT

Pressure, by itself, is a dimensional quantity (e.g., pounds per square foot, newtons per square meter). However, in Sections 1.7 and 1.8, we established the usefulness of certain dimensionless parameters such as M, Re, C_L. It makes sense, therefore, that a dimensionless pressure would also find use in aerodynamics. Such a quantity is the *pressure coefficient* C_p, first introduced in Section 1.5 and defined as

$$\boxed{C_p \equiv \frac{p - p_\infty}{q_\infty}} \tag{3.36}$$

where $$q_\infty = \tfrac{1}{2}\rho_\infty V_\infty^2$$

The definition given in Equation (3.36) is just that—a definition. It is used throughout aerodynamics, from incompressible to hypersonic flow. In the aerodynamic literature, it is very common to find pressures given in terms of C_p rather than the pressure itself. Indeed, the pressure coefficient is another similarity parameter that can be added to the list started in Sections 1.7 and 1.8.

For *incompressible flow,* C_p can be expressed in terms of velocity only. Consider the flow over an aerodynamic body immersed in a freestream with pressure p_∞ and velocity V_∞. Pick an arbitrary point in the flow where the pressure and velocity are p and V, respectively. From Bernoulli's equation,

$$p_\infty + \tfrac{1}{2}\rho V_\infty^2 = p + \tfrac{1}{2}\rho V^2$$

or $$p - p_\infty = \tfrac{1}{2}\rho \left(V_\infty^2 - V^2\right) \tag{3.37}$$

Substituting Equation (3.37) into (3.36), we have

$$C_p = \frac{p - p_\infty}{q_\infty} = \frac{\tfrac{1}{2}\rho \left(V_\infty^2 - V^2\right)}{\tfrac{1}{2}\rho V_\infty^2}$$

or

$$\boxed{C_p = 1 - \left(\frac{V}{V_\infty}\right)^2} \tag{3.38}$$

Equation (3.38) is a useful expression for the pressure coefficient; however, note that the form of Equation (3.38) holds for incompressible flow only.

Note from Equation (3.38) that the pressure coefficient at a stagnation point (where $V = 0$) in an incompressible flow is always equal to 1.0. This is the highest allowable value of C_p anywhere in the flow field. (For compressible flows, C_p at a stagnation point is greater than 1.0, as shown in Chapter 14.) Also, keep in mind that in regions of the flow where $V > V_\infty$ or $p < p_\infty$, C_p will be a negative value.

Another interesting property of the pressure coefficient can be seen by rearranging the definition given by Equation (3.36), as follows:

$$p = p_\infty + q_\infty C_p$$

Clearly, the value of C_p tells us how much p differs from p_∞ in multiples of the dynamic pressure. That is, if $C_p = 1$ (the value at a stagnation point in an incompressible flow), then $p = p_\infty + q_\infty$, or the local pressure is "one times" the dynamic pressure above freestream static pressure. If $C_p = -3$, then $p = p_\infty - 3q_\infty$, or the local pressure is three times the dynamic pressure below freestream static pressure.

EXAMPLE 3.11

Consider an airfoil in a flow with a freestream velocity of 150 ft/s. The velocity at a given point on the airfoil is 225 ft/s. Calculate the pressure coefficient at this point.

■ Solution

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2 = 1 - \left(\frac{225}{150}\right)^2 = \boxed{-1.25}$$

EXAMPLE 3.12

Consider the airplane model in Example 3.4. When it is at a high angle of attack, slightly less than that when C_L becomes a maximum, the peak (negative) pressure coefficient which occurs at a certain point on the airfoil surface is -5.3. Assuming inviscid, incompressible flow, calculate the velocity at this point when (*a*) $V_\infty = 80$ ft/s and (*b*) $V_\infty = 300$ ft/s.

■ Solution

Using Equation (3.38), we have

a. $V = \sqrt{V_\infty^2(1 - C_p)} = \sqrt{(80)^2[1 - (-5.3)]} = \boxed{200.8 \text{ ft/s}}$

b. $V = \sqrt{V_\infty^2(1 - C_p)} = \sqrt{(300)^2[1 - (-5.3)]} = \boxed{753 \text{ ft/s}}$

The above example illustrates two aspects of such a flow, as follows:

1. Consider a given point on the airfoil surface. The C_p is *given* at this point and, from the statement of the problem. C_p is obviously *unchanged* when the velocity is increased from 80 to 300 ft/s. Why? The answer underscores part of our discussion on dimensional analysis in Section 1.7, namely, C_p

should depend only on the Mach number, Reynolds number, shape and orientation of the body, and location on the body. For the low-speed inviscid flow considered here, the Mach number and Reynolds number are not in the picture. For this type of flow, the variation of C_p is a function *only* of location on the surface of the body, and the body shape and orientation. Hence, C_p will *not* change with V_∞ or ρ_∞ as long as the flow can be considered inviscid and incompressible. For such a flow, once the C_p distribution over the body has been determined by some means, the *same* C_p distribution will exist for all freestream values of V_∞ and ρ_∞.

2. In part (*b*) of Example 3.12, the velocity at the point where C_p is a peak (negative) value is a large value, namely, 753 ft/s. Is Equation (3.38) valid for this case? The answer is essentially *no*. Equation (3.38) assumes incompressible flow. The speed of sound at standard sea level is 1117 ft/s; hence, the freestream Mach number is $300/1117 = 0.269$. A flow where the local Mach number is less than 0.3 can be assumed to be essentially incompressible. Hence, the freestream Mach number satisfies this criterion. On the other hand, the flow rapidly expands over the top surface of the airfoil and accelerates to a velocity of 753 ft/s at the point of minimum pressure (the point of peak negative C_p). In the expansion, the speed of sound *decreases*. (We will find out why in Part 3.) Hence, at the point of minimum pressure, the local Mach number is *greater* than $\frac{753}{1117} = 0.674$. That is, the flow has expanded to such a high local Mach number that it is no longer incompressible. Therefore, the answer given in part (*b*) of Example 3.12 is not correct. (We will learn how to calculate the correct value in Part 3.) There is an interesting point to be made here. Just because a model is being tested in a low-speed, subsonic wind tunnel, it does not mean that the assumption of incompressible flow will hold for all aspects of the flow field. As we see here, in some regions of the flow field around a body, the flow can achieve such high local Mach numbers that it must be considered as compressible.

3.6 CONDITION ON VELOCITY FOR INCOMPRESSIBLE FLOW

Consulting our chapter road map in Figure 3.4, we have completed the left branch dealing with Bernoulli's equation. We now begin a more general consideration of incompressible flow, given by the center branch in Figure 3.4. However, before introducing Laplace's equation, it is important to establish a basic condition on velocity in an incompressible flow, as follows.

First, consider the physical definition of incompressible flow, namely, $\rho =$ constant. Since ρ is the mass per unit volume and ρ is constant, then a fluid element of fixed mass moving through an incompressible flow field must also have a fixed, constant volume. Recall Equation (2.32), which shows that $\nabla \cdot \mathbf{V}$ is physically the time rate of change of the volume of a moving fluid element per

unit volume. However, for an incompressible flow, we have just stated that the volume of a fluid element is constant [e.g., in Equation (2.32), $D(\delta\mathcal{V})/Dt \equiv 0$]. Therefore, for an incompressible flow,

$$\boxed{\nabla \cdot \mathbf{V} = 0} \tag{3.39}$$

The fact that the divergence of velocity is zero for an incompressible flow can also be shown directly from the continuity equation, Equation (2.52):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho\mathbf{V} = 0 \tag{2.52}$$

For incompressible flow, ρ = constant. Hence, $\partial\rho/\partial t = 0$ and $\nabla \cdot (\rho\mathbf{V}) = \rho\nabla \cdot \mathbf{V}$. Equation (2.52) then becomes

$$0 + \rho\nabla \cdot \mathbf{V} = 0$$

or

$$\nabla \cdot \mathbf{V} = 0$$

which is precisely Equation (3.39).

3.7 GOVERNING EQUATION FOR IRROTATIONAL, INCOMPRESSIBLE FLOW: LAPLACE'S EQUATION

We have seen in Section 3.6 that the principle of mass conservation for an incompressible flow can take the form of Equation (3.39):

$$\nabla \cdot \mathbf{V} = 0 \tag{3.39}$$

In addition, for an irrotational flow we have seen in Section 2.15 that a velocity potential ϕ can be defined such that [from Equation (2.154)]

$$\mathbf{V} = \nabla\phi \tag{2.154}$$

Therefore, for a flow that is both incompressible and irrotational, Equations (3.39) and (2.154) can be combined to yield

$$\nabla \cdot (\nabla\phi) = 0$$

or

$$\boxed{\nabla^2\phi = 0} \tag{3.40}$$

Equation (3.40) is *Laplace's equation*—one of the most famous and extensively studied equations in mathematical physics. Solutions of Laplace's equation are called *harmonic functions,* for which there is a huge bulk of existing literature. Therefore, it is most fortuitous that incompressible, irrotational flow is described by Laplace's equation, for which numerous solutions exist and are well understood.

For convenience, Laplace's equation is written below in terms of the three common orthogonal coordinate systems employed in Section 2.2:

Cartesian coordinates: $\phi = \phi(x, y, z)$

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0 \tag{3.41}$$

Cylindrical coordinates: $\phi = \phi(r, \theta, z)$

$$\nabla^2\phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} + \frac{\partial^2\phi}{\partial z^2} = 0 \tag{3.42}$$

Spherical coordinates: $\phi = \phi(r, \theta, \Phi)$

$$\nabla^2\phi = \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(r^2\sin\theta\frac{\partial\phi}{\partial r}\right) + \frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\phi}{\partial\theta}\right) + \frac{\partial}{\partial\Phi}\left(\frac{1}{\sin\theta}\frac{\partial\phi}{\partial\Phi}\right)\right] = 0 \tag{3.43}$$

Recall from Section 2.14 that, for a two-dimensional incompressible flow, a stream function ψ can be defined such that, from Equations (2.150*a* and *b*),

$$u = \frac{\partial\psi}{\partial y} \tag{2.150a}$$

$$v = -\frac{\partial\psi}{\partial x} \tag{2.150b}$$

The continuity equation, $\nabla \cdot \mathbf{V} = 0$, expressed in cartesian coordinates, is

$$\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{3.44}$$

Substituting Equations (2.150*a* and *b*) into (3.44), we obtain

$$\frac{\partial}{\partial x}\left(\frac{\partial\psi}{\partial y}\right) + \frac{\partial}{\partial y}\left(-\frac{\partial\psi}{\partial x}\right) = \frac{\partial^2\psi}{\partial x\,\partial y} - \frac{\partial^2\psi}{\partial y\,\partial x} = 0 \tag{3.45}$$

Since mathematically $\partial^2\psi/\partial x\,\partial y = \partial^2\psi/\partial y\,\partial x$, we see from Equation (3.45) that ψ automatically satisfies the continuity equation. Indeed, the very definition and use of ψ is a statement of the conservation of mass, and therefore Equations (2.150*a* and *b*) can be used in place of the continuity equation itself. If, in addition, the incompressible flow is irrotational, we have, from the irrotationality condition stated in Equation (2.131),

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \tag{2.131}$$

Substituting Equations (2.150*a* and *b*) into (2.131), we have

$$\frac{\partial}{\partial x}\left(-\frac{\partial\psi}{\partial x}\right) - \frac{\partial}{\partial y}\left(\frac{\partial\psi}{\partial y}\right) = 0$$

or
$$\boxed{\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0} \tag{3.46}$$

which is Laplace's equation. Therefore, the stream function also satisfies Laplace's equation, along with ϕ.

From Equations (3.40) and (3.46), we make the following obvious and important conclusions:

1. Any irrotational, incompressible flow has a velocity potential and stream function (for two-dimensional flow) that both satisfy Laplace's equation.
2. Conversely, any solution of Laplace's equation represents the velocity potential or stream function (two-dimensional) for an irrotational, incompressible flow.

Note that Laplace's equation is a second-order linear partial differential equation. The fact that it is *linear* is particularly important, because the sum of any particular solutions of a linear differential equation is also a solution of the equation. For example, if $\phi_1, \phi_2, \phi_3, \ldots, \phi_n$ represent n separate solutions of Equation (3.40), then the sum

$$\phi = \phi_1 + \phi_2 + \cdots + \phi_n$$

is also a solution of Equation (3.40). Since irrotational, incompressible flow is governed by Laplace's equation and Laplace's equation is linear, we conclude that a complicated flow pattern for an irrotational, incompressible flow can be synthesized by adding together a number of elementary flows that are also irrotational and incompressible. Indeed, this establishes the grand strategy for the remainder of our discussions on inviscid, incompressible flow. We develop flow-field solutions for several different elementary flows, which by themselves may not seem to be practical flows in real life. However, we then proceed to add (i.e., superimpose) these elementary flows in different ways such that the resulting flow fields do pertain to practical problems.

Before proceeding further, consider the irrotational, incompressible flow fields over different aerodynamic shapes, such as a sphere, cone, or airplane wing. Clearly, each flow is going to be distinctly different; the streamlines and pressure distribution over a sphere are quite different from those over a cone. However, these different flows are all governed by the same equation, namely, $\nabla^2 \phi = 0$. How, then, do we obtain different flows for the different bodies? The answer is found in the *boundary conditions*. Although the governing equation for the different flows is the same, the boundary conditions for the equation must conform to the different geometric shapes, and hence yield different flow-field solutions. Boundary conditions are therefore of vital concern in aerodynamic analysis. Let us examine the nature of boundary conditions further.

Consider the external aerodynamic flow over a stationary body, such as the airfoil sketched in Figure 3.18. The flow is bounded by (1) the freestream flow

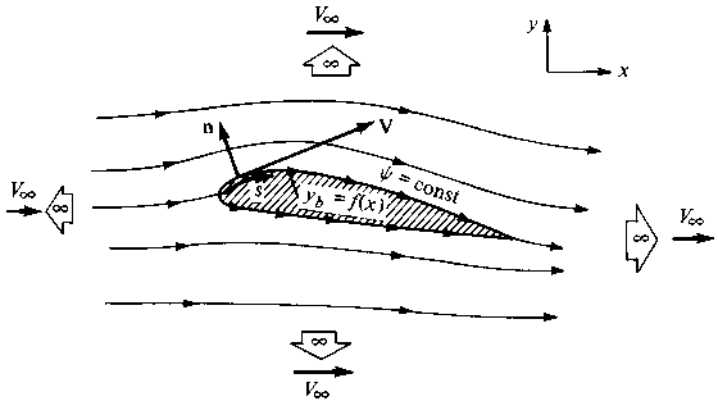


Figure 3.18 Boundary conditions at infinity and on a body; inviscid flow.

that occurs (theoretically) an infinite distance away from the body and (2) the surface of the body itself. Therefore, two sets of boundary conditions apply as follows.

3.7.1 Infinity Boundary Conditions

Far away from the body (toward infinity), in all directions, the flow approaches the uniform freestream conditions. Let V_∞ be aligned with the x direction as shown in Figure 3.18. Hence, at infinity,

$$u = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} = V_\infty \tag{3.47a}$$

$$v = \frac{\partial \phi}{\partial y} = \frac{\partial \psi}{\partial x} = 0 \tag{3.47b}$$

Equations (3.47*a* and *b*) are the *boundary conditions on velocity at infinity*. They apply at an infinite distance from the body in all directions, above and below, and to the left and right of the body, as indicated in Figure 3.18.

3.7.2 Wall Boundary Conditions

If the body in Figure 3.18 has a solid surface, then it is impossible for the flow to penetrate the surface. Instead, if the flow is viscous, the influence of friction between the fluid and the solid surface creates a zero velocity at the surface. Such viscous flows are discussed in Chapters 15 to 20. In contrast, for inviscid flows the velocity at the surface can be finite, but because the flow cannot penetrate the surface, the velocity vector must be *tangent* to the surface. This "wall tangency" condition is illustrated in Figure 3.18, which shows **V** tangent to the body surface. If the flow is tangent to the surface, then the component of velocity *normal* to the surface must be zero. Let **n** be a unit vector normal to the surface as shown in

Figure 3.18. The wall boundary condition can be written as

$$\mathbf{V} \cdot \mathbf{n} = (\nabla \phi) \cdot \mathbf{n} = 0 \tag{3.48a}$$

or

$$\frac{\partial \phi}{\partial n} = 0 \tag{3.48b}$$

Equation (3.48*a* or *b*) gives the boundary condition for velocity at the wall; it is expressed in terms of ϕ. If we are dealing with ψ rather than ϕ, then the wall boundary condition is

$$\frac{\partial \psi}{\partial s} = 0 \tag{3.48c}$$

where s is the distance measured along the body surface, as shown in Figure 3.18. Note that the body contour is a streamline of the flow, as also shown in Figure 3.18. Recall that $\psi =$ constant is the equation of a streamline. Thus, if the shape of the body in Figure 3.18 is given by $y_b = f(x)$, then

$$\psi_{\text{surface}} = \psi_{y=y_b} = \text{const} \tag{3.48d}$$

is an alternative expression for the boundary condition given in Equation (3.48*c*).

If we are dealing with neither ϕ nor ψ, but rather with the velocity components u and v themselves, then the wall boundary condition is obtained from the equation of a streamline, Equation (2.118), evaluated at the body surface; that is,

$$\boxed{\frac{dy_b}{dx} = \left(\frac{v}{u}\right)_{\text{surface}}} \tag{3.48e}$$

Equation (3.48*e*) states simply that the body surface is a streamline of the flow. The form given in Equation (3.48*e*) for the flow tangency condition at the body surface is used for all inviscid flows, incompressible to hypersonic, and does not depend on the formulation of the problem in terms of ϕ or ψ (or $\bar{\psi}$).

3.8 INTERIM SUMMARY

Reflecting on our previous discussions, the general approach to the solution of irrotational, incompressible flows can be summarized as follows:

1. Solve Laplace's equation for ϕ [Equation (3.40)] or ψ [Equation (3.46)] along with the proper boundary conditions [such as Equations (3.47) and (3.48)]. These solutions are usually in the form of a sum of elementary solutions (to be discussed in the following sections).
2. Obtain the flow velocity from $\mathbf{V} = \nabla \phi$ or $u = \partial \psi / \partial y$ and $v = -\partial \psi / \partial x$.
3. Obtain the pressure from Bernoulli's equation, $p + \frac{1}{2}\rho V^2 = p_\infty + \frac{1}{2}\rho V_\infty^2$, where p_∞ and V_∞ are known freestream conditions.

Since $\mathbf{V}$ and p are the primary dependent variables for an incompressible flow, steps 1 to 3 are all that we need to solve a given problem as long as the flow is incompressible and irrotational.

3.9 UNIFORM FLOW: OUR FIRST ELEMENTARY FLOW

In this section, we present the first of a series of elementary incompressible flows that later will be superimposed to synthesize more complex incompressible flows. For the remainder of this chapter and in Chapter 4, we deal with two-dimensional steady flows; three-dimensional steady flows are treated in Chapters 5 and 6.

Consider a uniform flow with velocity V_∞ oriented in the positive x direction, as sketched in Figure 3.19. It is easily shown (see Problem 3.8) that a uniform flow is a physically possible incompressible flow (i.e., it satisfies $\nabla \cdot \mathbf{V} = 0$) and that it is irrotational (i.e., it satisfies $\nabla \times \mathbf{V} = 0$). Hence, a velocity potential for uniform flow can be obtained such that $\nabla\phi = \mathbf{V}$. Examining Figure 3.19, and recalling Equation (2.156), we have

$$\frac{\partial \phi}{\partial x} = u = V_\infty \tag{3.49a}$$

and

$$\frac{\partial \phi}{\partial y} = v = 0 \tag{3.49b}$$

Integrating Equation (3.49a) with respect to x, we have

$$\phi = V_\infty x + f(y) \tag{3.50}$$

where $f(y)$ is a function of y only. Integrating Equation (3.49b) with respect to y, we obtain

$$\phi = \text{const} + g(x) \tag{3.51}$$

where $g(x)$ is a function of x only. In Equations (3.50) and (3.51), ϕ is the same function; hence, by comparing these equations, $g(x)$ must be $V_\infty x$, and $f(y)$ must be constant. Thus,

$$\phi = V_\infty x + \text{const} \tag{3.52}$$

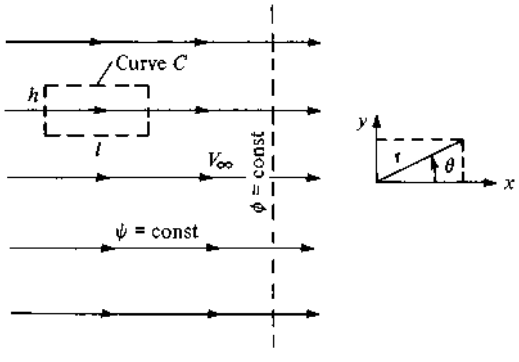


Figure 3.19 Uniform flow.

Note that in a practical aerodynamic problem, the actual value of ϕ is not significant; rather, ϕ is always used to obtain the velocity by differentiation; that is, $\nabla\phi = \mathbf{V}$. Since the derivative of a constant is zero, we can drop the constant from Equation (3.52) without any loss of rigor. Hence, Equation (3.52) can be written as

$$\boxed{\phi = V_\infty x} \tag{3.53}$$

Equation (3.53) is the velocity potential for a uniform flow with velocity V_∞ oriented in the positive x direction. Note that the derivation of Equation (3.53) does not depend on the assumption of incompressibility; it applies to any uniform flow, compressible or incompressible.

Consider the incompressible stream function ψ. From Figure 3.19 and Equations (2.150*a* and *b*), we have

$$\frac{\partial \psi}{\partial y} = u = V_\infty \tag{3.54a}$$

and

$$\frac{\partial \psi}{\partial x} = -v = 0 \tag{3.54b}$$

Integrating Equation (3.54*a*) with respect to y and Equation (3.54*b*) with respect to x, and comparing the results, we obtain

$$\boxed{\psi = V_\infty y} \tag{3.55}$$

Equation (3.55) is the stream function for an incompressible uniform flow oriented in the positive x direction.

From Section 2.14, the equation of a streamline is given by $\psi =$ constant. Therefore, from Equation (3.55), the streamlines for the uniform flow are given by $\psi = V_\infty y =$ constant. Because V_∞ is itself constant, the streamlines are thus given mathematically as $y =$ constant (i.e., as lines of constant y). This result is consistent with Figure 3.19, which shows the streamlines as horizontal lines (i.e., as lines of constant y). Also, note from Equation (3.53) that the equipotential lines are lines of constant x, as shown by the dashed line in Figure 3.19. Consistent with our discussion in Section 2.16, note that the lines of $\psi =$ constant and $\phi =$ constant are mutually perpendicular.

Equations (3.53) and (3.55) can be expressed in terms of polar coordinates, where $x = r\cos\theta$ and $y = r\sin\theta$, as shown in Figure 3.19. Hence,

$$\boxed{\phi = V_\infty r\cos\theta} \tag{3.56}$$

and

$$\boxed{\psi = V_\infty r\sin\theta} \tag{3.57}$$

Consider the circulation in a uniform flow. The definition of circulation is given by

$$\Gamma \equiv -\oint_C \mathbf{V} \cdot \mathbf{ds} \tag{2.136}$$

Let the closed curve C in Equation (2.136) be the rectangle shown at the left of Figure 3.19; h and l are the lengths of the vertical and horizontal sides, respectively, of the rectangle. Then

$$\oint_C \mathbf{V} \cdot \mathbf{ds} = -V_\infty l - 0(h) + V_\infty l + 0(h) = 0$$

or

$$\Gamma = 0 \tag{3.58}$$

Equation (3.58) is true for any arbitrary closed curve in the uniform flow. To show this, note that $\mathbf{V}_\infty$ is constant in both magnitude and direction, and hence

$$\Gamma = -\oint_C \mathbf{V} \cdot \mathbf{ds} = -\mathbf{V}_\infty \cdot \oint_C \mathbf{ds} = \mathbf{V}_\infty \cdot 0 = 0$$

because the line integral of **ds** around a closed curve is identically zero. Therefore, from Equation (3.58), we state that *circulation around any closed curve in a uniform flow is zero.*

The above result is consistent with Equation (2.137), which states that

$$\Gamma = -\iint_S (\nabla \times \mathbf{V}) \cdot \mathbf{dS} \tag{2.137}$$

We stated earlier that a uniform flow is irrotational; that is, $\nabla \times \mathbf{V} = 0$ everywhere. Hence, Equation (2.137) yields $\Gamma = 0$.

Note that Equations (3.53) and (3.55) satisfy Laplace's equation [see Equation (3.41)], which can be easily proved by simple substitution. Therefore, uniform flow is a viable elementary flow for use in building more complex flows.

3.10 SOURCE FLOW: OUR SECOND ELEMENTARY FLOW

Consider a two-dimensional, incompressible flow where all the streamlines are straight lines emanating from a central point O, as shown at the left of Figure 3.20. Moreover, let the velocity along each of the streamlines vary inversely with distance from point O. Such a flow is called a *source flow*. Examining Figure 3.20, we see that the velocity components in the radial and tangential directions are V_r and V_θ, respectively, where $V_\theta = 0$. The coordinate system in Figure 3.20 is a cylindrical coordinate system, with the z axis perpendicular to the page. (Note that polar coordinates are simply the cylindrical coordinates r and θ confined to a single plane given by $z =$ constant.) It is easily shown (see Problem 3.9) that (1) source flow is a physically possible incompressible flow, that is, $\nabla \cdot \mathbf{V} = 0$, at every point *except* the origin, where $\nabla \cdot \mathbf{V}$ becomes infinite, and (2) source flow is *irrotational* at every point.

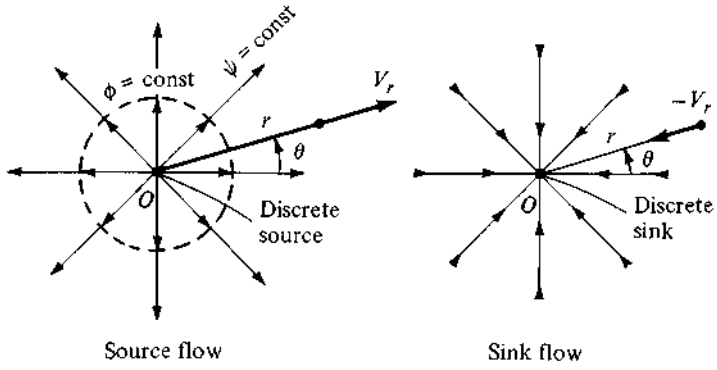


Figure 3.20 Source and sink flows.

In a source flow, the streamlines are directed *away from* the origin, as shown at the left of Figure 3.20. The opposite case is that of a *sink flow,* where by definition the streamlines are directed *toward* the origin, as shown at the right of Figure 3.20. For sink flow, the streamlines are still radial lines from a common origin, along which the flow velocity varies inversely with distance from point O. Indeed, a sink flow is simply a negative source flow.

The flows in Figure 3.20 have an alternate, somewhat philosophical interpretation. Consider the origin, point O, as a *discrete* source or sink. Moreover, interpret the radial flow surrounding the origin as simply being *induced* by the presence of the discrete source or sink at the origin (much like a magnetic field is induced in the space surrounding a current-carrying wire). Recall that, for a source flow, $\nabla \cdot \mathbf{V} = 0$ everywhere except at the origin, where it is infinite. Thus, the origin is a *singular point,* and we can interpret this singular point as a discrete source or sink of a given strength, with a corresponding induced flow field about the point. This interpretation is very convenient and is used frequently. Other types of singularities, such as doublets and vortices, are introduced in subsequent sections. Indeed, the irrotational, incompressible flow field about an arbitrary body can be visualized as a flow induced by a proper distribution of such singularities over the surface of the body. This concept is fundamental to many theoretical solutions of incompressible flow over airfoils and other aerodynamic shapes, and it is the very heart of modern numerical techniques for the solution of such flows. You will obtain a greater appreciation for the concept of distributed singularities for the solution of incompressible flow in Chapters 4 through 6. At this stage, however, simply visualize a discrete source (or sink) as a singularity that induces the flows shown in Figure 3.20.

Let us look more closely at the velocity field induced by a source or sink. By definition, the velocity is inversely proportional to the radial distance r. As stated earlier, this velocity variation is a physically possible flow, because it yields $\nabla \cdot \mathbf{V} = 0$. Moreover, it is the *only* such velocity variation for which the relation

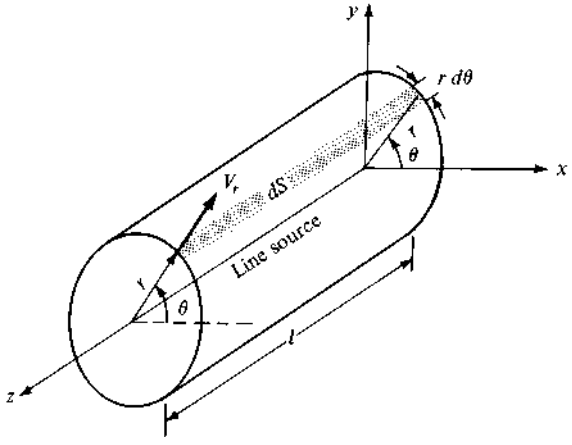


Figure 3.21 Volume flow rate from a line source.

$\nabla \cdot \mathbf{V} = 0$ is satisfied for the radial flows shown in Figure 3.20. Hence,

$$\mathbf{V}_r = \frac{c}{r} \tag{3.59a}$$

and

$$\mathbf{V}_\theta = 0 \tag{3.59b}$$

where c is constant. The value of the constant is related to the volume flow from the source, as follows. In Figure 3.20, consider a depth of length l perpendicular to the page, that is, a length l along the z axis. This is sketched in three-dimensional perspective in Figure 3.21. In Figure 3.21, we can visualize an entire line of sources along the z axis, of which the source O is just part. Therefore, in a two-dimensional flow, the discrete source, sketched in Figure 3.20, is simply a single point on the line source shown in Figure 3.21. The two-dimensional flow shown in Figure 3.20 is the same in any plane perpendicular to the z axis, that is, for any plane given by $z =$ constant. Consider the mass flow across the surface of the cylinder of radius r and height l as shown in Figure 3.21. The elemental mass flow across the surface element **dS** shown in Figure 3.21 is $\rho \mathbf{V} \cdot \mathbf{dS} = \rho V_r (r\, d\theta)(l)$. Hence, noting that V_r is the same value at any θ location for the fixed radius r, the total mass flow across the surface of the cylinder is

$$\dot{m} = \int_0^{2\pi} \rho V_r (r\, d\theta) l = \rho r l V_r \int_0^{2\pi} d\theta = 2\pi r l \rho V_r \tag{3.60}$$

Since ρ is defined as the mass per unit volume and $\dot{m}$ is mass per second, then $\dot{m}/\rho$ is the volume flow per second. Denote this rate of volume flow by $\dot{v}$. Thus, from Equation (3.60), we have

$$\dot{v} = \frac{\dot{m}}{\rho} = 2\pi r l V_r \tag{3.61}$$

Moreover, the rate of volume flow per unit length along the cylinder is $\dot{v}/l$. Denote this volume flow rate per unit length (which is the same as per unit depth perpendicular to the page in Figure 3.20) as Λ. Hence, from Equation 3.61, we obtain

$$\Lambda = \frac{\dot{v}}{l} = 2\pi r V_r$$

or

$$\boxed{V_r = \frac{\Lambda}{2\pi r}} \tag{3.62}$$

Hence, comparing Equations (3.59*a*) and (3.62), we see that the constant in Equation (3.59*a*) is $c = \Lambda/2\pi$. In Equation (3.62), Λ defines the *source strength,* it is physically the rate of volume flow from the source, per unit depth perpendicular to the page of Figure 3.20. Typical units of Λ are square meters per second or square feet per second. In Equation (3.62), a positive value of Λ represents a source, whereas a negative value represents a sink.

The velocity potential for a source can be obtained as follows. From Equations (2.157), (3.59*b*), and (3.62),

$$\frac{\partial \phi}{\partial r} = V_r = \frac{\Lambda}{2\pi r} \tag{3.63}$$

and

$$\frac{1}{r}\frac{\partial \phi}{\partial \theta} = V_\theta = 0 \tag{3.64}$$

Integrating Equation (3.63) with respect to r, we have

$$\phi = \frac{\Lambda}{2\pi} \ln r + f(\theta) \tag{3.65}$$

Integrating Equation (3.64) with respect to θ, we have

$$\phi = \text{const} + f(r) \tag{3.66}$$

Comparing Equations (3.65) and (3.66), we see that $f(r) = (\Lambda/2\pi) \ln r$ and $f(\theta) =$ constant. As explained in Section 3.9, the constant can be dropped without loss of rigor, and hence Equation (3.65) yields

$$\boxed{\phi = \frac{\Lambda}{2\pi} \ln r} \tag{3.67}$$

Equation (3.67) is the velocity potential for a two-dimensional source flow.

The stream function can be obtained as follows. From Equations (2.151*a* and *b*), (3.59*b*), and (3.62),

$$\frac{1}{r}\frac{\partial \psi}{\partial \theta} = V_r = \frac{\Lambda}{2\pi r} \tag{3.68}$$

and

$$-\frac{\partial \psi}{\partial r} = V_\theta = 0 \tag{3.69}$$

Integrating Equation (3.68) with respect to θ, we obtain

$$\psi = \frac{\Lambda}{2\pi}\theta + f(r) \tag{3.70}$$

Integrating Equation (3.69) with respect to r, we have

$$\psi = \text{const} + f(\theta) \tag{3.71}$$

Comparing Equations (3.70) and (3.71) and dropping the constant, we obtain

$$\boxed{\psi = \frac{\Lambda}{2\pi}\theta} \tag{3.72}$$

Equation (3.72) is the stream function for a two-dimensional source flow.

The equation of the streamlines can be obtained by setting Equation (3.72) equal to a constant:

$$\psi = \frac{\Lambda}{2\pi}\theta = \text{const} \tag{3.73}$$

From Equation (3.73), we see that θ = constant, which, in polar coordinates, is the equation of a straight line from the origin. Hence, Equation (3.73) is consistent with the picture of the source flow sketched in Figure 3.20. Moreover, Equation (3.67) gives an equipotential line as r = constant, that is, a circle with its center at the origin, as shown by the dashed line in Figure 3.20. Once again, we see that streamlines and equipotential lines are mutually perpendicular.

To evaluate the circulation for source flow, recall the $\nabla \times \mathbf{V} = 0$ everywhere. In turn, from Equation (2.137),

$$\Gamma = -\iint_S (\nabla \times \mathbf{V}) \cdot \mathbf{dS} = 0$$

for any closed curve C chosen in the flow field. Hence, as in the case of uniform flow discussed in Section 3.9, there is no circulation associated with the source flow.

It is straightforward to show that Equations (3.67) and (3.72) satisfy Laplace's equation, simply by substitution into $\nabla^2\phi = 0$ and $\nabla^2\psi = 0$ written in terms of cylindrical coordinates [see Equation (3.42)]. Therefore, source flow is a viable elementary flow for use in building more complex flows.

3.11 COMBINATION OF A UNIFORM FLOW WITH A SOURCE AND SINK

Consider a polar coordinate system with a source of strength Λ located at the origin. Superimpose on this flow a uniform stream with velocity V_∞ moving from left to right, as sketched in Figure 3.22. The stream function for the resulting flow is the sum of Equations (3.57) and (3.72):

$$\psi = V_\infty r \sin\theta + \frac{\Lambda}{2\pi}\theta \tag{3.74}$$

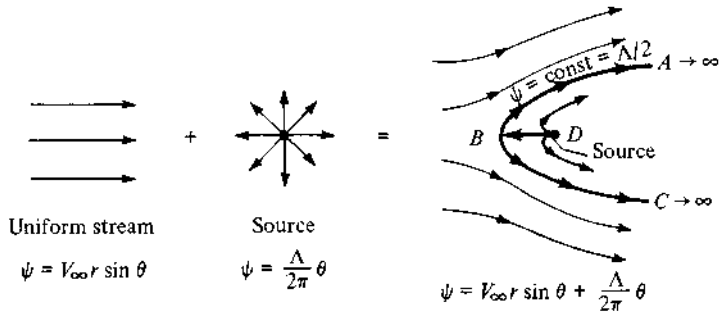


Figure 3.22 Superposition of a uniform flow and a source; flow over a semi-infinite body.

Since both Equations (3.57) and (3.72) are solutions of Laplace's equation, we know that Equation (3.74) also satisfies Laplace's equation; that is, Equation (3.74) describes a viable irrotational, incompressible flow. The streamlines of the combined flow are obtained from Equation (3.74) as

$$\psi = V_\infty r \sin\theta + \frac{\Lambda}{2\pi}\theta = \text{const} \tag{3.75}$$

The resulting streamline shapes from Equation (3.75) are sketched at the right of Figure 3.22. The source is located at point D. The velocity field is obtained by differentiating Equation (3.75):

$$V_r = \frac{1}{r}\frac{\partial \psi}{\partial \theta} = V_\infty \cos\theta + \frac{\Lambda}{2\pi r} \tag{3.76}$$

and

$$V_\theta = -\frac{\partial \psi}{\partial r} = -V_\infty \sin\theta \tag{3.77}$$

Note from Section 3.10 that the radial velocity from a source is $\Lambda/2\pi r$, and from Section 3.9 the component of the freestream velocity in the radial direction is $V_\infty \cos\theta$. Hence, Equation (3.76) is simply the direct sum of the two velocity fields—a result that is consistent with the linear nature of Laplace's equation. Therefore, not only can we add the values of ϕ or ψ to obtain more complex solutions, we can add their derivatives, that is, the velocities, as well.

The stagnation points in the flow can be obtained by setting Equations (3.76) and (3.77) equal to zero:

$$V_\infty \cos\theta + \frac{\Lambda}{2\pi r} = 0 \tag{3.78}$$

and

$$V_\infty \sin\theta = 0 \tag{3.79}$$

Solving for r and θ, we find that one stagnation point exists, located at $(r, \theta) = (\Lambda/2\pi V_\infty, \pi)$, which is labeled as point B in Figure 3.22. That is, the stagnation point is a distance $(\Lambda/2\pi V_\infty)$ directly upstream of the source. From this result, the

distance DB clearly grows smaller if V_∞ is increased and larger if Λ is increased—trends that also make sense based on intuition. For example, looking at Figure 3.22, you would expect that as the source strength is increased, keeping V_∞ the same, the stagnation point B will be blown further upstream. Conversely, if V_∞ is increased, keeping the source strength the same, the stagnation point will be blown further downstream.

If the coordinates of the stagnation point at B are substituted into Equation (3.75), we obtain

$$\psi = V_\infty \frac{\Lambda}{2\pi V_\infty} \sin \pi + \frac{\Lambda}{2\pi} \pi = \text{const}$$

$$\psi = \frac{\Lambda}{2} = \text{const}$$

Hence, the streamline that goes through the stagnation point is described by $\psi = \Lambda/2$. This streamline is shown as curve ABC in Figure 3.22.

Examining Figure 3.22, we now come to an important conclusion. Since we are dealing with inviscid flow, where the velocity at the surface of a solid body is tangent to the body, then *any* streamline of the combined flow at the right of Figure 3.22 could be replaced by a solid surface of the same shape. In particular, consider the streamline ABC. Because it contains the stagnation point at B, the streamline ABC is a *dividing* streamline; that is, it separates the fluid coming from the freestream and the fluid emanating from the source. All the fluid outside ABC is from the freestream, and all the fluid inside ABC is from the source. Therefore, as far as the freestream is concerned, the entire region inside ABC could be replaced with a solid body of the same shape, and the external flow, that is, the flow from the freestream, would not feel the difference. The streamline $\psi = \Lambda/2$ extends downstream to infinity, forming a semi-infinite body. Therefore, we are led to the following important interpretation. If we want to construct the flow over a solid semi-infinite body described by the curve ABC as shown in Figure 3.22, then all we need to do is take a uniform stream with velocity V_∞ and add to it a source of strength Λ at point D. The resulting superposition will then represent the flow over the prescribed solid semi-infinite body of shape ABC. This illustrates the practicality of adding elementary flows to obtain a more complex flow over a body of interest.

The superposition illustrated in Figure 3.22 results in the flow over the semi-infinite body ABC. This is a half-body that stretches to infinity in the downstream direction (i.e., the body is not closed). However, if we take a sink of equal strength as the source and add it to the flow downstream of point D, then the resulting body shape will be closed. Let us examine this flow in more detail.

Consider a polar coordinate system with a source and sink placed a distance b to the left and right of the origin, respectively, as sketched in Figure 3.23. The strengths of the source and sink are Λ and $-\Lambda$, respectively (equal and opposite). In addition, superimpose a uniform stream with velocity V_∞, as shown in Figure 3.23. The stream function for the combined flow at any point P with

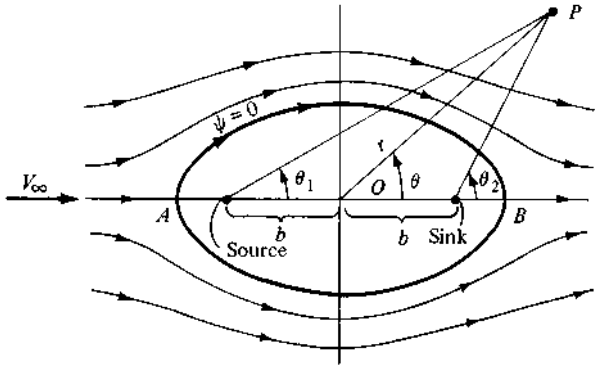


Figure 3.23 Superposition of a uniform flow and a source-sink pair; flow over a Rankine oval.

coordinates (r, θ) is obtained from Equations (3.57) and (3.72):

$$\psi = V_\infty r \sin\theta + \frac{\Lambda}{2\pi}\theta_1 - \frac{\Lambda}{2\pi}\theta_2$$

or

$$\psi = V_\infty r \sin\theta + \frac{\Lambda}{2\pi}(\theta_1 - \theta_2) \tag{3.80}$$

The velocity field is obtained by differentiating Equation (3.80) according to Equations (2.151*a* and *b*). Note from the geometry of Figure 3.23 that θ_1 and θ_2 in Equation (3.80) are functions of r, θ, and b. In turn, by setting $V = 0$, two stagnation points are found, namely, points A and B in Figure 3.23. These stagnation points are located such that (see Problem 3.13)

$$OA = OB = \sqrt{b^2 + \frac{\Lambda b}{\pi V_\infty}} \tag{3.81}$$

The equation of the streamlines is given by Equation (3.80) as

$$\psi = V_\infty r \sin\theta + \frac{\Lambda}{2\pi}(\theta_1 - \theta_2) = \text{const} \tag{3.82}$$

The equation of the specific streamline going through the stagnation points is obtained from Equation (3.82) by noting that $\theta = \theta_1 = \theta_2 = \pi$ at point A and $\theta = \theta_1 = \theta_2 = 0$ at point B. Hence, for the stagnation streamline, Equation (3.82) yields a value of zero for the constant. Thus, the stagnation streamline is given by $\psi = 0$, that is,

$$V_\infty r \sin\theta + \frac{\Lambda}{2\pi}(\theta_1 - \theta_2) = 0 \tag{3.83}$$

the equation of an oval, as sketched in Figure 3.23. Equation (3.83) is also the dividing streamline; all the flow from the source is consumed by the sink and is contained entirely inside the oval, whereas the flow outside the oval has originated with the uniform stream only. Therefore, in Figure 3.23, the region inside the oval

can be replaced by a solid body with the shape given by $\psi = 0$, and the region outside the oval can be interpreted as the inviscid, potential (irrotational), incompressible flow over the solid body. This problem was first solved in the nineteenth century by the famous Scottish engineer W. J. M. Rankine; hence, the shape given by Equation (3.83) and sketched in Figure 3.23 is called a *Rankine oval*.

3.12 DOUBLET FLOW: OUR THIRD ELEMENTARY FLOW

There is a special, degenerate case of a source-sink pair that leads to a singularity called a *doublet*. The doublet is frequently used in the theory of incompressible flow; the purpose of this section is to describe its properties.

Consider a source of strength Λ and a sink of equal (but opposite) strength $-\Lambda$ separated by a distance l, as shown in Figure 3.24*a*. At any point P in the flow, the stream function is

$$\psi = \frac{\Lambda}{2\pi}(\theta_1 - \theta_2) = -\frac{\Lambda}{2\pi}\Delta\theta \tag{3.84}$$

where $\Delta\theta = \theta_2 - \theta_1$ as seen from Figure 3.24*a*. Equation (3.84) is the stream function for a source-sink pair separated by the distance l.

Now in Figure 3.24*a*, let the distance l approach zero while the absolute magnitudes of the strengths of the source and sink increase in such a fashion that the product $l\Lambda$ remains constant. This limiting process is shown in Figure 3.24*b*. In the limit, as $l \to 0$ while $l\Lambda$ remains constant, we obtain a special flow pattern defined as a *doublet*. The *strength* of the doublet is denoted by κ and is defined as $\kappa \equiv l\Lambda$. The stream function for a doublet is obtained from Equation (3.84) as follows:

$$\psi = \lim_{\substack{l\to 0 \\ \kappa = l\Lambda = \text{const}}} \left(-\frac{\Lambda}{2\pi}d\theta\right) \tag{3.85}$$

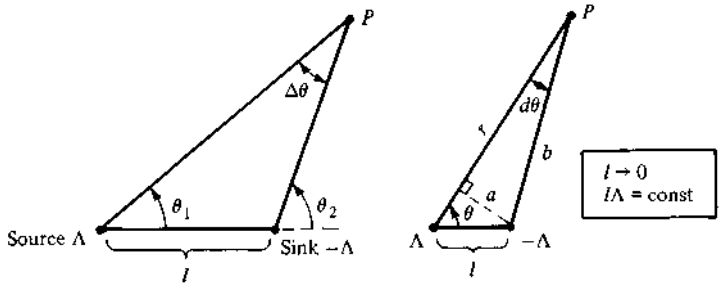


Figure 3.24 How a source-sink pair approaches a doublet in the limiting case.

where in the limit $\Delta\theta \to d\theta \to 0$. (Note that the source strength Λ approaches an infinite value in the limit.) In Figure 3.24*b*, let r and b denote the distances to point P from the source and sink, respectively. Draw a line from the sink perpendicular to r, and denote the length along this line by a. For an infinitesimal $d\theta$, the geometry of Figure 3.24*b* yields

$$a = l \sin\theta$$
$$b = r - l\cos\theta$$
$$d\theta = \frac{a}{b}$$

Hence,
$$d\theta = \frac{a}{b} = \frac{l\sin\theta}{r - l\cos\theta} \tag{3.86}$$

Substituting Equation (3.86) into (3.85), we have

$$\psi = \lim_{\substack{l\to 0 \\ \kappa=\text{const}}} \left(-\frac{\Lambda}{2\pi} \frac{l\sin\theta}{r - l\cos\theta} \right)$$

or
$$\psi = \lim_{\substack{l\to 0 \\ \kappa=\text{const}}} \left(-\frac{\kappa}{2\pi} \frac{\sin\theta}{r - l\cos\theta} \right)$$

or
$$\boxed{\psi = -\frac{\kappa}{2\pi}\frac{\sin\theta}{r}} \tag{3.87}$$

Equation (3.87) is the stream function for a doublet. In a similar fashion, the velocity potential for a doublet is given by (see Problem 3.14)

$$\boxed{\phi = \frac{\kappa}{2\pi}\frac{\cos\theta}{r}} \tag{3.88}$$

The streamlines of a doublet flow are obtained from Equation (3.87):

$$\psi = -\frac{\kappa}{2\pi}\frac{\sin\theta}{r} = \text{const} = c$$

or
$$r = -\frac{\kappa}{2\pi c}\sin\theta \tag{3.89}$$

Equation (3.89) gives the equation for the streamlines in doublet flow. Recall from analytic geometry that the following equation in polar coordinates

$$r = d\sin\theta \tag{3.90}$$

is a circle with a diameter d on the vertical axis and with the center located $d/2$ directly above the origin. Comparing Equations (3.89) and (3.90), we see that the streamlines for a doublet are a family of circles with diameter $\kappa/2\pi c$, as sketched in Figure 3.25. The different circles correspond to different values of the parameter c. Note that in Figure 3.24 we placed the source to the left of the sink; hence, in Figure 3.25 the direction of flow is out of the origin to

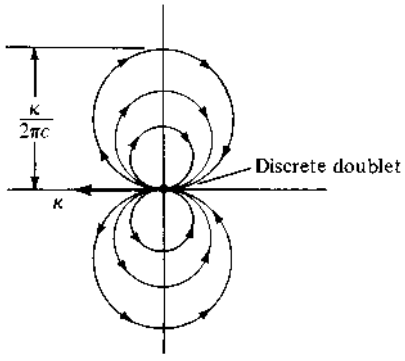


Figure 3.25 Doublet flow with strength κ.

the left and back into the origin from the right. In Figure 3.24, we could just as well have placed the sink to the left of the source. In such a case, the signs in Equations (3.87) and (3.88) would be reversed, and the flow in Figure 3.25 would be in the opposite direction. Therefore, a doublet has associated with it a sense of direction—the direction with which the flow moves around the circular streamlines. By convention, we designate the direction of the doublet by an arrow drawn from the sink to the source, as shown in Figure 3.25. In Figure 3.25, the arrow points to the left, which is consistent with the form of Equations (3.87) and (3.88). If the arrow would point to the right, the sense of rotation would be reversed, Equation (3.87) would have a positive sign, and Equation (3.88) would have a negative sign.

Returning to Figure 3.24, note that in the limit as $l \to 0$, the source and sink fall on top of each other. However, they do not extinguish each other, because the absolute magnitude of their strengths becomes infinitely large in the limit, and we have a singularity of strength $(\infty - \infty)$; this is an indeterminate form that can have a finite value.

As in the case of a source or sink, it is useful to interpret the doublet flow shown in Figure 3.25 as being *induced* by a discrete doublet of strength κ placed at the origin. Therefore, a doublet is a singularity that induces about it the double-lobed circular flow pattern shown in Figure 3.25.

3.13 NONLIFTING FLOW OVER A CIRCULAR CYLINDER

Consulting our road map given in Figure 3.4, we see that we are well into the third column, having already discussed uniform flow, sources and sinks, and doublets. Along the way, we have seen how the flow over a semi-infinite body can be simulated by the combination of a uniform flow with a source, and the flow over an oval-shaped body can be constructed by superimposing a uniform flow and a source-sink pair. In this section, we demonstrate that the combination of a uniform flow and a doublet produces the flow over a circular cylinder. A circular cylinder

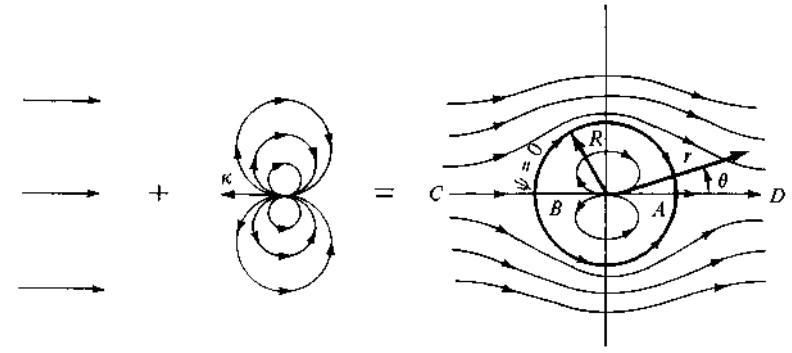


Figure 3.26 Superposition of a uniform flow and a doublet; nonlifting flow over a circular cylinder.

is one of the most basic geometric shapes available, and the study of the flow around such a cylinder is a classic problem in aerodynamics.

Consider the addition of a uniform flow with velocity V_∞ and a doublet of strength κ, as shown in Figure 3.26. The direction of the doublet is upstream, facing into the uniform flow. From Equations (3.57) and (3.87), the stream function for the combined flow is

$$\psi = V_\infty r \sin\theta - \frac{\kappa}{2\pi}\frac{\sin\theta}{r}$$

or

$$\psi = V_\infty r \sin\theta \left(1 - \frac{\kappa}{2\pi V_\infty r^2}\right) \tag{3.91}$$

Let $R^2 \equiv \kappa/2\pi V_\infty$. Then Equation (3.91) can be written as

$$\boxed{\psi = (V_\infty r \sin\theta)\left(1 - \frac{R^2}{r^2}\right)} \tag{3.92}$$

Equation (3.92) is the stream function for a uniform flow-doublet combination; it is also the stream function for the flow over a circular cylinder of radius R as shown in Figure 3.26 and as demonstrated below.

The velocity field is obtained by differentiating Equation (3.92), as follows:

$$V_r = \frac{1}{r}\frac{\partial\psi}{\partial\theta} = \frac{1}{r}(V_\infty r\cos\theta)\left(1 - \frac{R^2}{r^2}\right)$$

$$V_r = \left(1 - \frac{R^2}{r^2}\right) V_\infty \cos\theta \tag{3.93}$$

$$V_\theta = -\frac{\partial \psi}{\partial r} = -\left[(V_\infty r \sin\theta)\frac{2R^2}{r^3} + \left(1 - \frac{R^2}{r^2}\right)(V_\infty \sin\theta)\right]$$

$$V_\theta = -\left(1 + \frac{R^2}{r^2}\right) V_\infty \sin\theta \tag{3.94}$$

To locate the stagnation points, set Equations (3.93) and (3.94) equal to zero:

$$\left(1 - \frac{R^2}{r^2}\right) V_\infty \cos\theta = 0 \tag{3.95}$$

$$\left(1 + \frac{R^2}{r^2}\right) V_\infty \sin\theta = 0 \tag{3.96}$$

Simultaneously solving Equations (3.95) and (3.96) for r and θ, we find that there are two stagnation points, located at $(r, \theta) = (R, 0)$ and (R, π). These points are denoted as A and B, respectively, in Figure 3.26.

The equation of the streamline that passes through the stagnation point B is obtained by inserting the coordinates of B into Equation (3.92). For $r = R$ and $\theta = \pi$, Equation (3.92) yields $\psi = 0$. Similarly, inserting the coordinates of point A into Equation (3.92), we also find that $\psi = 0$. Hence, the same streamline goes through both stagnation points. Moreover, the equation of this streamline, from Equation (3.92), is

$$\psi = (V_\infty r \sin\theta)\left(1 - \frac{R^2}{r^2}\right) = 0 \tag{3.97}$$

Note that Equation (3.97) is satisfied by $r = R$ for all values of θ. However, recall that $R^2 \equiv \kappa/2\pi V_\infty$, which is a constant. Moreover, in polar coordinates, $r = \text{constant} = R$ is the equation of a circle of radius R with its center at the origin. Therefore, Equation (3.97) describes a circle with radius R, as shown in Figure 3.26. Moreover, Equation (3.97) is satisfied by $\theta = \pi$ and $\theta = 0$ for all values of r; hence, the entire horizontal axis through points A and B, extending infinitely far upstream and downstream, is part of the stagnation streamline.

Note that the $\psi = 0$ streamline, since it goes through the stagnation points, is the dividing streamline. That is, all the flow inside $\psi = 0$ (inside the circle) comes from the doublet, and all the flow outside $\psi = 0$ (outside the circle) comes from the uniform flow. Therefore, we can replace the flow inside the circle by a solid body, and the external flow will not know the difference. Consequently, the inviscid irrotational, incompressible flow over a circular cylinder of radius R can be synthesized by adding a uniform flow with velocity V_∞ and a doublet of strength κ, where R is related to V_∞ and κ through

$$R = \sqrt{\frac{\kappa}{2\pi V_\infty}} \tag{3.98}$$

Note from Equations (3.92) to (3.94) that the entire flow field is symmetrical about both the horizontal and vertical axes through the center of the cylinder, as

clearly seen by the streamline pattern sketched in Figure 3.26. Hence, the pressure distribution is also symmetrical about both axes. As a result, the pressure distribution over the top of the cylinder is exactly balanced by the pressure distribution over the bottom of the cylinder (i.e., *there is no net lift*). Similarly, the pressure distribution over the front of the cylinder is exactly balanced by the pressure distribution over the back of the cylinder (i.e., *there is no net drag*). In real life, the result of zero lift is easy to accept, but the result of zero drag makes no sense. We know that any aerodynamic body immersed in a real flow will experience a drag. This paradox between the theoretical result of zero drag, and the knowledge that in real life the drag is finite, was encountered in the year 1744 by the Frenchman Jean Le Rond d'Alembert—and it has been known as *d'Alembert's paradox* ever since. For d'Alembert and other fluid dynamic researchers during the eighteenth and nineteenth centuries, this paradox was unexplained and perplexing. Of course, today we know that the drag is due to viscous effects which generate frictional shear stress at the body surface and which cause the flow to separate from the surface on the back of the body, thus creating a large wake downstream of the body and destroying the symmetry of the flow about the vertical axis through the cylinder. These viscous effects are discussed in detail in Chapters 15 through 20. However, such viscous effects are not included in our present analysis of the inviscid flow over the cylinder. As a result, the inviscid theory predicts that the flow closes smoothly and completely behind the body, as sketched in Figure 3.26. It predicts no wake, and no asymmetries, resulting in the theoretical result of zero drag.

Let us quantify the above discussion. The velocity distribution on the surface of the cylinder is given by Equations (3.93) and (3.94) with $r = R$, resulting in

$$V_r = 0 \tag{3.99}$$

and

$$V_\theta = -2V_\infty \sin\theta \tag{3.100}$$

Note that at the surface of the cylinder, V_r is geometrically normal to the surface; hence, Equation (3.99) is consistent with the physical boundary condition that the component of velocity normal to a stationary solid surface must be zero. Equation (3.100) gives the tangential velocity, which is the full magnitude of velocity on the surface of the cylinder, that is, $V = V_\theta = -2V_\infty \sin\theta$ on the surface. The minus sign in Equation (3.100) is consistent with the sign convention in polar coordinates that V_θ is positive in the direction of increasing θ, that is, in the counterclockwise direction as shown in Figure 3.27. However, in Figure 3.26, the surface velocity for $0 \leq \theta \leq \pi$ is obviously in the opposite direction of increasing θ; hence, the minus sign in Equation (3.100) is proper. For $\pi \leq \theta \leq 2\pi$, the surface flow is in the same direction as increasing θ, but $\sin\theta$ is itself negative; hence, once again the minus sign in Equation (3.100) is proper. Note from Equation (3.100) that the velocity at the surface reaches a maximum value of $2V_\infty$ at the top and the bottom of the cylinder (where $\theta = \pi/2$ and $3\pi/2$, respectively), as shown in Figure 3.28. Indeed, these are the points of

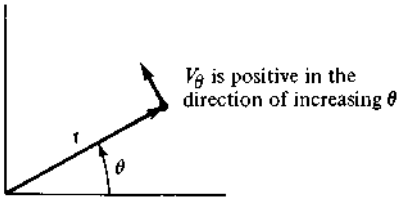


Figure 3.27 Sign convention for V_θ in polar coordinates.

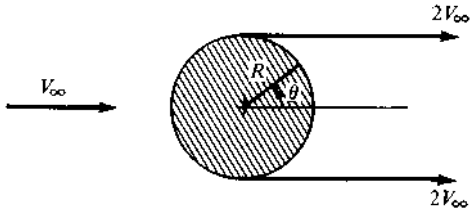


Figure 3.28 Maximum velocity in the flow over a circular cylinder.

maximum velocity for the entire flow field around the cylinder, as can be seen from Equations (3.93) and (3.94).

The pressure coefficient is given by Equation (3.38):

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2 \tag{3.38}$$

Combining Equations (3.100) and (3.38), we find that the surface pressure coefficient over a circular cylinder is

$$\boxed{C_p = 1 - 4\sin^2\theta} \tag{3.101}$$

Note that C_p varies from 1.0 at the stagnation points to -3.0 at the points of maximum velocity. The pressure coefficient distribution over the surface is sketched in Figure 3.29. The regions corresponding to the top and bottom halves of the cylinder are identified at the top of Figure 3.29. Clearly, the pressure distribution over the top half of the cylinder is equal to the pressure distribution over the bottom half, and hence the lift must be zero, as discussed earlier. Moreover, the regions corresponding to the front and rear halves of the cylinder are identified at the bottom of Figure 3.29. Clearly, the pressure distributions over the front and rear halves are the same, and hence the drag is theoretically zero, as also discussed previously. These results are confirmed by Equations (1.15) and (1.16). Since $c_f = 0$ (we are dealing with an inviscid flow), Equations (1.15) and (1.16)

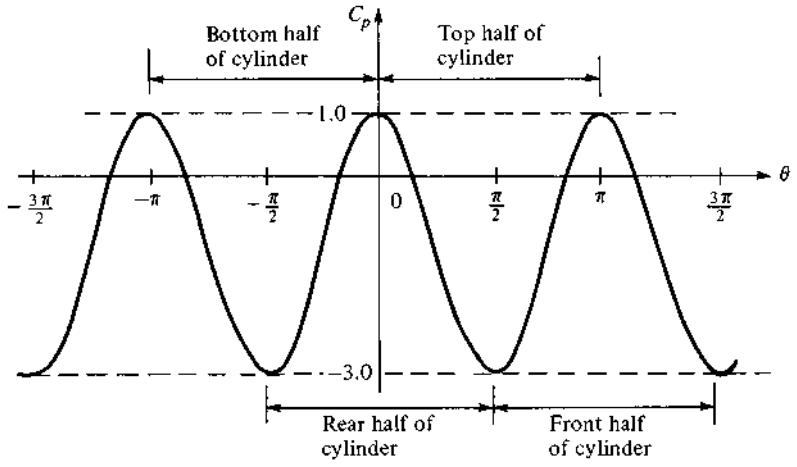


Figure 3.29 Pressure coefficient distribution over the surface of a circular cylinder; theoretical results for inviscid, incompressible flow.

become, respectively,

$$c_n = \frac{1}{c}\int_0^c (C_{p,l} - C_{p,u})\,dx \tag{3.102}$$

$$c_a = \frac{1}{c}\int_{\text{LE}}^{\text{TE}} (C_{p,u} - C_{p,l})\,dy \tag{3.103}$$

For the circular cylinder, the chord c is the horizontal diameter. From Figure 3.29, $C_{p,l} = C_{p,u}$ for corresponding stations measured along the chord, and hence the integrands in Equations (3.102) and (3.103) are identically zero, yielding $c_n = c_a = 0$. In turn, the lift and drag are zero, thus, again confirming our previous conclusions.

EXAMPLE 3.13

Consider the nonlifting flow over a circular cylinder. Calculate the locations on the surface of the cylinder where the surface pressure equals the freestream pressure.

■ Solution

When $p = p_\infty$, then $C_p = 0$. From Equation (3.101),

$$C_p = 0 = 1 - 4\sin^2\theta$$

Hence,

$$\sin\theta = \pm\tfrac{1}{2}$$

$$\boxed{\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ}$$

These points, as well as the stagnation points and points of minimum pressure, are illustrated in Figure 3.30. Note that at the stagnation point, where $C_p = 1$, the pressure is

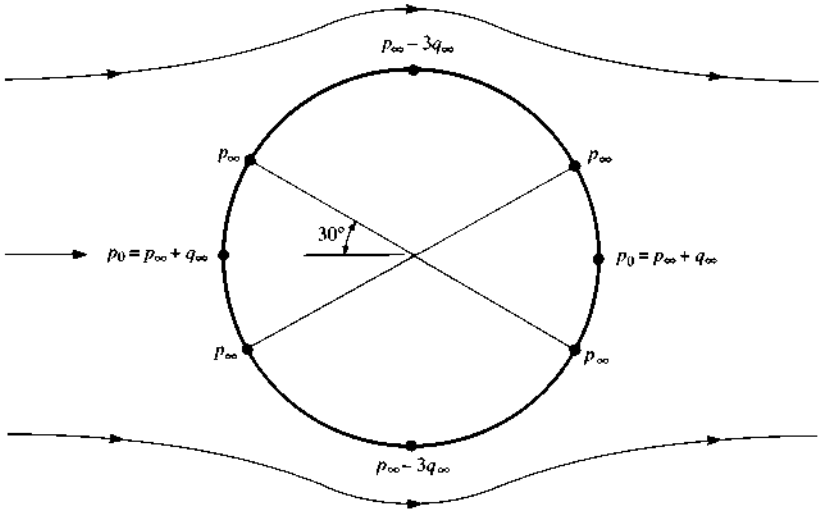


Figure 3.30 Values of pressure at various locations on the surface of a circular cylinder; nonlifting case.

$p_\infty + q_\infty$; the pressure decreases to p_∞ in the first 30° of expansion around the body, and the minimum pressure at the top and bottom of the cylinder, consistent with $C_p = -3$, is $p_\infty - 3q_\infty$.

EXAMPLE 3.14

In the nonlifting flow over a circular cylinder, consider the infinitesimally small fluid elements moving along the surface of the cylinder. Calculate the angular locations over the surface where the acceleration of the fluid elements are a local maximum and minimum. For the case where the radius of the cylinder is 1 m and the freestream flow velocity is 50 m/s, calculate the values of the local maximum and minimum accelerations.

■ Solution

From Equation (3.100) the local velocity of the fluid elements on the surface as a function of angular location θ is

$$V_\theta = -2V_\infty \sin\theta \tag{3.100}$$

The acceleration of the fluid elements is dV_θ/dt. From Equation (3.100),

$$\frac{dV_\theta}{dt} = -2V_\infty(\cos\theta)\frac{d\theta}{dt} \tag{E3.1}$$

Return to Figure 3.28. Let $d\theta$ be an incremental change in θ. The corresponding incremental distance on the cylinder surface subtended by $d\theta$ is ds, given by

$$ds = R\,d\theta$$

Hence,

$$\frac{d\theta}{dt} = \frac{1}{R}\frac{ds}{dt} \tag{E3.2}$$

Substituting Equation (E3.2) into (E3.1), we have

$$\frac{dV_\theta}{dt} = -2V_\infty(\cos\theta)\left(\frac{1}{R}\frac{ds}{dt}\right) \tag{E3.3}$$

For a fluid element moving over an incremental distance ds in time dt, its linear velocity is by definition ds/dt. Thus, $V_\theta \equiv ds/dt$. Replacing ds/dt in Equation (E3.3) with V_θ, we have

$$\frac{dV_\theta}{dt} = -2V_\infty(\cos\theta)\left(\frac{V_\theta}{R}\right) \tag{E3.4}$$

Substitute Equation (3.100) into (E3.4).

$$\frac{dV_\theta}{dt} = \frac{4V_\infty^2}{R}\sin\theta\cos\theta \tag{E3.5}$$

From the trigonometric identity

$$\sin 2\theta \equiv 2\sin\theta\cos\theta$$

Equation (E3.5) becomes

$$\frac{dV_\theta}{dt} = \frac{2V_\infty^2}{R}\sin 2\theta \tag{E3.6}$$

Equation (E3.6) gives the flow acceleration along the surface, dV_θ/dt, as a function of angular location θ along the surface. To find the θ locations at which the acceleration is a maximum or minimum, differentiate Equation (E3.6) with respect to θ, and set the result equal to zero.

$$\frac{d}{d\theta}\left(\frac{dV_\theta}{dt}\right) = \frac{4V_\infty^2}{R}\cos 2\theta = 0 \tag{E3.7}$$

Solving Equation (E3.7) for θ, we have the location where acceleration is either a local maximum or minimum:

$$\boxed{\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ} \tag{E3.8}$$

From Equation (E3.6), the values of the local flow acceleration at each one of these locations are respectively

$$\boxed{\frac{2V_\infty^2}{R}, -\frac{2V_\infty^2}{R}, \frac{2V_\infty^2}{R}, -\frac{2V_\infty^2}{R}} \tag{E3.9}$$

Interpretation: Return to Figures 3.27 and 3.28, and note our sign convention that θ is zero at the rearward stagnation point and increases in the *counterclockwise* direction; this

is the conventional polar coordinate system where θ increases in the counterclockwise direction, and we have followed this convention throughout the present chapter. That is, as θ sweeps *counterclockwise,* $\theta = 90^\circ$ is the top of the cylinder, $\theta = 180^\circ$ is the location of the forward stagnation point, and $\theta = 270^\circ$ is the bottom of the cylinder. The consistent sign convention for V_θ, as seen in Figure 3.27, is positive in the direction of increasing θ. The direction of V_∞ in Figure 3.28, however, is left to right. Hence the actual left-to-right flow over the *top* of the cylinder has a *negative* velocity because it is running *counter* to the positive direction of V_θ shown in Figure 3.27. The left-to-right flow over the bottom of the cylinder, however, has a positive velocity because it is running in the positive direction of V_θ. This is totally consistent with the result given for V_θ in Equation (3.100).

$$V_\theta = -2V_\infty \sin\theta \tag{3.100}$$

Over the top of the cylinder, where θ varies from 0° to 180°, Equation (3.100) gives a *negative* value of V_θ, whereas over the bottom of the cylinder, where θ varies from 180° to 360°, Equation (3.100) gives a *positive* value of V_θ. With this in mind, now consider the location and values of the local maximum and minimum acceleration as given by relations (E3.8) and (E3.9), respectively. At $\theta = 45^\circ$, the acceleration is a positive value; i.e., the time rate of change V_θ is positive. However, at $\theta = 45^\circ$, the velocity is a negative value, and with a positive time rate of change, the velocity is becoming less negative; i.e., the absolute value of V_θ is becoming *smaller*. The fluid element at $\theta = 45^\circ$ is *slowing down*. The point $\theta = 45^\circ$ is therefore a point of maximum *deceleration* (minimum acceleration). In contrast, at $\theta = 135^\circ$ the velocity is a negative value, but from (E3.9), the local acceleration is also negative. This means that at $\theta = 135^\circ$, with a negative time rate of change of velocity, the velocity itself is becoming more negative, and its absolute value is increasing. At $\theta = 135^\circ$, the fluid element is *speeding up,* and $\theta = 135^\circ$ is a point of maximum *acceleration*.

Over the bottom surface of the cylinder, V_θ is positive. At $\theta = 225^\circ$ (which is on the front face of the cylinder) the acceleration from (E3.9) is also positive; i.e., at $\theta = 225^\circ$ the fluid element is speeding up and hence $\theta = 225^\circ$ is a point of maximum *acceleration*. At $\theta = 315^\circ$ (which is on the back face of the cylinder) the acceleration from (E3.9) is negative; i.e., at $\theta = 315^\circ$ the fluid element is slowing down. Hence, $\theta = 315^\circ$ is a point of maximum deceleration.

In short, again examining Figure 3.28, the flow over the front face of the cylinder starts out at zero velocity at the front stagnation point ($\theta = 180^\circ$), accelerates to a maximum velocity of $2V_\infty$ at the top ($\theta = 90^\circ$) and bottom ($\theta = 270^\circ$) of the cylinder, with a maximum acceleration occurring at $\theta = 135^\circ$ and 225°. The flow then decelerates over the back face of the cylinder, coming to zero velocity at the rear stagnation point ($\theta = 0$), with maximum deceleration occurring at $\theta = 45^\circ$ and 315°. It is interesting, and it also makes intuitive sense, that the maximum acceleration and deceleration occur geometrically at the points on the cylinder surface halfway between the stagnation points and the points of maximum velocity at the top and bottom. We might have guessed this right from the beginning of this worked example. But frequently in physical science intuition can betray us, and it is necessary to make an analysis as we have done here to find the correct result.

Note: The interpretation given above is rather long and protracted. It was carried out in order to identify which of the positive or negative values in (E3.9) corresponded to an acceleration or deceleration, and at all times we had to keep our mind focused on the sign convention for the polar coordinate system shown in Figures 3.27 and 3.28. This polar coordinate system with θ increasing in the counterclockwise direction is standard mathematical convention, and we have chosen to use it throughout this chapter.

Finally, let us calculate the actual value of maximum acceleration and deceleration for $R = 1$ m and $V_\infty = 50$ m/s. From (E3.9),

$$\frac{2V_\infty^2}{R} = \frac{2(50)^2}{(1)} = \boxed{5000 \text{ m/s}^2}$$

Since the standard sea level value of the acceleration of gravity on earth is 9.8 ms/s^2, a fluid element flowing over the surface of the circular cylinder in this example experiences a mind-boggling maximum acceleration and deceleration of

$$\frac{5000}{9.8} = 510.2\ g$$

So once again, as we saw with Example 2.3, a fluid element in an otherwise seemingly benign flow field can experience tremendously large accelerations.

3.14 VORTEX FLOW: OUR FOURTH ELEMENTARY FLOW

Again, consulting our chapter road map in Figure 3.4, we have discussed three elementary flows—uniform flow, source flow, and doublet flow—and have superimposed these elementary flows to obtain the nonlifting flow over several body shapes, such as ovals and circular cylinders. In this section, we introduce our fourth, and last, elementary flow: vortex flow. In turn, in Sections 3.15 and 3.16, we see how the superposition of flows involving such vortices leads to cases with finite lift.

Consider a flow where all the streamlines are concentric circles about a given point, as sketched in Figure 3.31. Moreover, let the velocity along any given circular streamline be constant, but let it vary from one streamline to another inversely with distance from the common center. Such a flow is called a *vortex flow*. Examine Figure 3.31; the velocity components in the radial and tangential directions are V_r and V_θ, respectively, where $V_r = 0$ and $V_\theta = \text{constant}/r$. It is easily shown (try it yourself) that (1) vortex flow is a physically possible incompressible flow, that is, $\nabla \cdot \mathbf{V} = 0$ at every point, and (2) vortex flow is irrotational, that is, $\nabla \times \mathbf{V} = 0$, at every point except the origin.

From the definition of vortex flow, we have

$$V_\theta = \frac{\text{const}}{r} = \frac{C}{r} \tag{3.104}$$

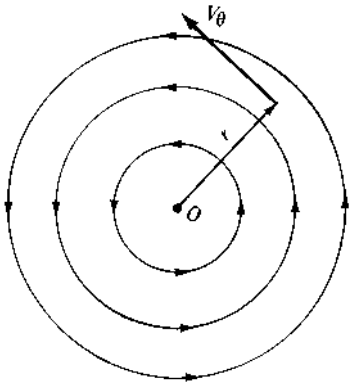


Figure 3.31 Vortex flow.

To evaluate the constant C, take the circulation around a given circular streamline of radius r:

$$\Gamma = -\oint_C \mathbf{V} \cdot \mathbf{ds} = -V_\theta (2\pi r)$$

or

$$\boxed{V_\theta = -\frac{\Gamma}{2\pi r}} \tag{3.105}$$

Comparing Equations (3.104) and (3.105), we see that

$$C = -\frac{\Gamma}{2\pi} \tag{3.106}$$

Therefore, for vortex flow, Equation (3.106) demonstrates that the circulation taken about all streamlines is the same value, namely, $\Gamma = -2\pi C$. By convention, Γ is called the *strength* of the vortex flow, and Equation (3.105) gives the velocity field for a vortex flow of strength Γ. Note from Equation (3.105) that V_θ is negative when Γ is positive; that is, a vortex of positive strength rotates in the *clockwise* direction. (This is a consequence of our sign convention on circulation defined in Section 2.13, namely, positive circulation is clockwise.)

We stated earlier that vortex flow is irrotational except at the origin. What happens at $r = 0$? What is the value of $\nabla \times \mathbf{V}$ at $r = 0$? To answer these questions, recall Equation (2.137) relating circulation to vorticity:

$$\Gamma = -\iint_S (\nabla \times \mathbf{V}) \cdot \mathbf{dS} \tag{2.137}$$

Combining Equations (3.106) and (2.137), we obtain

$$2\pi C = \iint_S (\nabla \times \mathbf{V}) \cdot \mathbf{dS} \tag{3.107}$$

Since we are dealing with two-dimensional flow, the flow sketched in Figure 3.31 takes place in the plane of the paper. Hence, in Equation (3.107), both $\nabla \times \mathbf{V}$ and **dS** are in the same direction, both perpendicular to the plane of the flow. Thus, Equation (3.107) can be written as

$$2\pi C = \iint_S (\nabla \times \mathbf{V}) \cdot \mathbf{dS} = \iint_S |\nabla \times \mathbf{V}| \, dS \tag{3.108}$$

In Equation (3.108), the surface integral is taken over the circular area inside the streamline along which the circulation $\Gamma = -2\pi C$ is evaluated. However, Γ is the same for all the circulation streamlines. In particular, choose a circle as close to the origin as we wish (i.e., let $r \to 0$). The circulation will still remain $\Gamma = -2\pi C$. However, the area inside this small circle around the origin will become infinitesimally small, and

$$\iint_S |\nabla \times \mathbf{V}| \, dS \to |\nabla \times \mathbf{V}| \, dS \tag{3.109}$$

Combining Equations (3.108) and (3.109), in the limit as $r \to 0$, we have

$$2\pi C = |\nabla \times \mathbf{V}| \, dS$$

or

$$|\nabla \times \mathbf{V}| = \frac{2\pi C}{dS} \tag{3.110}$$

However, as $r \to 0$, $dS \to 0$. Therefore, in the limit as $r \to 0$, from Equation (3.110), we have

$$|\nabla \times \mathbf{V}| \to \infty$$

Conclusion: Vortex flow is irrotational everywhere except at the point $r = 0$, where the vorticity is infinite. Therefore, the origin, $r = 0$, is a singular point in the flow field. We see that, along with sources, sinks, and doublets, the vortex flow contains a singularity. Hence, we can interpret the singularity itself, that is, point O in Figure 3.31, to be a point vortex which induces about it the circular vortex flow shown in Figure 3.31.

The velocity potential for vortex flow can be obtained as follows:

$$\frac{\partial \phi}{\partial r} = V_r = 0 \tag{3.111a}$$

$$\frac{1}{r}\frac{\partial \phi}{\partial \theta} = V_\theta = -\frac{\Gamma}{2\pi r} \tag{3.111b}$$

Integrating Equations (3.111*a* and *b*), we find

$$\boxed{\phi = -\frac{\Gamma}{2\pi}\theta} \tag{3.112}$$

Equation (3.112) is the velocity potential for vortex flow.

The stream function is determined in a similar manner:

$$\frac{1}{r}\frac{\partial \psi}{\partial \theta} = V_r = 0 \tag{3.113a}$$

$$-\frac{\partial \psi}{\partial r} = V_\theta = -\frac{\Gamma}{2\pi r} \tag{3.113b}$$

Integrating Equations (3.113*a* and *b*), we have

$$\boxed{\psi = \frac{\Gamma}{2\pi}\ln r} \tag{3.114}$$

Equation (3.114) is the stream function for vortex flow. Note that since $\psi =$ constant is the equation of the streamline, Equation (3.114) states that the streamlines of vortex flow are given by $r =$ constant (i.e., the streamlines are circles). Thus, Equation (3.114) is consistent with our definition of vortex flow. Also, note from Equation (3.112) that equipotential lines are given by $\theta =$ constant, that is, straight radial lines from the origin. Once again, we see that equipotential lines and streamlines are mutually perpendicular.

At this stage, we summarize the pertinent results for our four elementary flows in Table 3.1.

Table 3.1

Type of flow	Velocity	ϕ	ψ
Uniform flow in x direction	$u = V_\infty$	$V_\infty x$	$V_\infty y$
Source	$V_r = \dfrac{\Lambda}{2\pi r}$	$\dfrac{\Lambda}{2\pi}\ln r$	$\dfrac{\Lambda}{2\pi}\theta$
Vortex	$V_\theta = -\dfrac{\Gamma}{2\pi r}$	$-\dfrac{\Gamma}{2\pi}\theta$	$\dfrac{\Gamma}{2\pi}\ln r$
Doublet	$V_r = -\dfrac{\kappa}{2\pi}\dfrac{\cos\theta}{r^2}$	$\dfrac{\kappa}{2\pi}\dfrac{\cos\theta}{r}$	$-\dfrac{\kappa}{2\pi}\dfrac{\sin\theta}{r}$
	$V_\theta = -\dfrac{\kappa}{2\pi}\dfrac{\sin\theta}{r^2}$		

EXAMPLE 3.15

Consider the vortex flow discussed in this section. Imagine that you are standing at a location 20 feet from the center of the vortex, and you are feeling a 100 mi/h wind. What is the strength of the vortex?

■ Solution

From Equation (3.105),

$$V_\theta = -\frac{\Gamma}{2\pi r}$$

In this example, the direction of the 100 mi/h wind is not stipulated; for our purpose it is not relevant. We are interested only in the magnitude of the strength of the vortex. Hence, recalling that 88 ft/s = 60 mi/h, we have

$$|\Gamma| = 2\pi r V_\theta = 2\pi(20)(100)\frac{88}{60} = \boxed{1.843\,x\,10^4\ \text{ft}^2/\text{s}}$$

Comment: Actual numbers for circulation are not something frequently quoted. Therefore most aerodynamicists do not have a "feel" for the magnitude of Γ in most applications. In contrast, we do have a feel for more common properties such as velocity; we have a feel for what a 100 mi/h wind is like, especially anybody who has ventured outside in a tropical storm. The purpose of this example is to show us a number for Γ; in this case it is over 18,000 in the English engineering units of feet and seconds. In most cases we really do not care what the value of Γ is because it is immediately used to obtain more practical data, such as aerodynamic lift (as we will see in the next section).

3.15 LIFTING FLOW OVER A CYLINDER

In Section 3.13, we superimposed a uniform flow and a doublet to synthesize the flow over a circular cylinder, as shown in Figure 3.26. In addition, we proved that both the lift and drag were zero for such a flow. However, the streamline pattern shown at the right of Figure 3.26 is not the only flow that is theoretically possible around a circular cylinder. It *is* the only flow that is consistent with zero lift. However, there are other possible flow patterns around a circular cylinder—different flow patterns that result in a nonzero lift on the cylinder. Such lifting flows are discussed in this section.

Now you might be hesitant at this moment, perplexed by the question as to how a lift could possibly be exerted on a circular cylinder. Is not the body perfectly symmetric, and would not this geometry always result in a symmetric flow field with a consequent zero lift, as we have already discussed? You might be so perplexed that you run down to the laboratory, place a stationary cylinder in a low-speed tunnel, and measure the lift. To your satisfaction, you measure no lift, and you walk away muttering that the subject of this section is ridiculous—there

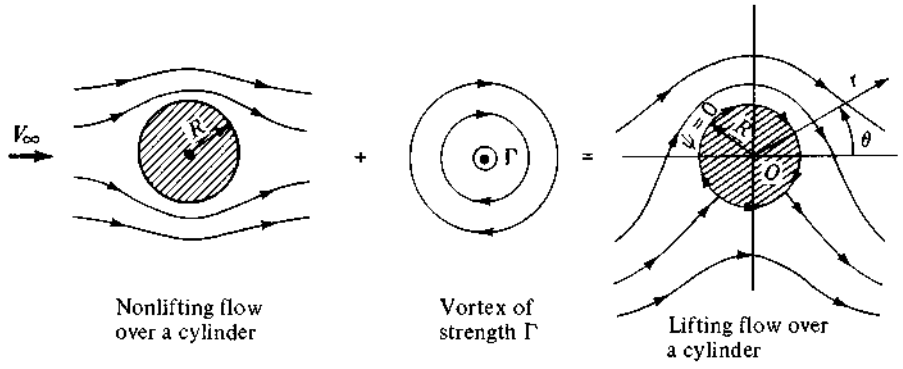


Figure 3.32 The synthesis of lifting flow over a circular cylinder.

is no lift on the cylinder. However, go back to the wind tunnel, and this time run a test with the cylinder *spinning* about its axis at relatively high revolutions per minute. This time you measure a *finite* lift. Also, by this time you might be thinking of other situations: spin on a baseball causes it to curve, and spin on a golfball causes it to hook or slice. Clearly, in real life there are nonsymmetric aerodynamic forces acting on these symmetric, spinning bodies. So, maybe the subject matter of this section is not so ridiculous after all. Indeed, as you will soon appreciate, the concept of lifting flow over a cylinder will start us on a journey which leads directly to the theory of the lift generated by airfoils, as discussed in Chapter 4.

Consider the flow synthesized by the addition of the nonlifting flow over a cylinder and a vortex of strength Γ, as shown in Figure 3.32. The stream function for nonlifting flow over a circular cylinder of radius R is given by Equation (3.92):

$$\psi_1 = (V_\infty r \sin\theta)\left(1 - \frac{R^2}{r^2}\right) \tag{3.92}$$

The stream function for a vortex of strength Γ is given by Equation (3.114). Recall that the stream function is determined within an arbitrary constant; hence, Equation (3.114) can be written as

$$\psi_2 = \frac{\Gamma}{2\pi}\ln r + \text{ const} \tag{3.115}$$

Since the value of the constant is arbitrary, let

$$\text{Const} = -\frac{\Gamma}{2\pi}\ln R \tag{3.116}$$

Combining Equations (3.115) and (3.116), we obtain

$$\psi_2 = \frac{\Gamma}{2\pi}\ln\frac{r}{R} \tag{3.117}$$

Equation (3.117) is the stream function for a vortex of strength Γ and is just as valid as Equation (3.114) obtained earlier; the only difference between these two equations is a constant of the value given by Equation (3.116).

The resulting stream function for the flow shown at the right of Figure 3.32 is

$$\psi = \psi_1 + \psi_2$$

or

$$\boxed{\psi = (V_\infty r \sin\theta)\left(1 - \frac{R^2}{r^2}\right) + \frac{\Gamma}{2\pi}\ln\frac{r}{R}} \tag{3.118}$$

From Equation (3.118), if $r = R$, then $\psi = 0$ for all values of θ. Since $\psi =$ constant is the equation of a streamline, $r = R$ is therefore a streamline of the flow, but $r = R$ is the equation of a circle of radius R. Hence, Equation (3.118) is a valid stream function for the inviscid, incompressible flow over a circular cylinder of radius R, as shown at the right of Figure 3.32. Indeed, our previous result given by Equation (3.92) is simply a special case of Equation (3.118) with $\Gamma = 0$.

The resulting streamline pattern given by Equation (3.118) is sketched at the right of Figure 3.32. Note that the streamlines are no longer symmetrical about the horizontal axis through point O, and you might suspect (correctly) that the cylinder will experience a resulting finite normal force. However, the streamlines are symmetrical about the vertical axis through O, and as a result the drag will be zero, as we prove shortly. Note also that because a vortex of strength Γ has been added to the flow, the circulation about the cylinder is now finite and equal to Γ.

The velocity field can be obtained by differentiating Equation (3.118). An equally direct method of obtaining the velocities is to add the velocity field of a vortex to the velocity field of the nonlifting cylinder. (Recall that because of the linearity of the flow, the velocity components of the superimposed elementary flows add directly.) Hence, from Equations (3.93) and (3.94) for nonlifting flow over a cylinder of radius R, and Equations (3.111*a* and *b*) for vortex flow, we have, for the lifting flow over a cylinder of radius R,

$$V_r = \left(1 - \frac{R^2}{r^2}\right) V_\infty \cos\theta \tag{3.119}$$

$$V_\theta = -\left(1 + \frac{R^2}{r^2}\right) V_\infty \sin\theta - \frac{\Gamma}{2\pi r} \tag{3.120}$$

To locate the stagnation points in the flow, set $V_r = V_\theta = 0$ in Equations (3.119) and (3.120) and solve for the resulting coordinates (r, θ):

$$V_r = \left(1 - \frac{R^2}{r^2}\right) V_\infty \cos\theta = 0 \tag{3.121}$$

$$V_\theta = -\left(1 + \frac{R^2}{r^2}\right) V_\infty \sin\theta - \frac{\Gamma}{2\pi r} = 0 \tag{3.122}$$

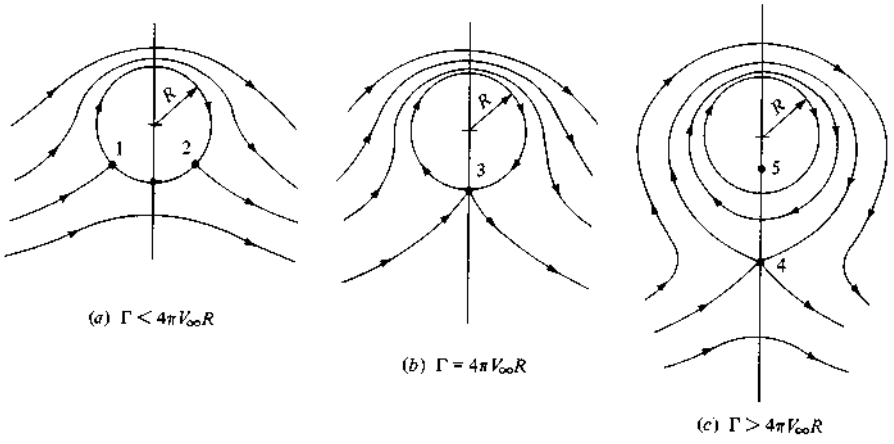


Figure 3.33 Stagnation points for the lifting flow over a circular cylinder.

From Equation (3.121), $r = R$. Substituting this result into Equation (3.122) and solving for θ, we obtain

$$\theta = \arcsin\left(-\frac{\Gamma}{4\pi V_\infty R}\right) \tag{3.123}$$

Since Γ is a positive number, from Equation (3.123) θ must be in the third and fourth quadrants. That is, there can be two stagnation points on the bottom half of the circular cylinder, as shown by points 1 and 2 in Figure 3.33*a*. These points are located at (R, θ), where θ is given by Equation (3.123). However, this result is valid only when $\Gamma/4\pi V_\infty R < 1$. If $\Gamma/4\pi V_\infty R > 1$, then Equation (3.123) has no meaning. If $\Gamma/4\pi V_\infty R = 1$, there is only one stagnation point on the surface of the cylinder, namely, point $(R, -\pi/2)$ labeled as point 3 in Figure 3.33*b*. For the case of $\Gamma/4\pi V_\infty R > 1$, return to Equation (3.121). We saw earlier that it is satisfied by $r = R$; however, it is also satisfied by $\theta = \pi/2$ or $-\pi/2$. Substituting $\theta = -\pi/2$ into Equation (3.122), and solving for r, we have

$$r = \frac{\Gamma}{4\pi V_\infty} \pm \sqrt{\left(\frac{\Gamma}{4\pi V_\infty}\right)^2 - R^2} \tag{3.124}$$

Hence, for $\Gamma/4\pi V_\infty R > 1$, there are two stagnation points, one inside and the other outside the cylinder, and both on the vertical axis, as shown by points 4 and 5 in Figure 3.33*c*. [How does one stagnation point fall *inside* the cylinder? Recall that $r = R$, or $\psi = 0$, is just one of the allowed streamlines of the flow. There is a theoretical flow inside the cylinder—flow that is issuing from the doublet at the origin superimposed with the vortex flow for $r < R$. The circular streamline $r = R$ is the dividing streamline between this flow and the flow from the freestream. Therefore, as before, we can replace the dividing streamline by

a solid body—our circular cylinder—and the *external* flow will not know the difference. Hence, although one stagnation point falls inside the body (point 5), we are not realistically concerned about it. Instead, from the point of view of flow over a solid cylinder of radius R, point 4 is the only meaningful stagnation point for the case $\Gamma/4\pi V_\infty R > 1$.]

The results shown in Figure 3.33 can be visualized as follows. Consider the inviscid incompressible flow of given freestream velocity V_∞ over a cylinder of given radius R. If there is no circulation (i.e., if $\Gamma = 0$), the flow is given by the sketch at the right of Figure 3.26, with horizontally opposed stagnation points A and B. Now assume that a circulation is imposed on the flow, such that $\Gamma < 4\pi V_\infty R$. The flow sketched in Figure 3.33*a* will result; the two stagnation points will move to the lower surface of the cylinder as shown by points 1 and 2. Assume that Γ is further increased until $\Gamma = 4\pi V_\infty R$. The flow sketched in Figure 3.33*b* will result, with only one stagnation point at the bottom of the cylinder, as shown by point 3. When Γ is increased still further such that $\Gamma > 4\pi V_\infty R$, the flow sketched in Figure 3.33*c* will result. The stagnation point will lift from the cylinder's surface and will appear in the flow directly below the cylinder, as shown by point 4.

From the above discussion, Γ is clearly a parameter that can be chosen freely. There is no single value of Γ that "solves" the flow over a circular cylinder; rather, the circulation can be any value. Therefore, for the incompressible flow over a circular cylinder, there are an infinite number of possible potential flow solutions, corresponding to the infinite choices for values of Γ. This statement is not limited to flow over circular cylinders, but rather, it is a general statement that holds for the incompressible potential flow over all smooth two-dimensional bodies. We return to these ideas in subsequent sections.

From the symmetry, or lack of it, in the flows sketched in Figures 3.32 and 3.33, we intuitively concluded earlier that a finite normal force (lift) must exist on the body but that the drag is zero; that is, d'Alembert's paradox still prevails. Let us quantify these statements by calculating expressions for lift and drag, as follows.

The velocity on the surface of the cylinder is given by Equation (3.120) with $r = R$:

$$V = V_\theta = -2V_\infty \sin\theta - \frac{\Gamma}{2\pi R} \tag{3.125}$$

In turn, the pressure coefficient is obtained by substituting Equation (3.125) into Equation (3.38):

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2 = 1 - \left(-2\sin\theta - \frac{\Gamma}{2\pi R V_\infty}\right)^2$$

or

$$C_p = 1 - \left[4\sin^2\theta + \frac{2\Gamma\sin\theta}{\pi R V_\infty} + \left(\frac{\Gamma}{2\pi R V_\infty}\right)^2\right] \tag{3.126}$$

In Section 1.5, we discussed in detail how the aerodynamic force coefficients can be obtained by integrating the pressure coefficient and skin friction coefficient over the surface. For inviscid flow, $c_f = 0$. Hence, the drag coefficient c_d is given by Equation (1.16) as

$$c_d = c_a = \frac{1}{c}\int_{\text{LE}}^{\text{TE}} (C_{p,u} - C_{p,l})\, dy$$

or

$$c_d = \frac{1}{c}\int_{\text{LE}}^{\text{TE}} C_{p,u}\, dy - \frac{1}{c}\int_{\text{LE}}^{\text{TE}} C_{p,l}\, dy \tag{3.127}$$

Converting Equation (3.127) to polar coordinates, we note that

$$y = R\sin\theta \qquad dy = R\cos\theta\, d\theta \tag{3.128}$$

Substituting Equation (3.128) into (3.127), and noting that $c = 2R$, we have

$$c_d = \frac{1}{2}\int_{\pi}^{0} C_{p,u}\cos\theta\, d\theta - \frac{1}{2}\int_{\pi}^{2\pi} C_{p,l}\cos\theta\, d\theta \tag{3.129}$$

The limits of integration in Equation (3.129) are explained as follows. In the first integral, we are integrating from the leading edge (the front point of the cylinder), moving over the *top* surface of the cylinder. Hence, θ is equal to π at the leading edge and, moving over the top surface, *decreases* to 0 at the trailing edge. In the second integral, we are integrating from the leading edge to the trailing edge while moving over the *bottom* surface of the cylinder. Hence, θ is equal to π at the leading edge and, moving over the bottom surface, *increases* to 2π at the trailing edge. In Equation (3.129), both $C_{p,u}$ and $C_{p,l}$ are given by the same analytic expression for C_p, namely, Equation (3.126). Hence, Equation (3.129) can be written as

$$c_d = -\frac{1}{2}\int_{0}^{\pi} C_p\cos\theta\, d\theta - \frac{1}{2}\int_{\pi}^{2\pi} C_p\cos\theta\, d\theta$$

or

$$c_d = -\frac{1}{2}\int_{0}^{2\pi} C_p\cos\theta\, d\theta \tag{3.130}$$

Substituting Equation (3.126) into (3.130), and noting that

$$\int_{0}^{2\pi} \cos\theta\, d\theta = 0 \tag{3.131a}$$

$$\int_{0}^{2\pi} \sin^2\theta\cos\theta\, d\theta = 0 \tag{3.131b}$$

$$\int_{0}^{2\pi} \sin\theta\cos\theta\, d\theta = 0 \tag{3.131c}$$

we immediately obtain

$$\boxed{c_d = 0} \tag{3.132}$$

Equation (3.132) confirms our intuitive statements made earlier. The drag on a cylinder in an inviscid, incompressible flow is zero, regardless of whether or not the flow has circulation about the cylinder.

The lift on the cylinder can be evaluated in a similar manner as follows. From Equation (1.15) with $c_f = 0$,

$$c_l = c_n = \frac{1}{c}\int_0^c C_{p,l}\,dx - \frac{1}{c}\int_0^c C_{p,u}\,dx \tag{3.133}$$

Converting to polar coordinates, we obtain

$$x = R\cos\theta \qquad dx = -R\sin\theta\,d\theta \tag{3.134}$$

Substituting Equation (3.134) into (3.133), we have

$$c_l = -\frac{1}{2}\int_\pi^{2\pi} C_{p,l}\sin\theta\,d\theta + \frac{1}{2}\int_\pi^0 C_{p,u}\sin\theta\,d\theta \tag{3.135}$$

Again, noting that $C_{p,l}$ and $C_{p,u}$ are both given by the same analytic expression, namely, Equation (3.126), Equation (3.135) becomes

$$c_l = -\frac{1}{2}\int_0^{2\pi} C_p\sin\theta\,d\theta \tag{3.136}$$

Substituting Equation (3.126) into (3.136), and noting that

$$\int_0^{2\pi}\sin\theta\,d\theta = 0 \tag{3.137a}$$

$$\int_0^{2\pi}\sin^3\theta\,d\theta = 0 \tag{3.137b}$$

$$\int_0^{2\pi}\sin^2\theta\,d\theta = \pi \tag{3.137c}$$

we immediately obtain

$$c_l = \frac{\Gamma}{RV_\infty} \tag{3.138}$$

From the definition of c_l (see Section 1.5), the lift per unit span L' can be obtained from

$$L' = q_\infty S c_l = \tfrac{1}{2}\rho_\infty V_\infty^2 S c_l \tag{3.139}$$

Here, the planform area $S = 2R(1)$. Therefore, combining Equations (3.138) and (3.139), we have

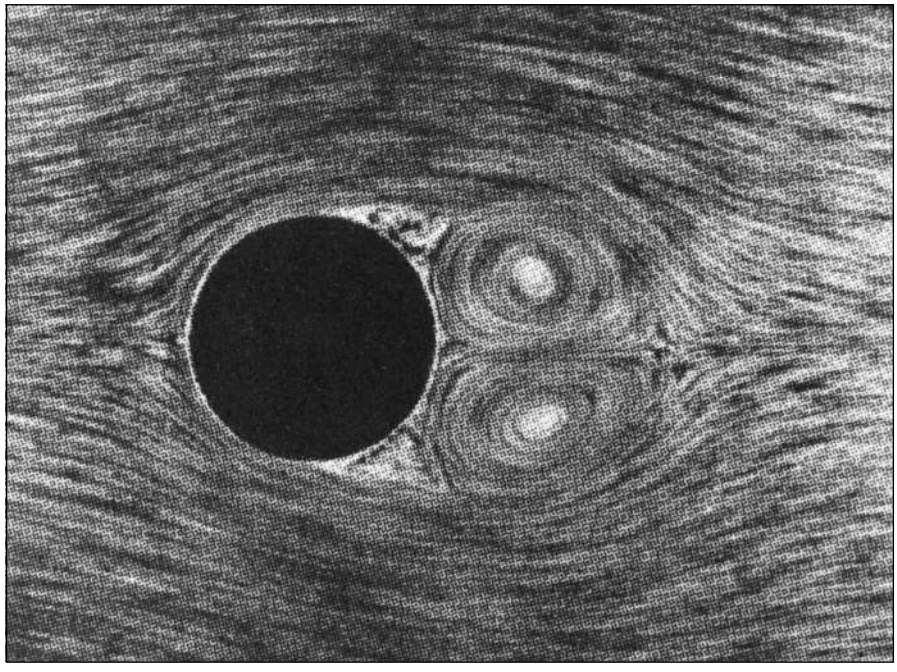
$$L' = \frac{1}{2}\rho_\infty V_\infty^2 2R \frac{\Gamma}{RV_\infty}$$

or

$$\boxed{L' = \rho_\infty V_\infty \Gamma} \tag{3.140}$$

Equation (3.140) gives the lift per unit span for a circular cylinder with circulation Γ. It is a remarkably simple result, and it states that *the lift per unit span is directly proportional to circulation*. Equation (3.140) is a powerful relation in theoretical aerodynamics. It is called the *Kutta-Joukowski theorem,* named after the German mathematician M. Wilhelm Kutta (1867–1944) and the Russian physicist Nikolai E. Joukowski (1847–1921), who independently obtained it during the first decade of 20th century. We will have more to say about the Kutta-Joukowski theorem in Section 3.16.

What are the connections between the above theoretical results and real life? As stated earlier, the prediction of zero drag is totally erroneous—viscous effects cause skin friction and flow separation which always produce a finite drag. The inviscid flow treated in this chapter simply does not model the proper physics for drag calculations. On the other hand, the prediction of lift via Equation (3.140) is quite realistic. Let us return to the wind-tunnel experiments mentioned at the beginning of this chapter. If a stationary, nonspinning cylinder is placed in a low-speed wind tunnel, the flow field will appear as shown in Figure 3.34*a*. The streamlines over the front of the cylinder are similar to theoretical predictions, as sketched at the right of Figure 3.26. However, because of viscous effects, the flow separates over the rear of the cylinder, creating a recirculating flow in the wake downstream of the body. This separated flow greatly contributes to the finite drag measured for the cylinder. On the other hand, Figure 3.34*a* shows a reasonably symmetric flow about the horizontal axis, and the measurement of lift is essentially zero. Now let us *spin* the cylinder in a clockwise direction about its axis. The resulting flow fields are shown in Figure 3.34*b* and *c*. For a moderate amount of spin (Figure 3.34*b*), the stagnation points move to the lower part of the cylinder, similar to the theoretical flow sketched in Figure 3.33*a*. If the spin is sufficiently increased (Figure 3.34*c*), the stagnation point lifts off the surface, similar to the theoretical flow sketched in Figure 3.33*c*. And what is most important, a *finite lift* is measured for the spinning cylinder in the wind tunnel. What is happening here? Why does spinning the cylinder produce lift? In actuality, the friction between the fluid and the surface of the cylinder tends to drag the fluid near the surface in the same direction as the rotational motion. Superimposed on top of the usual nonspinning flow, this "extra" velocity contribution creates a higher-than-usual velocity at the top of the cylinder and a lower-than-usual velocity at the bottom, as sketched in Figure 3.35. These velocities are assumed to be just outside the viscous boundary layer on the surface. Recall from Bernoulli's equation that as the velocity increases, the pressure decreases. Hence, from Figure 3.35, the pressure



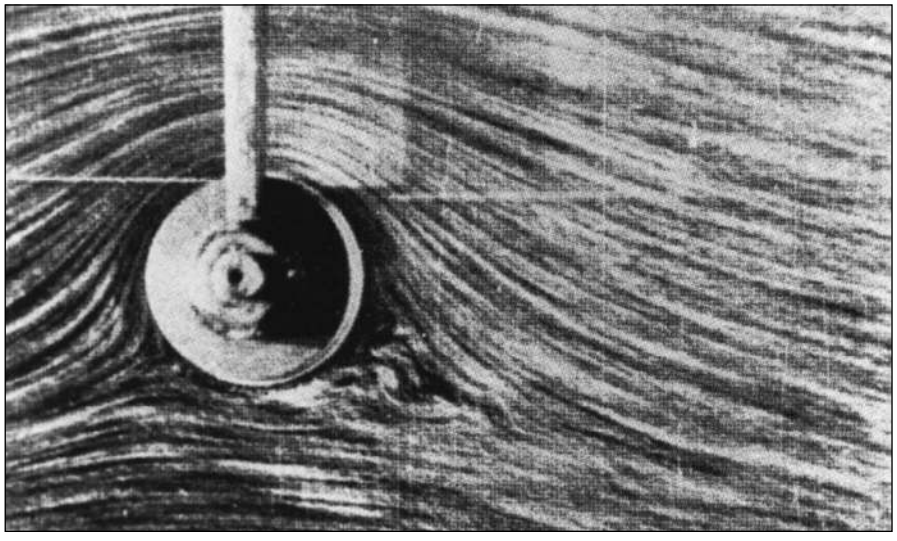
(*a*)

Figure 3.34 These flow-field pictures were obtained in water, where aluminum filings were scattered on the surface to show the direction of the streamlines. (*a*) Shown above is the case for the nonspinning cylinder. *(Source: Prandtl and Tietjens, Reference 8.)*

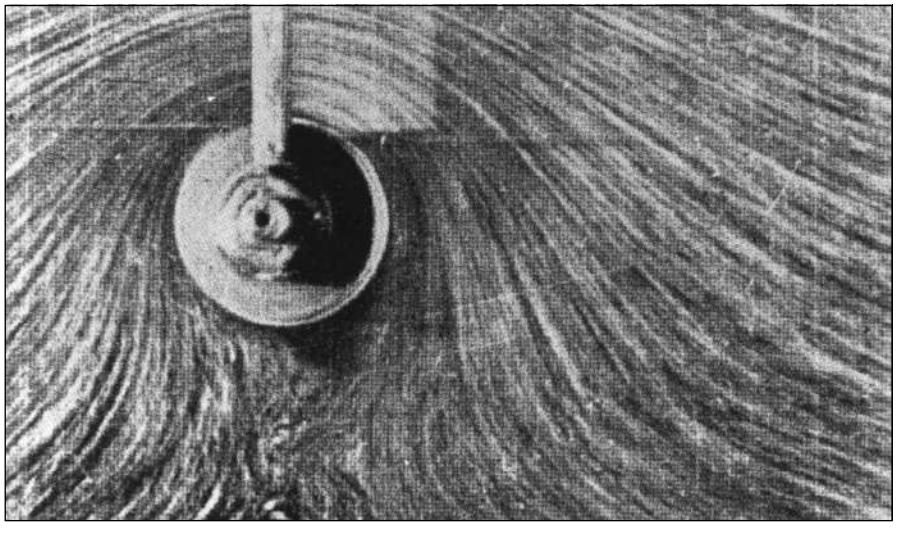
on the top of the cylinder is lower than on the bottom. This pressure imbalance creates a net upward force, that is, a finite lift. Therefore, the theoretical prediction embodied in Equation (3.140) that the flow over a circular cylinder can produce a finite lift is verified by experimental observation.

The general ideas discussed above concerning the generation of lift on a spinning circular cylinder in a wind tunnel also apply to a spinning sphere. This explains why a baseball pitcher can throw a curve and how a golfer can hit a hook or slice—all of which are due to nonsymmetric flows about the spinning bodies, and hence the generation of an aerodynamic force perpendicular to the body's angular velocity vector. This phenomenon is called the *Magnus effect,* named after the German engineer who first observed and explained it in Berlin in 1852.

It is interesting to note that a rapidly spinning cylinder can produce a much higher lift than an airplane wing of the same planform area; however, the drag on the cylinder is also much higher than a well-designed wing. As a result, the Magnus effect is not employed for powered flight. On the other hand, in the 1920s, the German engineer Anton Flettner replaced the sail on a boat with a rotating



(*b*)



(*c*)

Figure 3.34 (continued) These flow-field pictures were obtained in water, where aluminum filings were scattered on the surface to show the direction of the streamlines. (*b*) Spinning cylinder: peripheral velocity of the surface $= 3V_\infty$. (*c*) Spinning cylinder: peripheral velocity of the surface $= 6V_\infty$. *(Source: Prandtl and Tietjens, Reference 8.)*

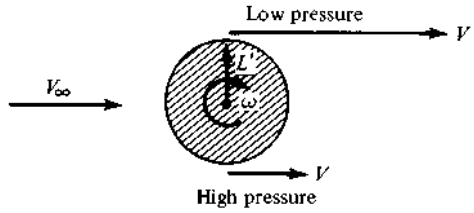


Figure 3.35 Creation of lift on a spinning cylinder.

circular cylinder with its axis vertical to the deck. In combination with the wind, this spinning cylinder provided propulsion for the boat. Moreover, by the action of two cylinders in tandem and rotating in opposite directions, Flettner was able to turn the boat around. Flettner's device was a technical success, but an economic failure because the maintenance on the machinery to spin the cylinders at the necessary high rotational speeds was too costly. Today, the Magnus effect has an important influence on the performance of spinning missiles; indeed, a certain amount of modern high-speed aerodynamic research has focused on the Magnus forces on spinning bodies for missile applications.

EXAMPLE 3.16

Consider the lifting flow over a circular cylinder. The lift coefficient is 5. Calculate the peak (negative) pressure coefficient.

■ Solution

Examining Figure 3.32, note that the maximum velocity for the *nonlifting* flow over a cylinder is $2V_\infty$ and that it occurs at the top and bottom points on the cylinder. When the vortex in Figure 3.32 is added to the flow field, the direction of the vortex velocity is in the *same* direction as the flow on the top of the cylinder, but opposes the flow on the bottom of the cylinder. Hence, the maximum velocity for the lifting case occurs at the *top* of the cylinder and is equal to the sum of the nonlifting value, $-2V_\infty$, and the vortex, $-\Gamma/2\pi R$. (*Note:* We are still following the usual sign convention; since the velocity on the top of the cylinder is in the opposite direction of increasing θ for the polar coordinate system, the velocity magnitudes here are negative.) Hence,

$$V = -2V_\infty - \frac{\Gamma}{2\pi R} \tag{E.1}$$

The lift coefficient and Γ are related through Equation (3.138):

$$c_l = \frac{\Gamma}{RV_\infty} = 5$$

Hence,

$$\frac{\Gamma}{R} = 5V_\infty \tag{E.2}$$

Substituting Equation (E.2) into (E.1), we have

$$V = -2V_\infty - \frac{5}{2\pi}V_\infty = -2.796V_\infty \tag{E.3}$$

Substituting Equation (E.3) into Equation (3.38), we obtain

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2 = 1 - (2.796)^2 = \boxed{-6.82}$$

This example is designed in part to make the following point. Recall that, for an inviscid, incompressible flow, the distribution of C_p over the surface of a body depends only on the shape and orientation of the body—the flow properties such as velocity and density are irrelevant here. Recall Equation (3.101), which gives C_p as a function of θ only, namely, $C_p = 1 - 4\sin^2\theta$. However, for the case of lifting flow, the distribution of C_p over the surface is a function of one additional parameter—namely, the lift coefficient. Clearly, in this example, only the value of c_l is given. However, this is powerful enough to define the flow uniquely; the value of C_p at any point on the surface follows directly from the value of lift coefficient, as demonstrated in the above problem.

EXAMPLE 3.17

For the flow field in Example 3.16, calculate the location of the stagnation points and the points on the cylinder where the pressure equals freestream static pressure.

■ Solution

From Equation (3.123), the stagnation points are given by

$$\theta = \arcsin\left(-\frac{\Gamma}{4\pi V_\infty R}\right)$$

From Example 3.16,

$$\frac{\Gamma}{RV_\infty} = 5$$

Thus,

$$\theta = \arcsin\left(-\frac{5}{4\pi}\right) = \boxed{203.4^\circ \quad \text{and} \quad 336.6^\circ}$$

To find the locations where $p = p_\infty$, first construct a formula for the pressure coefficient on the cylinder surface:

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2$$

where

$$V = -2V_\infty \sin\theta - \frac{\Gamma}{2\pi R}$$

Thus,

$$C_p = 1 - \left(-2\sin\theta - \frac{\Gamma}{2\pi R}\right)^2$$

$$= 1 - 4\sin^2\theta - \frac{2\Gamma\sin\theta}{\pi R V_\infty} - \left(\frac{\Gamma}{2\pi R V_\infty}\right)^2$$

From Example 3.16, $\Gamma/RV_\infty = 5$. Thus,

$$C_p = 1 - 4\sin^2\theta - \frac{10}{\pi}\sin\theta - \left(\frac{5}{2\pi}\right)^2$$

$$= 0.367 - 3.183\sin\theta - 4\sin^2\theta$$

A check on this equation can be obtained by calculating C_p at $\theta = 90^\circ$ and seeing if it agrees with the result obtained in Example 3.16. For $\theta = 90^\circ$, we have

$$C_p = 0.367 - 3.183 - 4 = \boxed{-6.82}$$

This is the same result as in Example 3.16; the equation checks.

To find the values of θ where $p = p_\infty$, set $C_p = 0$:

$$0 = 0.367 - 3.183\sin\theta - 4\sin^2\theta$$

From the quadratic formula,

$$\sin\theta = \frac{3.183 \pm \sqrt{(3.183)^2 + 5.872}}{-8} = \boxed{-0.897 \quad \text{or} \quad 0.102}$$

Hence, $\theta = 243.8^\circ \quad \text{and} \quad 296.23^\circ$

Also, $\theta = 5.85^\circ \quad \text{and} \quad 174.1^\circ$

There are four points on the circular cylinder where $p = p_\infty$. These are sketched in Figure 3.36, along with the stagnation point locations. As shown in Example 3.16, the minimum pressure occurs at the top of the cylinder and is equal to $p_\infty - 6.82q_\infty$. A *local* minimum pressure occurs at the bottom of the cylinder, where $\theta = 3\pi/2$. This local minimum is given by

$$C_p = 0.367 - 3.183\sin\frac{3\pi}{2} - 4\sin^2\frac{3\pi}{2}$$

$$= 0.367 + 3.183 - 4 = \boxed{-0.45}$$

Hence, at the bottom of the cylinder, $p = p_\infty - 0.45q_\infty$.

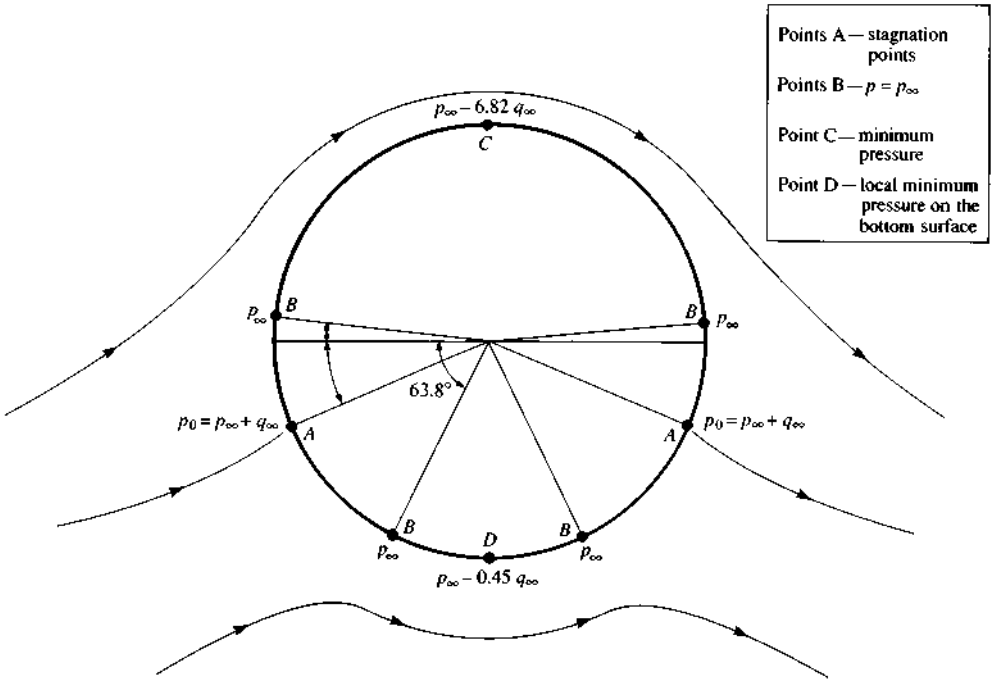


Figure 3.36 Values of pressure at various locations on the surface of a circular cylinder; lifting case with finite circulation. The values of pressure correspond to the case discussed in Example 3.17.

EXAMPLE 3.18

Consider the lifting flow over a circular cylinder with a diameter of 0.5 m. The freestream velocity is 25 m/s, and the maximum velocity on the surface of the cylinder is 75 m/s. The freestream conditions are those for a standard altitude of 3 km. Calculate the lift per unit span on the cylinder.

■ Solution

From Appendix D, at an altitude of 3 km, $\rho = 0.90926$ kg/m^3. The maximum velocity occurs at the top of the cylinder, where $\theta = 90°$, From Equation (3.125),

$$V_\theta = -2V_\infty \sin\theta - \frac{\Gamma}{2\pi R}$$

At $\theta = 90°$

$$V_\theta = -2V_\infty - \frac{\Gamma}{2\pi R}$$

or,

$$\Gamma = -2\pi R(V_\theta + 2V_\infty)$$

Recalling our sign convention that Γ is positive in the clockwise direction, and V_θ is negative in the clockwise direction (reflect again on Figure 3.32), we have

$$V_\theta = -75 \text{m/s}$$

Hence,
$$\Gamma = -2\pi R(V_\theta + 2V_\infty) = -2\pi(0.25)[-75 + 2(25)]$$

$$\Gamma = -2\pi(0.25)(-25) = 39.27\,\text{m}^2/\text{s}$$

From Equation (3.140), the lift per unit span is

$$L' = \rho_\infty V_\infty \Gamma$$

$$L' = (0.90926)(25)(39.27) = \boxed{892.7 \text{ N/m}}$$

Comment: In the course of this example, we have calculated a value for circulation; $\Gamma = 39.27$ m^2/s. In English engineering units, this is

$$\Gamma = 39.27 \frac{\text{m}^2}{\text{s}} \left(\frac{3.28 \text{ ft}}{1 \text{ m}} \right)^2 = 422.5 \frac{\text{ft}^2}{\text{s}}.$$

In Example 3.15 we commented on the numerical value of circulation; in that case we obtained a value of 18,430 ft^2/s, much larger than in the present example. But we are still seeing values of Γ in the hundreds, if not in the thousands—just an interesting observation.

3.16 THE KUTTA-JOUKOWSKI THEOREM AND THE GENERATION OF LIFT

Although the result given by Equation (3.140) was derived for a circular cylinder, it applies in general to cylindrical bodies of arbitrary cross section. For example, consider the incompressible flow over an airfoil section, as sketched in Figure 3.37. Let curve A be any curve in the flow *enclosing* the airfoil. If the airfoil is producing lift, the velocity field around the airfoil will be such that the line integral of velocity around A will be finite, that is, the circulation

$$\Gamma \equiv \oint_A \mathbf{V} \cdot \mathbf{ds}$$

is finite. In turn, the lift per unit span L' on the airfoil will be given by the *Kutta-Joukowski theorem,* as embodied in Equation (3.140):

$$\boxed{L' = \rho_\infty V_\infty \Gamma} \tag{3.140}$$

This result underscores the importance of the concept of circulation, defined in Section 2.13. The Kutta-Joukowski theorem states that lift per unit span on a two-dimensional body is directly proportional to the circulation around the body. Indeed, the concept of circulation is so important at this stage of our discussion that you should reread Section 2.13 before proceeding further.

The general derivation of Equation (3.140) for bodies of arbitrary cross section can be carried out using the method of complex variables. Such mathematics

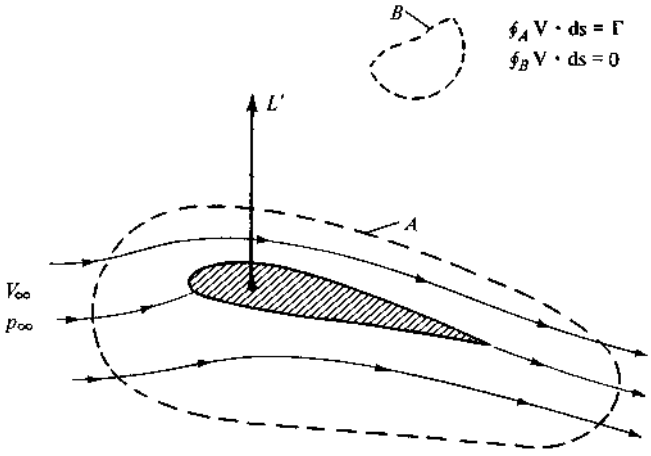


Figure 3.37 Circulation around a lifting airfoil.

is beyond the scope of this book. (It can be shown that arbitrary functions of complex variables are general solutions of Laplace's equation, which in turn governs incompressible potential flow. Hence, more advanced treatments of such flows utilize the mathematics of complex variables as an important tool. See Reference 9 for a particularly lucid treatment of inviscid, incompressible flow at a more advanced level.)

In Section 3.15, the lifting flow over a circular cylinder was synthesized by superimposing a uniform flow, a doublet, and a vortex. Recall that all three elementary flows are irrotational at all points, except for the vortex, which has infinite vorticity at the origin. Therefore, the lifting flow over a cylinder as shown in Figure 3.33 is irrotational at every point except at the origin. If we take the circulation around any curve *not* enclosing the origin, we obtain from Equation (2.137) the result that $\Gamma = 0$. It is only when we choose a curve that encloses the origin, where $\nabla \times \mathbf{V}$ is infinite, that Equation (2.137) yields a finite Γ, equal to the strength of the vortex. The same can be said about the flow over the airfoil in Figure 3.37. As we show in Chapter 4, the flow *outside* the airfoil is irrotational, and the circulation around any closed curve *not* enclosing the airfoil (such as curve *B* in Figure 3.37) is consequently zero. On the other hand, we also show in Chapter 4 that the flow over an airfoil is synthesized by distributing vortices either on the surface or inside the airfoil. These vortices have the usual singularities in $\nabla \times \mathbf{V}$, and therefore, if we choose a curve that encloses the airfoil (such as curve *A* in Figure 3.37), Equation (2.137) yields a finite value of Γ, equal to the *sum* of the vortex strengths distributed on or inside the airfoil. The important point here is that, in the Kutta-Joukowski theorem, the value of Γ used in Equation (3.140) must be evaluated around a closed curve that *encloses the body;* the curve can be otherwise arbitrary, but it must have the body inside it.

At this stage, let us pause and assess our thoughts. The approach we have discussed above—the definition of circulation and the use of Equation (3.140) to

obtain the lift—is the essence of the *circulation theory of lift* in aerodynamics. Its development at the turn of the twentieth century created a breakthrough in aerodynamics. However, let us keep things in perspective. The circulation theory of lift is an *alternative* way of thinking about the generation of lift on an aerodynamic body. Keep in mind that the true physical sources of aerodynamic force on a body are the pressure and shear stress distributions exerted on the surface of the body, as explained in Section 1.5. The Kutta-Joukowski theorem is simply an alternative way of expressing the *consequences* of the surface pressure distribution; it is a mathematical expression that is consistent with the special tools we have developed for the analysis of inviscid, incompressible flow. Indeed, recall that Equation (3.140) was derived in Section 3.15 by integrating the pressure distribution over the surface. Therefore, it is not quite proper to say that circulation "causes" lift. Rather, lift is "caused" by the net imbalance of the surface pressure distribution, and circulation is simply a defined quantity determined from the same pressures. The relation between the surface pressure distribution (which produces lift L') and circulation is given by Equation (3.140). However, in the theory of incompressible, potential flow, it is generally much easier to determine the circulation around the body rather than calculate the detailed surface pressure distribution. Therein lies the power of the circulation theory of lift.

Consequently, the theoretical analysis of lift on two-dimensional bodies in incompressible, inviscid flow focuses on the calculation of the circulation about the body. Once Γ is obtained, then the lift per unit span follows directly from the Kutta-Joukowski theorem. As a result, in subsequent sections we constantly address the question: How can we calculate the circulation for a given body in a given incompressible, inviscid flow?

3.17 NONLIFTING FLOWS OVER ARBITRARY BODIES: THE NUMERICAL SOURCE PANEL METHOD

In this section, we return to the consideration of nonlifting flows. Recall that we have already dealt with the nonlifting flows over a semi-infinite body and a Rankine oval and both the nonlifting and the lifting flows over a circular cylinder. For those cases, we added our elementary flows in certain ways and discovered that the dividing streamlines turned out to fit the shapes of such special bodies. However, this indirect method of starting with a given combination of elementary flows and seeing what body shape comes out of it can hardly be used in a practical sense for bodies of arbitrary shape. For example, consider the airfoil in Figure 3.37. Do we know in advance the correct combination of elementary flows to synthesize the flow over this specified body? The answer is no. Rather, what we want is a direct method; that is, let us *specify* the shape of an arbitrary body and *solve* for the distribution of singularities which, in combination with a uniform stream, produce the flow over the given body. The purpose of this section is to present such a direct method, limited for the present to nonlifting flows. We consider a numerical method appropriate for solution on a high-speed digital computer. The technique

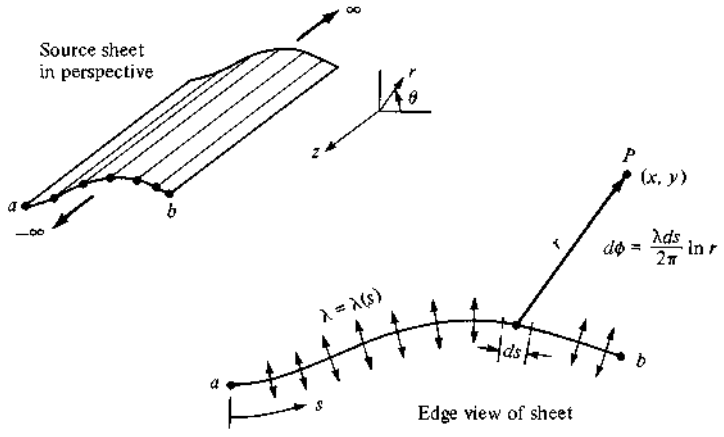


Figure 3.38 Source sheet.

is called the *source panel method,* which, since the late 1960s, has become a standard aerodynamic tool in industry and most research laboratories. In fact, the numerical solution of potential flows by both source and vortex panel techniques has revolutionized the analysis of low-speed flows. We return to various numerical panel techniques in Chapters 4 through 6. As a modern student of aerodynamics, it is necessary for you to become familiar with the fundamentals of such panel methods. The purpose of the present section is to introduce the basic ideas of the source panel method, which is a technique for the numerical solution of nonlifting flows over arbitrary bodies.

First, let us extend the concept of a source or sink introduced in Section 3.10. In that section, we dealt with a single line source, as sketched in Figure 3.21. Now imagine that we have an infinite number of such line sources side by side, where the strength of each line source is infinitesimally small. These side-by-side line sources form a *source sheet,* as shown in perspective in the upper left of Figure 3.38. If we look along the series of line sources (looking along the z axis in Figure 3.38), the source sheet will appear as sketched at the lower right of Figure 3.38. Here, we are looking at an edge view of the sheet; the line sources are all perpendicular to the page. Let s be the distance measured along the source sheet in the edge view. Define $\lambda = \lambda(s)$ to be the *source strength per unit length along* s. [To keep things in perspective, recall from Section 3.10 that the strength of a single line source Λ was defined as the volume flow rate per unit depth, that is, per unit length in the z direction. Typical units for Λ are square meters per second or square feet per second. In turn, the strength of a source sheet $\lambda(s)$ is the volume flow rate per unit depth (in the z direction) and per unit length (in the s direction). Typical units for λ are meters per second or feet per second.] Therefore, the strength of an infinitesimal portion ds of the sheet, as shown in Figure 3.38, is $\lambda\, ds$. This small section of the source sheet can be treated as a distinct source of strength $\lambda\, ds$. Now consider point P in the flow, located a distance r from ds; the cartesian coordinates of P are (x, y). The small section

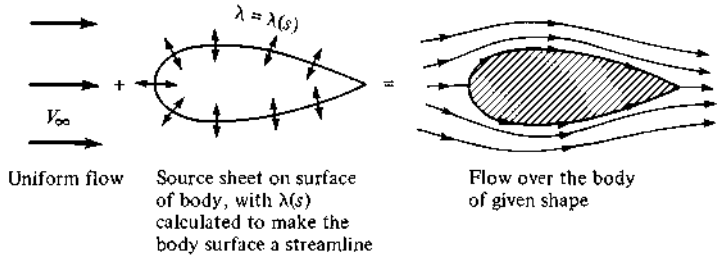


Figure 3.39 Superposition of a uniform flow and a source sheet on a body of given shape, to produce the flow over the body.

of the source sheet of strength $\lambda\, ds$ induces an infinitesimally small potential $d\phi$ at point P. From Equation (3.67), $d\phi$ is given by

$$d\phi = \frac{\lambda\, ds}{2\pi} \ln r \tag{3.141}$$

The complete velocity potential at point P, induced by the entire source sheet from a to b, is obtained by integrating Equation (3.141):

$$\phi(x, y) = \int_a^b \frac{\lambda\, ds}{2\pi} \ln r \tag{3.142}$$

Note that, in general, $\lambda(s)$ can change from positive to negative along the sheet; that is, the "source" sheet is really a combination of line sources and line sinks.

Next, consider a given body of arbitrary shape in a flow with freestream velocity V_∞, as shown in Figure 3.39. Let us cover the surface of the prescribed body with a source sheet, where the strength $\lambda(s)$ varies in such a fashion that the combined action of the uniform flow and the source sheet makes the airfoil surface a streamline of the flow. Our problem now becomes one of finding the appropriate $\lambda(s)$. The solution of this problem is carried out numerically, as follows.

Let us approximate the source sheet by a series of straight panels, as shown in Figure 3.40. Moreover, let the source strength λ per unit length be constant over a given panel, but allow it to vary from one panel to the next. That is, if there are a total of n panels, the source panel strengths per unit length are $\lambda_1, \lambda_2, \ldots, \lambda_j \ldots, \lambda_n$. These panel strengths are unknown; the main thrust of the panel technique is to solve for λ_j, $j = 1$ to n, such that the body surface becomes a streamline of the flow. This boundary condition is imposed numerically by defining the midpoint of each panel to be a *control point* and by determining the λ_j's such that the normal component of the flow velocity is zero at each control point. Let us now quantify this strategy.

Let P be a point located at (x, y) in the flow, and let r_{pj} be the distance from any point on the jth panel to P, as shown in Figure 3.40. The velocity potential induced at P due to the jth panel $\Delta\phi_j$ is, from Equation (3.142),

$$\Delta\phi_j = \frac{\lambda_j}{2\pi} \int_j \ln r_{pj}\, ds_j \tag{3.143}$$

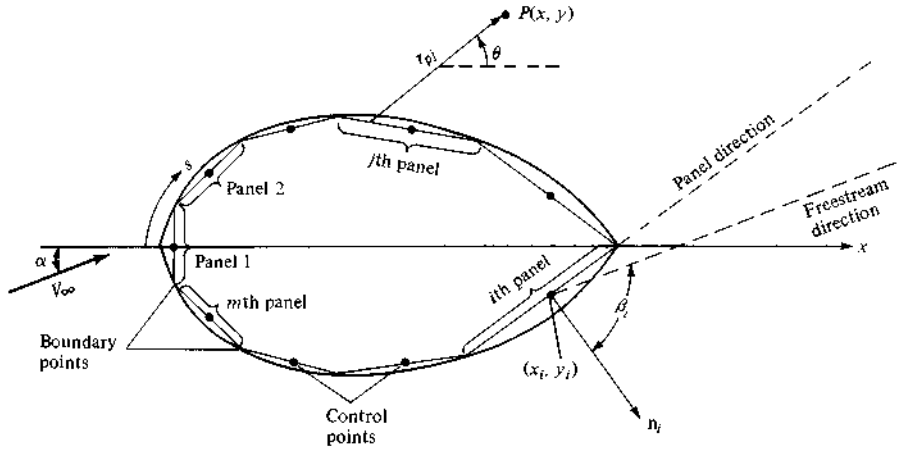


Figure 3.40 Source panel distribution over the surface of a body of arbitrary shape.

In Equation (3.143), λ_j is constant over the jth panel, and the integral is taken over the jth panel only. In turn, the *potential at P* due to *all* the panels is Equation (3.143) summed over all the panels:

$$\phi(P) = \sum_{j=1}^{n} \Delta\phi_j = \sum_{j=1}^{n} \frac{\lambda_j}{2\pi} \int_j \ln r_{pj}\, ds_j \tag{3.144}$$

In Equation (3.144), the distance r_{pj} is given by

$$r_{pj} = \sqrt{(x - x_j)^2 + (y - y_j)^2} \tag{3.145}$$

where (x_j, y_j) are coordinates along the surface of the jth panel. Since point P is just an arbitrary point in the flow, let us put P at the control point of the ith panel. Let the coordinates of this control point be given by (x_i, y_i), as shown in Figure 3.40. Then Equations (3.144) and (3.145) become

$$\phi(x_i, y_i) = \sum_{j=1}^{n} \frac{\lambda_j}{2\pi} \int_j \ln r_{ij}\, ds_j \tag{3.146}$$

and

$$r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{3.147}$$

Equation (3.146) is physically the contribution of *all* the panels to the potential at the control point of the ith panel.

Recall that the boundary condition is applied at the control points; that is, the normal component of the flow velocity is zero at the control points. To evaluate this component, first consider the component of freestream velocity perpendicular to the panel. Let $\mathbf{n}_i$ be the unit vector normal to the ith panel, directed out of the body, as shown in Figure 3.40. Also, note that the slope of the ith panel is $(dy/dx)_i$. In general, the freestream velocity will be at some incidence angle α to the x axis,

as shown in Figure 3.40. Therefore, inspection of the geometry of Figure 3.40 reveals that the component of $\mathbf{V}_\infty$ normal to the ith panel is

$$V_{\infty,n} = \mathbf{V}_\infty \cdot \mathbf{n}_i = V_\infty \cos \beta_i \tag{3.148}$$

where β_i is the angle between $\mathbf{V}_\infty$ and $\mathbf{n}_i$. Note that $V_{\infty,n}$ is positive when directed away from the body, and negative when directed toward the body.

The normal component of velocity induced at (x_i, y_i) by the source panels is, from Equation (3.146),

$$V_n = \frac{\partial}{\partial n_i}[\phi(x_i, y_i)] \tag{3.149}$$

where the derivative is taken in the direction of the outward unit normal vector, and hence, again, V_n is positive when directed away from the body. When the derivative in Equation (3.149) is carried out, r_{ij} appears in the denominator. Consequently, a singular point arises on the ith panel because when $j = i$, at the control point itself $r_{ij} = 0$. It can be shown that when $j = i$, the contribution to the derivative is simply $\lambda_i/2$. Hence, Equation (3.149) combined with Equation (3.146) becomes

$$V_n = \frac{\lambda_i}{2} + \sum_{\substack{j=1 \\ (j \neq 1)}}^{n} \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial n_i}(\ln r_{ij})\, ds_j \tag{3.150}$$

In Equation (3.150), the first term $\lambda_i/2$ is the normal velocity induced at the ith control point by the ith panel itself, and the summation is the normal velocity induced at the ith control point by all the other panels.

The normal component of the flow velocity at the ith control point is the sum of that due to the freestream [Equation (3.148)] and that due to the source panels [Equation (3.150)]. The boundary condition states that this sum must be zero:

$$V_{\infty,n} + V_n = 0 \tag{3.151}$$

Substituting Equations (3.148) and (3.150) into (3.151), we obtain

$$\frac{\lambda_i}{2} + \sum_{\substack{j=1 \\ (j \neq 1)}}^{n} \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial n_i}(\ln r_{ij})\, ds_j + V_\infty \cos \beta_i = 0 \tag{3.152}$$

Equation (3.152) is the crux of the source panel method. The values of the integrals in Equation (3.152) depend simply on the panel geometry; they are not properties of the flow. Let $I_{i,j}$ be the value of this integral when the control point is on the ith panel and the integral is over the jth panel. Then Equation (3.152) can be written as

$$\frac{\lambda_i}{2} + \sum_{\substack{j=1 \\ (j \neq 1)}}^{n} \frac{\lambda_j}{2\pi} I_{i,j} + V_\infty \cos \beta_i = 0 \tag{3.153}$$

Equation (3.153) is a linear *algebraic* equation with n unknowns $\lambda_1, \lambda_2, \ldots, \lambda_n$. It represents the flow boundary condition evaluated at the control point of the ith panel. Now apply the boundary condition to the control points of *all* the panels; that is, in Equation (3.153), let $i = 1, 2, \ldots, n$. The results will be a system of n

linear algebraic equations with n unknowns ($\lambda_1, \lambda_2, \ldots, \lambda_n$), which can be solved simultaneously by conventional numerical methods.

Look what has happened! After solving the system of equations represented by Equation (3.153) with $i = 1, 2, \ldots, n$, we now have the distribution of source panel strengths which, in an appropriate fashion, cause the body surface in Figure 3.40 to be a streamline of the flow. This approximation can be made more accurate by increasing the number of panels, hence more closely representing the source sheet of continuously varying strength $\lambda(s)$ shown in Figure 3.39. Indeed, the accuracy of the source panel method is amazingly good; a circular cylinder can be accurately represented by as few as 8 panels, and most airfoil shapes, by 50 to 100 panels. (For an airfoil, it is desirable to cover the leading-edge region with a number of small panels to represent accurately the rapid surface curvature and to use larger panels over the relatively flat portions of the body. Note that, in general, all the panels in Figure 3.40 can be different lengths.)

Once the λ_i's ($i = 1, 2, \ldots, n$) are obtained, the velocity *tangent* to the surface at each control point can be calculated as follows. Let s be the distance along the body surface, measured positive from front to rear, as shown in Figure 3.40. The component of freestream velocity tangent to the surface is

$$V_{\infty,s} = V_\infty \sin \beta_i \tag{3.154}$$

The tangential velocity V_s at the control point of the ith panel induced by all the panels is obtained by differentiating Equation (3.146) with respect to s:

$$V_s = \frac{\partial \phi}{\partial s} = \sum_{j=1}^{n} \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial s} (\ln r_{ij})\, ds_j \tag{3.155}$$

[The tangential velocity on a flat source panel induced by the panel itself is zero; hence, in Equation (3.155), the term corresponding to $j = i$ is zero. This is easily seen by intuition, because the panel can only emit volume flow from its surface in a direction perpendicular to the panel itself.] The total surface velocity at the ith control point V_i is the sum of the contribution from the freestream [Equation (3.154)] and from the source panels [Equation (3.155)]:

$$V_i = V_{\infty,s} + V_s = V_\infty \sin \beta_i + \sum_{j=1}^{n} \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial s} (\ln r_{ij})\, ds_j \tag{3.156}$$

In turn, the pressure coefficient at the ith control point is obtained from Equation (3.38):

$$C_{p,i} = 1 - \left(\frac{V_i}{V_\infty} \right)^2$$

In this fashion, the source panel method gives the pressure distribution over the surface of a nonlifting body of arbitrary shape.

When you carry out a source panel solution as described above, the accuracy of your results can be tested as follows. Let S_j be the length of the jth panel. Recall that λ_j is the strength of the jth panel *per unit length*. Hence, the strength of the jth panel itself is $\lambda_j S_j$. For a closed body, such as in Figure 3.40, the *sum*

of all the source and sink strengths must be zero, or else the body itself would be adding or absorbing mass from the flow—an impossible situation for the case we are considering here. Hence, the values of the λ_j's obtained above should obey the relation

$$\sum_{j=1}^{n} \lambda_j S_j = 0 \tag{3.157}$$

Equation (3.157) provides an independent check on the accuracy of the numerical results.

EXAMPLE 3.19

Calculate the pressure coefficient distribution around a circular cylinder using the source panel technique.

■ Solution

We choose to cover the body with eight panels of equal length, as shown in Figure 3.41. This choice is arbitrary; however, experience has shown that, for the case of a circular cylinder, the arrangement shown in Figure 3.41 provides sufficient accuracy. The panels are numbered from 1 to 8, and the control points are shown by the dots in the center of each panel.

Let us evaluate the integrals $I_{i,j}$ which appear in Equation (3.153). Consider Figure 3.42, which illustrates two arbitrary chosen panels. In Figure 3.42, (x_i, y_i) are the coordinates of the control point of the ith panel and (x_j, y_j) are the running coordinates over the entire jth panel. The coordinates of the boundary points for the ith panel are (X_i, Y_i) and (X_{i+1}, Y_{i+1}); similarly, the coordinates of the boundary points for the jth panel are (X_j, Y_j) and (X_{j+1}, Y_{j+1}). In this problem, $\mathbf{V}_\infty$ is in the x direction; hence, the angles between the x axis and the unit vectors $\mathbf{n}_i$ and $\boldsymbol{n}_j$ are β_i and β_j, respectively. Note that, in general, both β_i and β_j vary from 0 to 2π. Recall that the integral $I_{i,j}$ is

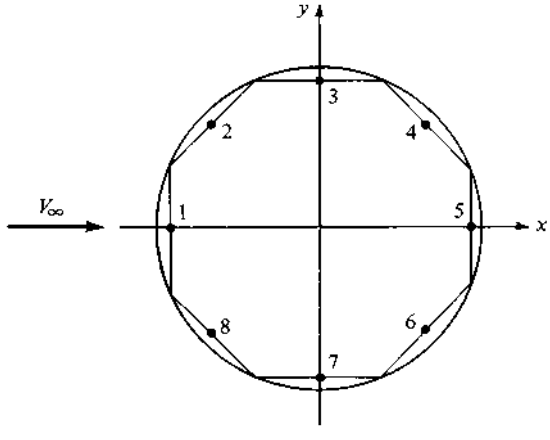


Figure 3.41 Source panel distribution around a circular cylinder.

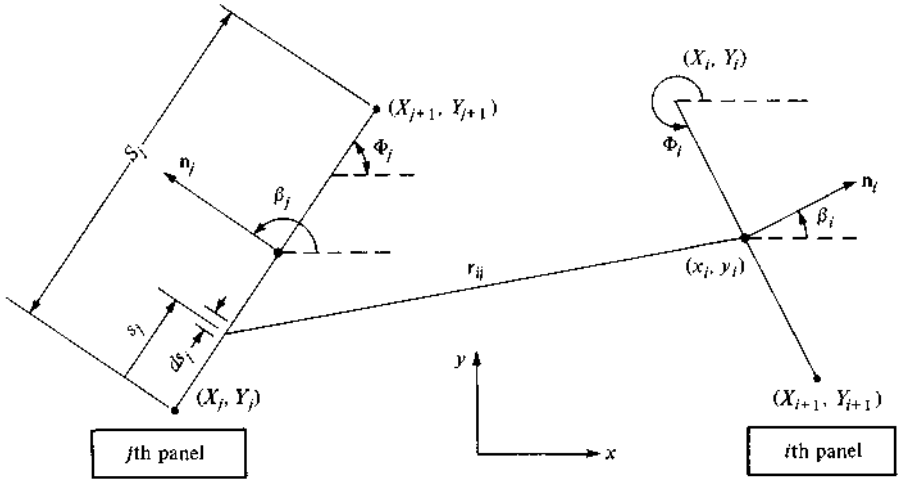


Figure 3.42 Geometry required for the evaluation of I_{ij}.

evaluated at the ith control point and the integral is taken over the complete jth panel:

$$I_{i,j} = \int_j \frac{\partial}{\partial n_i}(\ln r_{ij})\, ds_j \tag{3.158}$$

Since
$$r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

then
$$\frac{\partial}{\partial n_i}(\ln r_{ij}) = \frac{1}{r_{ij}} \frac{\partial r_{ij}}{\partial n_i}$$

$$= \frac{1}{r_{ij}} \frac{1}{2}[(x_i - x_j)^2 + (y_i - y_j)^2]^{-1/2}$$

$$\times \left[2(x_i - x_j)\frac{dx_i}{dn_i} + 2(y_i - y_j)\frac{dy_i}{dn_i}\right]$$

or
$$\frac{\partial}{\partial n_i}(\ln r_{ij}) = \frac{(x_i - x_j)\cos\beta_i + (y_i - y_j)\sin\beta_i}{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{3.159}$$

Note in Figure 3.42 that Φ_i and Φ_j are angles measured in the counterclockwise direction from the x axis to the bottom of each panel. From this geometry,

$$\beta_i = \Phi_i + \frac{\pi}{2}$$

Hence,
$$\sin\beta_i = \cos\Phi_i \tag{3.160a}$$

$$\cos\beta_i = -\sin\Phi_i \tag{3.160b}$$

Also, from the geometry of Figure 3.38, we have

$$x_j = X_j + s_j \cos\Phi_j \tag{3.161a}$$

and
$$y_j = Y_j + s_j \sin\Phi_j \tag{3.161b}$$

Substituting Equations (3.159) to (3.161) into (3.158), we obtain

$$I_{i,j} = \int_0^{S_j} \frac{Cs_j + D}{s_j^2 + 2As_j + B} ds_j \tag{3.162}$$

where

$$A = -(x_i - X_j)\cos\Phi_j - (y_i - Y_j)\sin\Phi_j$$
$$B = (x_i - X_j)^2 + (y_i - Y_j)^2$$
$$C = \sin(\Phi_i - \Phi_j)$$
$$D = (y_i - Y_j)\cos\Phi_i - (x_i - X_j)\sin\Phi_i$$
$$S_j = \sqrt{(X_{j+1} - X_j)^2 + (Y_{j+1} - Y_j)^2}$$

Letting $$E = \sqrt{B - A^2} = (x_i - X_j)\sin\Phi_j - (y_i - Y_j)\cos\Phi_j$$

we obtain an expression for Equation (3.162) from any standard table of integrals:

$$I_{i,j} = \frac{C}{2}\ln\left(\frac{S_j^2 + 2AS_j + B}{B}\right) + \frac{D - AC}{E}\left(\tan^{-1}\frac{S_j + A}{E} - \tan^{-1}\frac{A}{E}\right) \tag{3.163}$$

Equation (3.163) is a general expression for two arbitrarily oriented panels; it is not restricted to the case of a circular cylinder.

We now apply Equation (3.163) to the circular cylinder shown in Figure 3.41. For purposes of illustration, let us choose panel 4 as the ith panel and panel 2 as the jth panel; that is, let us calculate $I_{4,2}$. From the geometry of Figure 3.41, assuming a unit radius for the cylinder, we see that

$$X_j = -0.9239 \quad X_{j+1} = -0.3827 \quad Y_j = 0.3827$$
$$Y_{j+1} = 0.9239 \quad \Phi_i = 315^\circ \quad \Phi_j = 45^\circ$$
$$x_i = 0.6533 \quad y_i = 0.6533$$

Hence, substituting these numbers into the above formulas, we obtain

$$A = -1.3065 \quad B = 2.5607 \quad C = -1 \quad D = 1.3065$$
$$S_j = 0.7654 \quad E = 0.9239$$

Inserting the above values into Equation (3.163), we obtain

$$I_{4,2} = 0.4018$$

Return to Figures 3.41 and 3.42. If we now choose panel 1 as the jth panel, keeping panel 4 as the ith panel, we obtain, by means of a similar calculation, $I_{4,1} = 0.4074$. Similarly, $I_{4,3} = 0.3528$, $I_{4,5} = 0.3528$, $I_{4,6} = 0.4018$, $I_{4,7} = 0.4074$, and $I_{4,8} = 0.4084$.

Return to Equation (3.153), which is evaluated for the ith panel in Figures 3.40 and 3.42. Written for panel 4, Equation (3.153) becomes (after multiplying each term by 2 and

noting that $\beta_i = 45^\circ$ for panel 4)

$$0.4074\lambda_1 + 0.4018\lambda_2 + 0.3528\lambda_3 + \pi\lambda_4 + 0.3528\lambda_5 + 0.4018\lambda_6 + 0.4074\lambda_7 + 0.4084\lambda_8 = -0.7071\, 2\pi V_\infty \tag{3.164}$$

Equation (3.164) is a linear algebraic equation in terms of the eight unknowns, $\lambda_1, \lambda_2, \ldots, \lambda_8$. If we now evaluate Equation (3.153) for each of the seven other panels, we obtain a total of eight equations, including Equation (3.164), which can be solved simultaneously for the eight unknown λ's. The results are

$$\lambda_1/2\pi V_\infty = 0.3765 \quad \lambda_2/2\pi V_\infty = 0.2662 \quad \lambda_3/2\pi V_\infty = 0$$
$$\lambda_4/2\pi V_\infty = -0.2662 \quad \lambda_5/2\pi V_\infty = -0.3765 \quad \lambda_6/2\pi V_\infty = -0.2662$$
$$\lambda_7/2\pi V_\infty = 0 \quad \lambda_8/2\pi V_\infty = 0.2662$$

Note the symmetrical distribution of the λ's, which is to be expected for the nonlifting circular cylinder. Also, as a check on the above solution, return to Equation (3.157). Since each panel in Figure 3.41 has the same length, Equation (3.157) can be written simply as

$$\sum_{j=1}^{n} \lambda_j = 0$$

Substituting the values for the λ's obtained into Equation (3.157), we see that the equation is identically satisfied.

The velocity at the control point of the ith panel can be obtained from Equation (3.156). In that equation, the integral over the jth panel is a geometric quantity that is evaluated in a similar manner as before. The result is

$$\int_j \frac{\partial}{\partial s}(\ln r_{ij})ds_j = \frac{D - AC}{2E} \ln \frac{S_j^2 + 2AS_j + B}{B} - C\left(\tan^{-1}\frac{S_j + A}{E} - \tan^{-1}\frac{A}{E}\right) \tag{3.165}$$

With the integrals in Equation (3.156) evaluated by Equation (3.165), and with the values for $\lambda_1, \lambda_2, \ldots, \lambda_8$ obtained above inserted into Equation (3.156), we obtain the velocities $V_1, V_2, \ldots, V_8$. In turn, the pressure coefficients $C_{p,1}, C_{p,2}, \ldots, C_{p,8}$ are obtained directly from

$$C_{p,i} = 1 - \left(\frac{V_i}{V_\infty}\right)^2$$

Results for the pressure coefficients obtained from this calculation are compared with the exact analytical result, Equation (3.101) in Figure 3.43. Amazingly enough, in spite of the relatively crude paneling shown in Figure 3.41, the numerical pressure coefficient results are excellent.

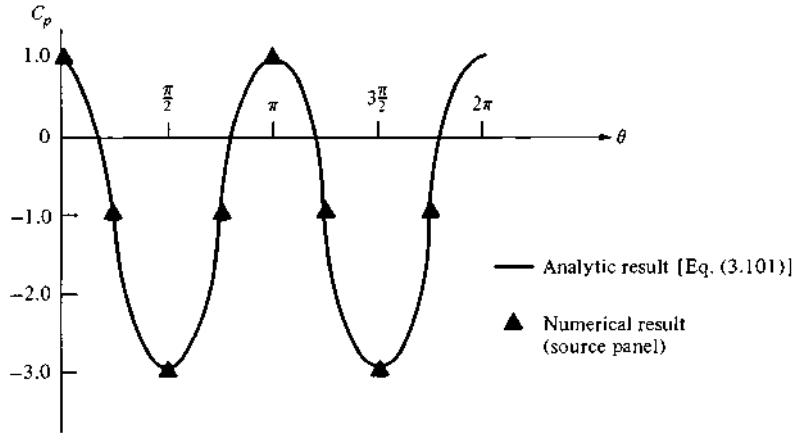


Figure 3.43 Pressure distribution over a circular cylinder; comparison of the source panel results and theory.

3.18 APPLIED AERODYNAMICS: THE FLOW OVER A CIRCULAR CYLINDER—THE REAL CASE

The inviscid, incompressible flow over a circular cylinder was treated in Section 3.13. The resulting theoretical streamlines are sketched in Figure 3.26, characterized by a symmetrical pattern where the flow "closes in" behind the cylinder. As a result, the pressure distribution over the front of the cylinder is the same as that over the rear (see Figure 3.29). This leads to the theoretical result that the pressure drag is zero—d'Alembert's paradox.

The real flow over a circular cylinder is quite different from that studied in Section 3.13, the difference due to the influence of friction. Moreover, the drag coefficient for the real flow over a cylinder is certainly not zero. For a *viscous* incompressible flow, the results of dimensional analysis (Section 1.7) clearly demonstrate that the drag coefficient is a function of the Reynolds number. The variation of $C_D = f(\text{Re})$ for a circular cylinder is shown in Figure 3.44, which is based on a wealth of experimental data. Here, $\text{Re} = (\rho_\infty V_\infty d)/\mu_\infty$, where d is the diameter of the cylinder. Note that C_D is very large for the extremely small values of $\text{Re} < 1$, but decreases monotonically until $\text{Re} \approx 300{,}000$. At this Reynolds number, there is a precipitous drop of C_D from a value near 1 to about 0.3, then a slight recovery to about 0.6 for $\text{Re} = 10^7$. (*Note:* These results are consistent with the comparison shown in Figure 1.54*d* and *e*, contrasting C_D for a circular cylinder at low and high Re.) What causes this precipitous drop in C_D when the Reynolds number reaches about 300,000? A detailed answer must await our discussion of viscous flow in Chapter 4, and later in Part 4. However, we state now that the phenomenon is caused by a sudden transition of laminar flow within the boundary layer at the lower values of Re to a turbulent boundary

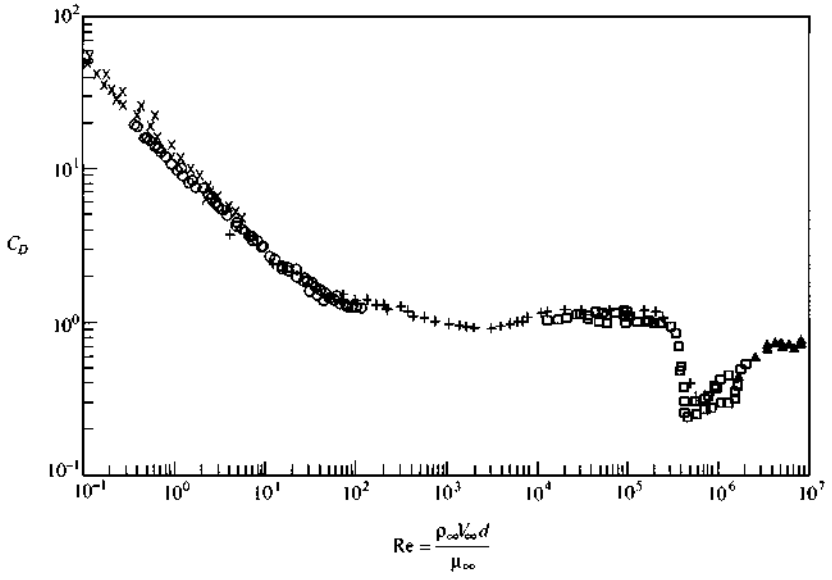


Figure 3.44 Variation of cylinder-drag coefficient with Reynolds number. (*Source:* Experimental data as compiled in Panton, Ronald, *Incompressible Flow,* Wiley-Interscience, New York, 1984.)

layer at the higher values of Re. Why does a turbulent boundary layer result in a smaller C_D for this case? Stay tuned; the answer is given in Chapter 4.

The variation of C_D shown in Figure 3.44 across a range of Re from 10^{-1} to 10^7 is accompanied by tremendous variations in the qualitative aspects of the flow field, as itemized, and as sketched in Figure 3.45.

1. For very low values of Re, say, $0 < \mathrm{Re} < 4$, the streamlines are almost (but not exactly) symmetrical, and the flow is attached, as sketched in Figure 3.45*a*. This regime of viscous flow is called *Stokes flow*; it is characterized by a near balance of pressure forces with friction forces acting on any given fluid element; the flow velocity is so low that inertia effects are very small. A photograph of this type of flow is shown in Figure 3.46, which shows the flow of water around a circular cylinder where $\mathrm{Re} = 1.54$. The streamlines are made visible by aluminum powder on the surface, along with a time exposure of the film.
2. For $4 < \mathrm{Re} < 40$, the flow becomes separated on the back of the cylinder, forming two distinct, stable vortices that remain in the position shown in Figure 3.45*b*. A photograph of this type of flow is given in Figure 3.47, where $\mathrm{Re} = 26$.
3. As Re is increased above 40, the flow behind the cylinder becomes unstable; the vortices which were in a fixed position in Figure 3.45*b* now are alternately shed from the body in a regular fashion and flow downstream.

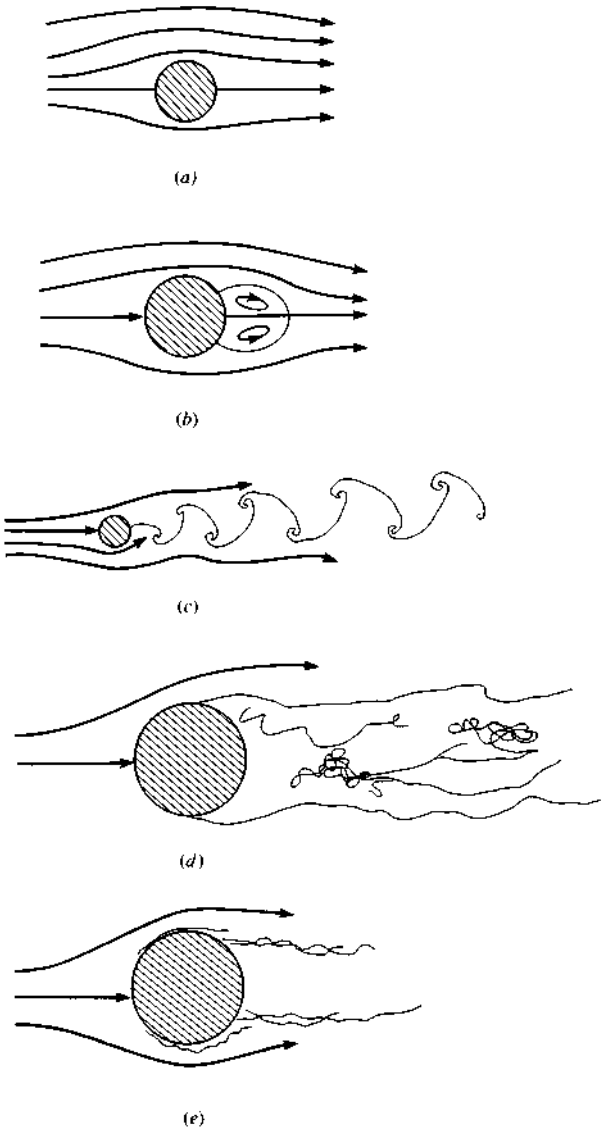


Figure 3.45 Various types of flow over a circular cylinder. (*Source:* Panton, Ronald, *Incompressible Flow,* Wiley-Interscience, New York, 1984.)

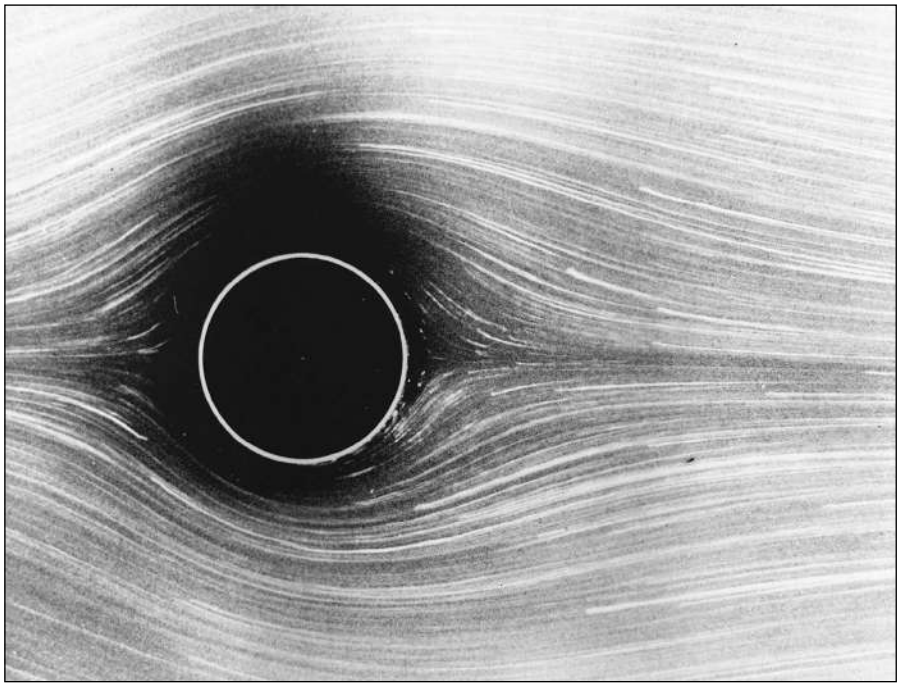


Figure 3.46 Flow over a circular cylinder. Re = 1.54. (*Source:* Photograph by Sadatoshi Taneda, from Van Dyke, Milton, *An Album of Fluid Motion,* The Parabolic Press, Stanford, Calif., 1982.)



Figure 3.47 Flow over a circular cylinder. Re = 26. (*Source:* Photograph by Sadatoshi Taneda, from Van Dyke, Milton, *An Album of Fluid Motion,* The Parabolic Press, Stanford, Calif., 1982.)

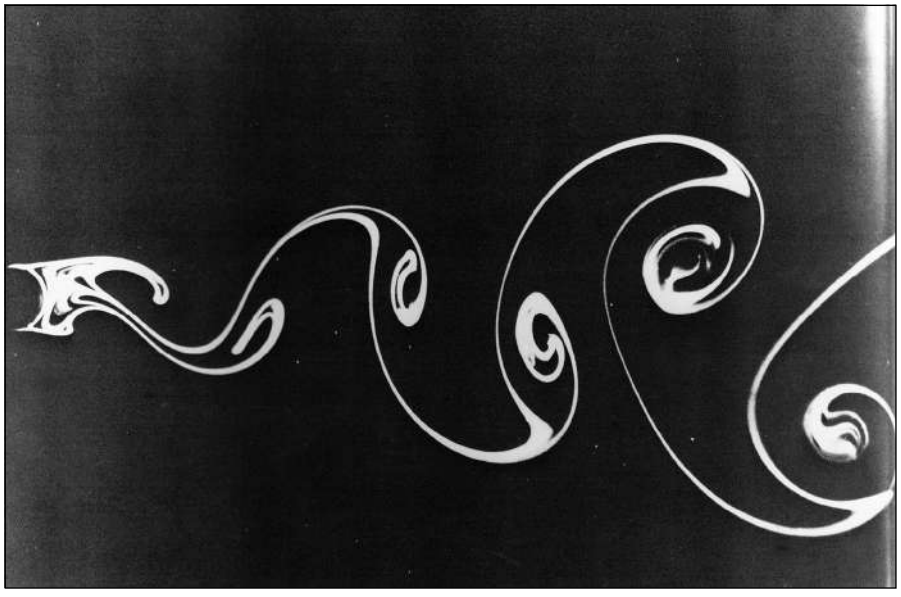


Figure 3.48 Flow over a circular cylinder. Re = 140. A Karman vortex street exists behind the cylinder at this Reynolds number. (*Source:* Photograph by Sadatoshi Taneda, from Van Dyke, Milton, *An Album of Fluid Motion,* The Parabolic Press, Stanford, Calif., 1982.)

This flow is sketched in Figure 3.45*c*. A photograph of this type of flow is shown in Figure 3.48, where Re = 140. This is a water flow where the streaklines are made visible by the electrolytic precipitation method. (In this method, metal plating on the cylinder surface acts as an anode, white particles are precipitated by electrolysis near the anode, and these particles subsequently flow downstream, forming a *streakline*. The definition of a streakline is given in Section 2.11.) The alternately shed vortex pattern shown in Figures 3.45*c* and 3.48 is called a *Karman vortex street,* named after Theodor von Kármán, who began to study and analyze this pattern in 1911 while at Göttingen University in Germany. (von Karman subsequently had a long and very distinguished career in aerodynamics, moving to the California Institute of Technology in 1930, and becoming America's best-known aerodynamicist in the mid-twentieth century. An autobiography of von Karman was published in 1967; see Reference 49. This reference is "must" reading for anyone interested in a riveting perspective on the history of aerodynamics in the twentieth century.)

4. As the Reynolds number is increased to large numbers, the Karman vortex street becomes turbulent and begins to metamorphose into a distinct wake. The laminar boundary layer on the cylinder separates from the surface on the forward face, at a point about 80° from the stagnation point. This is sketched in Figure 3.45*d*. The value of the Reynolds number for this flow is

on the order of 10^5. Note, from Figure 3.44, that C_D is a relatively constant value near unity for $10^3 < \text{Re} < 3 \times 10^5$.

5. For $3 \times 10^5 < \text{Re} < 3 \times 10^6$, the separation of the laminar boundary layer still takes place on the forward face of the cylinder. However, in the free shear layer over the top of the separated region, transition to turbulent flow takes place. The flow then reattaches on the back face of the cylinder, but separates again at about 120° around the body measured from the stagnation point. This flow is sketched in Figure 3.45*e*. This transition to turbulent flow, and the corresponding thinner wake (comparing Figure 3.45*e* with Figure 3.45*d*), reduces the pressure drag on the cylinder and is responsible for the precipitous drop in C_D at $\text{Re} = 3 \times 10^5$ shown in Figure 3.44. (More details on this phenomenon are covered in Chapter 4 and Part 4.)
6. For $\text{Re} > 3 \times 10^6$, the boundary layer transits directly to turbulent flow at some point on the forward face, and the boundary layer remains totally attached over the surface until it separates at an angular location slightly less than 120° on the back surface. For this regime of flow, C_D actually increases slightly with increasing Re because the separation points on the back surface begin to move closer to the top and bottom of the cylinder, producing a fatter wake, and hence larger pressure drag.

In summary, from the photographs and sketches in this section, we see that the real flow over a circular cylinder is dominated by friction effects, namely, the separation of the flow over the rearward face of the cylinder. In turn, a finite pressure drag is created on the cylinder, and d'Alembert's paradox is resolved.

Let us examine the production of drag more closely. The theoretical pressure distribution over the surface of a cylinder in an inviscid, incompressible flow was given in Figure 3.29. In contrast, several real pressure distributions based on experimental measurements for different Reynolds numbers are shown in Figure 3.49, and are compared with the theoretical inviscid flow results obtained in Section 3.13. Note that theory and experiment agree well on the forward face of the cylinder, but that dramatic differences occur over the rearward face. The theoretical results show the pressure decreasing around the forward face from the initial total pressure at the stagnation point, reaching a minimum pressure at the top and bottom of the cylinder ($\theta = 90°$ and 270°), and then increasing again over the rearward face, recovering to the total pressure at the rear stagnation point. In contrast, in the real case where flow separation occurs, the pressures are relatively constant in the separated region over the rearward face and have values slightly less than freestream pressure. (In regions of separated flow, the pressure frequently exhibits a nearly constant value.) In the separated region over the rearward face, the pressure clearly does not recover to the higher values that exist on the front face. There is a net imbalance of the pressure distribution between the front and back faces, with the pressures on the front being higher than on the back, and this imbalance produces the drag on the cylinder.

Return to Figure 3.44, and examine again the variation of C_D as a function of Re. The regimes associated with the very low Reynolds numbers, such as

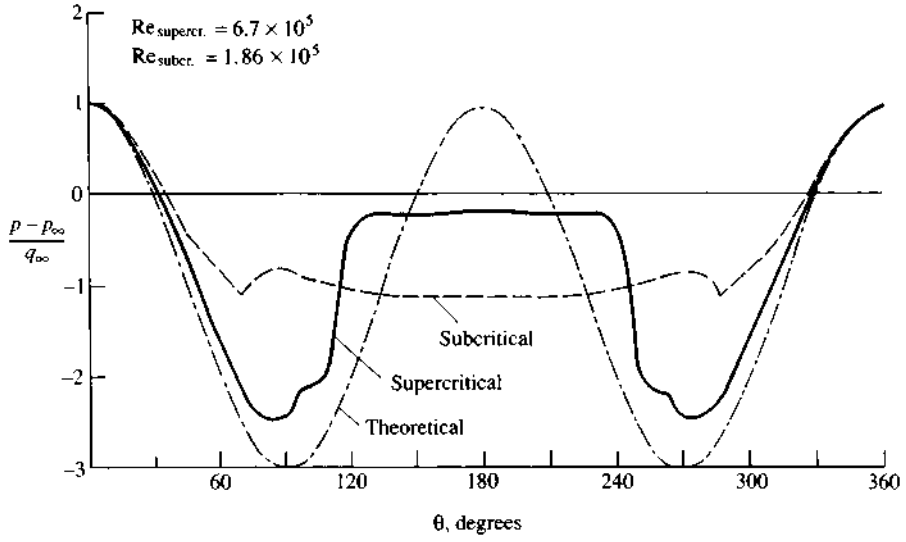


Figure 3.49 Pressure distribution over a circular cylinder in low-speed flow. Comparison of the theoretical pressure distribution with two experimental pressure distributions—one for a subcritical Re and the other for a supercritical Re.

Stokes flow for Re $\approx$ 1, are usually of no interest to aeronautical applications. For example, consider a circular cylinder in an airflow of 30 m/s (about 100 ft/s, or 68 mi/h) at standard sea level conditions, where $\rho_\infty = 1.23$ kg/m^3 and $\mu_\infty = 1.79 \times 10^{-5}$ kg/(m $\cdot$ s). The smaller the diameter of the cylinder, the smaller will be the Reynolds number. *Question:* What is the required cylinder diameter in order to have Re = 1? The answer is obtained from

$$\text{Re} = \frac{\rho_\infty V_\infty d}{\mu_\infty} = 1$$

Hence,

$$d = \frac{\mu_\infty}{\rho_\infty V_\infty} = \frac{1.79 \times 10^{-5}}{(1.23)(30)} = 4 \times 10^{-7} \text{ m}$$

To have Re = 1 for the above conditions, the diameter of the cylinder would have to be extremely small; note that the value of $d = 4 \times 10^{-7}$ m is only slightly larger than the mean free path at standard sea level, which is 6.6×10^{-8} m. (See Section 1.10 for the definition of the mean free path.) Clearly, Reynolds numbers on the order of unity are of little practical aerodynamic importance.

If this is so, then what values of Re for the flow over cylinders are of practical importance? For one such example, consider the wing wires between the upper and lower wings on a World War I biplane, such as the SPAD XIII shown in Figure 3.50. The diameter of these wires is about $\frac{3}{32}$ in, or 0.0024 m. The top speed of the SPAD was 130 mi/h, or 57.8 m/s. For this velocity at standard sea



Figure 3.50 The French SPAD XIII, an example of a strut-and-wire biplane from World War I. Captain Eddie Rickenbacker is shown at the front of the airplane. *(Courtesy of U.S. Air Force.)*

level, we have

$$\text{Re} = \frac{\rho_\infty V_\infty d}{\mu_\infty} = \frac{(1.23)(57.8)(0.0024)}{1.79 \times 10^{-5}} = 9532$$

With this value of Re, we are beginning to enter the world of practical aerodynamics for the flow over cylinders. It is interesting to note that, from Figure 3.44, $C_D = 1$ for the wires on the SPAD. In terms of airplane aerodynamics, this is a high drag coefficient for any component of an aircraft. Indeed, the bracing wires used on biplanes of the World War I era were a source of high drag for the aircraft, so much so that early in the war, bracing wire with a symmetric airfoil-like cross section was utilized to help reduce this drag. Such wire was developed by the British at the Royal Aircraft Factory at Farnborough, and was first tested experimentally as early as 1914 on an SE-4 biplane. Interestingly enough, the SPAD used ordinary round wire, and in spite of this was the fastest of all World War I aircraft.

This author was struck by another example of the effect of cylinder drag while traveling in Charleston, South Carolina, shortly after hurricane Hugo devastated the area on September 28, 1989. Traveling north out of Charleston on U.S. Route 17, near the small fishing town of McClellanville, one passes through the Francis Marion National Forest. This forest was virtually destroyed by the hurricane; 60-ft pine trees were snapped off near their base, and approximately

8 out of every 10 trees were down. The sight bore an eerie resemblance to scenes from the battlefields in France during World War I. What type of force can destroy an entire forest in this fashion? To answer this question, we note that the Weather Bureau measured wind gusts as high as 175 mi/h during the hurricane. Let us approximate the wind force on a typical 60-ft pine tree by the aerodynamic drag on a cylinder of a length of 60 ft and a diameter of 5 ft. Since $V = 175$ mi/h $=$ 256.7 ft/s, $\rho_\infty = 0.002377$ slug/ft^3, and $\mu_\infty = 3.7373 \times 10^{-7}$ slug/(ft $\cdot$ s), then the Reynolds number is

$$\text{Re} = \frac{\rho_\infty V_\infty d}{\mu_\infty} = \frac{(0.002377)(256.7)(5)}{3.7373 \times 10^{-7}} = 8.16 \times 10^6$$

Examining Figure 3.44, we see that $C_D = 0.7$. Since C_D is based on the drag per unit length of the cylinder as well as the projected frontal area, we have for the total drag exerted on an entire tree that is 60 ft tall

$$\begin{aligned} D &= q_\infty S C_D = \tfrac{1}{2}\rho_\infty V_\infty^2 (d)(60) C_D \\ &= \tfrac{1}{2}(0.002377)(256.7)^2(5)(60)(0.7) = 16{,}446 \text{ lb} \end{aligned}$$

a 16,000 lb force on the tree—it is no wonder a whole forest was destroyed. (In the above analysis, we neglected the end effects of the flow over the end of the vertical cylinder. Moreover, we did not correct the standard sea level density for the local reduction in barometric pressure experienced inside a hurricane. However, these are relatively small effects in comparison to the overall force on the cylinder.) The aerodynamics of a tree, and especially that of a forest, are more sophisticated than discussed here. Indeed, the aerodynamics of trees have been studied experimentally with trees actually mounted in a wind tunnel.[6]

3.19 HISTORICAL NOTE: BERNOULLI AND EULER—THE ORIGINS OF THEORETICAL FLUID DYNAMICS

Bernoulli's equation, expressed by Equations (3.14) and (3.15), is historically the most famous equation in fluid dynamics. Moreover, we derived Bernoulli's equation from the general momentum equation in partial differential equation form. The momentum equation is just one of the three fundamental equations of fluid dynamics—the others being continuity and energy. These equations are derived and discussed in Chapter 2 and applied to an incompressible flow in Chapter 3. Where did these equations first originate? How old are they, and who is responsible for them? Considering the fact that all of fluid dynamics in general, and aerodynamics in particular, is built on these fundamental equations, it is important to pause for a moment and examine their historical roots.

[6] For more details, see the interesting discussion on forest aerodynamics in the book by John E. Allen entitled *Aerodynamics, The Science of Air in Motion,* McGraw-Hill, New York, 1982.

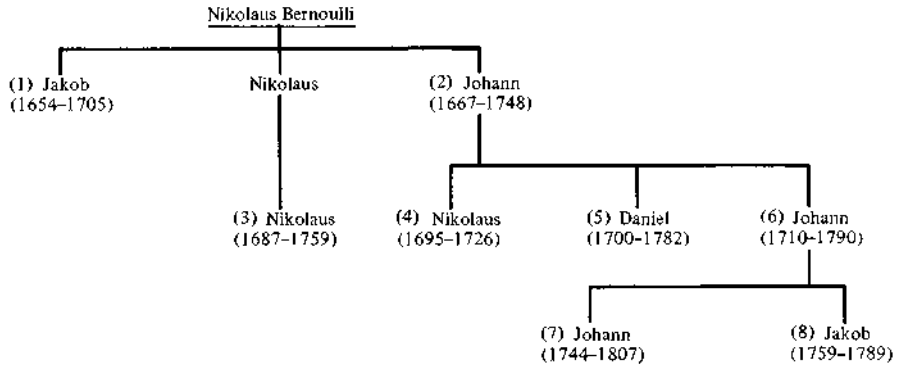


Figure 3.51 Bernoulli's family tree.

As discussed in Section 1.1, Isaac Newton, in his *Principia* of 1687, was the first to establish on a rational basis the relationships between force, momentum, and acceleration. Although he tried, he was unable to apply these concepts properly to a moving fluid. The real foundations of theoretical fluid dynamics were not laid until the next century—developed by a triumvirate consisting of Daniel Bernoulli, Leonhard Euler, and Jean Le Rond d'Alembert.

First, consider Bernoulli. Actually, we must consider the whole family of Bernoulli's because Daniel Bernoulli was a member of a prestigious family that dominated European mathematics and physics during the early part of the eighteenth century. Figure 3.51 is a portion of the Bernoulli family tree. It starts with Nikolaus Bernoulli, who was a successful merchant and druggist in Basel, Switzerland, during the seventeenth century. With one eye on this family tree, let us simply list some of the subsequent members of this highly accomplished family:

1. Jakob—Daniel's uncle. Mathematician and physicist, he was professor of mathematics at the University of Basel. He made major contributions to the development of calculus and coined the term "integral."
2. Johann—Daniel's father. He was a professor of mathematics at Groningen, Netherlands, and later at the University of Basel. He taught the famous French mathematician L'Hospital the elements of calculus, and after the death of Newton in 1727 he was considered Europe's leading mathematician at that time.
3. Nikolaus—Daniel's cousin. He studied mathematics under his uncles and held a master's degree in mathematics and a doctor of jurisprudence.
4. Nikolaus—Daniel's brother. He was Johann's favorite son. He held a master of arts degree, and assisted with much of Johann's correspondence to Newton and Liebniz concerning the development of calculus.
5. Daniel himself—to be discussed below.

6. Johann—Daniel's other brother. He succeeded his father in the Chair of Mathematics at Basel and won the prize of the Paris Academy four times for his work.
7. Johann—Daniel's nephew. A gifted child, he earned the master of jurisprudence at the age of 14. When he was 20, he was invited by Frederick II to reorganize the astronomical observatory at the Berlin Academy.
8. Jakob—Daniel's other nephew. He graduated in jurisprudence but worked in mathematics and physics. He was appointed to the Academy in St. Petersburg, Russia, but he had a promising career prematurely that ended when he drowned in the river Neva at the age of 30.

With such a family pedigree, Daniel Bernoulli was destined for success.

Daniel Bernoulli was born in Groningen, Netherlands, on February 8, 1700. His father, Johann, was a professor at Groningen but returned to Basel, Switzerland, in 1705 to occupy the Chair of Mathematics which had been vacated by the death of Jacob Bernoulli. At the University of Basel, Daniel obtained a master's degree in 1716 in philosophy and logic. He went on to study medicine in Basel, Heidelburg, and Strasbourg, obtaining his Ph.D. in anatomy and botany in 1721. During these studies, he maintained an active interest in mathematics. He followed this interest by moving briefly to Venice, where he published an important work entitled *Exercitationes Mathematicae* in 1724. This earned him much attention and resulted in his winning the prize awarded by the Paris Academy—the first of 10 he was eventually to receive. In 1725, Daniel moved to St. Petersburg, Russia, to join the academy. The St. Petersburg Academy had gained a substantial reputation for scholarship and intellectual accomplishment at that time. During the next 8 years, Bernoulli experienced his most creative period. While at St. Petersburg, he wrote his famous book *Hydrodynamica,* completed in 1734, but not published until 1738. In 1733, Daniel returned to Basel to occupy the Chair of Anatomy and Botany, and in 1750 moved to the Chair of Physics created exclusively for him. He continued to write, give very popular and well-attended lectures in physics, and make contributions to mathematics and physics until his death in Basel on March 17, 1782.

Daniel Bernoulli was famous in his own time. He was a member of virtually all the existing learned societies and academies, such as Bologna, St. Petersburg, Berlin, Paris, London, Bern, Turin, Zurich, and Mannheim. His importance to fluid dynamics is centered on his book *Hydrodynamica* (1738). (With this book, Daniel introduced the term "hydrodynamics" to the literature.) In this book, he ranged over such topics as jet propulsion, manometers, and flow in pipes. Of most importance, he attempted to obtain a relationship between pressure and velocity. Unfortunately, his derivation was somewhat obscure, and Bernoulli's equation, ascribed by history to Daniel via his *Hydrodynamica,* is not to be found in this book, at least not in the form we see it today [such as Equations (3.14) and (3.15)]. The propriety of Equations (3.14) and (3.15) is further complicated by his father, Johann, who also published a book in 1743 entitled *Hydraulica*. It is clear from this latter book that the father understood Bernoulli's theorem

better than his son; Daniel thought of pressure strictly in terms of the height of a manometer column, whereas Johann had the more fundamental understanding that pressure was a force acting on the fluid. (It is interesting to note that Johann Bernoulli was a person of some sensitivity and irritability, with an overpowering drive for recognition. He tried to undercut the impact of Daniel's *Hydrodynamica* by predating the publication date of *Hydraulica* to 1728, to make it appear to have been the first of the two. There was little love lost between son and father.)

During Daniel Bernoulli's most productive years, partial differential equations had not yet been introduced into mathematics and physics; hence, he could not approach the derivation of Bernoulli's equation in the same fashion as we have in Section 3.2. The introduction of partial differential equations to mathematical physics was due to d'Alembert in 1747. d'Alembert's role in fluid mechanics is detailed in Section 3.20. Suffice it to say here that his contributions were equally if not more important than Bernoulli's, and d'Alembert represents the second member of the triumvirate which molded the foundations of theoretical fluid dynamics in the eighteenth century.

The third and probably pivotal member of this triumvirate was Leonhard Euler. He was a giant among the eighteenth-century mathematicians and scientists. As a result of his contributions, his name is associated with numerous equations and techniques, for example, the Euler numerical solution of ordinary differential equations, eulerian angles in geometry, and the momentum equations for inviscid fluid flow [see Equation (3.12)].

Leonhard Euler was born on April 15, 1707, in Basel, Switzerland. His father was a Protestant minister who enjoyed mathematics as a pastime. Therefore, Euler grew up in a family atmosphere that encouraged intellectual activity. At the age of 13, Euler entered the University of Basel which at that time had about 100 students and 19 professors. One of those professors was Johann Bernoulli, who tutored Euler in mathematics. Three years later, Euler received his master's degree in philosophy.

It is interesting that three of the people most responsible for the early development of theoretical fluid dynamics—Johann and Daniel Bernoulli and Euler—lived in the same town of Basel, were associated with the same university, and were contemporaries. Indeed, Euler and the Bernoullis were close and respected friends—so much that, when Daniel Bernoulli moved to teach and study at the St. Petersburg Academy in 1725, he was able to convince the academy to hire Euler as well. At this invitation, Euler left Basel for Russia; he never returned to Switzerland, although he remained a Swiss citizen throughout his life.

Euler's interaction with Daniel Bernoulli in the development of fluid mechanics grew strong during these years at St. Petersburg. It was here that Euler conceived of pressure as a point property that can vary from point to point throughout a fluid and obtained a differential equation relating pressure and velocity, that is, *Euler's equation* given by Equation (3.12). In turn, Euler integrated the differential equation to obtain, for the first time in history, Bernoulli's equation in the form of Equations (3.14) and (3.15). Hence, we see that Bernoulli's equation is really a misnomer; credit for it is legitimately shared by Euler.

When Daniel Bernoulli returned to Basel in 1733, Euler succeeded him at St. Petersburg as a professor of physics. Euler was a dynamic and prolific man; by 1741 he had prepared 90 papers for publication and written the two-volume book *Mechanica*. The atmosphere surrounding St. Petersburg was conducive to such achievement. Euler wrote in 1749: "I and all others who had the good fortune to be for some time with the Russian Imperial Academy cannot but acknowledge that we owe everything which we are and possess to the favorable conditions which we had there."

However, in 1740, political unrest in St. Petersburg caused Euler to leave for the Berlin Society of Sciences, at that time just formed by Frederick the Great. Euler lived in Berlin for the next 25 years, where he transformed the society into a major academy. In Berlin, Euler continued his dynamic mode of working, preparing at least 380 papers for publication. Here, as a competitor with d'Alembert (see Section 3.20), Euler formulated the basis for mathematical physics.

In 1766, after a major disagreement with Frederick the Great over some financial aspects of the academy, Euler moved back to St. Petersburg. This second period of his life in Russia became one of physical suffering. In that same year, he became blind in one eye after a short illness. An operation in 1771 resulted in restoration of his sight, but only for a few days. He did not take proper precautions after the operation, and within a few days, he was completely blind. However, with the help of others, he continued his work. His mind was sharp as ever, and his spirit did not diminish. His literary output even increased—about half of his total papers were written after 1765!

On September 18, 1783, Euler conducted business as usual—giving a mathematics lesson, making calculations of the motion of balloons, and discussing with friends the planet of Uranus, which had recently been discovered. At about 5 P.M., he suffered a brain hemorrhage. His only words before losing consciousness were "I am dying." By 11 P.M., one of the greatest minds in history had ceased to exist.

With the lives of Bernoulli, Euler, and d'Alembert (see Section 3.20) as background, let us now trace the geneology of the basic equations of fluid dynamics. For example, consider the continuity equation in the form of Equation (2.52). Although Newton had postulated the obvious fact that the mass of a specified object was constant, this principle was not appropriately applied to fluid mechanics until 1749. In this year, d'Alembert gave a paper in Paris, entitled "Essai d'une nouvelle theorie de la resistance des fluides," in which he formulated differential equations for the conservation of mass in special applications to plane and axisymmetric flows. Euler took d'Alembert's results and, 8 years later, generalized them in a series of three basic papers on fluid mechanics. In these papers, Euler published, for the first time in history, the continuity equation in the form of Equation (2.52) and the momentum equations in the form of Equations (2.113*a* and *c*), without the viscous terms. Hence, two of the three basic conservation equations used today in modern fluid dynamics were well established long before the American Revolutionary War—such equations were contemporary with the time of George Washington and Thomas Jefferson!

The origin of the energy equation in the form of Equation (2.96) without viscous terms has its roots in the development of thermodynamics in the nineteenth century. Its precise first use is obscure and is buried somewhere in the rapid development of physical science in the nineteenth century.

The purpose of this section has been to give you some feeling for the historical development of the fundamental equations of fluid dynamics. Maybe we can appreciate these equations more when we recognize that they have been with us for quite some time and that they are the product of much thought from some of the greatest minds of the eighteenth century.

3.20 HISTORICAL NOTE: D'ALEMBERT AND HIS PARADOX

You can well imagine the frustration that Jean le Rond d'Alembert felt in 1744 when, in a paper entitled "Traite de l'equilibre et des mouvements de fluids pour servir de suite au traite de dynamique," he obtained the result of zero drag for the inviscid, incompressible flow over a closed two-dimensional body. Using different approaches, d'Alembert encountered this result again in 1752 in his paper entitled "Essai sur la resistance" and again in 1768 in his "Opuscules mathematiques." In this last paper can be found the quote given at the beginning of Chapter 15; in essence, he had given up trying to explain the cause of this paradox. Even though the prediction of fluid-dynamic drag was a very important problem in d'Alembert's time, and in spite of the number of great minds that addressed it, the fact that viscosity is responsible for drag was not appreciated. Instead, d'Alembert's analyses used momentum principles in a frictionless flow, and quite naturally he found that the flow field closed smoothly around the downstream portion of the bodies, resulting in zero drag. Who was this man, d'Alembert? Considering the role his paradox played in the development of fluid dynamics, it is worth our time to take a closer look at the man himself.

d'Alembert was born illegitimately in Paris on November 17, 1717. His mother was Madame De Tenun, a famous salon hostess of that time, and his father was Chevalier Destouches-Canon, a cavalry officer. d'Alembert was immediately abandoned by his mother (she was an ex-nun who was afraid of being forcibly returned to the convent). However, his father quickly arranged for a home for d'Alembert—with a family of modest means named Rousseau. d'Alembert lived with this family for the next 47 years. Under the support of his father, d'Alembert was educated at the College de Quatre-Nations, where he studied law and medicine, and later turned to mathematics. For the remainder of his life, d'Alembert would consider himself a mathematician. By a program of self-study, d'Alembert learned the works of Newton and the Bernoullis. His early mathematics caught the attention of the Paris Academy of Sciences, of which he became a member in 1741. d'Alembert published frequently and sometimes rather hastily, in order to be in print before his competition. However, he made substantial contributions to the science of his time. For example, he was (1) the first to formulate the wave equation of classical physics, (2) the first to express the concept of a partial

differential equation, (3) the first to solve a partial differential equation—he used separation of variables—and (4) the first to express the differential equations of fluid dynamics in terms of a field. His contemporary, Leonhard Euler (see Sections 1.1 and 3.18) later expanded greatly on these equations and was responsible for developing them into a truly rational approach for fluid-dynamic analysis.

During the course of his life, d'Alembert became interested in many scientific and mathematical subjects, including vibrations, wave motion, and celestial mechanics. In the 1750s, he had the honored position of science editor for the *Encyclopedia*—a major French intellectual endeavor of the eighteenth century which attempted to compile all existing knowledge into a large series of books. As he grew older, he also wrote papers on nonscientific subjects, mainly musical structure, law, and religion.

In 1765, d'Alembert became very ill. He was helped to recover by the nursing of Mlle. Julie de Lespinasse, the woman who was d'Alembert's only love throughout his life. Although he never married, d'Alembert lived with Julie de Lespinasse until she died in 1776. d'Alembert had always been a charming gentleman, renowned for his intelligence, gaiety, and considerable conversational ability. However, after Mlle. de Lespinasse's death, he became frustrated and morose—living a life of despair. He died in this condition on October 29, 1783, in Paris.

d'Alembert was one of the great mathematicians and physicists of the eighteenth century. He maintained active communications and dialogue with both Bernoulli and Euler and ranks with them as one of the founders of modern fluid dynamics. This, then, is the man behind the paradox, which has existed as an integral part of fluid dynamics for the past two centuries.

3.21 SUMMARY

Return to the road map given in Figure 3.4. Examine each block of the road map to remind yourself of the route we have taken in this discussion of the fundamentals of inviscid, incompressible flow. Before proceeding further, make certain that you feel comfortable with the detailed material represented by each block, and how each block is related to the overall flow of ideas and concepts.

For your convenience, some of the highlights of this chapter are summarized next:

Bernoulli's equation

$$p + \tfrac{1}{2}\rho V^2 = \text{const}$$

(*a*) Applies to inviscid, incompressible flows only.

(*b*) Holds along a streamline for a rotational flow.

(*c*) Holds at every point throughout an irrotational flow.

(*d*) In the form given above, body forces (such as gravity) are neglected, and steady flow is assumed.

Quasi-one-dimensional continuity equation

$$\rho AV = \text{const} \qquad \text{(for compressible flow)}$$
$$AV = \text{const} \qquad \text{(for incompressible flow)}$$

From a measurement of the Pitot pressure p_0 and static pressure p_1, the velocity of an incompressible flow is given by

$$V_1 = \sqrt{\frac{2(p_0 - p_1)}{\rho}} \tag{3.34}$$

Pressure coefficient

Definition:

$$C_p = \frac{p - p_\infty}{q_\infty} \tag{3.36}$$

where dynamic pressure is $q_\infty \equiv \frac{1}{2}\rho_\infty V_\infty^2$.
For incompressible steady flow with no friction:

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2 \tag{3.38}$$

Governing equations

$$\nabla \cdot \mathbf{V} = 0 \qquad \text{(condition of incompressibility)} \tag{3.39}$$

$$\nabla^2 \phi = 0 \qquad \text{(Laplace's equation; holds for irrotational, incompressible flow)} \tag{3.40}$$

or $$\nabla^2 \psi = 0 \tag{3.46}$$

Boundary conditions

$$u = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} = V_\infty$$
$$v = \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x} = 0 \qquad \text{at infinity}$$

$$\mathbf{V} \cdot \mathbf{n} = 0 \qquad \text{at body (flow tangency condition)}$$

Elementary flows

(*a*) Uniform flow:

$$\phi = V_\infty x = V_\infty r \cos\theta \tag{3.53}$$

$$\psi = V_\infty y = V_\infty r \sin\theta \tag{3.55}$$

(*b*) Source flow:

$$\phi = \frac{\Lambda}{2\pi} \ln r \tag{3.67}$$

$$\psi = \frac{\Lambda}{2\pi}\theta \tag{3.72}$$

$$V_r = \frac{\Lambda}{2\pi r} \qquad V_\theta = 0 \tag{3.62}$$

(*c*) Doublet flow:

$$\phi = \frac{\kappa}{2\pi}\frac{\cos\theta}{r} \tag{3.88}$$

$$\psi = -\frac{\kappa}{2\pi}\frac{\sin\theta}{r} \tag{3.87}$$

(*d*) Vortex flow:

$$\phi = -\frac{\Gamma}{2\pi}\theta \tag{3.112}$$

$$\psi = \frac{\Gamma}{2\pi} \ln r \tag{3.114}$$

$$V_\theta = -\frac{\Gamma}{2\pi r} \qquad V_r = 0 \tag{3.105}$$

Inviscid flow over a cylinder

(*a*) Nonlifting (uniform flow and doublet)

$$\psi = (V_\infty r \sin\theta)\left(1 - \frac{R^2}{r^2}\right) \tag{3.92}$$

where R = radius of cylinder $= \kappa/2\pi V_\infty$.

Surface velocity: $V_\infty = -2V_\infty \sin\theta$ (3.100)

Surface pressure coefficient: $C_p = 1 - 4\sin^2\theta$ (3.101)

$$L = D = 0$$

(*b*) Lifting (uniform flow + doublet + vortex)

$$\psi = (V_\infty r \sin\theta)\left(1 - \frac{R^2}{r^2}\right) + \frac{\Gamma}{2\pi} \ln \frac{r}{R} \tag{3.118}$$

(*continued*)

Surface velocity: $$V_\theta = -2V_\infty \sin\theta - \frac{\Gamma}{2\pi R} \tag{3.125}$$

$$L' = \rho_\infty V_\infty \Gamma \qquad \text{(lift per unit span)} \tag{3.140}$$

$$D = 0$$

Kutta-Joukowski theorem
For a closed two-dimensional body of arbitrary shape, the lift per unit span is $L' = \rho_\infty V_\infty \Gamma$.

Source panel method
This is a numerical method for calculating the nonlifting flow over bodies of arbitrary shape. Governing equations:

$$\frac{\lambda_i}{2} + \sum_{\substack{j=1 \\ (j\neq 1)}}^{n} \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial n_i}(\ln r_{ij})\, ds_j + V_\infty \cos\beta_i = 0 \qquad (i = 1, 2, \ldots, n) \tag{3.152}$$

3.22 PROBLEMS

Note: All the following problems assume an inviscid, incompressible flow. Also, standard sea level density and pressure are 1.23 kg/m^3 (0.002377 slug/ft^3) and 1.01×10^5 N/m^2 (2116 lb/ft^2), respectively.

3.1 For an irrotational flow, show that Bernoulli's equation holds between *any* points in the flow, not just along a streamline.

3.2 Consider a venturi with a throat-to-inlet area ratio of 0.8, mounted on the side of an airplane fuselage. The airplane is in flight at standard sea level. If the static pressure at the throat is 2100 lb/ft^2, calculate the velocity of the airplane.

3.3 Consider a venturi with a small hole drilled in the side of the throat. This hole is connected via a tube to a closed reservoir. The purpose of the venturi is to create a vacuum in the reservoir when the venturi is placed in an airstream. (The *vacuum* is defined as the pressure difference *below* the outside ambient pressure.) The venturi has a throat-to-inlet area ratio of 0.85. Calculate the maximum vacuum obtainable in the reservoir when the venturi is placed in an airstream of 90 m/s at standard sea level conditions.

3.4 Consider a low-speed open-circuit subsonic wind tunnel with an inlet-to-throat area ratio of 12. The tunnel is turned on, and the pressure

difference between the inlet (the settling chamber) and the test section is read as a height difference of 10 cm on a U-tube mercury manometer. (The density of liquid mercury is 1.36×10^4 kg/m^3.) Calculate the velocity of the air in the test section.

3.5 Assume that a Pitot tube is inserted into the test-section flow of the wind tunnel in Problem 3.4. The tunnel test section is completely sealed from the outside ambient pressure. Calculate the pressure measured by the Pitot tube, assuming the static pressure at the tunnel inlet is atmospheric.

3.6 A Pitot tube on an airplane flying at standard sea level reads 1.07×10^5 N/m^2. What is the velocity of the airplane?

3.7 At a given point on the surface of the wing of the airplane in Problem 3.6, the flow velocity is 130 m/s. Calculate the pressure coefficient at this point.

3.8 Consider a uniform flow with velocity V_∞. Show that this flow is a physically possible incompressible flow and that it is irrotational.

3.9 Show that a source flow is a physically possible incompressible flow everywhere except at the origin. Also show that it is irrotational everywhere.

3.10 Prove that the velocity potential and the stream function for a uniform flow, Equations (3.53) and (3.55), respectively, satisfy Laplace's equation.

3.11 Prove that the velocity potential and the stream function for a source flow, Equations (3.67) and (3.72), respectively, satisfy Laplace's equation.

3.12 Consider the flow over a semi-infinite body as discussed in Section 3.11. If V_∞ is the velocity of the uniform stream, and the stagnation point is 1 ft upstream of the source:

a. Draw the resulting semi-infinite body to scale on graph paper.

b. Plot the pressure coefficient distribution over the body; that is, plot C_p versus distance along the centerline of the body.

3.13 Derive Equation (3.81). *Hint:* Make use of the symmetry of the flow field shown in Figure 3.18; that is, start with the knowledge that the stagnation points must lie on the axis aligned with the direction of V_∞.

3.14 Derive the velocity potential for a doublet; that is, derive Equation (3.88). *Hint:* The easiest method is to start with Equation (3.87) for the stream function and extract the velocity potential.

3.15 Consider the nonlifting flow over a circular cylinder. Derive an expression for the pressure coefficient at an arbitrary point (r, θ) in this flow, and show that it reduces to Equation (3.101) on the surface of the cylinder.

3.16 Consider the nonlifting flow over a circular cylinder of a given radius, where $V_\infty = 20$ ft/s. If V_∞ is doubled, that is, $V_\infty = 40$ ft/s, does the shape of the streamlines change? Explain.

3.17 Consider the lifting flow over a circular cylinder of a given radius and with a given circulation. If V_∞ is doubled, keeping the circulation the same, does the shape of the streamlines change? Explain.

3.18 The lift on a spinning circular cylinder in a freestream with a velocity of 30 m/s and at standard sea level conditions is 6 N/m of span. Calculate the circulation around the cylinder.

3.19 A typical World War I biplane fighter (such as the French SPAD shown in Figure 3.50) has a number of vertical interwing struts and diagonal bracing wires. Assume for a given airplane that the total length for the vertical struts (summed together) is 25 ft, and that the struts are cylindrical with a diameter of 2 in. Assume also that the total length of the bracing wires is 80 ft, with a cylindrical diameter of $\frac{3}{32}$ in. Calculate the drag (in pounds) contributed by these struts and bracing wires when the airplane is flying at 120 mi/h at standard sea level. Compare this component of drag with the total zero-lift drag for the airplane, for which the total wing area is 230 ft^2 and the zero-lift drag coefficient is 0.036.

3.20 The Kutta-Joukowski theorem, Equation (3.140), was derived exactly for the case of the lifting cylinder. In Section 3.16 it is stated without proof that Equation (3.140) also applies in general to a two-dimensional body of arbitrary shape. Although this general result can be proven mathematically, it also can be accepted by making a physical argument as well. Make this physical argument by drawing a closed curve around the body where the closed curve is very far away from the body, so far away that in perspective the body becomes a very small speck in the middle of the domain enclosed by the closed curve.

3.21 Consider the streamlines over a circular cylinder as sketched at the right of Figure 3.26. Single out the first three streamlines flowing over the top of the cylinder. Designate each streamline by its stream function, ψ_1, ψ_2, and ψ_3. The first streamline wets the surface of the cylinder; designate $\psi_1 = 0$. The streamline above that is Ψ_2, and the next one above that is ψ_3. Assume the streamlines start out in the free stream equally spaced. Hence, the volume flow rates between the streamlines are the same. The streamline ψ_2 passes through the point $(1.2R,\pi/2)$ directly above the top of the cylinder. Calculate the location of the point directly above the top of the cylinder through which the streamline ψ_3 flows. Comment on the spacing between the streamlines directly above the top.

3.22 Consider the flow field over a circular cylinder mounted perpendicular to the flow in the test section of a low-speed subsonic wind tunnel. At standard sea-level conditions, if the flow velocity at some region of the flow field exceeds about 250 mi/h, compressibility begins to have an effect in that region. Calculate the velocity of the flow in the test section of the wind tunnel above which compressibility effects begin to become important, i.e., above which we can not accurately assume totally incompressible flow over the cylinder for the wind tunnel tests.

3.23 Prove that the flow field specified in Example 2.1 is not incompressible; i.e., it is a compressible flow as stated without proof in Example 2.1.

CHAPTER 4

Incompressible Flow over Airfoils

Of the many problems now engaging attention, the following are considered of immediate importance and will be considered by the committee as rapidly as funds can be secured for the purpose. . . . The evolution of more efficient wing sections of practical form, embodying suitable dimensions for an economical structure, with moderate travel of the center-of-pressure and still affording a large range of angle-of-attack combined with efficient action.

From the first Annual Report of the NACA, 1915

PREVIEW BOX

Imagine that you have just been given an airfoil of a particular shape at a certain angle of attack to a given low-speed flow, and you have been asked to obtain the lift (or more importantly, the lift coefficient) of the airfoil. What do you do (besides panicking)? Your first inclination might be to make a model of the airfoil, put it in a low-speed wind tunnel, and measure the lift coefficient. This is indeed what aerodynamicists have been doing for more than 100 years. The early part of this chapter discusses such experimental measurements of airfoil properties in low-speed wind tunnels. These measurements give you an immediate feel for airfoil lift, drag, and moment coefficients as a function of angle of attack. The experimental measurements give you a fast track towards obtaining a comfortable and practical understanding as to how airfoils behave. That is what the first three sections of this chapter are all about.

Most of the rest of this chapter, however, deals with our second inclination as to how to obtain the airfoil properties, namely, to *calculate* them. This is a horse of a different color. You will be introduced to the elegant circulation theory of lift—the crowning jewel of inviscid, incompressible flow theory for the calculation of lift. At the turn of the twentieth century, the circulation theory of lift was a breakthrough

in the theoretical prediction of lift. In this chapter, we first apply this theory to thin airfoils at small angles of attack; thin airfoil theory was developed in Germany during World War I and is by far the most tractable means of obtaining analytical solutions for lift and moments on an airfoil. But, as we shall see, thin airfoil theory, as its name implies, holds only for thin airfoils at small angles of attack. This is not as restrictive as it seems, however, because many airplanes over the past years have relatively thin airfoils, and cruise at relatively small angles of attack. Thin airfoil theory gives us a lot of practical results, plus the intellectual gratification of carrying through some elegant theoretical thinking—give it a good read and I think you will like it.

Since the 1960s, the advent and development of the high-speed digital computer allowed detailed numerical solutions based on the circulation theory of lift, solutions for the lift on a body of arbitrary shape and thickness at any angle of attack, subject of course to the assumption of inviscid potential flow. These numerical solutions, an extension of the panel solutions discussed in Section 3.17, are discussed towards the end of this chapter—they are the "gold standard" for low-speed, inviscid-flow airfoil calculations, and are used throughout the aeronautical industry and by many aeronautical research and development laboratories. The concept of panel solutions is an inspired numerical application of the circulation theory of lift, and it has opened the door to the analysis of practically any airfoil shape at any angle of attack.

Airfoils come in many different shapes. An historical sequence of airfoil shapes through 1935 is shown in Figure 4.1. Beginning in 1938, the National Advisory Committee for Aeronautics (NACA) developed a revolutionary series of airfoil shapes designed to encourage laminar flow in the boundary layer over the airfoil, hence dramatically reducing skin friction drag on the airfoil; the shape of a representative laminar-flow airfoil is given in Figure 4.2. Although these shapes never produced the desired amount of laminar flow in practice, by a stroke of serendipity they proved to be excellent high-speed airfoils for jet-powered airplanes after 1945. Beginning in 1965 National Aeronautics and Space Administration (NASA)

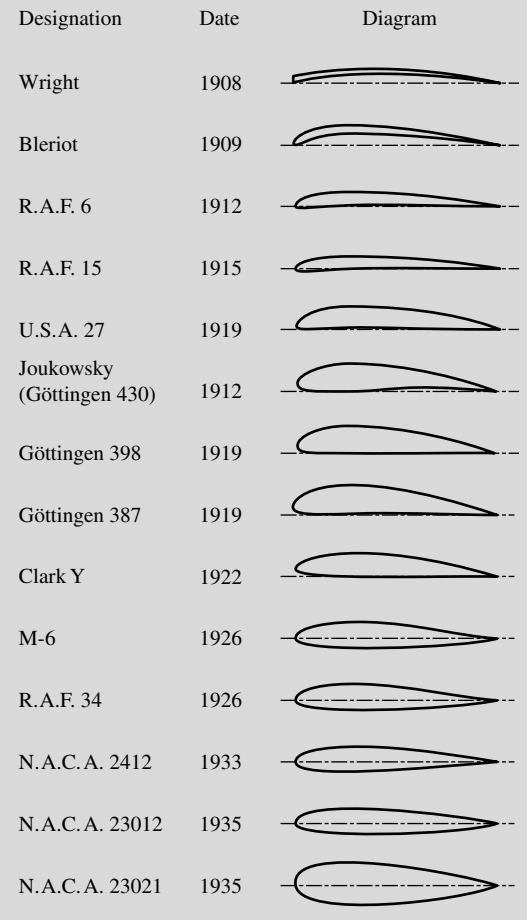


Figure 4.1 Historical sequence of airfoil sections. (Millikan, Clark B., *Aerodynamics of the Airplane,* Wiley, 1941.)

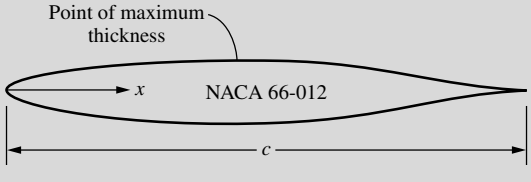


Figure 4.2 Laminar-flow airfoil shape.



Figure 4.3 Supercritical airfoil shape.

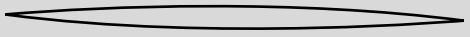




Figure 4.4 Supersonic airfoil shapes.

developed another series of revolutionary airfoil shapes, supercritical airfoils, designed for efficient flight near Mach one; a typical supercritical airfoil shape is shown in Figure 4.3. Classic airfoil shapes for supersonic flow are shown in Figure 4.4; note the very slender profiles with sharp leading edges. All the airfoil shapes shown in Figure 4.1 through 4.4 were designed for specific purposes in their time and have been used on untold numbers of different airplanes. Today, the proper design of new airfoil shapes is more important than ever. Using numerical techniques, aircraft companies usually custom-design the airfoil shapes for new airplanes, shapes that best fit the design requirements for the specific airplane. This chapter is exclusively devoted to the study of airfoils; it discusses the fundamental aspects of airfoil aerodynamics—aspects that are the heart of airfoil design and performance.

Figure 4.5 shows an airplane in flight, sustained in the air by the aerodynamic action of its wing. Airplane wings are made up of airfoil shapes. The first step in understanding the aerodynamics of wings is to understand the aerodynamics of airfoils. Airfoil aerodynamics is important stuff—it is the stuff of this chapter. Moreover, it is truly interesting. It is fun to visualize the flow over an airfoil and to learn how to calculate the resulting lift on the airfoil. Read on and enjoy.



Figure 4.5 DeHaviland DHC-6 Twin Otter.
© PhotoDisc/Alamy RF

4.1 INTRODUCTION

With the advent of successful powered flight at the turn of the twentieth century, the importance of aerodynamics ballooned almost overnight. In turn, interest grew in the understanding of the aerodynamic action of such lifting surfaces as fixed wings on airplanes and, later, rotors on helicopters. In the period 1912–1918, the analysis of airplane wings took a giant step forward when Ludwig Prandtl and his colleagues at Göttingen, Germany, showed that the aerodynamic consideration of wings could be split into two parts: (1) the study of the

section of a wing—an airfoil—and (2) the modification of such airfoil properties to account for the complete, finite wing. This approach is still used today; indeed, the theoretical calculation and experimental measurement of modern airfoil properties have been a major part of the aeronautics research carried out by the National Aeronautics and Space Administration (NASA) in the 1970s and 1980s. (See Chapter 5 of Reference 2 for a historical sketch on airfoil development and Reference 10 for a description of modern airfoil research.) Following Prandtl's philosophy, the present chapter deals exclusively with airfoils, whereas Chapter 5 treats the case of a complete, finite wing. Therefore, in this chapter and Chapter 5, we make a major excursion into aerodynamics as applied to airplanes.

What is an airfoil? Consider a wing as drawn in perspective in Figure 4.6. The wing extends in the y direction (the span direction). The freestream velocity V_∞ is parallel to the xz plane. Any section of the wing cut by a plane parallel to the xz plane is called an *airfoil*. The purpose of this chapter is to present theoretical methods for the calculation of airfoil properties. In most of this chapter we will deal with inviscid flow, which does not lead to predictions of airfoil drag; indeed, d'Alembert's paradox says that the drag on an airfoil is zero—clearly not a realistic answer. We will have to wait until Section 4.12 and Chapter 15 and a discussion of viscous flow before predictions of drag can be made. However, the lift and moments on the airfoil are due mainly to the pressure distribution, which (below the stall) is dictated by inviscid flow. Therefore, this chapter concentrates on the theoretical prediction of airfoil lift and moments.

The road map for this chapter is given in Figure 4.7. After some initial discussion on airfoil nomenclature and characteristics, we present two approaches to low-speed airfoil theory. One is the classical thin airfoil theory developed during the period 1910–1920 (the right-hand branch of Figure 4.7). The other is the modern numerical approach for arbitrary airfoils using vortex panels

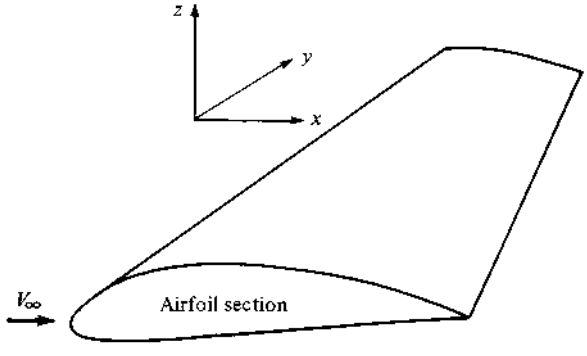


Figure 4.6 Definition of an airfoil.

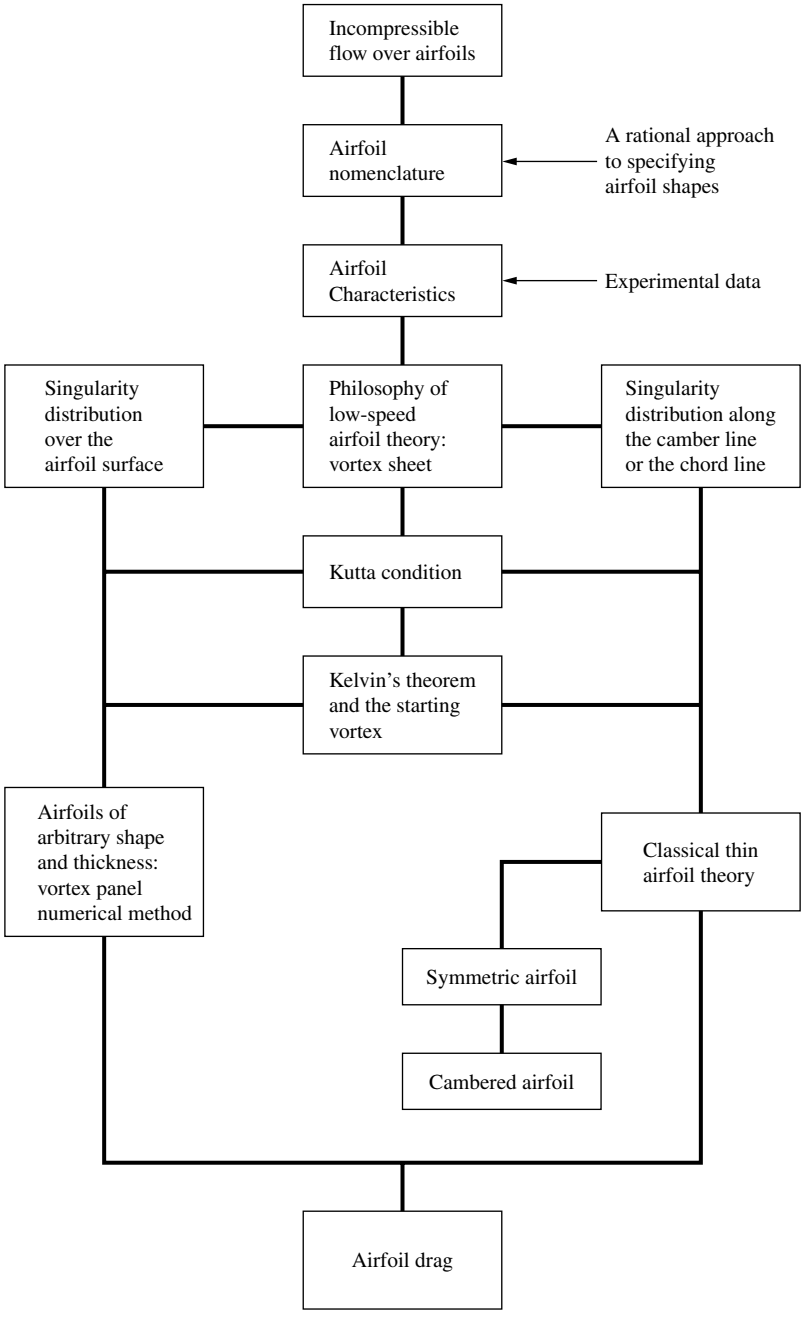


Figure 4.7 Road map for Chapter 4.

(the left-hand branch of Figure 4.7). Please refer to this road map as you work your way through this chapter.

4.2 AIRFOIL NOMENCLATURE

The first patented airfoil shapes were developed by Horatio F. Phillips in 1884. Phillips was an Englishman who carried out the first serious wind-tunnel experiments on airfoils. In 1902, the Wright brothers conducted their own airfoil tests in a wind tunnel, developing relatively efficient shapes which contributed to their successful first flight on December 17, 1903 (see Section 1.1). Clearly, in the early days of powered flight, airfoil design was basically customized and personalized. However, in the early 1930s, NACA—the forerunner of NASA—embarked on a series of definitive airfoil experiments using airfoil shapes that were constructed rationally and systematically. Many of these NACA airfoils are in common use today. Therefore, in this chapter we follow the nomenclature established by the NACA; such nomenclature is now a well-known standard.

Consider the airfoil sketched in Figure 4.8. The *mean camber line* is the locus of points halfway between the upper and lower surfaces as measured perpendicular to the mean camber line itself. The most forward and rearward points of the mean camber line are the *leading and trailing edges,* respectively. The straight line connecting the leading and trailing edges is the *chord line* of the airfoil, and the precise distance from the leading to the trailing edge measured along the chord line is simply designated the *chord c* of the airfoil. The *camber* is the maximum distance between the mean camber line and the chord line, measured perpendicular to the chord line. The *thickness* is the distance between the upper and lower surfaces, also measured perpendicular to the chord line. The shape of the airfoil at the leading edge is usually circular, with a leading-edge radius of approximately $0.02c$. The shapes of all standard NACA airfoils are generated by specifying the shape of the mean camber line and then wrapping a specified symmetrical thickness distribution around the mean camber line.

The force-and-moment system on an airfoil was discussed in Section 1.5, and the relative wind, angle of attack, lift, and drag were defined in Figure 1.16. You should review these considerations before proceeding further.

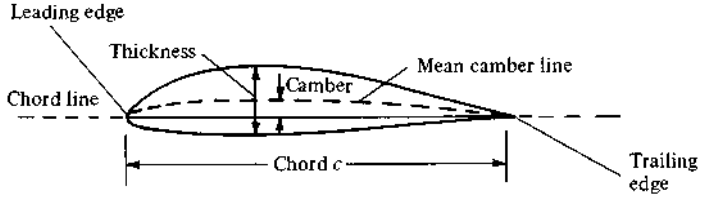


Figure 4.8 Airfoil nomenclature.

The NACA identified different airfoil shapes with a logical numbering system. For example, the first family of NACA airfoils, developed in the 1930s, was the "four-digit" series, such as the NACA 2412 airfoil. Here, the first digit is the maximum camber in hundredths of chord, the second digit is the location of maximum camber along the chord from the leading edge in tenths of chord, and the last two digits give the maximum thickness in hundredths of chord. For the NACA 2412 airfoil, the maximum camber is $0.02c$ located at $0.4c$ from the leading edge, and the maximum thickness is $0.12c$. It is common practice to state these numbers in percent of chord, that is, 2 percent camber at 40 percent chord, with 12 percent thickness. An airfoil with no camber, that is, with the camber line and chord line coincident, is called a *symmetric airfoil*. Clearly, the shape of a symmetric airfoil is the same above and below the chord line. For example, the NACA 0012 airfoil is a symmetric airfoil with a maximum thickness of 12 percent.

The second family of NACA airfoils was the "five-digit" series, such as the NACA 23012 airfoil. Here, the first digit when multiplied by $\frac{3}{2}$ gives the design lift coefficient[1] in tenths, the next two digits when divided by 2 give the location of maximum camber along the chord from the leading edge in hundredths of chord, and the final two digits give the maximum thickness in hundredths of chord. For the NACA 23012 airfoil, the design lift coefficient is 0.3, the location of maximum camber is at $0.15c$, and the airfoil has 12 percent maximum thickness.

One of the most widely used family of NACA airfoils is the "6-series" laminar flow airfoils, developed during World War II. An example is the NACA 65-218. Here, the first digit simply identifies the series, the second gives the location of minimum pressure in tenths of chord from the leading edge (for the basic symmetric thickness distribution at zero lift), the third digit is the design lift coefficient in tenths, and the last two digits give the maximum thickness in hundredths of chord. For the NACA 65-218 airfoil, the 6 is the series designation, the minimum pressure occurs at $0.5c$ for the basic symmetric thickness distribution at zero lift, the design lift coefficient is 0.2, and the airfoil is 18 percent thick.

The complete NACA airfoil numbering system is given in Reference 11. Indeed, Reference 11 is a definitive presentation of the classic NACA airfoil work up to 1949. It contains a discussion of airfoil theory, its application, coordinates for the shape of NACA airfoils, and a huge bulk of experimental data for these airfoils. This author strongly encourages you to read Reference 11 for a thorough presentation of airfoil characteristics.

As a matter of interest, the following is a short partial listing of airplanes currently in service that use standard NACA airfoils.

[1] The design lift coefficient is the theoretical lift coefficient for the airfoil when the angle of attack is such that the slope of the mean camber line at the leading edge is parallel to the freestream velocity.

Airplane	Airfoil
Beechcraft Sundowner	NACA 63A415
Beechcraft Bonanza	NACA 23016.5 (at root) NACA 23012 (at tip)
Cessna 150	NACA 2412
Fairchild A-10	NACA 6716 (at root) NACA 6713 (at tip)
Gates Learjet 24D	NACA 64A109
General Dynamics F-16	NACA 64A204
Lockheed C-5 Galaxy	NACA 0012 (modified)

In addition, many of the large aircraft companies today design their own special-purpose airfoils; for example, the Boeing 727, 737, 747, 757, 767, and 777 all have specially designed Boeing airfoils. Such capability is made possible by modern airfoil design computer programs utilizing either panel techniques or direct numerical finite-difference solutions of the governing partial differential equations for the flow field. (Such equations are developed in Chapter 2.)

4.3 AIRFOIL CHARACTERISTICS

Before discussing the theoretical calculation of airfoil properties, let us examine some typical results. During the 1930s and 1940s, the NACA carried out numerous measurements of the lift, drag, and moment coefficients on the standard NACA airfoils. These experiments were performed at low speeds in a wind tunnel where the constant-chord wing spanned the entire test section from one sidewall to the other. In this fashion, the flow "sees" a wing without wing tips—a so-called infinite wing, which theoretically stretches to infinity along the span (in the y direction in Figure 4.6). Because the airfoil section is the same at any spanwise location along the infinite wing, the properties of the airfoil and the infinite wing are identical. Hence, airfoil data are frequently called infinite wing data. (In contrast, we see in Chapter 5 that the properties of a finite wing are somewhat different from its airfoil properties.)

The typical variation of lift coefficient with angle of attack for an airfoil is sketched in Figure 4.9. At low-to-moderate angles of attack, c_l varies *linearly* with α; the slope of this straight line is denoted by a_0 and is called the *lift slope*. In this region, the flow moves smoothly over the airfoil and is attached over most of the surface, as shown in the streamline picture at the left of Figure 4.9. However, as α becomes large, the flow tends to separate from the top surface of the airfoil, creating a large wake of relatively "dead air" behind the airfoil as shown at the right of Figure 4.9. Inside this separated region, the flow is recirculating, and part of the flow is actually moving in a direction opposite to the freestream—so-called reversed flow. (Refer also to Figure 1.42.) This separated flow is due to viscous effects and is discussed in Section 4.12 and Chapter 15. The consequence of this separated flow at high α is a precipitous decrease in lift and a large increase in drag; under such conditions the airfoil is said to be *stalled*. The maximum

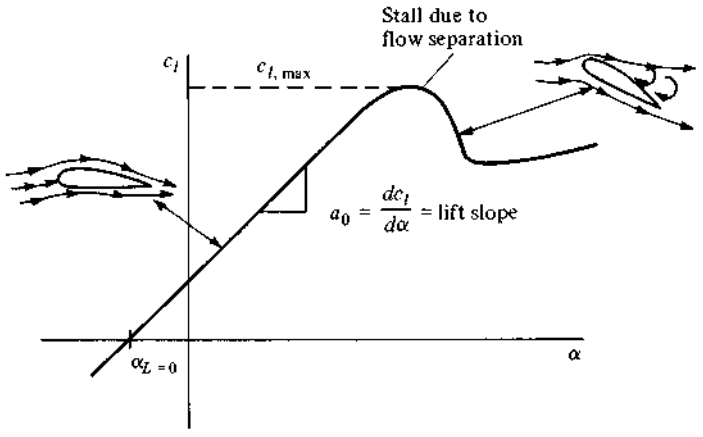


Figure 4.9 Schematic of lift-coefficient variation with angle of attack for an airfoil.

value of c_l, which occurs just prior to the stall, is denoted by $c_{l,\max}$; it is one of the most important aspects of airfoil performance, because it determines the stalling speed of an airplane. The higher is $c_{l,\max}$, the lower is the stalling speed. A great deal of modern airfoil research has been directed toward increasing $c_{l,\max}$. Again examining Figure 4.9, we see that c_l increases linearly with α until flow separation begins to have an effect. Then the curve becomes nonlinear, c_l reaches a maximum value, and finally the airfoil stalls. At the other extreme of the curve, noting Figure 4.9, the lift at $\alpha = 0$ is finite; indeed, the lift goes to zero only when the airfoil is pitched to some negative angle of attack. The value of α when lift equals zero is called the *zero-lift angle of attack* and is denoted by $\alpha_{L=0}$. For a symmetric airfoil, $\alpha_{L=0} = 0$, whereas for all airfoils with positive camber (camber *above* the chord line), $\alpha_{L=0}$ is a negative value, usually on the order of -2 or $-3°$.

The inviscid flow airfoil theory discussed in this chapter allows us to predict the lift slope a_0 and $\alpha_{L=0}$ for a given airfoil. It does not allow us to calculate $c_{l,\max}$, which is a difficult viscous flow problem, to be discussed in Chapters 15 to 20.

Experimental results for lift and moment coefficients for the NACA 2412 airfoil are given in Figure 4.10. Here, the moment coefficient is taken about the quarter-chord point. Recall from Section 1.6 that the force-and-moment system on an airfoil can be transferred to any convenient point; however, the quarter-chord point is commonly used. (Refresh your mind on this concept by reviewing Section 1.6, especially Figure 1.25.) Also shown in Figure 4.10 are theoretical results to be discussed later. Note that the experimental data are given for two different Reynolds numbers. The lift slope a_0 is not influenced by Re; however, $c_{l,\max}$ is dependent upon Re. This makes sense, because $c_{l,\max}$ is governed by viscous effects, and Re is a similarity parameter that governs the strength of inertia forces relative to viscous forces in the flow. [See Section 1.7 and Equation (1.35).]

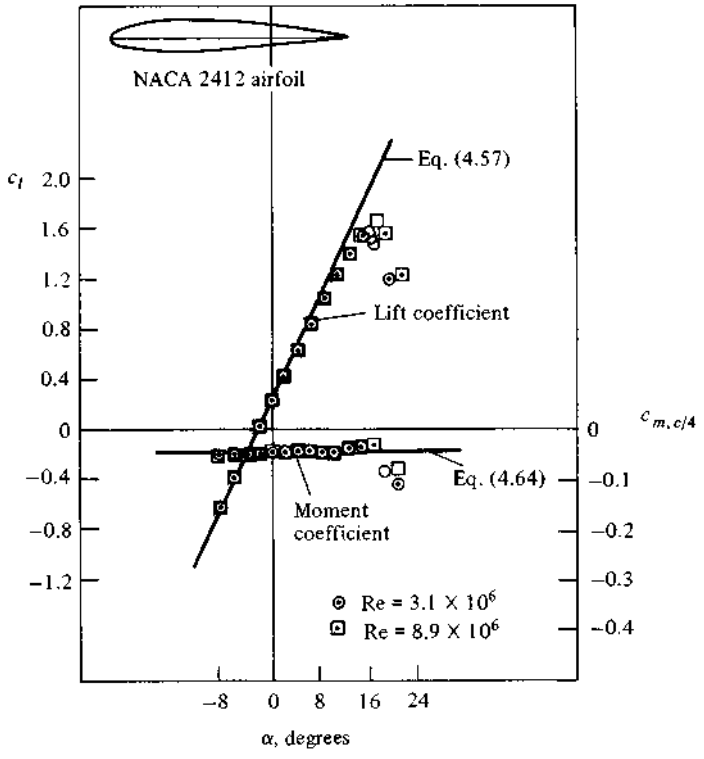


Figure 4.10 Experimental data for lift coefficient and moment coefficient about the quarter-chord point for an NACA 2412 airfoil. *(Source: Data obtained from Abbott and von Doenhoff, Reference 11.)* Also shown is a comparison with theory described in Section 4.8.

The moment coefficient is also insensitive to Re except at large α. The NACA 2412 airfoil is a commonly used airfoil, and the results given in Figure 4.10 are quite typical of airfoil characteristics. For example, note from Figure 4.10 that $\alpha_{L=0} = -2.1^\circ$, $c_{l,\max} \approx 1.6$, and the stall occurs at $\alpha \approx 16^\circ$.

This chapter deals primarily with airfoil theory for an inviscid, incompressible flow; such theory is incapable of predicting airfoil drag, as noted earlier. However, for the sake of completeness, experimental data for the drag coefficient c_d for the NACA 2412 airfoil are given in Figure 4.11 as a function of the angle of attack.[2] The physical source of this drag coefficient is both skin friction drag and pressure drag due to flow separation (so-called form drag). The sum of these two

[2] In many references, such as Reference 11, it is common to plot c_d versus c_l, rather than versus α. A plot of c_d versus c_l is called a *drag polar*. For the sake of consistency with Figure 4.10, we choose to plot c_d versus α here.

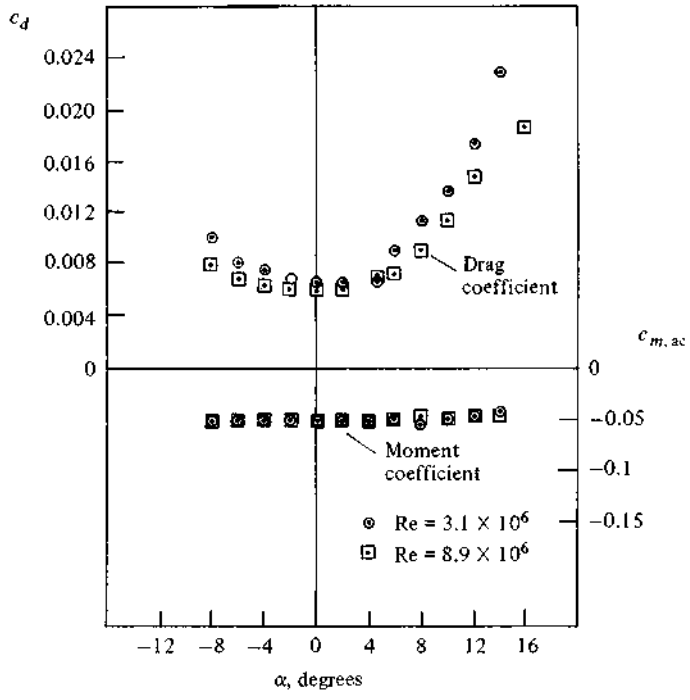


Figure 4.11 Experimental data for profile drag coefficient and moment coefficient about the aerodynamic center for the NACA 2412 airfoil. *(Source: Data obtained from Abbott and von Doenhoff, Reference 11.)*

effects yields the *profile* drag coefficient c_d for the airfoil, which is plotted in Figure 4.11. Note that c_d is sensitive to Re, which is to be expected since both skin friction and flow separation are viscous effects. Again, we must wait until Section 4.12 and Chapters 15 to 20 to obtain some tools for theoretically predicting c_d.

Also plotted in Figure 4.11 is the moment coefficient about the aerodynamic center $c_{m,\text{ac}}$. In general, moments on an airfoil are a function of α. However, there is one point on the airfoil about which the moment is independent of angle of attack; such a point is defined as the *aerodynamic center*. Clearly, the data in Figure 4.11 illustrate a constant value for $c_{m,\text{ac}}$ over a wide range of α.

For an elementary but extensive discussion of airfoil and wing properties, see Chapter 5 of Reference 2.

EXAMPLE 4.1

Consider an NACA 2412 airfoil with a chord of 0.64 m in an airstream at standard sea level conditions. The freestream velocity is 70 m/s. The lift per unit span is 1254 N/m. Calculate the angle of attack and the drag per unit span.

■ Solution

At standard sea level, $\rho = 1.23$ kg/m^3:

$$q_\infty = \tfrac{1}{2}\rho_\infty V_\infty^2 = \tfrac{1}{2}(1.23)(70)^2 = 3013.5 \text{ N/m}^2$$

$$c_l = \frac{L'}{q_\infty S} = \frac{L'}{q_\infty c(1)} = \frac{1254}{3013.5(0.64)} = 0.65$$

From Figure 4.10, for $c_l = 0.65$, we obtain $\boxed{\alpha = 4^\circ}$.

To obtain the drag per unit span, we must use the data in Figure 4.11. However, since $c_d = f(\text{Re})$, let us calculate Re. At standard sea level, $\mu = 1.789 \times 10^{-5}$ kg/(m · s). Hence,

$$\text{Re} = \frac{\rho_\infty V_\infty c}{\mu_\infty} = \frac{1.23(70)(0.64)}{1.789 \times 10^{-5}} = 3.08 \times 10^6$$

Therefore, using the data for Re $= 3.1 \times 10^6$ in Figure 4.11, we find $c_d = 0.0068$. Thus,

$$D' = q_\infty S c_d = q_\infty c(1) c_d = 3013.5(0.64)(0.0068) = \boxed{13.1 \text{ N/m}}$$

EXAMPLE 4.2

For the airfoil and flow conditions given in Example 4.1, calculate the moment per unit span about the aerodynamic center.

■ Solution

From Figure 4.11, $c_{m,\text{ac}}$, which is independent of angle of attack, is -0.05. The moment per unit span about the aerodynamic center is

$$M'_{\text{ac}} = q_\infty S c c_{m,\text{ac}}$$

$$= (3013.5)(0.64)(0.64)(-0.05) = \boxed{-61.7 \text{ Nm}}$$

Recall the sign convention for aerodynamic moments introduced in Section 1.5, namely, that a negative moment, as obtained here, is a pitch-down moment, tending to reduce the angle of attack.

EXAMPLE 4.3

For the NACA 2412 airfoil, calculate and compare the lift-to-drag ratios at angles of attack of 0, 4, 8, and 12 degrees. The Reynolds number is 3.1×10^6.

■ Solution

The lift-to-drag ratio, L/D, is given by

$$\frac{L}{D} = \frac{q_\infty S c_\ell}{q_\infty S c_d} = \frac{c_\ell}{c_d}$$

From Figures 4.10 and 4.11, we have

α	C_ℓ	C_d	C_ℓ/C_d
0	0.25	0.0065	38.5
4	0.65	0.0070	93
8	1.08	0.0112	96
12	1.44	0.017	85

Note that, as the angle of attack increases, the lift-to-drag ratio first increases, reaches a maximum, and then decreases. The maximum lift-to-drag ratio, $(L/D)_{\max}$, is an important parameter in airfoil performance; it is a direct measure of aerodynamic efficiency. The higher the value of $(L/D)_{\max}$, the more efficient is the airfoil. The values of L/D for airfoils are quite large numbers in comparison to that for a complete airplane. Due to the extra drag associated with all parts of the airplane, values of $(L/D)_{\max}$ for real airplanes are on the order of 10 to 20.

4.4 PHILOSOPHY OF THEORETICAL SOLUTIONS FOR LOW-SPEED FLOW OVER AIRFOILS: THE VORTEX SHEET

In Section 3.14, the concept of vortex flow was introduced; refer to Figure 3.31 for a schematic of the flow induced by a point vortex of strength Γ located at a given point O. (Recall that Figure 3.31, with its counterclockwise flow, corresponds to a negative value of Γ. By convention, a positive Γ induces a clockwise flow.) Let us now expand our concept of a point vortex. Referring to Figure 3.31, imagine a straight line perpendicular to the page, going through point O, and extending to infinity both out of and into the page. This line is a straight *vortex filament* of strength Γ. A straight vortex filament is drawn in perspective in Figure 4.12. (Here, we show a clockwise flow, which corresponds to a positive value of Γ.) The flow induced in any plane perpendicular to the straight vortex filament by the filament itself is identical to that induced by a point vortex of strength Γ; that is, in Figure 4.12, the flows in the planes perpendicular to the vortex filament at O and O' are identical to each other and are identical to the flow induced by a point vortex of strength Γ. Indeed, the point vortex described in Section 3.14 is simply a section of a straight vortex filament.

In Section 3.17, we introduced the concept of a source sheet, which is an infinite number of line sources side by side, with the strength of each line source being infinitesimally small. For vortex flow, consider an analogous situation. Imagine an infinite number of straight vortex filaments side by side, where the strength of each filament is infinitesimally small. These side-by-side vortex filaments form a *vortex sheet,* as shown in perspective in the upper left of Figure 4.13. If we look along the series of vortex filaments (looking along the y axis in Figure 4.13), the vortex sheet will appear as sketched at the lower right of Figure 4.13. Here, we are looking at an edge view of the sheet; the vortex filaments are all perpendicular

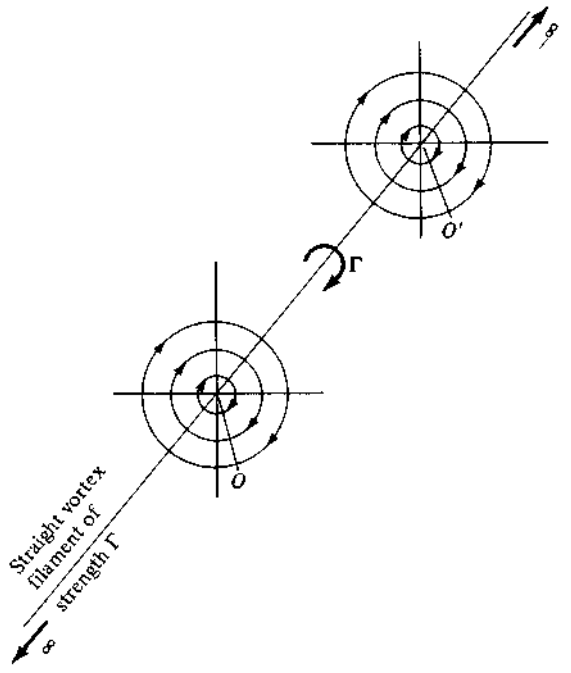


Figure 4.12 Vortex filament.

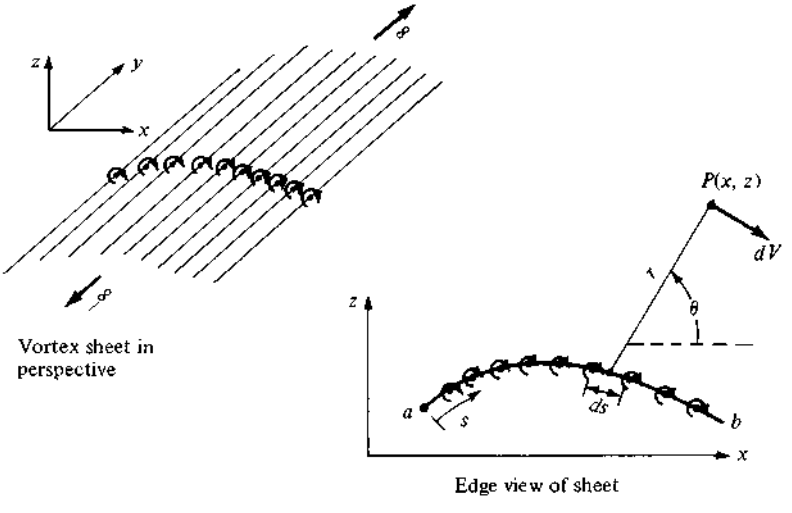


Figure 4.13 Vortex sheet.

to the page. Let s be the distance measured along the vortex sheet in the edge view. Define $\gamma = \gamma(s)$ as the strength of the vortex sheet, per unit length along s. Thus, the strength of an infinitesimal portion ds of the sheet is $\gamma\, ds$. This small section of the vortex sheet can be treated as a distinct vortex of strength $\gamma\, ds$. Now consider point P in the flow, located a distance r from ds; the Cartesian coordinates of P are (x, z). The small section of the vortex sheet of strength $\gamma\, ds$ induces an infinitesimally small velocity dV at point P. From Equation (3.105), dV is given by

$$dV = -\frac{\gamma\, ds}{2\pi r} \tag{4.1}$$

and is in a direction perpendicular to r, as shown in Figure 4.13. The velocity at P induced by the entire vortex sheet is the summation of Equation (4.1) from point a to point b. Note that dV, which is perpendicular to r, changes direction at point P as we sum from a to b; hence, the incremental velocities induced at P by different sections of the vortex sheet must be added vectorally. Because of this, it is sometimes more convenient to deal with the velocity potential. Again referring to Figure 4.13, the increment in velocity potential $d\phi$ induced at point P by the elemental vortex $\gamma\, ds$ is, from Equation (3.112),

$$d\phi = -\frac{\gamma\, ds}{2\pi}\theta \tag{4.2}$$

In turn, the velocity potential at P due to the entire vortex sheet from a to b is

$$\boxed{\phi(x, z) = -\frac{1}{2\pi}\int_a^b \theta\gamma\, ds} \tag{4.3}$$

Equation (4.1) is particularly useful for our discussion of classical thin airfoil theory, whereas Equation (4.3) is important for the numerical vortex panel method.

Recall from Section 3.14 that the circulation Γ around a point vortex is equal to the strength of the vortex. Similarly, the circulation around the vortex sheet in Figure 4.13 is the sum of the strengths of the elemental vortices; that is

$$\boxed{\Gamma = \int_a^b \gamma\, ds} \tag{4.4}$$

Recall that the source sheet introduced in Section 3.17 has a discontinuous change in the direction of the *normal* component of velocity across the sheet (from Figure 3.38, note that the normal component of velocity changes direction by 180° in crossing the sheet), whereas the tangential component of velocity is the same immediately above and below the source sheet. In contrast, for a vortex sheet, there is a discontinuous change in the tangential component of velocity across the sheet, whereas the normal component of velocity is preserved across the sheet. This change in tangential velocity across the vortex sheet is related to the strength of the sheet as follows. Consider a vortex sheet as sketched in

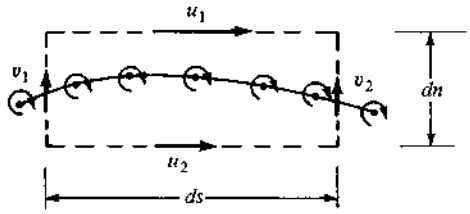


Figure 4.14 Tangential velocity jump across a vortex sheet.

Figure 4.14. Consider the rectangular dashed path enclosing a section of the sheet of length ds. The velocity components tangential to the top and bottom of this rectangular path are u_1 and u_2, respectively, and the velocity components tangential to the left and right sides are v_1 and v_2, respectively. The top and bottom of the path are separated by the distance dn. From the definition of circulation given by Equation (2.136), the circulation around the dashed path is

$$\Gamma = -(v_2\,dn - u_1\,ds - v_1\,dn + u_2\,ds)$$

or

$$\Gamma = (u_1 - u_2)\,ds + (v_1 - v_2)\,dn \tag{4.5}$$

However, since the strength of the vortex sheet contained inside the dashed path is $\gamma\,ds$, we also have

$$\Gamma = \gamma\,ds \tag{4.6}$$

Therefore, from Equations (4.5) and (4.6),

$$\gamma\,ds = (u_1 - u_2)\,ds + (v_1 - v_2)\,dn \tag{4.7}$$

Let the top and bottom of the dashed line approach the vortex sheet; that is, let $dn \to 0$. In the limit, u_1 and u_2 become the velocity components tangential to the vortex sheet immediately above and below the sheet, respectively, and Equation (4.7) becomes

$$\gamma\,ds = (u_1 - u_2)\,ds$$

or

$$\boxed{\gamma = u_1 - u_2} \tag{4.8}$$

Equation (4.8) is important; it states that *the local jump in tangential velocity across the vortex sheet is equal to the local sheet strength.*

We have now defined and discussed the properties of a vortex sheet. The concept of a vortex sheet is instrumental in the analysis of the low-speed characteristics of an airfoil. A philosophy of airfoil theory of inviscid, incompressible flow is as follows. Consider an airfoil of arbitrary shape and thickness in a freestream with velocity V_∞, as sketched in Figure 4.15. Replace the airfoil surface with a vortex sheet of variable strength $\gamma(s)$, as also shown in Figure 4.15. Calculate the variation of γ as a function of s such that the induced velocity field from the vortex sheet when added to the uniform velocity of magnitude V_∞ will make

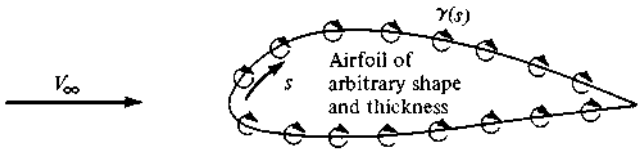


Figure 4.15 Simulation of an arbitrary airfoil by distributing a vortex sheet over the airfoil surface.

the vortex sheet (hence the airfoil surface) a *streamline of the flow*. In turn, the circulation around the airfoil will be given by

$$\Gamma = \int \gamma \, ds$$

where the integral is taken around the complete surface of the airfoil. Finally, the resulting lift is given by the Kutta-Joukowski theorem:

$$L' = \rho_\infty V_\infty \Gamma$$

This philosophy is not new. It was first espoused by Ludwig Prandtl and his colleagues at Göttingen, Germany, during the period 1912–1922. However, no general analytical solution for $\gamma = \gamma(s)$ exists for an airfoil of arbitrary shape and thickness. Rather, the strength of the vortex sheet must be found numerically, and the practical implementation of the above philosophy had to wait until the 1960s with the advent of large digital computers. Today, the above philosophy is the foundation of the modern vortex panel method, to be discussed in Section 4.9.

The concept of replacing the airfoil surface in Figure 4.15 with a vortex sheet is more than just a mathematical device; it also has physical significance. In real life, there is a thin boundary layer on the surface, due to the action of friction between the surface and the airflow (see Figures 1.41 and 1.46). This boundary layer is a highly viscous region in which the large velocity gradients produce substantial vorticity; that is, $\nabla \times \mathbf{V}$ is finite within the boundary layer. (Review Section 2.12 for a discussion of vorticity.) Hence, in real life, there is a distribution of vorticity along the airfoil surface due to viscous effects, and our philosophy of replacing the airfoil surface with a vortex sheet (such as in Figure 4.15) can be construed as a way of modeling this effect in an inviscid flow.[3]

Imagine that the airfoil in Figure 4.15 is made very thin. If you were to stand back and look at such a thin airfoil from a distance, the portions of the vortex sheet

[3] It is interesting to note that some recent research by NASA is hinting that even as complex a problem as flow separation, heretofore thought to be a completely viscous-dominated phenomenon, may in reality be an inviscid-dominated flow which requires only a rotational flow. For example, some inviscid flow-field numerical solutions for flow over a circular cylinder, when vorticity is introduced either by means of a nonuniform freestream or a curved shock wave, are accurately predicting the separated flow on the rearward side of the cylinder. However, as exciting as these results may be, they are too preliminary to be emphasized in this book. We continue to talk about flow separation in Chapters 15 to 20 as being a viscous-dominated effect, until definitely proved otherwise. This recent research is mentioned here only as another example of the physical connection between vorticity, vortex sheets, viscosity, and real life.

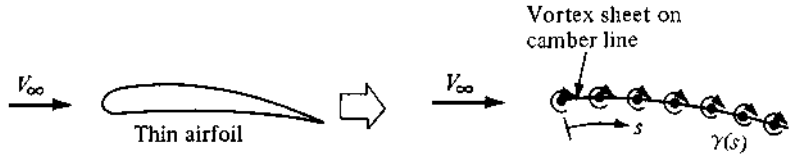


Figure 4.16 Thin airfoil approximation.

on the top and bottom surface of the airfoil would almost coincide. This gives rise to a method of approximating a thin airfoil by replacing it with a single vortex sheet distributed over the camber line of the airfoil, as sketched in Figure 4.16. The strength of this vortex sheet $\gamma(s)$ is calculated such that, in combination with the freestream, the camber line becomes a streamline of the flow. Although the approach shown in Figure 4.16 is approximate in comparison with the case shown in Figure 4.15, it has the advantage of yielding a closed-form analytical solution. This philosophy of thin airfoil theory was first developed by Max Munk, a colleague of Prandtl, in 1922 (see Reference 12). It is discussed in Sections 4.7 and 4.8.

4.5 THE KUTTA CONDITION

The lifting flow over a circular cylinder was discussed in Section 3.15, where we observed that an infinite number of potential flow solutions were possible, corresponding to the infinite choice of Γ. For example, Figure 3.33 illustrates three different flows over the cylinder, corresponding to three different values of Γ. The same situation applies to the potential flow over an airfoil; for a given airfoil at a given angle of attack, there are an infinite number of valid theoretical solutions, corresponding to an infinite choice of Γ. For example, Figure 4.17 illustrates two different flows over the same airfoil at the same angle of attack but with different values of Γ. At first, this may seem to pose a dilemma. We know from experience that a given airfoil at a given angle of attack produces a single value of lift (e.g., see Figure 4.10). So, although there is an infinite number of possible potential flow solutions, nature knows how to pick a particular solution. Clearly, the philosophy discussed in the previous section is not complete—we need an additional condition that *fixes* Γ for a given airfoil at a given α.

To attempt to find this condition, let us examine some experimental results for the development of the flow field around an airfoil which is set into motion from an initial state of rest. Figure 4.18 shows a series of classic photographs of the flow over an airfoil, taken from Prandtl and Tietjens (Reference 8). In Figure 4.18*a*, the flow has just started, and the flow pattern is just beginning to develop around the airfoil. In these early moments of development, the flow tries to curl around the sharp trailing edge from the bottom surface to the top surface, similar to the sketch shown at the left of Figure 4.17. However, more advanced considerations of inviscid, incompressible flow (see, e.g., Reference 9) show the theoretical result that the velocity becomes infinitely large at a sharp corner. Hence, the type

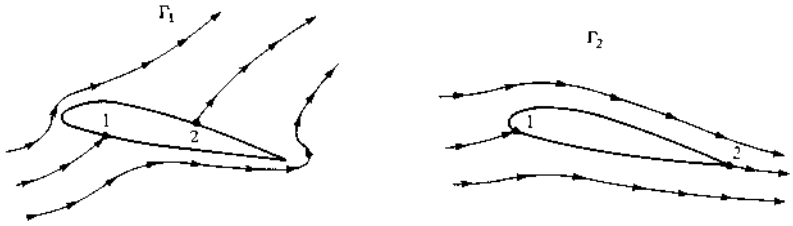
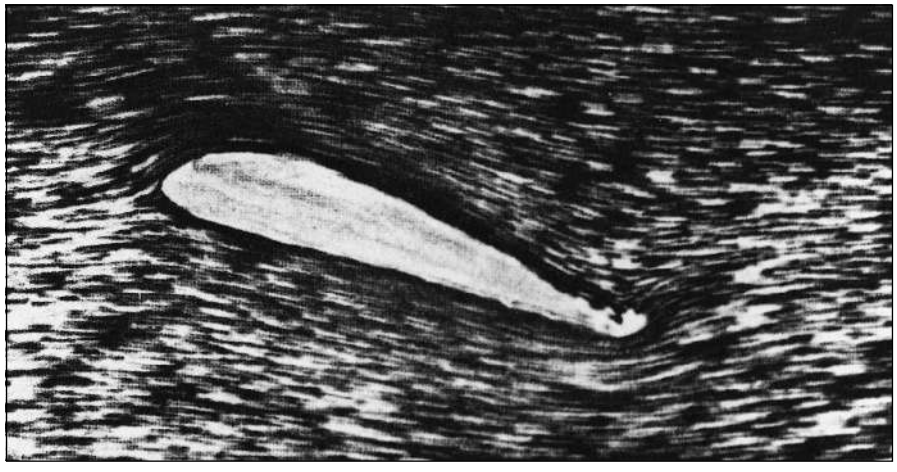
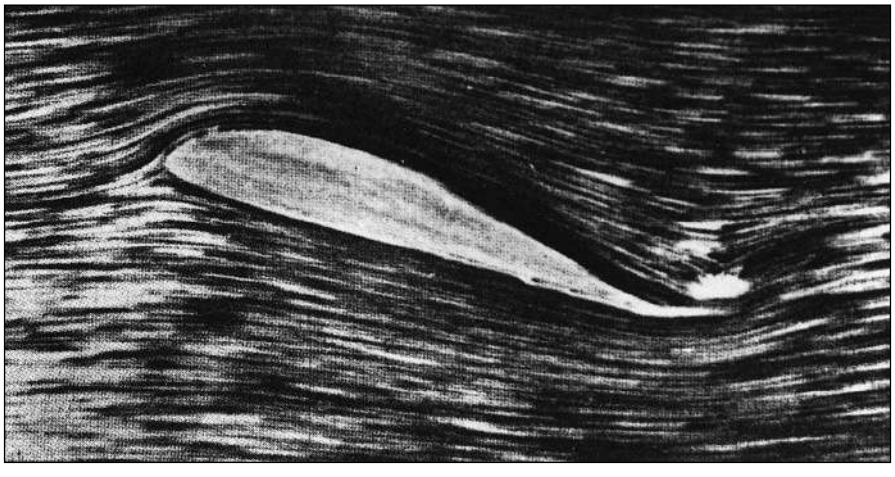


Figure 4.17 Effect of different values of circulation on the potential flow over a given airfoil at a given angle of attack. Points 1 and 2 are stagnation points.

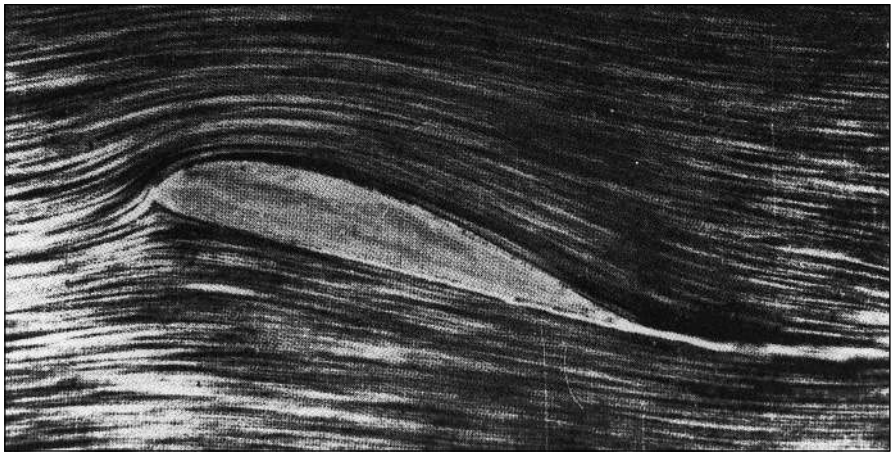


(*a*)



(*b*)

Figure 4.18 The development of steady flow over an airfoil; the airfoil is impulsively started from rest and attains a steady velocity through the fluid. (*a*) A moment just after starting. (*b*) An intermediate time. *(Source: Prandtl and Tiejens, Reference 8.)*



(*c*)

Figure 4.18 (continued) The development of steady flow over an airfoil; the airfoil is impulsively started from rest and attains a steady velocity through the fluid. (*c*) The final steady flow. *(Source: Prandtl and Tiejens, Reference 8.)*

of flow sketched at the left of Figure 4.17, and shown in Figure 4.18*a*, is not tolerated very long by nature. Rather, as the real flow develops over the airfoil, the stagnation point on the upper surface (point 2 in Figure 4.17) moves toward the trailing edge. Figure 4.18*b* shows this intermediate stage. Finally, after the initial transient process dies out, the steady flow shown in Figure 4.18*c* is reached. This photograph demonstrates that the flow is *smoothly leaving the top and the bottom surfaces of the airfoil at the trailing edge*. This flow pattern is sketched at the right of Figure 4.17 and represents the type of pattern to be expected for the steady flow over an airfoil.

Reflecting on Figures 4.17 and 4.18, we emphasize again that in establishing the steady flow over a given airfoil at a given angle of attack, nature adopts that particular value of circulation (Γ_2 in Figure 4.17) which results in the flow leaving smoothly at the trailing edge. This observation was first made and used in a theoretical analysis by the German mathematician M. Wilhelm Kutta in 1902. Therefore, it has become known as the *Kutta condition*.

In order to apply the Kutta condition in a theoretical analysis, we need to be more precise about the nature of the flow at the trailing edge. The trailing edge can have a finite angle, as shown in Figures 4.17 and 4.18 and as sketched at the left of Figure 4.19, or it can be cusped, as shown at the right of Figure 4.19. First, consider the trailing edge with a finite angle, as shown at the left of Figure 4.19. Denote the velocities along the top surface and the bottom surface as V_1 and V_2, respectively. V_1 is parallel to the top surface at point a, and V_2 is parallel to the bottom surface at point a. For the finite-angle trailing edge, if these velocities were finite at point a, then we would have two velocities in two different directions at

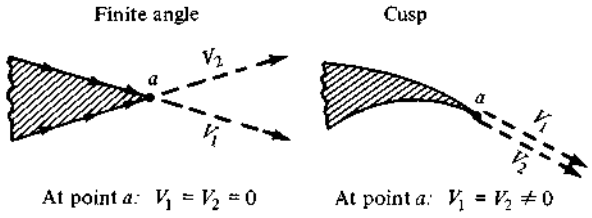


Figure 4.19 Different possible shapes of the trailing edge and their relation to the Kutta condition.

the same point, as shown at the left of Figure 4.19. However, this is not physically possible, and the only recourse is for both V_1 and V_2 to be zero at point a. That is, for the finite trailing edge, point a is a stagnation point, where $V_1 = V_2 = 0$. In contrast, for the cusped trailing edge shown at the right of Figure 4.19, V_1 and V_2 are in the same direction at point a, and hence both V_1 and V_2 can be finite. However, the pressure at point a, p_2, is a single, unique value, and Bernoulli's equation applied at both the top and bottom surfaces immediately adjacent to point a yields

$$p_a + \tfrac{1}{2}\rho V_1^2 = p_a + \tfrac{1}{2}\rho V_2^2$$

or

$$V_1 = V_2$$

Hence, for the cusped trailing edge, we see that the velocities leaving the top and bottom surfaces of the airfoil at the trailing edge are finite and equal in magnitude and direction.

We can summarize the statement of the Kutta condition as follows:

1. For a given airfoil at a given angle of attack, the value of Γ around the airfoil is such that the flow leaves the trailing edge smoothly.
2. If the trailing-edge angle is finite, then the trailing edge is a stagnation point.
3. If the trailing edge is cusped, then the velocities leaving the top and bottom surfaces at the trailing edge are finite and equal in magnitude and direction.

Consider again the philosophy of simulating the airfoil with vortex sheets placed either on the surface or on the camber line, as discussed in Section 4.4. The strength of such a vortex sheet is variable along the sheet and is denoted by $\gamma(s)$. The statement of the Kutta condition in terms of the vortex sheet is as follows. At the trailing edge (TE), from Equation (4.8), we have

$$\gamma(\text{TE}) = \gamma(a) = V_1 - V_2 \tag{4.9}$$

However, for the finite-angle trailing edge, $V_1 = V_2 = 0$; hence, from Equation (4.9), $\gamma(\text{TE}) = 0$. For the cusped trailing edge, $V_1 = V_2 \neq 0$; hence, from Equation (4.9), we again obtain the result that $\gamma(\text{TE}) = 0$. Therefore, the Kutta

condition expressed in terms of the strength of the vortex sheet is

$$\boxed{\gamma(\text{TE}) = 0} \tag{4.10}$$

4.5.1 Without Friction Could We Have Lift?

In Section 1.5 we emphasized that the resultant aerodynamic force on a body immersed in a flow is due to the net integrated effect of the pressure and shear stress distributions over the body surface. Moreover, in Section 4.1 we noted that lift on an airfoil is primarily due to the surface pressure distribution, and that shear stress has virtually no effect on lift. It is easy to see why. Look at the airfoil shapes in Figures 4.17 and 4.18, for example. Recall that pressure acts *normal* to the surface, and for these airfoils the direction of this normal pressure is essentially in the vertical direction, that is, the lift direction. In contrast the shear stress acts *tangential* to the surface, and for these airfoils the direction of this tangential shear stress is mainly in the horizontal direction, that is, the drag direction. Hence, pressure is the dominant player in the generation of lift, and shear stress has a negligible effect on lift. It is for this reason that the lift on an airfoil below the stall can be accurately predicted by *inviscid* theories such as that discussed in this chapter.

However, if we lived in a perfectly inviscid world, an airfoil could not produce lift. Indeed, the presence of friction is the very reason why we have lift. These sound like strange, even contradictory statements to our discussion in the preceding paragraph. What is going on here? The answer is that in real life, the *way* that nature insures that the flow will leave smoothly at the trailing edge, that is, the mechanism that nature uses to choose the flow shown in Figure 4.18*c*, is that the viscous boundary layer remains attached to the surface all the way to the trailing edge. *Nature enforces the Kutta condition by means of friction*. If there were no boundary layer (i.e., no friction), there would be no physical mechanism in the real world to achieve the Kutta condition.

So we are led to the most ironic situation that lift, which is created by the surface pressure distribution—an inviscid phenomenon, would not exist in a frictionless (inviscid) world. In this regard, we can say that without friction we could not have lift. However, we say this in the informed manner as discussed above.

4.6 KELVIN'S CIRCULATION THEOREM AND THE STARTING VORTEX

In this section, we put the finishing touch to the overall philosophy of airfoil theory before developing the quantitative aspects of the theory itself in subsequent sections. This section also ties up a loose end introduced by the Kutta condition described in the previous section. Specifically, the Kutta condition states that the circulation around an airfoil is just the right value to ensure that the flow smoothly leaves the trailing edge. *Question:* How does nature generate this circulation?

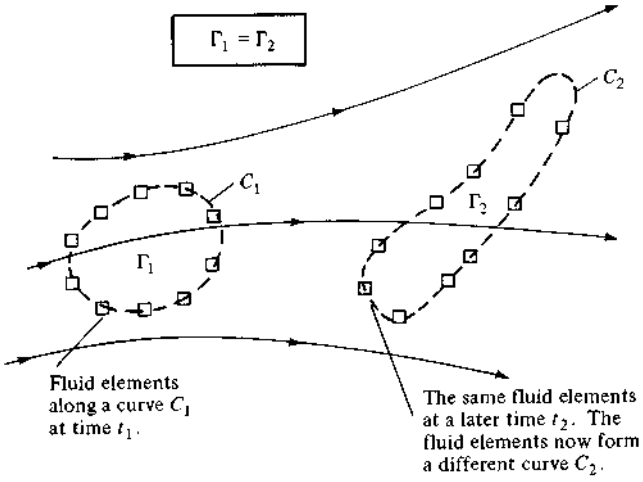


Figure 4.20 Kelvin's theorem.

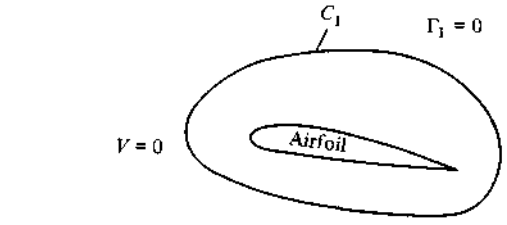
Does it come from nowhere, or is circulation somehow conserved over the whole flow field? Let us examine these matters more closely.

Consider an arbitrary inviscid, incompressible flow as sketched in Figure 4.20. Assume that all body forces **f** are zero. Choose an arbitrary curve C_1 and identify the fluid elements that are on this curve at a given instant in time t_1. Also, by definition the circulation around curve C_1 is $\Gamma_1 = -\int_{C_1} \mathbf{V} \cdot \mathbf{ds}$. Now let these specific fluid elements move downstream. At some later time, t_2, these *same* fluid elements will form another curve C_2, around which the circulation is $\Gamma_2 = -\int_{C_2} \mathbf{V} \cdot \mathbf{ds}$. For the conditions stated above, we can readily show that $\Gamma_1 = \Gamma_2$. In fact, since we are following a set of specific fluid elements, we can state that circulation around a closed curve formed by a set of contiguous fluid elements remains constant as the fluid elements move throughout the flow. Recall from Section 2.9 that the substantial derivative gives the time rate of change following a given fluid element. Hence, a mathematical statement of the above discussion is simply

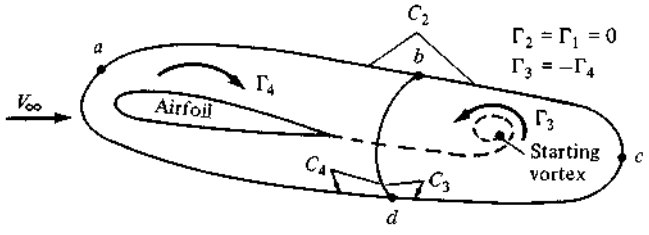
$$\boxed{\frac{D\Gamma}{Dt} = 0} \tag{4.11}$$

which says that the time rate of change of circulation around a closed curve consisting of the same fluid elements is zero. Equation (4.11) along with its supporting discussion is called *Kelvin's circulation theorem*.[4] Its derivation from

[4] Kelvin's theorem also holds for an inviscid compressible flow in the special case where $\rho = \rho(p)$; that is, the density is some single-valued function of pressure. Such is the case for isentropic flow, to be treated in later chapters.



(*a*) Fluid at rest relative to the airfoil



(*b*) Picture some moments after the start of the flow

Figure 4.21 The creation of the starting vortex and the resulting generation of circulation around the airfoil.

first principles is left as Problem 4.3. Also, recall our definition and discussion of a vortex sheet in Section 4.4. An interesting consequence of Kelvin's circulation theorem is proof that a stream surface which is a vortex sheet at some instant in time remains a vortex sheet for all times.

Kelvin's theorem helps to explain the generation of circulation around an airfoil, as follows. Consider an airfoil in a fluid at rest, as shown in Figure 4.21*a*. Because $\mathbf{V} = 0$ everywhere, the circulation around curve C_1 is zero. Now start the flow in motion over the airfoil. Initially, the flow will tend to curl around the trailing edge, as explained in Section 4.5 and illustrated at the left of Figure 4.17. In so doing, the velocity at the trailing edge theoretically becomes infinite. In real life, the velocity tends toward a very large finite number. Consequently, during the very first moments after the flow is started, a thin region of very large velocity gradients (and therefore high vorticity) is formed at the trailing edge. This high-vorticity region is fixed to the same fluid elements, and consequently it is flushed downstream as the fluid elements begin to move downstream from the trailing edge. As it moves downstream, this thin sheet of intense vorticity is unstable, and it tends to roll up and form a picture similar to a point vortex. This vortex is called the *starting vortex* and is sketched in Figure 4.21*b*. After the flow around the airfoil has come to a steady state where the flow leaves the trailing edge smoothly (the Kutta condition), the high velocity gradients at the trailing edge disappear and vorticity

is no longer produced at that point. However, the starting vortex has already been formed during the starting process, and it moves steadily downstream with the flow forever after. Figure 4.21*b* shows the flow field sometime after steady flow has been achieved over the airfoil, with the starting vortex somewhere downstream. The fluid elements that initially made up curve C_1 in Figure 4.21*a* have moved downstream and now make up curve C_2, which is the complete circuit *abcda* shown in Figure 4.21*b*. Thus, from Kelvin's theorem, the circulation Γ_2 around curve C_2 (which encloses both the airfoil and the starting vortex) is the same as that around curve C_1, namely, zero. $\Gamma_2 = \Gamma_1 = 0$. Now let us subdivide C_2 into two loops by making the cut *bd*, thus forming curves C_3 (circuit *bcdb*) and C_4 (circuit *abda*). Curve C_3 encloses the starting vortex, and curve C_4 encloses the airfoil. The circulation Γ_3 around curve C_3 is due to the starting vortex; by inspecting Figure 4.21*b*, we see that Γ_3 is in the counterclockwise direction (i.e., a negative value). The circulation around curve C_4 enclosing the airfoil is Γ_4. Since the cut *bd* is common to both C_3 and C_4, the sum of the circulations around C_3 and C_4 is simply equal to the circulation around C_2:

$$\Gamma_3 + \Gamma_4 = \Gamma_2$$

However, we have already established that $\Gamma_2 = 0$. Hence,

$$\Gamma_4 = -\Gamma_3$$

that is, the circulation around the airfoil is equal and opposite to the circulation around the starting vortex.

This brings us to the summary as well as the crux of this section. As the flow over an airfoil is started, the large velocity gradients at the sharp trailing edge result in the formation of a region of intense vorticity which rolls up downstream of the trailing edge, forming the starting vortex. This starting vortex has associated with it a counterclockwise circulation. Therefore, as an equal-and-opposite reaction, a clockwise circulation around the airfoil is generated. As the starting process continues, vorticity from the trailing edge is constantly fed into the starting vortex, making it stronger with a consequent larger counterclockwise circulation. In turn, the clockwise circulation around the airfoil becomes stronger, making the flow at the trailing edge more closely approach the Kutta condition, thus weakening the vorticity shed from the trailing edge. Finally, the starting vortex builds up to just the right strength such that the equal-and-opposite clockwise circulation around the airfoil leads to smooth flow from the trailing edge (the Kutta condition is exactly satisfied). When this happens, the vorticity shed from the trailing edge becomes zero, the starting vortex no longer grows in strength, and a steady circulation exists around the airfoil.

EXAMPLE 4.4

For the NACA 2412 airfoil at the conditions given in Example 4.1, calculate the strength of the steady-state starting vortex.

■ Solution

From the given conditions in Example 4.1,

$$L' = 1254 \text{ N/m}$$

$$V_\infty = 70 \text{ m/s}$$

$$\rho_\infty = 1.23 \text{ kg/m}^3$$

From the Kutta-Joukowski theorem, Equation (3.140),

$$L' = \rho_\infty V_\infty \Gamma$$

the circulation associated with the flow over the airfoil is

$$\Gamma = \frac{L'}{\rho_\infty V_\infty} = \frac{1254}{(1.23)(70)} = 14.56 \ \frac{\text{m}^2}{\text{s}}$$

Referring to Figure 4.21, the steady-state starting vortex has strength equal and opposite to the circulation around the airfoil. Hence,

$$\boxed{\text{Strength of starting vortex} = -14.56 \ \frac{\text{m}^2}{\text{s}}}$$

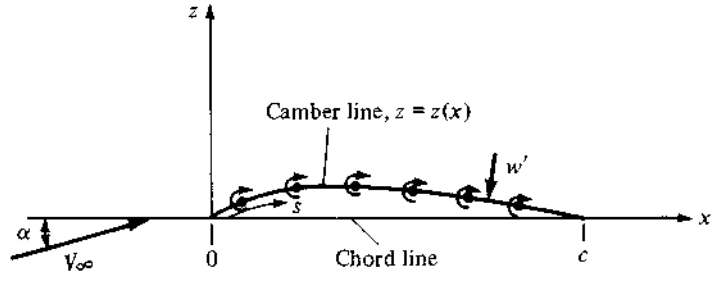
Note: For practical calculations in aerodynamics, an actual number for circulation is rarely needed. Circulation is a mathematical quantity defined by Equation (2.136), and it is an essential theoretical concept within the framework of the circulation theory of lift. For example, in Section 4.7, an analytical expression for circulation is derived as Equation (4.30), and then immediately inserted into the Kutta-Joukowski theorem, Equation (4.31), yielding a formula for the lift coefficient, Equation (4.33). Nowhere do we need to calculate an actual number for Γ. In the present example, however, the *strength* of the starting vortex is indeed given by its circulation, and hence to compare the strengths of various starting vortices, calculating numbers for Γ is relevant. Even this can be considered only an academic exercise. In this author's experience, no practical aerodynamic calculation requires the strength of a starting vortex. The starting vortex is simply a theoretical construct that is consistent with the generation of circulation around a lifting two-dimensional body.

4.7 CLASSICAL THIN AIRFOIL THEORY: THE SYMMETRIC AIRFOIL

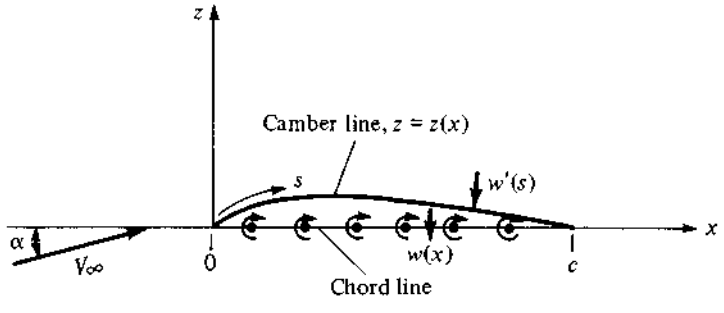
Some experimentally observed characteristics of airfoils and a philosophy for the theoretical prediction of these characteristics have been discussed in the preceding sections. Referring to our chapter road map in Figure 4.7, we have now completed the central branch. In this section, we move to the right-hand branch of Figure 4.7, namely, a quantitative development of thin airfoil theory. The basic equations necessary for the calculation of airfoil lift and moments are established in this section, with an application to symmetric airfoils. The case of cambered airfoils will be treated in Section 4.8.

For the time being, we deal with *thin* airfoils; for such a case, the airfoil can be simulated by a vortex sheet placed along the camber line, as discussed in Section 4.4. Our purpose is to calculate the variation of $\gamma(s)$ such that the camber line becomes a streamline of the flow and such that the Kutta condition is satisfied at the trailing edge; that is, $\gamma(\text{TE}) = 0$ [see Equation (4.10)]. Once we have found the particular $\gamma(s)$ that satisfies these conditions, then the total circulation Γ around the airfoil is found by integrating $\gamma(s)$ from the leading edge to the trailing edge. In turn, the lift is calculated from Γ via the Kutta-Joukowski theorem.

Consider a vortex sheet placed on the camber line of an airfoil, as sketched in Figure 4.22*a*. The freestream velocity is V_∞, and the airfoil is at the angle of attack α. The x axis is oriented along the chord line, and the z axis is perpendicular to the chord. The distance measured along the camber line is denoted by s. The shape of the camber line is given by $z = z(x)$. The chord length is c. In Figure 4.22*a*, w' is the component of velocity normal to the camber line induced by the vortex sheet; $w' = w'(s)$. For a thin airfoil, we rationalized in Section 4.4 that the distribution of a vortex sheet over the surface of the airfoil, when viewed from a distance, looks almost the same as a vortex sheet placed on the camber line. Let us stand back once again and view Figure 4.22*a* from a distance. If the airfoil



(*a*) Vortex sheet on the camber line



(*b*) Vortex sheet on the chord line

Figure 4.22 Placement of the vortex sheet for thin airfoil analysis.

is thin, the camber line is close to the chord line, and viewed from a distance, the vortex sheet appears to fall approximately on the chord line. Therefore, once again, let us reorient our thinking and place the vortex sheet on the chord line, as sketched in Figure 4.22*b*. Here, $\gamma = \gamma(x)$. We still wish the camber line to be a streamline of the flow, and $\gamma = \gamma(x)$ is calculated to satisfy this condition as well as the Kutta condition $\gamma(c) = 0$. That is, the strength of the vortex sheet on the chord line is determined such that the camber line (not the chord line) is a streamline.

For the camber line to be a streamline, the component of velocity normal to the camber line must be zero at all points along the camber line. The velocity at any point in the flow is the sum of the uniform freestream velocity and the velocity induced by the vortex sheet. Let $V_{\infty,n}$ be the component of the freestream velocity normal to the camber line. Thus, for the camber line to be a streamline,

$$V_{\infty,n} + w'(s) = 0 \tag{4.12}$$

at every point along the camber line.

An expression for $V_{\infty,n}$ in Equation (4.12) is obtained by the inspection of Figure 4.23. At any point P on the camber line, where the slope of the camber line is dz/dx, the geometry of Figure 4.23 yields

$$V_{\infty,n} = V_\infty \sin\left[\alpha + \tan^{-1}\left(-\frac{dz}{dx}\right)\right] \tag{4.13}$$

For a thin airfoil at small angle of attack, both α and $\tan^{-1}(-dz/dx)$ are small values. Using the approximation that $\sin\theta \approx \tan\theta \approx \theta$ for small θ, where θ is in

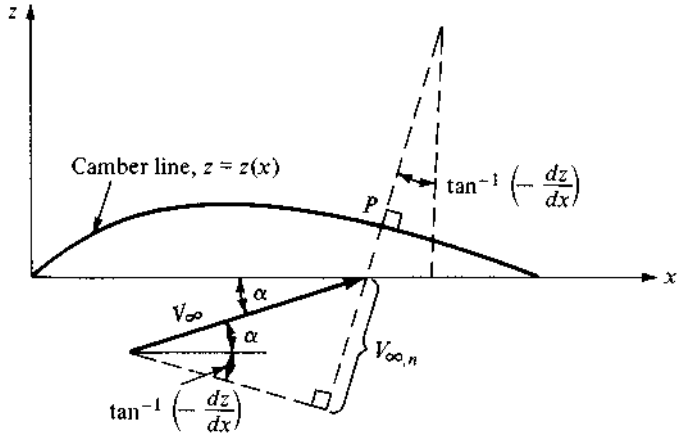


Figure 4.23 Determination of the component of freestream velocity normal to the camber line.

radians, Equation (4.13) reduces to

$$V_{\infty,n} = V_\infty \left(\alpha - \frac{dz}{dx} \right) \tag{4.14}$$

Equation (4.14) gives the expression for $V_{\infty,n}$ to be used in Equation (4.12). Keep in mind that, in Equation (4.14), α is in radians.

Returning to Equation (4.12), let us develop an expression for $w'(s)$ in terms of the strength of the vortex sheet. Refer again to Figure 4.22*b*. Here, the vortex sheet is along the chord line, and $w'(s)$ is the component of velocity normal to the camber line induced by the vortex sheet. Let $w(x)$ denote the component of velocity normal to the *chord line* induced by the vortex sheet, as also shown in Figure 4.22*b*. If the airfoil is thin, the camber line is close to the chord line, and it is consistent with thin airfoil theory to make the approximation that

$$w'(s) \approx w(x) \tag{4.15}$$

An expression for $w(x)$ in terms of the strength of the vortex sheet is easily obtained from Equation (4.1), as follows. Consider Figure 4.24, which shows the vortex sheet along the chord line. We wish to calculate the value of $w(x)$ at the location x. Consider an elemental vortex of strength $\gamma\, d\xi$ located at a distance ξ from the origin along the chord line, as shown in Figure 4.24. The strength of the vortex sheet γ varies with the distance along the chord; that is, $\gamma = \gamma(\xi)$. The velocity dw at point x induced by the elemental vortex at point ξ is given by Equation (4.1) as

$$dw = -\frac{\gamma(\xi)\, d\xi}{2\pi(x - \xi)} \tag{4.16}$$

In turn, the velocity $w(x)$ induced at point x by *all* the elemental vortices along the chord line is obtained by integrating Equation (4.16) from the leading edge ($\xi = 0$) to the trailing edge ($\xi = c$):

$$w(x) = -\int_0^c \frac{\gamma(\xi)\, d\xi}{2\pi(x - \xi)} \tag{4.17}$$

Combined with the approximation stated by Equation (4.15), Equation (4.17) gives the expression for $w'(s)$ to be used in Equation (4.12).

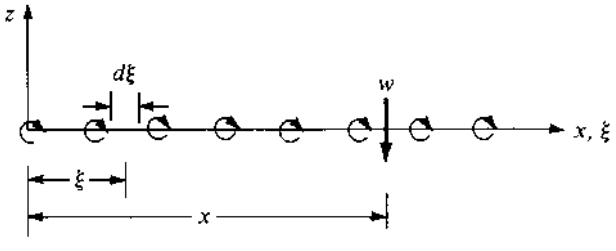


Figure 4.24 Calculation of the induced velocity at the chord line.

Recall that Equation (4.12) is the boundary condition necessary for the camber line to be a streamline. Substituting Equations (4.14), (4.15), and (4.17) into (4.12), we obtain

$$V_\infty \left(\alpha - \frac{dz}{dx} \right) - \int_0^c \frac{\gamma(\xi)\, d\xi}{2\pi(x - \xi)} = 0$$

or

$$\boxed{\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi)\, d\xi}{x - \xi} = V_\infty \left(\alpha - \frac{dz}{dx} \right)} \tag{4.18}$$

the *fundamental equation of thin airfoil theory;* it is simply a statement that the camber line is a streamline of the flow.

Note that Equation (4.18) is written at a given point x on the chord line, and that dz/dx is evaluated at that point x. The variable ξ is simply a dummy variable of integration which varies from 0 to c along the chord line, as shown in Figure 4.24. The vortex strength $\gamma = \gamma(\xi)$ is a variable along the chord line. For a given airfoil at a given angle of attack, both α and dz/dx are known values in Equation (4.18). Indeed, the only unknown in Equation (4.18) is the vortex strength $\gamma(\xi)$. Hence, Equation (4.18) is an integral equation, the solution of which yields the variation of $\gamma(\xi)$ such that the camber line is a streamline of the flow. The central problem of thin airfoil theory is to solve Equation (4.18) for $\gamma(\xi)$, subject to the Kutta condition, namely, $\gamma(c) = 0$.

In this section, we treat the case of a symmetric airfoil. As stated in Section 4.2, a symmetric airfoil has no camber; the camber line is coincident with the chord line. Hence, for this case, $dz/dx = 0$, and Equation (4.18) becomes

$$\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi)\, d\xi}{x - \xi} = V_\infty \alpha \tag{4.19}$$

In essence, within the framework of thin airfoil theory, a symmetric airfoil is treated the same as a flat plate; note that our theoretical development does not account for the airfoil thickness distribution. Equation (4.19) is an *exact* expression for the inviscid, incompressible flow over a flat plate at a small angle of attack.

To help deal with the integral in Equations (4.18) and (4.19), let us transform ξ into θ via the following transformation:

$$\xi = \frac{c}{2}(1 - \cos\theta) \tag{4.20}$$

Since x is a fixed point in Equations (4.18) and (4.19), it corresponds to a particular value of θ, namely, θ_0, such that

$$x = \frac{c}{2}(1 - \cos\theta_0) \tag{4.21}$$

Also, from Equation (4.20),

$$d\xi = \frac{c}{2} \sin\theta\, d\theta \tag{4.22}$$

Substituting Equations (4.20) to (4.22) into (4.19), and noting that the limits of integration become $\theta = 0$ at the leading edge (where $\xi = 0$) and $\theta = \pi$ at the trailing edge (where $\xi = c$), we obtain

$$\frac{1}{2\pi}\int_0^\pi \frac{\gamma(\theta)\sin\theta\, d\theta}{\cos\theta - \cos\theta_0} = V_\infty \alpha \tag{4.23}$$

A rigorous solution of Equation (4.23) for $\gamma(\theta)$ can be obtained from the mathematical theory of integral equations, which is beyond the scope of this book. Instead, we simply state that the solution is

$$\boxed{\gamma(\theta) = 2\alpha V_\infty \frac{(1+\cos\theta)}{\sin\theta}} \tag{4.24}$$

We can verify this solution by substituting Equation (4.24) into (4.23) yielding

$$\frac{1}{2\pi}\int_0^\pi \frac{\gamma(\theta)\sin\theta\, d\theta}{\cos\theta - \cos\theta_0} = \frac{V_\infty \alpha}{\pi}\int_0^\pi \frac{(1+\cos\theta)\, d\theta}{\cos\theta - \cos\theta_0} \tag{4.25}$$

The following standard integral appears frequently in airfoil theory and is derived in Appendix E of Reference 9:

$$\int_0^\pi \frac{\cos n\theta\, d\theta}{\cos\theta - \cos\theta_0} = \frac{\pi \sin n\theta_0}{\sin\theta_0} \tag{4.26}$$

Using Equation (4.26) in the right-hand side of Equation (4.25), we find that

$$\frac{V_\infty \alpha}{\pi}\int_0^\pi \frac{(1+\cos\theta)\, d\theta}{\cos\theta - \cos\theta_0} = \frac{V_\infty \alpha}{\pi}\left(\int_0^\pi \frac{d\theta}{\cos\theta - \cos\theta_0} + \int_0^\pi \frac{\cos\theta\, d\theta}{\cos\theta - \cos\theta_0}\right)$$

$$= \frac{V_\infty \alpha}{\pi}(0 + \pi) = V_\infty \alpha \tag{4.27}$$

Substituting Equation (4.27) into (4.25), we have

$$\frac{1}{2\pi}\int_0^\pi \frac{\gamma(\theta)\sin\theta\, d\theta}{\cos\theta - \cos\theta_0} = V_\infty \alpha$$

which is identical to Equation (4.23). Hence, we have shown that Equation (4.24) is indeed the solution to Equation (4.23). Also, note that at the trailing edge, where $\theta = \pi$, Equation (4.24) yields

$$\gamma(\pi) = 2\alpha V_\infty \frac{0}{0}$$

which is an indeterminant form. However, using L'Hospital's rule on Equation (4.24),

$$\gamma(\pi) = 2\alpha V_\infty \frac{-\sin\pi}{\cos\pi} = 0$$

Thus, Equation (4.24) also satisfies the Kutta condition.

We are now in a position to calculate the lift coefficient for a thin, symmetric airfoil. The total circulation around the airfoil is

$$\Gamma = \int_0^c \gamma(\xi)\, d\xi \tag{4.28}$$

Using Equations (4.20) and (4.22), Equation (4.28) transforms to

$$\Gamma = \frac{c}{2} \int_0^\pi \gamma(\theta) \sin\theta\, d\theta \tag{4.29}$$

Substituting Equation (4.24) into (4.29), we obtain

$$\Gamma = \alpha c V_\infty \int_0^\pi (1 + \cos\theta)\, d\theta = \pi \alpha c V_\infty \tag{4.30}$$

Substituting Equation (4.30) into the Kutta-Joukowski theorem, we find that the lift per unit span is

$$L' = \rho_\infty V_\infty \Gamma = \pi \alpha c \rho_\infty V_\infty^2 \tag{4.31}$$

The lift coefficient is

$$c_l = \frac{L'}{q_\infty S} \tag{4.32}$$

where $S = c(1)$

Substituting Equation (4.31) into (4.32), we have

$$c_l = \frac{\pi \alpha c \rho_\infty V_\infty^2}{\frac{1}{2} \rho_\infty V_\infty^2 c(1)}$$

or

$$\boxed{c_l = 2\pi\alpha} \tag{4.33}$$

and

$$\boxed{\text{Lift slope} = \frac{dc_l}{d\alpha} = 2\pi} \tag{4.34}$$

Equations (4.33) and (4.34) are important results; they state the theoretical result that the lift coefficient is *linearly proportional to angle of attack,* which is supported by the experimental results discussed in Section 4.3. They also state that the theoretical lift slope is equal to 2π rad^{-1}, which is 0.11 degree^{-1}. The experimental lift coefficient data for an NACA 0012 symmetric airfoil are given in Figure 4.25; note that Equation (4.33) accurately predicts c_l over a large range of angle of attack. (The NACA 0012 airfoil section is commonly used on airplane tails and helicopter blades.)

The moment about the leading edge can be calculated as follows. Consider the elemental vortex of strength $\gamma(\xi)\, d\xi$ located a distance ξ from the leading edge, as sketched in Figure 4.26. The circulation associated with this elemental vortex is $d\Gamma = \gamma(\xi)\, d\xi$. In turn, the increment of lift dL contributed by the elemental vortex is $dL = \rho_\infty V_\infty\, d\Gamma$. This increment of lift creates a moment

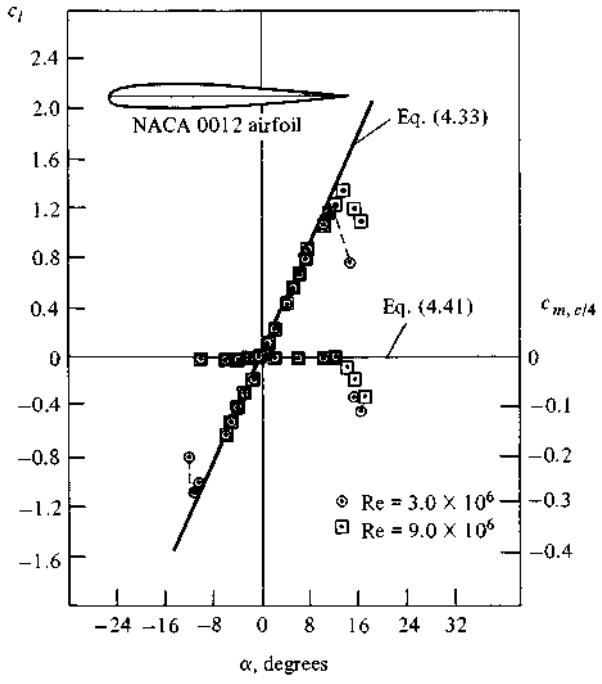


Figure 4.25 Comparison between theory and experiment for the lift and moment coefficients for an NACA 0012 airfoil. *(Source: Abbott and von Doenhoff, Reference 11.)*

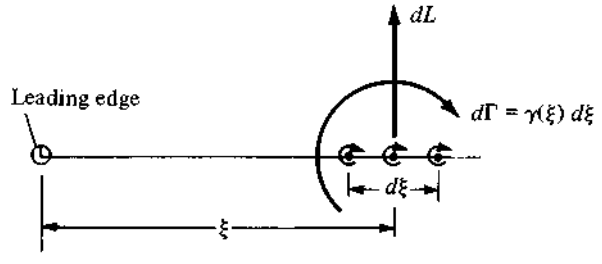


Figure 4.26 Calculation of moments about the leading edge.

about the leading edge $dM = -\xi(dL)$. The total moment about the leading edge (LE) (per unit span) due to the entire vortex sheet is therefore

$$M'_{\text{LE}} = -\int_0^c \xi(dL) = -\rho_\infty V_\infty \int_0^c \xi\gamma(\xi)\, d\xi \tag{4.35}$$

Transforming Equation (4.35) via Equations (4.20) and (4.22), and performing the integration, we obtain (the details are left for Problem 4.4):

$$M'_{\text{LE}} = -q_\infty c^2 \frac{\pi\alpha}{2} \tag{4.36}$$

The moment coefficient is

$$c_{m,\text{le}} = \frac{M'_{\text{LE}}}{q_\infty S c}$$

where $S = c(1)$. Hence,

$$c_{m,\text{le}} = \frac{M'_{\text{LE}}}{q_\infty c^2} = -\frac{\pi\alpha}{2} \tag{4.37}$$

However, from Equation (4.33),

$$\pi\alpha = \frac{c_l}{2} \tag{4.38}$$

Combining Equations (4.37) and (4.38), we obtain

$$\boxed{c_{m,\text{le}} = -\frac{c_l}{4}} \tag{4.39}$$

From Equation (1.22), the moment coefficient about the quarter-chord point is

$$c_{m,c/4} = c_{m,\text{le}} + \frac{c_l}{4} \tag{4.40}$$

Combining Equations (4.39) and (4.40), we have

$$\boxed{c_{m,c/4} = 0} \tag{4.41}$$

In Section 1.6, a definition is given for the center of pressure as that point about which the moments are zero. Clearly, Equation (4.41) demonstrates the theoretical result that the *center of pressure is at the quarter-chord point for a symmetric airfoil*.

By the definition given in Section 4.3, that point on an airfoil where moments are independent of angle of attack is called the aerodynamic center. From Equation (4.41), the moment about the quarter chord is zero for all values of α. Hence, for a symmetric airfoil, we have the theoretical result that the *quarter-chord point is both the center of pressure and the aerodynamic center*.

The theoretical result for $c_{m,c/4} = 0$ in Equation (4.41) is supported by the experimental data given in Figure 4.25. Also, note that the experimental value of $c_{m,c/4}$ is constant over a wide range of α, thus demonstrating that the real aerodynamic center is essentially at the quarter chord.

Let us summarize the above results. The essence of thin airfoil theory is to find a distribution of vortex sheet strength along the chord line that will make the camber line a streamline of the flow while satisfying the Kutta condition $\gamma(\text{TE}) = 0$. Such a vortex distribution is obtained by solving Equation (4.18) for $\gamma(\xi)$, or in terms of the transformed independent variable θ, solving Equation (4.23) for $\gamma(\theta)$ [recall that Equation (4.23) is written for a symmetric airfoil]. The resulting vortex distribution $\gamma(\theta)$ for a symmetric airfoil is given by Equation (4.24). In turn, this vortex distribution, when inserted into the Kutta-Joukowski theorem,

gives the following important theoretical results for a symmetric airfoil:

1. $c_l = 2\pi\alpha$.
2. Lift slope $= 2\pi$.
3. The center of pressure and the aerodynamic center are both located at the quarter-chord point.

EXAMPLE 4.5

Consider a thin flat plate at 5 deg. angle of attack. Calculate the: (*a*) lift coefficient, (*b*) moment coefficient about the leading edge, (*c*) moment coefficient about the quarter-chord point, and (*d*) moment coefficient about the trailing edge.

■ Solution

Recall that the results obtained in Section 4.7, although couched in terms of a thin symmetric airfoil, apply in particular to a flat plate with zero thickness.

(*a*) From Equation (4.33),

$$c_\ell = 2\pi\alpha$$

where α is in radians

$$\alpha = \frac{5}{57.3} = 0.0873 \text{ rad}$$

$$c_\ell = 2\pi(0.0873) = \boxed{0.5485}$$

(*b*) From Equation (4.39)

$$c_{m,\ell e} = -\frac{c_\ell}{4} = -\frac{0.5485}{4} = \boxed{-0.137}$$

(*c*) From Equation (4.41)

$$c_{m,c/4} = \boxed{0}$$

(*d*) Figure 4.27 is a sketch of the force and moment system on the plate. We place the lift at the quarter-chord point, along with the moment about the quarter-chord point.

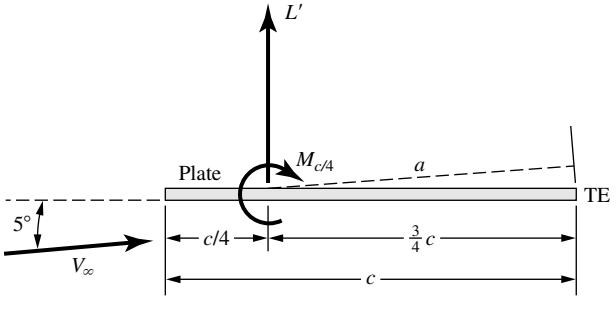


Figure 4.27 Flat plate at 5-degree angle of attack.

This represents the force and moment system on the plate. Recall from the discussion in Section 1.6 that the force and moment system can be represented by the lift acting through any point on the plate, and giving the moment about that point. Here, for convenience, we place the lift at the quarter-chord point.

The lift acts perpendicular to V_∞. (Part of the statement of the Kutta-Joukowski theorem given by Equation (3.140) is that the direction of the force associated with the circulation Γ is perpendicular to V_∞.) From Figure 4.27, the moment arm from L' to the trailing edge is the length a, where

$$a = \left(\tfrac{3}{4}c\right)\cos\alpha = \left(\tfrac{3}{4}c\right)\cos 5^\circ$$

One of the assumptions of thin airfoil theory is that the angle of attack is small, and hence we can assume that $\cos\alpha \approx 1$. Therefore, the moment arm from the point of action of the lift to the trailing edge is reasonably given by $\frac{3}{4}c$. (Note that, in the previous Figure 4.26, the assumption of small α is already implicit because the moment arm is drawn parallel to the plate.)

Examining Figure 4.27, the moment about the trailing edge is

$$M'_{\text{te}} = \left(\frac{3}{4}c\right) L' + M'_{c/4}$$

$$c_{m,\text{te}} = \frac{M'_{\text{te}}}{q_\infty c^2} = \left(\frac{3}{4}c\right)\frac{L'}{q_\infty c^2} + \frac{M'_{c/4}}{q_\infty c^2}$$

$$c_{m,\text{te}} = \frac{3}{4}c_\ell + c_{m,c/4}$$

Since $c_{m,c/4} = 0$ we have

$$c_{m,\text{te}} = \frac{3}{4}c_\ell$$

$$c_{m,\text{te}} = \frac{3}{4}(0.5485) = \boxed{0.411}$$

4.8 THE CAMBERED AIRFOIL

Thin airfoil theory for a cambered airfoil is a generalization of the method for a symmetric airfoil discussed in Section 4.7. To treat the cambered airfoil, return to Equation (4.18):

$$\frac{1}{2\pi}\int_0^c \frac{\gamma(\xi)\,d\xi}{x-\xi} = V_\infty\left(\alpha - \frac{dz}{dx}\right) \tag{4.18}$$

For a cambered airfoil, dz/dx is finite, and this makes the analysis more elaborate than in the case of a symmetric airfoil, where $dz/dx = 0$. Once again, let us transform Equation (4.18) via Equations (4.20) to (4.22), obtaining

$$\frac{1}{2\pi}\int_0^\pi \frac{\gamma(\theta)\sin\theta\,d\theta}{\cos\theta - \cos\theta_0} = V_\infty\left(\alpha - \frac{dz}{dx}\right) \tag{4.42}$$

We wish to obtain a solution for $\gamma(\theta)$ from Equation (4.42), subject to the Kutta condition $\gamma(\pi) = 0$. Such a solution for $\gamma(\theta)$ will make the camber line a streamline of the flow. However, as before a rigorous solution of Equation (4.42) for $\gamma(\theta)$ is beyond the scope of this book. Rather, the result is stated below:

$$\gamma(\theta) = 2V_\infty \left(A_0 \frac{1 + \cos\theta}{\sin\theta} + \sum_{n=1}^{\infty} A_n \sin n\theta \right) \tag{4.43}$$

Note that the above expression for $\gamma(\theta)$ consists of a leading term very similar to Equation (4.24) for a symmetric airfoil, plus a Fourier sine series with coefficients A_n. The values of A_n depend on the shape of the camber line dz/dx, and A_0 depends on both dz/dx and α, as shown below.

The coefficients A_0 and $A_n (n = 1, 2, 3, \ldots)$ in Equation (4.43) must be specific values in order that the camber line be a streamline of the flow. To find these specific values, substitute Equations (4.43) into Equation (4.42):

$$\frac{1}{\pi} \int_0^\pi \frac{A_0(1 + \cos\theta)\, d\theta}{\cos\theta - \cos\theta_0} + \frac{1}{\pi} \sum_{n=1}^{\infty} \int_0^\pi \frac{A_n \sin n\theta \sin\theta\, d\theta}{\cos\theta - \cos\theta_0} = \alpha - \frac{dz}{dx} \tag{4.44}$$

The first integral can be evaluated from the standard form given in Equation (4.26). The remaining integrals can be obtained from another standard form, which is derived in Appendix E of Reference 9, and which is given below:

$$\int_0^\pi \frac{\sin n\theta \sin\theta\, d\theta}{\cos\theta - \cos\theta_0} = -\pi \cos n\theta_0 \tag{4.45}$$

Hence, using Equations (4.26) and (4.45), we can reduce Equation (4.44) to

$$A_0 - \sum_{n=1}^{\infty} A_n \cos n\theta_0 = \alpha - \frac{dz}{dx}$$

or

$$\frac{dz}{dx} = (\alpha - A_0) + \sum_{n=1}^{\infty} A_n \cos n\theta_0 \tag{4.46}$$

Recall that Equation (4.46) was obtained directly from Equation (4.42), which is the transformed version of the fundamental equation of thin airfoil theory, Equation (4.18). Furthermore, recall that Equation (4.18) is evaluated at a given point x along the chord line, as sketched in Figure 4.24. Hence, Equation (4.46) is also evaluated at the given point x; here, dz/dx and θ_0 correspond to the same point x on the chord line. Also, recall that dz/dx is a function of θ_0, where $x = (c/2)(1 - \cos\theta_0)$ from Equation (4.21).

Examine Equation (4.46) closely. It is in the form of a Fourier cosine series expansion for the function of dz/dx. In general, the Fourier cosine series representation of a function $f(\theta)$ over an interval $0 \le \theta \le \pi$ is given by

$$f(\theta) = B_0 + \sum_{n=1}^{\infty} B_n \cos n\theta \tag{4.47}$$

where, from Fourier analysis, the coefficients B_0 and B_n are given by

$$B_0 = \frac{1}{\pi} \int_0^\pi f(\theta)\, d\theta \tag{4.48}$$

and

$$B_n = \frac{2}{\pi} \int_0^\pi f(\theta) \cos n\theta\, d\theta \tag{4.49}$$

(See, e.g., page 217 of Reference 6.) In Equation (4.46), the function dz/dx is analogous to $f(\theta)$ in the general form given in Equation (4.47). Thus, from Equations (4.48) and (4.49), the coefficients in Equation (4.46) are given by

$$\alpha - A_0 = \frac{1}{\pi} \int_0^\pi \frac{dz}{dx}\, d\theta_0$$

or

$$A_0 = \alpha - \frac{1}{\pi} \int_0^\pi \frac{dz}{dx}\, d\theta_0 \tag{4.50}$$

and

$$A_n = \frac{2}{\pi} \int_0^\pi \frac{dz}{dx} \cos n\theta_0\, d\theta_0 \tag{4.51}$$

Keep in mind that in the above, dz/dx is a function of θ_0. Note from Equation (4.50) that A_0 depends on both α and the shape of the camber line (through dz/dx), whereas from Equation (4.51) the values of A_n depend only on the shape of the camber line.

Pause for a moment and think about what we have done. We are considering the flow over a cambered airfoil of given shape dz/dx at a given angle of attack α. In order to make the camber line a streamline of the flow, the strength of the vortex sheet along the chord line must have the distribution $\gamma(\theta)$ given by Equation (4.43), where the coefficients A_0 and A_n are given by Equations (4.50) and (4.51), respectively. Also, note that Equation (4.43) satisfies the Kutta condition $\gamma(\pi) = 0$. Actual numbers for A_0 and A_n can be obtained for a given shape airfoil at a given angle of attack simply by carrying out the integrations indicated in Equations (4.50) and (4.51). For an example of such calculations applied to an NACA 23012 airfoil, see Example 4.5 at the end of this section. Also, note that when $dz/dx = 0$, Equation (4.43) reduces to Equation (4.24) for a symmetric airfoil. Hence, the symmetric airfoil is a special case of Equation (4.43).

Let us now obtain expressions for the aerodynamic coefficients for a cambered airfoil. The total circulation due to the entire vortex sheet from the leading edge to the trailing edge is

$$\Gamma = \int_0^c \gamma(\xi)\, d\xi = \frac{c}{2} \int_0^\pi \gamma(\theta) \sin\theta\, d\theta \tag{4.52}$$

Substituting Equation (4.43) for $\gamma(\theta)$ into Equation (4.52), we obtain

$$\Gamma = cV_\infty \left[A_0 \int_0^\pi (1 + \cos\theta)\, d\theta + \sum_{n=1}^{\infty} A_n \int_0^\pi \sin n\theta \sin\theta\, d\theta \right] \tag{4.53}$$

From any standard table of integrals,

$$\int_0^\pi (1 + \cos\theta)\, d\theta = \pi$$

and

$$\int_0^\pi \sin n\theta \sin\theta\, d\theta = \begin{cases} \pi/2 & \text{for } n = 1 \\ 0 & \text{for } n \neq 1 \end{cases}$$

Hence, Equation (4.53) becomes

$$\Gamma = cV_\infty \left(\pi A_0 + \frac{\pi}{2} A_1 \right) \tag{4.54}$$

From Equation (4.54), the lift per unit span is

$$L' = \rho_\infty V_\infty \Gamma = \rho_\infty V_\infty^2 c \left(\pi A_0 + \frac{\pi}{2} A_1 \right) \tag{4.55}$$

In turn, Equation (4.55) leads to the lift coefficient in the form

$$c_l = \frac{L'}{\frac{1}{2}\rho_\infty V_\infty^2 c(1)} = \pi(2A_0 + A_1) \tag{4.56}$$

Recall that the coefficients A_0 and A_1 in Equation (4.56) are given by Equations (4.50) and (4.51), respectively. Hence, Equation (4.56) becomes

$$\boxed{c_l = 2\pi \left[\alpha + \frac{1}{\pi} \int_0^\pi \frac{dz}{dx} (\cos\theta_0 - 1)\, d\theta_0 \right]} \tag{4.57}$$

and

$$\boxed{\text{Lift slope} \equiv \frac{dc_l}{d\alpha} = 2\pi} \tag{4.58}$$

Equations (4.57) and (4.58) are important results. Note that, as in the case of the symmetric airfoil, the theoretical lift slope for a cambered airfoil is 2π. It is a general result from thin airfoil theory that $dc_l/d\alpha = 2\pi$ for any shape airfoil. However, the expression for c_l itself differs between a symmetric and a cambered airfoil, the difference being the integral term in Equation (4.57). This integral term has physical significance, as follows. Return to Figure 4.9, which illustrates the lift curve for an airfoil. The angle of zero lift is denoted by $\alpha_{L=0}$ and is a negative value. From the geometry shown in Figure 4.9, clearly

$$c_l = \frac{dc_l}{d\alpha} (\alpha - \alpha_{L=0}) \tag{4.59}$$

Substituting Equation (4.58) into (4.59), we have

$$c_l = 2\pi(\alpha - \alpha_{L=0}) \tag{4.60}$$

Comparing Equations (4.60) and (4.57), we see that the integral term in Equation (4.57) is simply the negative of the zero-lift angle; that is

$$\boxed{\alpha_{L=0} = -\frac{1}{\pi}\int_0^\pi \frac{dz}{dx}(\cos\theta_0 - 1)\,d\theta_0} \tag{4.61}$$

Hence, from Equation (4.61), thin airfoil theory provides a means to predict the angle of zero lift. Note that Equation (4.61) yields $\alpha_{L=0} = 0$ for a symmetric airfoil, which is consistent with the results shown in Figure 4.25. Also, note that the more highly cambered the airfoil, the larger will be the absolute magnitude of $\alpha_{L=0}$.

Returning to Figure 4.26, the moment about the leading edge can be obtained by substituting $\gamma(\theta)$ from Equation (4.43) into the transformed version of Equation (4.35). The details are left for Problem 4.9. The result for the moment coefficient is

$$c_{m,\text{le}} = -\frac{\pi}{2}\left(A_0 + A_1 - \frac{A_2}{2}\right) \tag{4.62}$$

Substituting Equation (4.56) into (4.62), we have

$$\boxed{c_{m,\text{le}} = -\left[\frac{c_l}{4} + \frac{\pi}{4}(A_1 - A_2)\right]} \tag{4.63}$$

Note that, for $dz/dx = 0$, $A_1 = A_2 = 0$ and Equation (4.63) reduces to Equation (4.39) for a symmetric airfoil.

The moment coefficient about the quarter chord can be obtained by substituting Equation (4.63) into (4.40), yielding

$$\boxed{c_{m,c/4} = \frac{\pi}{4}(A_2 - A_1)} \tag{4.64}$$

Unlike the symmetric airfoil, where $c_{m,c/4} = 0$, Equation (4.64) demonstrates that $c_{m,c/4}$ is finite for a cambered airfoil. Therefore, the quarter chord is *not* the center of pressure for a cambered airfoil. However, note that A_1 and A_2 depend only on the shape of the camber line and do not involve the angle of attack. Hence, from Equation (4.64), $c_{m,c/4}$ is *independent* of α. Thus, the quarter-chord point is the *theoretical location* of the aerodynamic center for a cambered airfoil.

The location of the center of pressure can be obtained from Equation (1.21):

$$x_{\text{cp}} = -\frac{M'_{\text{LE}}}{L'} = -\frac{c_{m,\text{le}}c}{c_l} \tag{4.65}$$

Substituting Equation (4.63) into (4.65), we obtain

$$\boxed{x_{\text{cp}} = \frac{c}{4}\left[1 + \frac{\pi}{c_l}(A_1 - A_2)\right]} \tag{4.66}$$

Equation (4.66) demonstrates that the center of pressure for a cambered airfoil varies with the lift coefficient. Hence, as the angle of attack changes, the center of pressure also changes. Indeed, as the lift approaches zero, x_{cp} moves toward infinity; that is, it leaves the airfoil. For this reason, the center of pressure is not always a convenient point at which to draw the force system on an airfoil. Rather, the force-and-moment system on an airfoil is more conveniently considered at the aerodynamic center. (Return to Figure 1.25 and the discussion at the end of Section 1.6 for the referencing of the force-and-moment system on an airfoil.)

EXAMPLE 4.6

Consider an NACA 23012 airfoil. The mean camber line for this airfoil is given by

$$\frac{z}{c} = 2.6595\left[\left(\frac{x}{c}\right)^3 - 0.6075\left(\frac{x}{c}\right)^2 + 0.1147\left(\frac{x}{c}\right)\right] \qquad \text{for } 0 \le \frac{x}{c} \le 0.2025$$

and

$$\frac{z}{c} = 0.02208\left(1 - \frac{x}{c}\right) \qquad \text{for } 0.2025 \le \frac{x}{c} \le 1.0$$

Calculate (*a*) the angle of attack at zero lift, (*b*) the lift coefficient when $\alpha = 4^\circ$, (*c*) the moment coefficient about the quarter chord, and (*d*) the location of the center of pressure in terms of x_{cp}/c, when $\alpha = 4^\circ$. Compare the results with experimental data.

■ Solution

We will need dz/dx. From the given shape of the mean camber line, this is

$$\frac{dz}{dx} = 2.6595\left[3\left(\frac{x}{c}\right)^2 - 1.215\left(\frac{x}{c}\right) + 0.1147\right] \qquad \text{for } 0 \le \frac{x}{c} \le 0.2025$$

and

$$\frac{dz}{dx} = -0.02208 \qquad \text{for } 0.2025 \le \frac{x}{c} \le 1.0$$

Transforming from x to θ, where $x = (c/2)(1 - \cos\theta)$, we have

$$\frac{dz}{dx} = 2.6595\left[\frac{3}{4}(1 - 2\cos\theta + \cos^2\theta) - 0.6075(1 - \cos\theta) + 0.1147\right]$$

or

$$= 0.6840 - 2.3736\cos\theta + 1.995\cos^2\theta \qquad \text{for } 0 \le \theta \le 0.9335 \text{ rad}$$

and

$$= -0.02208 \qquad \text{for } 0.9335 \le \theta \le \pi$$

(*a*) From Equation (4.61),

$$\alpha_{L=0} = -\frac{1}{\pi}\int_0^\pi \frac{dz}{dx}(\cos\theta - 1)\,d\theta$$

(*Note:* For simplicity, we have dropped the subscript zero from θ; in Equation (4.61), θ_0 is the variable of integration—it can just as well be symbolized as θ for the variable of integration.) Substituting the equation for dz/dx into Equation (4.61), we have

$$\alpha_{L=0} = -\frac{1}{\pi}\int_0^{0.9335}(-0.6840 + 3.0576\cos\theta - 4.3686\cos^2\theta + 1.995\cos^3\theta)\,d\theta - \frac{1}{\pi}\int_{0.9335}^{\pi}(0.02208 - 0.02208\cos\theta)\,d\theta \tag{E.1}$$

From a table of integrals, we see that

$$\int \cos\theta\, d\theta = \sin\theta$$

$$\int \cos^2\theta\, d\theta = \tfrac{1}{2}\sin\theta\cos\theta + \tfrac{1}{2}\theta$$

$$\int \cos^3\theta\, d\theta = \tfrac{1}{3}\sin\theta(\cos^2\theta + 2)$$

Hence, Equation (E.1) becomes

$$\begin{aligned}\alpha_{L=0} = -\frac{1}{\pi}[&-2.8683\theta + 3.0576\sin\theta - 2.1843\sin\theta\cos\theta \\ &+ 0.665\sin\theta(\cos^2\theta + 2)]_0^{0.9335} \\ &- \frac{1}{\pi}[0.02208\theta - 0.02208\sin\theta]_{0.9335}^{\pi}\end{aligned}$$

Hence, $$\alpha_{L=0} = -\frac{1}{\pi}(-0.0065 + 0.0665) = -0.0191 \text{ rad}$$

or $$\boxed{\alpha_{L=0} = -1.09^\circ}$$

(b) $\alpha = 4^\circ = 0.0698$ rad
From Equation (4.60),

$$c_l = 2\pi(\alpha - \alpha_{L=0}) = 2\pi(0.0698 + 0.0191) = \boxed{0.559}$$

(c) The value of $c_{m,c/4}$ is obtained from Equation (4.64). For this, we need the two Fourier coefficients A_1 and A_2. From Equation (4.51),

$$A_1 = \frac{2}{\pi}\int_0^{\pi}\frac{dz}{dx}\cos\theta\, d\theta$$

$$\begin{aligned}A_1 &= \frac{2}{\pi}\int_0^{0.9335}(0.6840\cos\theta - 2.3736\cos^2\theta + 1.995\cos^3\theta)\, d\theta \\ &\quad + \frac{2}{\pi}\int_{0.9335}^{\pi}(-0.02208\cos\theta)\, d\theta \\ &= \frac{2}{\pi}[0.6840\sin\theta - 1.1868\sin\theta\cos\theta - 1.1868\theta + 0.665\sin\theta(\cos^2\theta + 2)]_0^{0.9335} \\ &\quad + \frac{2}{\pi}[-0.02208\sin\theta]_{0.09335}^{\pi} \\ &= \frac{2}{\pi}(0.1322 + 0.0177) = 0.0954\end{aligned}$$

From Equation (4.51),

$$A_2 = \frac{2}{\pi}\int_0^{\pi} \frac{dz}{dx}\cos 2\theta\, d\theta = \frac{2}{\pi}\int_0^{\pi} \frac{dz}{dx}(2\cos^2\theta - 1)\, d\theta$$

$$= \frac{2}{\pi}\int_0^{0.9335} (-0.6840 + 2.3736\cos\theta - 0.627\cos^2\theta - 4.747\cos^3\theta + 3.99\cos^4\theta)\, d\theta + \frac{2}{\pi}\int_{0.9335}^{\pi} (0.02208 - 0.0446\cos^2\theta)\, d\theta$$

Note:

$$\int \cos^4\theta\, d\theta = \tfrac{1}{4}\cos^3\theta\sin\theta + \tfrac{3}{8}(\sin\theta\cos\theta + \theta)$$

Thus,

$$A_2 = \frac{2}{\pi}\left\{-0.6840\theta + 2.3736\sin\theta - 0.628\left(\frac{1}{2}\right)(\sin\theta\cos\theta + \theta) - 4.747\left(\frac{1}{3}\right)\sin\theta(\cos^2\theta + 2) + 3.99\left[\frac{1}{4}\cos^3\sin\theta + \frac{3}{8}(\sin\theta\cos\theta + \theta)\right]\right\}_0^{0.9335} + \frac{2}{\pi}\left[0.02208\theta - 0.0446\left(\frac{1}{2}\right)(\sin\theta\cos\theta + \theta)\right]_{0.9335}^{\pi}$$

$$= \frac{2}{\pi}(0.11384 + 0.01056) = 0.0792$$

From Equation (4.64)

$$c_{m,c/4} = \frac{\pi}{4}(A_2 - A_1) = \frac{\pi}{4}(0.0792 - 0.0954)$$

$$\boxed{c_{m,c/4} = -0.0127}$$

(*d*) From Equation (4.66)

$$x_{\text{cp}} = \frac{c}{4}\left[1 + \frac{\pi}{c_l}(A_1 - A_2)\right]$$

Hence,

$$\frac{x_{\text{cp}}}{c} = \frac{1}{4}\left[1 + \frac{\pi}{0.559}(0.0954 - 0.0792)\right] = 0.273$$

Comparison with Experimental Data The data for the NACA 23012 airfoil are shown in Figure 4.28. From this, we make the following tabulation:

	Calculated	Experiment
$\alpha_{L=0}$	-1.09°	-1.1°
c_l (at $\alpha = 4^\circ$)	0.559	0.55
$c_{m,c/4}$	−0.0127	−0.01

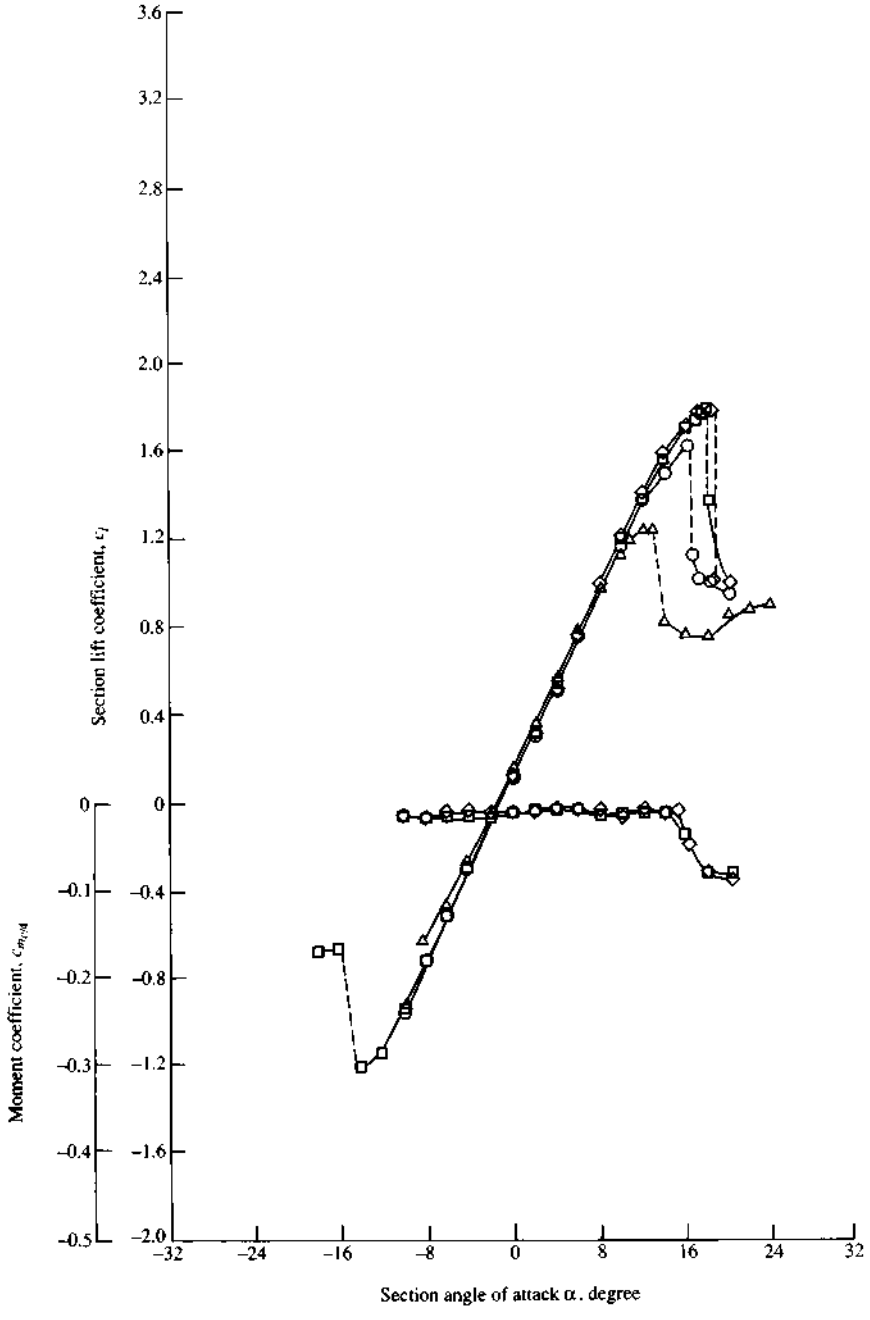


Figure 4.28 Lift- and moment-coefficient data for an NACA 23012 airfoil, for comparison with the theoretical results obtained in Example 4.6.

Note that the results from thin airfoil theory for a cambered airfoil agree very well with the experimental data. Recall that excellent agreement between thin airfoil theory for a symmetric airfoil and experimental data has already been shown in Figure 4.25. Hence, all of the work we have done in this section to develop thin airfoil theory is certainly worth the effort. Moreover, this illustrates that the development of thin airfoil theory in the early 1900s was a crowning achievement in theoretical aerodynamics and validates the mathematical approach of replacing the chord line of the airfoil with a vortex sheet, with the flow tangency condition evaluated along the mean camber line.

This brings to an end our introduction to classical thin airfoil theory. Returning to our road map in Figure 4.7, we have now completed the right-hand branch.

4.9 THE AERODYNAMIC CENTER: ADDITIONAL CONSIDERATIONS

The definition of the aerodynamic center is given in Section 4.3; it is that point on a body about which the aerodynamically generated moment is *independent of angle of attack*. At first thought, it is hard to imagine that such a point could exist. However, the moment coefficient data in Figure 4.11, which are constant with angle of attack, experimentally prove the existence of the aerodynamic center. Moreover, thin airfoil theory as derived in Sections 4.7 and 4.8 clearly shows that, within the assumptions embodied in the theory, not only does the aerodynamic center exist but that it is located at the quarter-chord point on the airfoil. Therefore, to Figure 1.24 which illustrates three different ways of stating the force and moment system on an airfoil, we can now add a fourth way, namely, the specification of the lift and drag acting through the aerodynamic center, and the value of the moment about the aerodynamic center. This is illustrated in Figure 4.29.

For most conventional airfoils, the aerodynamic center is close to, but not necessarily exactly at, the quarter-chord point. Given data for the shape of the

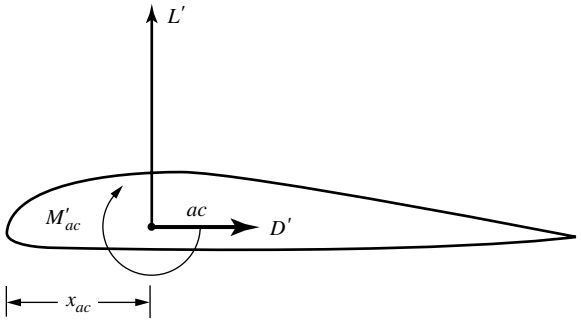


Figure 4.29 Lift, drag, and moments about the aerodynamic center.

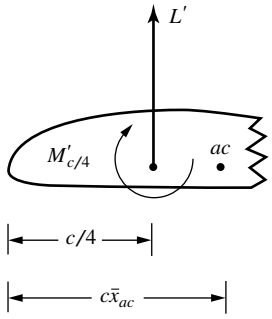


Figure 4.30 Lift and moments about the quarter-chord point, and a sketch useful for locating the aerodynamic center.

lift coefficient curve and the moment coefficient curve taken around an arbitrary point, we can calculate the location of the aerodynamic center as follows. Consider the lift and moment system taken about the quarter-chord point, as shown in Figure 4.30. We designate the location of the aerodynamic center by $c\bar{x}_{\text{ac}}$ measured from the leading edge. Here, $\bar{x}_{\text{ac}}$ is the location of the aerodynamic center as a fraction of the chord length c. Taking moments about the aerodynamic center designated by ac in Figure 4.30, we have

$$M'_{\text{ac}} = L'(c\bar{x}_{\text{ac}} - c/4) + M'_{c/4} \tag{4.67}$$

Dividing Equation (4.67) by $q_\infty Sc$, we have

$$\frac{M'_{\text{ac}}}{q_\infty Sc} = \frac{L'}{q_\infty S}(\bar{x}_{\text{ac}} - 0.25) + \frac{M'_{c/4}}{q_\infty Sc}$$

or

$$c_{m,\text{ac}} = c_l(\bar{x}_{\text{ac}} - 0.25) + c_{m,c/4} \tag{4.68}$$

Differentiating Equation (4.68) with respect to angle of attack α, we have

$$\frac{dc_{m,\text{ac}}}{d\alpha} = \frac{dc_l}{d\alpha}(\bar{x}_{\text{ac}} - 0.25) + \frac{dc_{m,c/4}}{d\alpha} \tag{4.69}$$

However, in Equation (4.69), $dc_{m,\text{ac}}/d\alpha$ is zero by definition of the aerodynamic center. Hence, Equation (4.69) becomes

$$0 = \frac{dc_l}{d\alpha}(\bar{x}_{\text{ac}} - 0.25) + \frac{dc_{m,c/4}}{d\alpha} \tag{4.70}$$

For airfoils below the stalling angle of attack, the slopes of the lift coefficient and moment coefficient curves are constant. Designating these slopes by

$$\frac{dc_l}{d\alpha} \equiv a_0; \quad \frac{dc_{m,c/4}}{d\alpha} \equiv m_0$$

Equation (4.70) becomes

$$0 = a_0(\bar{x}_{ac} - 0.25) + m_0$$

or

$$\boxed{\bar{x}_{ac} = -\frac{m_0}{a_0} + 0.25} \tag{4.71}$$

Hence, Equation (4.71) proves that, for a body with linear lift and moment curves, that is, where a_0 and m_0 are fixed values, the aerodynamic center exists as a fixed point on the airfoil. Moreover, Equation (4.71) allows the calculation of the location of this point.

EXAMPLE 4.7

Consider the NACA 23012 airfoil studied in Example 4.6. Experimental data for this airfoil is plotted in Figure 4.28, and can be obtained from Reference 11. It shows that, at $\alpha = 4^\circ$, $c_l = 0.55$ and $c_{m,c/4} = -0.005$. The zero-lift angle of attack is -1.1°. Also, at $\alpha = -4^\circ$, $c_{m,c/4} = -0.0125$. (Note that the "experimental" value of $c_{m,c/4} = -0.01$ tabulated at the end of Example 4.6 is an average value over a range of angle of attack. Since the calculated value of $c_{m,c/4}$ from thin airfoil theory states that the quarter-chord point is the aerodynamic center, it makes sense in Example 4.6 to compare the calculated $c_{m,c/4}$ with an experimental value averaged over a range of angle of attack. However, in the present example, because $c_{m,c/4}$ in reality varies with angle of attack, we use the actual data at two different angles of attack.) From the given information, calculate the location of the aerodynamic center for the NACA 23012 airfoil.

■ Solution

Since $c_l = 0.55$ at $\alpha = 4^\circ$ and $c_l = 0$ at $\alpha = -1.1^\circ$, the lift slope is

$$a_0 = \frac{0.55 - 0}{4 - (-1.1)} = 0.1078 \text{ per degree}$$

The slope of the moment coefficient curve is

$$m_0 = \frac{-0.005 - (-0.0125)}{4 - (-4)} = 9.375 \times 10^{-4} \text{ per degree}$$

From Equation (4.71),

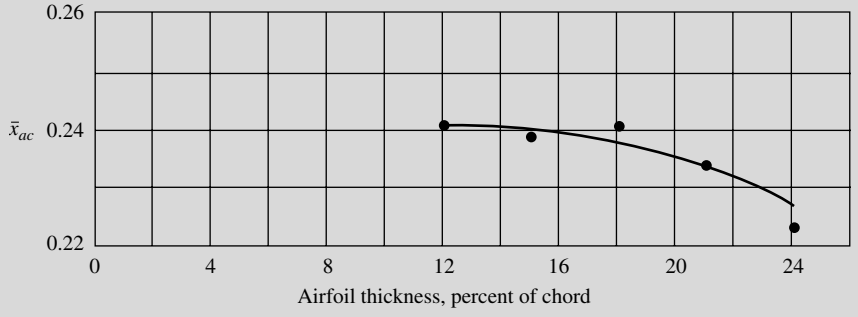
$$\bar{x}_{ac} = -\frac{m_0}{a_0} + 0.25 = -\frac{9.375 \times 10^{-4}}{0.1078} + 0.25 = \boxed{0.241}$$

The result agrees exactly with the measured value quoted on page 183 of Abbott and Von Doenhoff (Reference 11).

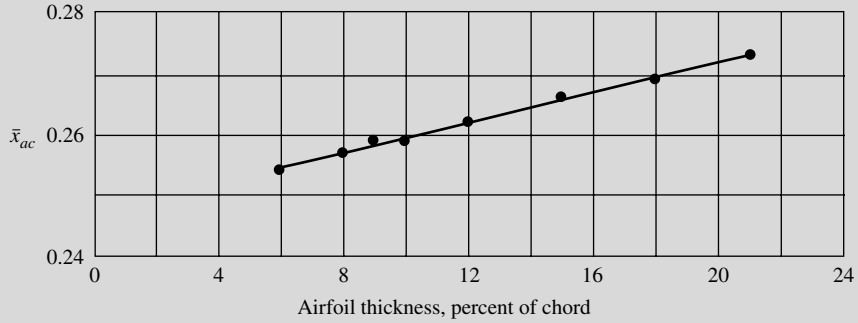
DESIGN BOX

The result of Example 4.7 shows that the aerodynamic center for the NACA 23012 airfoil is located *ahead* of, but very close to, the quarter-chord point. For some other families of airfoils, the aerodynamic center is located *behind,* but similarly close to, the quarter-chord point. For a given airfoil family, the location of the aerodynamic center depends on the airfoil thickness, as shown in Figure 4.31. The variation of $\bar{x}_{ac}$ with thickness for the NACA 230XX family is given in Figure 4.31*a*. Here, the aerodynamic center is ahead of the quarter-chord point, and becomes progressively farther ahead as the airfoil thickness is increased. In contrast, the variation of $\bar{x}_{ac}$ with thickness for the NACA 64-2XX family is given in Figure 4.31*b*. Here, the aerodynamic center is behind the quarter-chord point, and becomes progressively farther behind as the airfoil thickness is increased.

From the point of view of purely aerodynamics, the existence of the aerodynamic center is interesting, but the specification of the force and moment system on the airfoil by placing the lift and drag at the aerodynamic center and giving the value of M'_{ac} as illustrated in Figure 4.29, is not more useful than placing the lift and drag at any other point on the airfoil and giving the value of M' at that point, such as shown in Figure 1.25. However, in flight dynamics, and in particular the consideration of the stability and control of flight vehicles, placing the lift and drag at, and



(*a*) **NACA 230XX Airfoil**



(*b*) **NACA 64-2XX Airfoil**

Figure 4.31 Variation of the location of the aerodynamic center with airfoil thickness. (*a*) NACA 230XX airfoil. (*b*) NACA 64-2XX airfoil.

dealing with the moment about, the aerodynamic center, is particularly convenient. The fact that M_{ac} for a flight vehicle is independent of angle of attack simplifies the analysis of the stability and control characteristics, and the use of the aerodynamic center therefore becomes important in airplane design. In the design process, it is important to know where the aerodynamic centers of the various components of the aircraft (wing, tail, fuselage, etc.) are located, and above all the location of the aerodynamic center for the complete flight vehicle. It is for this reason that we have placed extra emphasis on the aerodynamic center in Section 4.9. For an introduction to stability and control see Chapter 7 of the author's book *Introduction to Flight,* 5th edition, McGraw-Hill, Boston, 2005. For more information about the aerodynamic center, and its use in airplane design, see the author's book *Aircraft Performance and Design*, McGraw-Hill, Boston, 1999.

4.10 LIFTING FLOWS OVER ARBITRARY BODIES: THE VORTEX PANEL NUMERICAL METHOD

The thin airfoil theory described in Sections 4.7 and 4.8 is just what it says—it applies only to thin airfoils at small angles of attack. (Make certain that you understand exactly where in the development of thin airfoil theory these assumptions are made and the reasons for making them.) The advantage of thin airfoil theory is that closed-form expressions are obtained for the aerodynamic coefficients. Moreover, the results compare favorably with experimental data for airfoils of about 12 percent thickness or less. However, the airfoils on many low-speed airplanes are thicker than 12 percent. Moreover, we are frequently interested in high angles of attack, such as occur during takeoff and landing. Finally, we are sometimes concerned with the generation of aerodynamic lift on other body shapes, such as automobiles or submarines. Hence, thin airfoil theory is quite restrictive when we consider the whole spectrum of aerodynamic applications. We need a method that allows us to calculate the aerodynamic characteristics of bodies of arbitrary shape, thickness, and orientation. Such a method is described in this section. Specifically, we treat the vortex panel method, which is a numerical technique that has come into widespread use since the early 1970s. In reference to our road map in Figure 4.7, we now move to the left-hand branch. Also, since this chapter deals with airfoils, we limit our attention to two-dimensional bodies.

The vortex panel method is directly analogous to the source panel method described in Section 3.17. However, because a source has zero circulation, source panels are useful only for nonlifting cases. In contrast, vortices have circulation, and hence vortex panels can be used for lifting cases. (Because of the similarities between source and vortex panel methods, return to Section 3.17 and review the basic philosophy of the source panel method before proceeding further.)

The philosophy of covering a body *surface* with a vortex sheet of such a strength to make the surface a streamline of the flow was discussed in Section 4.4 We then went on to simplify this idea by placing the vortex sheet on the camber line of the airfoil as shown in Figure 4.16, thus establishing the basis for thin airfoil theory. We now return to the original idea of wrapping the vortex sheet over the complete surface of the body, as shown in Figure 4.15. We wish to find

$\gamma(s)$ such that the body surface becomes a streamline of the flow. There exists no closed-form analytical solution for $\gamma(s)$; rather, the solution must be obtained numerically. This is the purpose of the vortex panel method.

Let us approximate the vortex sheet shown in Figure 4.15 by a series of straight panels, as shown earlier in Figure 3.40. (In Chapter 3, Figure 3.40 was used to discuss source panels; here, we use the same sketch for discussion of vortex panels.) Let the vortex strength $\gamma(s)$ per unit length be constant over a given panel, but allow it to vary from one panel to the next. That is, for the n panels shown in Figure 3.40, the vortex panel strengths per unit length are $\gamma_1, \gamma_2, \ldots, \gamma_j, \ldots, \gamma_n$. These panel strengths are unknowns; the main thrust of the panel technique is to solve for γ_j, $j = 1$ to n, such that the body surface becomes a streamline of the flow and such that the Kutta condition is satisfied. As explained in Section 3.17, the midpoint of each panel is a control point at which the boundary condition is applied; that is, at each control point, the normal component of the flow velocity is zero.

Let P be a point located at (x, y) in the flow, and let r_{pj} be the distance from any point on the jth panel to P, as shown in Figure 3.40. The radius r_{pj} makes the angle θ_{pj} with respect to the x axis. The velocity potential induced at P due to the jth panel, $\Delta\phi_j$, is, from Equation (4.3),

$$\Delta\phi_j = -\frac{1}{2\pi}\int_j \theta_{pj}\gamma_j\, ds_j \tag{4.72}$$

In Equation (4.72), γ_j is constant over the jth panel, and the integral is taken over the jth panel only. The angle θ_{pj} is given by

$$\theta_{pj} = \tan^{-1}\frac{y - y_j}{x - x_j} \tag{4.73}$$

In turn, the potential at P due to *all* the panels is Equation (4.72) summed over all the panels:

$$\phi(P) = \sum_{j=1}^{n}\phi_j = -\sum_{j=1}^{n}\frac{\gamma_j}{2\pi}\int_j \theta_{pj}\, ds_j \tag{4.74}$$

Since point P is just an arbitrary point in the flow, let us put P at the control point of the ith panel shown in Figure 3.40. The coordinates of this control point are (x_i, y_i). Then Equations (4.73) and (4.74) become

$$\theta_{ij} = \tan^{-1}\frac{y_i - y_j}{x_i - x_j}$$

and

$$\phi(x_i, y_i) = -\sum_{j=1}^{n}\frac{\gamma_j}{2\pi}\int_j \theta_{ij}\, ds_j \tag{4.75}$$

Equation (4.75) is physically the contribution of *all* the panels to the potential at the control point of the ith panel.

At the control points, the normal component of the velocity is zero; this velocity is the superposition of the uniform flow velocity and the velocity induced by all the vortex panels. The component of V_∞ normal to the ith panel is given by Equation (3.148):

$$V_{\infty,n} = V_\infty \cos \beta_i \tag{3.148}$$

The normal component of velocity induced at (x_i, y_i) by the vortex panels is

$$V_n = \frac{\partial}{\partial n_i}[\phi(x_i, y_i)] \tag{4.76}$$

Combining Equations (4.75) and (4.76), we have

$$V_n = -\sum_{j=1}^{n} \frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_i} ds_j \tag{4.77}$$

where the summation is over all the panels. The normal component of the flow velocity at the ith control point is the sum of that due to the freestream [Equation (3.148)] and that due to the vortex panels [Equation (4.77)]. The boundary condition states that this sum must be zero:

$$V_{\infty,n} + V_n = 0 \tag{4.78}$$

Substituting Equations (3.148) and (4.77) into (4.78), we obtain

$$V_\infty \cos \beta_i - \sum_{j=1}^{n} \frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_i} ds_j = 0 \tag{4.79}$$

Equation (4.79) is the crux of the vortex panel method. The values of the integrals in Equation (4.79) depend simply on the panel geometry; they are not properties of the flow. Let $J_{i,j}$ be the value of this integral when the control point is on the ith panel. Then Equation (4.79) can be written as

$$V_\infty \cos \beta_i - \sum_{j=1}^{n} \frac{\gamma_j}{2\pi} J_{i,j} = 0 \tag{4.80}$$

Equation (4.80) is a linear algebraic equation with n unknowns, $\gamma_1, \gamma_2, \ldots, \gamma_n$. It represents the flow boundary condition evaluated at the control point of the ith panel. If Equation (4.80) is applied to the control points of *all* the panels, we obtain a system of n linear equations with n unknowns.

To this point, we have been deliberately paralleling the discussion of the source panel method given in Section 3.17; however, the similarity stops here. For the source panel method, the n equations for the n unknown source strengths are routinely solved, giving the flow over a nonlifting body. In contrast, for the lifting case with vortex panels, in addition to the n equations given by Equation (4.80) applied at all the panels, we must also satisfy the Kutta condition. This can be done in several ways. For example, consider Figure 4.32, which illustrates a detail of the vortex panel distribution at the trailing edge. Note that the length of each

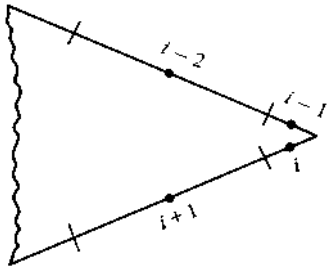


Figure 4.32 Vortex panels at the trailing edge.

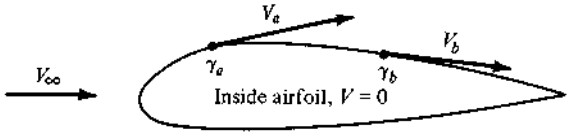


Figure 4.33 Airfoil as a solid body, with zero velocity inside the profile.

panel can be different; their length and distribution over the body are up to your discretion. Let the two panels at the trailing edge (panels i and $i-1$ in Figure 4.32) be very small. The Kutta condition is applied *precisely* at the trailing edge and is given by $\gamma(\text{TE}) = 0$. To approximate this numerically, if points i and $i-1$ are close enough to the trailing edge, we can write

$$\gamma_i = -\gamma_{i-1} \tag{4.81}$$

such that the strengths of the two vortex panels i and $i-1$ exactly cancel at the point where they touch at the trailing edge. Thus, in order to impose the Kutta condition on the solution of the flow, Equation (4.81) (or an equivalent expression) must be included. Note that Equation (4.80) evaluated at all the panels and Equation (4.81) constitute an *overdetermined* system of n unknowns with $n+1$ equations. Therefore, to obtain a determined system, Equation (4.80) is not evaluated at one of the control points on the body. That is, we choose to ignore one of the control points, and we evaluate Equation (4.80) at the other $n-1$ control points. This, in combination with Equation (4.81), now gives a system of n linear algebraic equations with n unknowns, which can be solved by standard techniques.

At this stage, we have conceptually obtained the values of $\gamma_1, \gamma_2, \ldots, \gamma_n$ which make the body surface a streamline of the flow and which also satisfy the Kutta condition. In turn, the flow velocity tangent to the surface can be obtained directly from γ. To see this more clearly, consider the airfoil shown in Figure 4.33. We are concerned only with the flow outside the airfoil and on its surface. Therefore, let the velocity be zero at every point *inside* the body, as shown in Figure 4.33. In particular, the velocity just inside the vortex sheet on the surface

is zero. This corresponds to $u_2 = 0$ in Equation (4.8). Hence, the velocity just outside the vortex sheet is, from Equation (4.8),

$$\gamma = u_1 - u_2 = u_1 - 0 = u_1$$

In Equation (4.8), u denotes the velocity tangential to the vortex sheet. In terms of the picture shown in Figure 4.33, we obtain $V_a = \gamma_a$ at point a, $V_b = \gamma_b$ at point b, etc. Therefore, *the local velocities tangential to the airfoil surface are equal to the local values of* γ. In turn, the local pressure distribution can be obtained from Bernoulli's equation.

The total circulation and the resulting lift are obtained as follows. Let s_j be the length of the jth panel. Then the circulation due to the jth panel is $\gamma_j s_j$. In turn, the total circulation due to all the panels is

$$\Gamma = \sum_{j=1}^{n} \gamma_j s_j \tag{4.82}$$

Hence, the lift per unit span is obtained from

$$L' = \rho_\infty V_\infty \sum_{j=1}^{n} \gamma_j s_j \tag{4.83}$$

The presentation in this section is intended to give only the general flavor of the vortex panel method. There are many variations of the method in use today, and you are encouraged to read the modern literature, especially as it appears in the *AIAA Journal* and the *Journal of Aircraft* since 1970. The vortex panel method as described in this section is termed a "first-order" method because it assumes a constant value of γ over a given panel. Although the method may appear to be straightforward, its numerical implementation can sometimes be frustrating. For example, the results for a given body are sensitive to the number of panels used, their various sizes, and the way they are distributed over the body surface (i.e., it is usually advantageous to place a large number of small panels near the leading and trailing edges of an airfoil and a smaller number of larger panels in the middle). The need to ignore one of the control points in order to have a determined system in n equations for n unknowns also introduces some arbitrariness in the numerical solution. Which control point do you ignore? Different choices sometimes yield different numerical answers for the distribution of γ over the surface. Moreover, the resulting numerical distributions for γ are not always smooth, but rather, they have oscillations from one panel to the next as a result of numerical inaccuracies. The problems mentioned above are usually overcome in different ways by different groups who have developed relatively sophisticated panel programs for practical use. For example, what is more common today is to use a *combination* of both source and vortex panels (source panels to basically simulate the airfoil thickness and vortex panels to introduce circulation) in a panel solution. This combination helps to mitigate some of the practical numerical problems just discussed. Again, you are encouraged to consult the literature for more information.

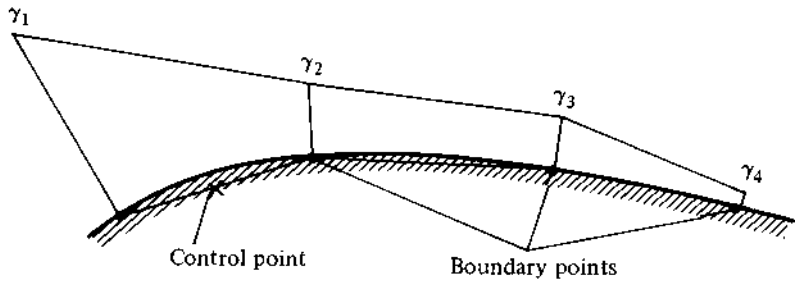


Figure 4.34 Linear distribution of γ over each panel—a second-order panel method.

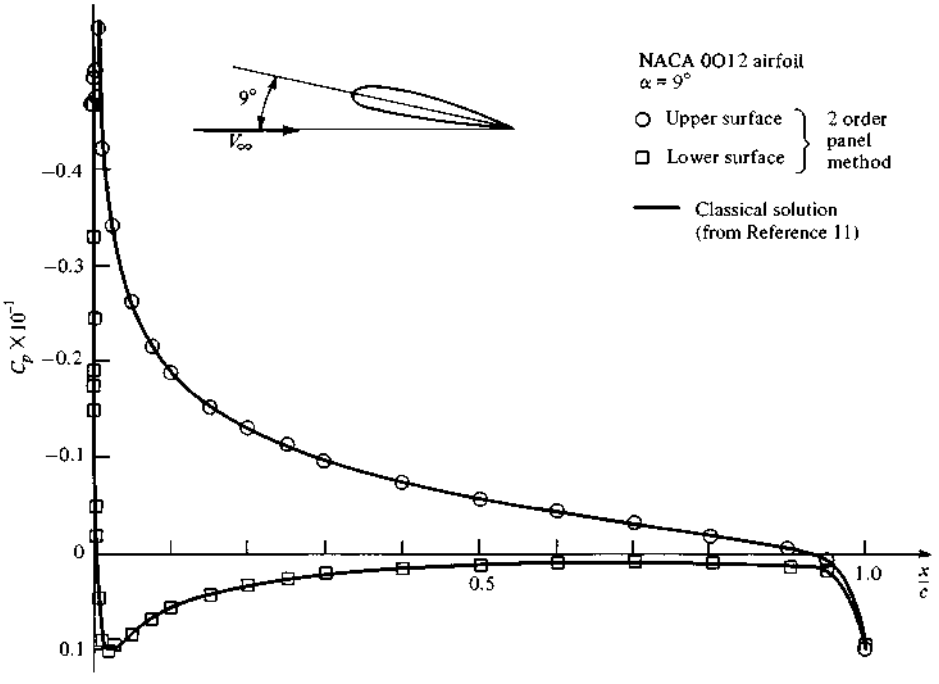


Figure 4.35 Pressure coefficient distribution over an NACA 0012 airfoil; comparison between second-order vortex panel method and NACA theoretical results from Reference 11. The numerical panel results were obtained by one of the author's graduate students, Dr. Tae-Hwan Cho.

Such accuracy problems have also encouraged the development of higher-order panel techniques. For example, a "second-order" panel method assumes a *linear* variation of γ over a given panel, as sketched in Figure 4.34. Here, the value of γ at the edges of each panel is matched to its neighbors, and the values γ_1, γ_2, γ_3, etc. at the *boundary points* become the unknowns to be solved. The flow-tangency boundary condition is still applied at the *control point* of each

panel, as before. Some results using a second-order vortex panel technique are given in Figure 4.35, which shows the distribution of pressure coefficients over the upper and lower surfaces of an NACA 0012 airfoil at a 9° angle of attack. The circles and squares are numerical results from a second-order vortex panel technique developed at the University of Maryland, and the solid lines are from NACA results given in Reference 11. Excellent agreement is obtained.

Again, you are encouraged to consult the literature before embarking on any serious panel solutions of your own. For example, Reference 14 is a classic paper on panel methods, and Reference 15 highlights many of the basic concepts of panel methods along with actual computer program statement listings for simple applications. Reference 66 is a modern compilation of papers, several of which deal with current panel techniques. Finally, Katz and Plotkin (Reference 67) give perhaps the most thorough discussion of panel techniques and their foundations to date.

4.11 MODERN LOW-SPEED AIRFOILS

The nomenclature and aerodynamic characteristics of standard NACA airfoils are discussed in Sections 4.2 and 4.3; before progressing further, you should review these sections in order to reinforce your knowledge of airfoil behavior, especially in light of our discussions on airfoil theory. Indeed, the purpose of this section is to provide a modern sequel to the airfoils discussed in Sections 4.2 and 4.3.

During the 1970s, NASA designed a series of low-speed airfoils that have performance superior to the earlier NACA airfoils. The standard NACA airfoils were based almost exclusively on experimental data obtained during the 1930s and 1940s. In contrast, the new NASA airfoils were designed on a computer using a numerical technique similar to the source and vortex panel methods discussed earlier, along with numerical predictions of the viscous flow behavior (skin friction and flow separation). Wind-tunnel tests were then conducted to verify the computer-designed profiles and to obtain the definitive airfoil properties. Out of this work first came the general aviation–Whitcomb [GA(W)-1] airfoil, which has since been redesignated the LS(1)-0417 airfoil. The shape of this airfoil is given in Figure 4.36, obtained from Reference 16. Note that it has a large leading-edge radius ($0.08c$ in comparison to the standard $0.02c$) in order to flatten the usual peak in pressure coefficient near the nose. Also, note that the bottom surface near the trailing edge is cusped in order to increase the camber and hence the aerodynamic

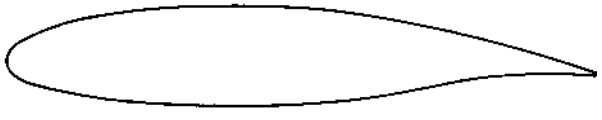


Figure 4.36 Profile for the NASA LS(1)-0417 airfoil. When first introduced, this airfoil was labeled the GA(W)-1 airfoil, a nomenclature which has now been superseded. *(From Reference 16.)*

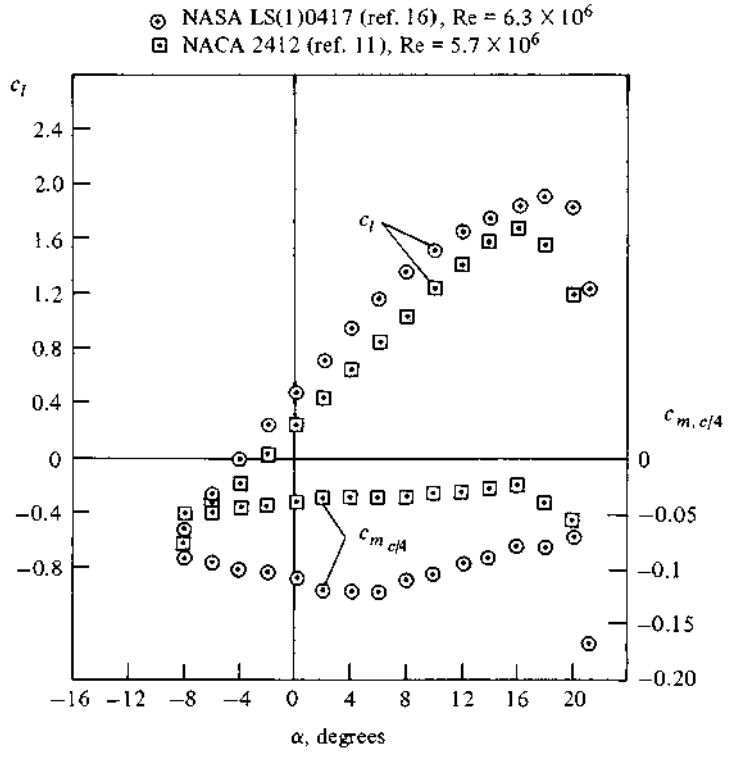


Figure 4.37 Comparison of the modern NASA LS(1)-0417 airfoil with the standard NACA 2412 airfoil.

loading in that region. Both design features tend to discourage flow separation over the top surface at high angle of attack, hence yielding higher values of the maximum lift coefficient. The experimentally measured lift and moment properties (from Reference 16) are given in Figure 4.37, where they are compared with the properties for an NACA 2412 airfoil, obtained from Reference 11. Note that $c_{l,\max}$ for the NASA LS(1)-0417 is considerably higher than for the NACA 2412.

The NASA LS(1)-0417 airfoil has a maximum thickness of 17 percent and a design lift coefficient of 0.4. Using the same camber line, NASA has extended this airfoil into a family of low-speed airfoils of different thicknesses, for example, the NASA LS(1)-0409 and the LS(1)-0413. (See Reference 17 for more details.) In comparison with the standard NACA airfoils having the same thicknesses, these new LS(1)-04xx airfoils all have:

1. Approximately 30 percent higher $c_{l,\max}$.
2. Approximately a 50 percent increase in the ratio of lift to drag (L/D) at a lift coefficient of 1.0. This value of $c_l = 1.0$ is typical of the climb lift coefficient for general aviation aircraft, and a high value of L/D greatly improves the climb performance. (See Reference 2 for a general

introduction to airplane performance and the importance of a high L/D ratio to airplane efficiency.)

It is interesting to note that the shape of the airfoil in Figure 4.36 is very similar to the supercritical airfoils to be discussed in Chapter 11. The development of the supercritical airfoil by NASA aerodynamicist Richard Whitcomb in 1965 resulted in a major improvement in airfoil drag behavior at high subsonic speeds, near Mach 1. The supercritical airfoil was a major breakthrough in high-speed aerodynamics. The LS(1)-0417 low-speed airfoil shown in Figure 4.36, first introduced as the GA(W)-1 airfoil, was a later spin-off from supercritical airfoil research. It is also interesting to note that the first production aircraft to use the NASA LS(1)-0417 airfoil was the Piper PA-38 Tomahawk, introduced in the late 1970s.

In summary, new airfoil development is alive and well in the aeronautics of the late twentieth century. Moreover, in contrast to the purely experimental development of the earlier airfoils, we now enjoy the benefit of powerful computer programs using panel methods and advanced viscous flow solutions for the design of new airfoils. Indeed, in the 1980s NASA established an official Airfoil Design Center at The Ohio State University, which services the entire general aviation industry with over 30 different computer programs for airfoil design and analysis. For additional information on such new low-speed airfoil development, you are urged to read Reference 16, which is the classic first publication dealing with these airfoils, as well as the concise review given in Reference 17.

DESIGN BOX

This chapter deals with incompressible flow over airfoils. Moreover, the analytical thin airfoil theory and the numerical panel methods discussed here are techniques for calculating the aerodynamic characteristics for a *given airfoil of specified shape*. Such an approach is frequently called the *direct problem,* wherein the shape of the body is given, and the surface pressure distribution (for example) is calculated. For design purposes, it is desirable to turn this process inside-out; it is desirable to specify the surface pressure distribution—a pressure distribution that will achieve enhanced airfoil performance—and calculate the shape of the airfoil that will produce the specified pressure distribution. This approach is called the *inverse problem*. Before the advent of the high-speed digital computer, and the concurrent rise of the discipline of computational fluid dynamics in the 1970s (see Section 2.17.2), the analytical solution of the inverse problem was difficult, and was not used by the practical airplane designer. Instead, for most of the airplanes designed before and during the twentieth century, the choice of an airfoil shape was based on reasonable experimental data (at best), and guesswork (at worst). This story is told in some detail in Reference 62. The design problem was made more comfortable with the introduction of the various families of NACA airfoils, beginning in the early 1930s. A logical method was used for the geometrical design of these airfoils, and definitive experimental data on the NACA airfoils were made available (such as shown in Figures 4.10, 4.11, and 4.28). For this reason, many airplanes designed during the middle of the twentieth century used standard NACA airfoil sections. Even today, the NACA airfoils are sometimes the most expeditious choice of the airplane designer, as indicated by the tabulation (by no means complete) in Section 4.2 of airplanes using such airfoils.

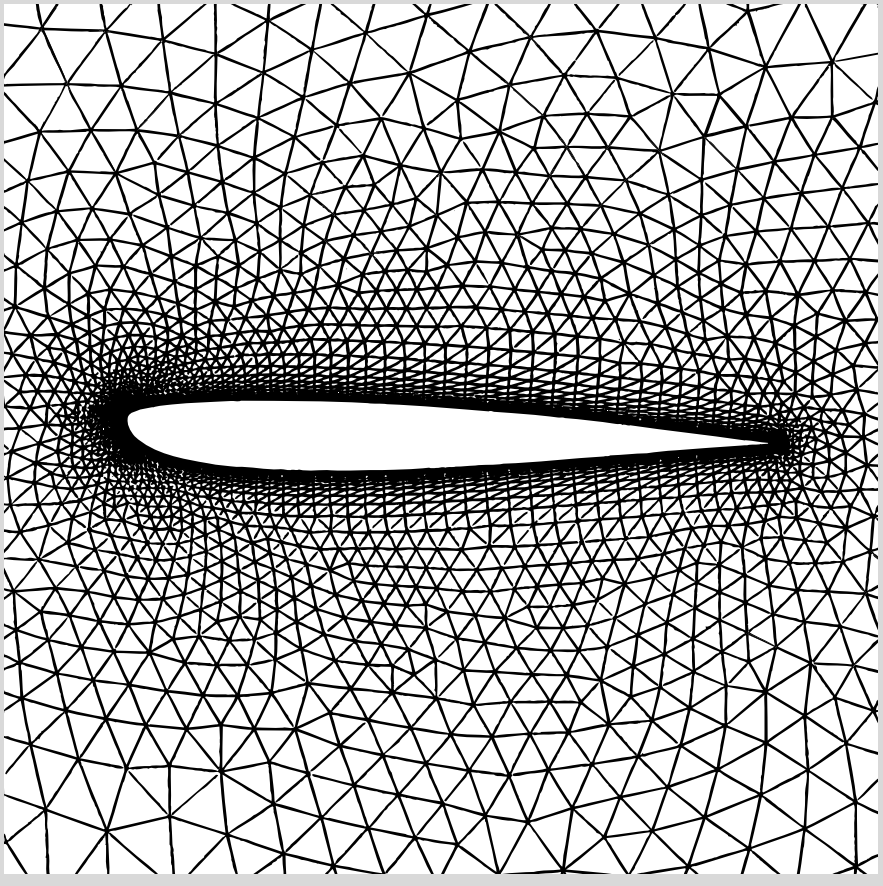
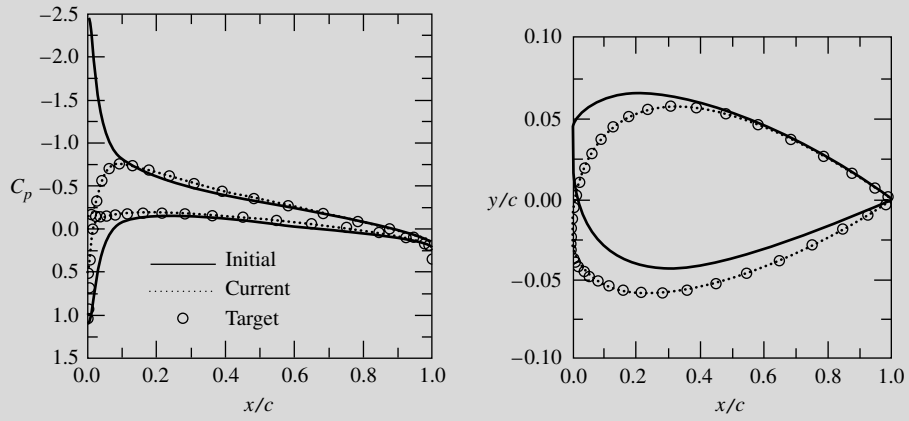


Figure 4.38 Unstructured mesh for the numerical calculation of the flow over an airfoil. *(Data Source: Anderson and Bonhaus, Reference 68.)*

However, today the power of computational fluid dynamics (CFD) is revolutionizing airfoil design and analysis. The inverse problem, and indeed the next step—the overall automated procedure that results in a completely optimized airfoil shape for a given design point—are being made tractable by CFD. An example of such work is illustrated in Figures 4.38 and 4.39, taken from the recent work of Kyle Anderson and Daryl Bonhaus (Reference 68). Here, CFD solutions of the continuity, momentum, and energy equations for a compressible, viscous flow (the Navier-Stokes equations, as denoted in Section 2.17.2) are carried out for the purpose of airfoil design. Using a finite volume CFD technique, and the grid shown in Figure 4.38, the inverse problem is solved. The *specified* pressure distribution over the top and bottom surfaces of the airfoil is given by the circles in Figure 4.39*a*. The optimization technique is iterative and requires starting with a pressure distribution that is not the desired, specified one; the initial distribution is given by the solid curves in Figure 4.39*a*, and the airfoil shape corresponding to this initial pressure distribution is shown by the solid curve in Figure 4.39*b*. (In Figure 4.39*b*, the airfoil shape appears distorted because an expanded scale is used for the ordinate.) After 10 design cycles, the optimized airfoil shape that supports



(*a*) **Pressure coefficient distributions** (*b*) **Airfoil shapes**

Figure 4.39 An example of airfoil optimized design using computational fluid dynamics. *(Data Source: Anderson and Bonhaus, Reference 68.)*

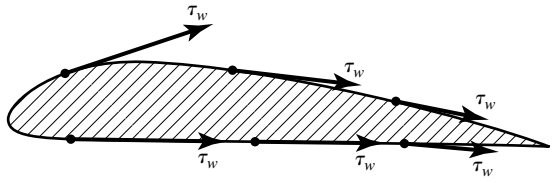
the specified pressure distribution is obtained, as given by the circles in Figure 4.39*b*. The initial airfoil shape is also shown in constant scale in Figure 4.38.

The results given in Figures 4.38 and 4.39 are shown here simply to provide the flavor of modern airfoil design and analysis. This is reflective of the wave of future airfoil design procedures, and you are encouraged to read the contemporary literature in order to keep up with this rapidly evolving field. However, keep in mind that the simpler analytical approach of thin airfoil theory discussed in the present chapter, and especially the simple practical results of this theory, will continue to be part of the whole "toolbox" of procedures to be used by the designer in the future. The fundamentals embodied in thin airfoil theory will continue to be part of the fundamentals of aerodynamics and will always be there as a partner with the modern CFD techniques.

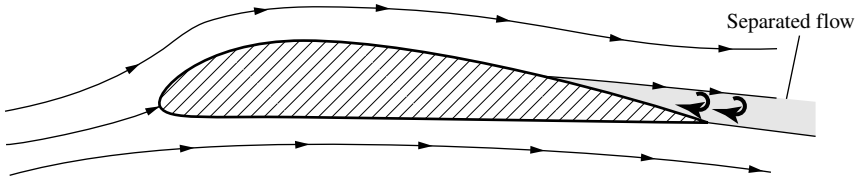
4.12 VISCOUS FLOW: AIRFOIL DRAG

This is another "stand-alone" viscous flow section in the same spirit as Section 1.11. It does not break the continuity of our discussions on inviscid flow; rather, it is designed to complement them. Before reading further, you are encouraged to review the introduction to boundary layers given in Section 1.11.

The lift on an airfoil is primarily due to the pressure distribution exerted on its surface; the shear stress distribution acting on the airfoil, when integrated in the lift direction, is usually negligible. The lift, therefore, can be accurately calculated assuming inviscid flow in conjunction with the Kutta condition at the trailing edge. When used to predict drag, however, this same approach yields zero drag, a result that goes against common sense, and is called *d'Alembert's paradox* after Jean le Rond d'Alembert, the eighteenth-century French mathematician and scientist who first made such drag calculations for inviscid flows over two-dimensional bodies (see Sections 3.13 and 3.20).



(*a*)



(*b*)

Figure 4.40 Subsonic airfoil drag is due to two components: (*a*) shear stress acting on the surface, and (*b*) pressure drag due to flow separation.

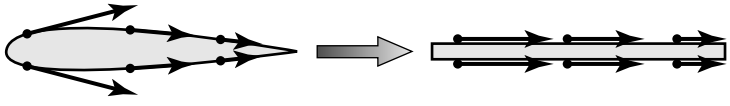


Figure 4.41 Estimation of skin-friction drag on an airfoil from that on a flat plate.

The paradox is immediately removed when viscosity (friction) is included in the flow. Indeed, viscosity in the flow is totally responsible for the aerodynamic drag on an airfoil. It acts through two mechanisms:

1. *Skin-friction drag,* due to the shear stress acting on the surface (Figure 4.40*a*), and
2. *Pressure drag due to flow separation,* sometimes called *form drag* (Figure 4.40*b*).

That shear stress creates drag is self-evident from Figure 4.40*a*. The pressure drag created by flow separation (Figure 4.40*b*) is a more subtle phenomenon and will be discussed towards the end of this section.

4.12.1 Estimating Skin-Friction Drag: Laminar Flow

As a first approximation, we assume that skin-friction drag on an airfoil is essentially the same as the skin-friction drag on a flat plate at zero angle of attack, as illustrated in Figure 4.41. Obviously, this approximation becomes more accurate the thinner the airfoil and the smaller the angle of attack. Consistent with the rest of this chapter, we will continue to deal with low-speed incompressible flow.

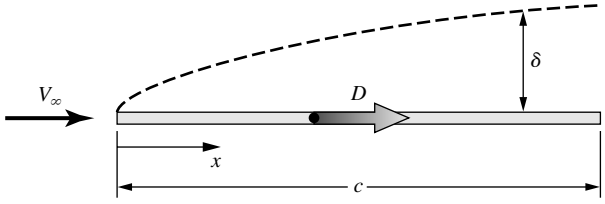


Figure 4.42 Total friction drag on a flat plate.

We first deal with the case of completely laminar flow over the airfoil (and hence the flat plate) in Figure 4.41. There is an exact analytical solution for the laminar boundary-layer flow over a flat plate. The details of this solution are given in Section 18.2, where we present boundary-layer theory in some detail. For the present section, we will use just the results of Section 18.2.

The boundary-layer thickness for incompressible laminar flow over a flat plate at zero angle of attack is given by Equation (18.23), repeated and renumbered below:

$$\delta = \frac{5.0x}{\sqrt{\text{Re}_x}} \tag{4.84}$$

where Re_x is the Reynolds number based on distance x measured from the leading edge (Figure 4.42),

$$\text{Re}_x = \frac{\rho_e V_\infty x}{\mu_\infty}$$

Note from Equation (4.84) that $\delta \propto \sqrt{x}$, that is, the boundary-layer thickness grows parabolically with distance from the leading edge.

The local shear stress, integrated over both the top and bottom surfaces of the flat plate shown in Figure 4.41, yields the net friction drag, D_f, on the plate, illustrated in Figure 4.42. To begin with, however, let us consider just *one* surface of the plate, either the top surface or the bottom surface. The shear stress distribution over the top surface is the same as that over the bottom surface. Let us choose the top surface. The integral of the shear stress over the top surface gives the net friction drag on that surface, $D_{f,\text{top}}$. Clearly, the net friction drag due to the shear stress integrated over the bottom surface, $D_{f,\text{bottom}}$, is the same value, $D_{f,\text{bottom}} = D_{f,\text{top}}$. Hence, the total skin friction drag, D_f, is

$$D_f = 2D_{f,\text{top}} = 2D_{f,\text{bottom}}$$

Define the skin-friction drag coefficient for the flow over *one* surface as

$$C_f \equiv \frac{D_{f,\text{top}}}{q_\infty S} = \frac{D_{f,\text{bottom}}}{q_\infty S} \tag{4.85}$$

The skin-friction drag coefficient is a function of the Reynolds number, and is given by Equation (18.22), repeated and renumbered below,

$$C_f = \frac{1.328}{\sqrt{\text{Re}_c}} \tag{4.86}$$

where Re_c is the Reynolds number based on the chord length c shown in Figure 4.42

$$\text{Re}_c = \frac{\rho_\infty V_\infty c}{\mu_\infty}$$

EXAMPLE 4.8

Consider the NACA 2412 airfoil, data for which is given in Figures 4.10 and 4.11. The data are given for two values of the Reynolds number based on chord length. For the case where $\text{Re}_c = 3.1 \times 10^6$, estimate: (*a*) the laminar boundary layer thickness at the trailing edge for a chord length of 1.5 m and (*b*) the net laminar skin-friction drag coefficient for the airfoil.

■ Solution

(*a*) From Equation (4.84) applied at the trailing edge, where $x = c$, we have

$$\delta = \frac{5.0c}{\sqrt{\text{Re}_c}} = \frac{(5.0)(1.5)}{\sqrt{3.1 \times 10^6}} = \boxed{0.00426 \text{ m}}$$

Notice how thin the boundary layer is; at the trailing edge, where its thickness is the largest, the boundary layer is only 0.426 cm thick.

(*b*) From Equation (4.86),

$$C_f = \frac{1.328}{\sqrt{\text{Re}_c}} = \frac{1.328}{\sqrt{3.1 \times 10^6}} = 7.54 \times 10^{-4}$$

Recall that the above result is for a single surface, either the top or bottom of the plate. Taking both surfaces into account:

$$\text{Net } C_f = 2(7.54 \times 10^{-4}) = \boxed{0.0015}$$

From the data in Figure 4.11, we see that at zero angle of attack for $\text{Re} = 3.1 \times 10^6$, the airfoil drag coefficient is 0.0068. This measured value is about 4.5 times higher than the value of 0.0015 we just calculated. But wait a moment! For the relatively high Reynolds number of 3.1×10^6, the boundary layer over the airfoil will be *turbulent,* not laminar. So our laminar flow calculation is not an appropriate estimate for the boundary layer thickness and the airfoil drag coefficient. Let us take the next step.

4.12.2 Estimating Skin-Friction Drag: Turbulent Flow

In contrast to the situation for laminar flow, there are no exact analytical solutions for turbulent flow. This sad state of affairs is discussed in Chapter 19. The analysis of any turbulent flow requires some amount of empirical data. All analyses of turbulent flow are approximate.

The analysis of the turbulent boundary layer over a flat plate is no exception. From Chapter 19 we lift the following approximate results for the incompressible turbulent flow over a flat plate. From Equation (19.1) repeated and

renumbered below:

$$\delta = \frac{0.37x}{\text{Re}_x^{1/5}} \tag{4.87}$$

and from Equation (19.2) repeated and renumbered below:

$$C_f = \frac{0.074}{\text{Re}_c^{1/5}} \tag{4.88}$$

We emphasize again that Equations (4.87) and (4.88) are only approximate results, and they represent only one set of results among a myriad of different turbulent flow analyses for the flat plate boundary layer. Nevertheless, Equations (4.87) and (4.88) give us some reasonable means to estimate the boundary-layer thickness and skin-friction drag coefficient for turbulent flow. Note that, in contrast to the inverse square root variation with Reynolds number for laminar flow, the turbulent flow results show an inverse fifth root variation with Reynolds number.

EXAMPLE 4.9

Repeat Example 4.8 assuming a turbulent boundary layer over the airfoil.

■ Solution

Once again we replace the airfoil with a flat plate at zero angle of attack.

(*a*) The boundary-layer thickness at the trailing edge, where $x = c$ and $\text{Re}_x = \text{Re}_c = 3.1 \times 10^6$, is given by Equation (4.87):

$$\delta = \frac{0.37x}{\text{Re}_x^{1/5}} = \frac{0.37(1.5)}{(3.1 \times 10^6)^{1/5}} = \boxed{0.0279 \text{ m}}$$

The turbulent boundary layer is still thin, 2.79 cm at the trailing edge, but by comparison is much thicker than the laminar boundary layer thickness of 0.426 cm from Example 4.8.

(*b*) The skin-friction drag coefficient (based on one side of the flat plate) is given by Equation (4.88):

$$C_f = \frac{0.074}{\text{Re}_c^{1/5}} = \frac{0.074}{(3.1 \times 10^6)^{1/5}} = 0.00372$$

The net skin-friction drag coefficient, taking into account both the top and bottom surfaces of the flat plate, is

$$\text{Net } C_f = 2(0.00372) = \boxed{0.00744}$$

This result is a factor of five larger than for the laminar boundary layer, and serves as an illustration of the considerable increase in skin friction caused by a turbulent boundary layer in comparison to that caused by a laminar boundary layer.

The result for the skin friction drag coefficient in Example 4.9 is larger than the measured drag coefficient of the airfoil of 0.0068, which is the *sum* of both